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A Bayesian Hierarchical Model of Individual Player Performance in the 2022 FIFA World Cup

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ABSTRACT

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This dissertation examines a Bayesian hierarchical model to assess individual player ability in the 2022 FIFA World Cup. This is a challenging task since player performance is affected not only by individual skill, but also by external factors such as team quality, opponent strength, weather conditions, and match context. Moreover, the tournament consists of a small number of matches, especially for players whose teams are eliminated at earlier stages. Using player data from the entire tournament, the analysis focuses on three main performance indicators: interceptions, shots and passes, which are modelled using either a Poisson or a negative binomial distribution. A negative binomial is more adequate when overdispersion is present. The intensity of an event includes player ability, team effects, and opponent strength in order to evaluate the contribution made by each player in a given match. The two models obtained for each indicator were compared, namely using Bayesian RMSE and MAE. When fitting the interceptions and shots models, convergence was achieved. Since fewer players were included for each of these models, and since interception and shot counts are relatively low, the models perform as expected and produce acceptable values of the Bayesian RMSE and MAE. This suggests that the modelling framework utilised in this dissertation is well suited to these two performance metrics and it provides reliable and stable estimates. However, fitting the passing model proved more complicated due to convergence issues encountered and higher than desirable Bayesian RMSE and MAE. This indicates that the proposed model is more suited for events where individual player ability is more easy to separate from the corresponding team ability (like interceptions and shots) and less suited for events (like passing) where individual player ability and corresponding team ability are more easily conflated.

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I used generative AI tools solely to assist with improving the clarity and presentation of the writing, while all work, analysis, and conclusions remain my own original work.

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List of Notation

Notation	***Description***
Y	A random variable
y	One observed value of Y
$\mathbf{y}$	All observed values of Y
$\mathbf{\Theta}$	Random vector of parameters
$\boldsymbol{\theta}$	Realisation of $\mathbf{\Theta}$
$\mathbf{\Phi}$	Random vector of hyperparameters
$\boldsymbol{\phi}$	Realisation of $\mathbf{\Phi}$
H	Latent scoring/tuning parameter
η	Realisation of H
φ	Dispersion parameter
n	Sample size
N	Total number of MCMC samples
B	Burn-in period
L	The post burn-in sample size of the chain

Chapter 1

Introduction

In the competitive world of football, staying ahead of competitors requires more than just talent and hard work. It requires a lot of planning and detailed analysis of the opponent, as well as the ability to change the game plan if the match is not going as planned. This section discusses the importance of sports analytics in football and how modelling can be applied in this context. Moreover, the history of the sport and the World Cup will be explored, the aim of the dissertation will be stated, and its structure will be presented.

1.1 History of the World Cup

International football began in 1872 with the first official match being between Scotland and England in that year. The FIFA World Cup is a football competition among the senior men's national teams from all across the world. The first World Cup took place in 1930 in Uruguay because the country was one of the strongest football nations at the time and was willing to cover the travel and accommodation costs of all participating teams, an offer no other nation could afford to make in those days. The tournament has since taken place every 4 years with the exception of 1942 and 1946 due to the Second World War. The qualification phase starts 3 years prior, to determine which teams qualify for the tournament phase. In the tournament phase, 32 teams compete for the title at different grounds in the host nation, and the tournament takes over a month. From 2026 onwards, the competition is scheduled to expand to 48 teams.

The competition has been won by eight national teams. Brazil have won the tournament five times and are the only nation to have participated in every edition of the World Cup. The other World Cup winners are Germany and Italy with four titles each, Argentina with 3 titles, France and Uruguay with two titles each and England and Spain with one title each.

1.2 Importance of Sports Analytics in Football

Sports analytics is all about using data and insights to improve athlete and team performance. The earliest use of sports analytics dates back to the 1970s when Bill James started using data to challenge scouting opinions in baseball. Coaches nowadays no longer use intuition or new methods to approach the opponent, but they use key data-driven insights to enhance training sessions, refine tactics and make more informed decisions. It has clearly become a crucial part of modern football, and teams that do not make use of sports analytics risk falling behind, as nearly all clubs rely on it to gain a competitive edge.

Sports analytics help coaches and staff within a football club analyse player performance using statistics such as passes completed, interceptions, expected goals (xG) and distance covered. These metrics help coaches identify strengths and areas needing improvement, allowing them to tailor training sessions to improve overall performance. Teams use data to study opponent's patterns; such as their formation, rotations and pressing styles. Coaches make use of this information to adjust their own game plan so the team is well prepared to face any opponent. They may change formations to exploit an opponent's weak side or to better protect areas where the opponent is particularly strong.

1.3 Importance of Player Performance Evaluation

Football teams feel the need to evaluate individual performances since player transfers involve major financial investments. Teams may purchase players and later sell them for double the original fee. However, if they are not careful in assessing which players truly suit their needs, they can end up losing significant amounts of money if the players bought do not perform. In the context of the FIFA World Cup, national federations evaluate all players of their own nationality carefully, regardless of their club's performance or strength. They then compare these players with other players of the same nationality playing in different leagues.

National federations must account for the differences between players who compete in highly demanding leagues and those who play in less intense leagues. Since tactics, styles of play, and levels of competition vary widely across leagues, it is naturally harder for players in tougher leagues to stand out statistically. To deal with this, federations attempt to use more sophisticated methods, such as video analysis, to assess performance, which helps analysts separate a player's individual skill from the influence of their club environment.

Since national team camps typically occur only four times a year, coaches have limited opportunities to observe players directly. Consequently, club-level performance becomes the main indicator of a player's quality, fitness, and tactical intelligence. National teams monitor a wide range of metrics, including passing accuracy, defensive actions, shots attempted, and distance covered, to assess both technique and physicality.

This data allows coaches to identify the best possible players available for selection. In addition, they assess each player's primary position and tactical role to determine whether they fit the team's preferred formation and whether they have the versatility to adapt to different positions when required.

After thoroughly evaluating those players who play at the top level and are of the adequate nationality, national federations choose the best 26 players who travel to the World Cup to represent their country. This process ensures that only the players deemed most capable and best suited to the team's needs are included in the final squad. This gives them the best possible chance of winning the entire tournament.

1.4 Aim of the Dissertation

The main aim of this dissertation is to develop and apply a Bayesian hierarchical model to evaluate individual player performance in the 2022 FIFA World Cup, focusing specifically on certain football performance metrics, which are quantitative measures used to assess different aspects of a player's contribution during a match, namely interceptions made, shots on goal and passes completed. The dataset used in this study was obtained from Kaggle, an online data science platform, and contains a lot of performance metrics for each player who took part in the World Cup [1].

The analysis is focused on players who participated in at least 50 percent of their team's total minutes, as including those with limited playing time may introduce noise and reduce the reliability of the model's results and estimates. A combination of individual and team characteristics will be considered to produce an overall prediction for each player in the World Cup. These predictions are derived from football performance metrics available in the dataset, including interceptions, shots and passes completed, which serve as relevant indicators of underlying player skill.

1.5 Structure of the Dissertation

The dissertation proceeds with a literature review, which examines papers that are somewhat similar to this analysis. Three types of papers are studied. First, Bayesian papers that analyse different sports are reviewed. Second, Bayesian papers that focus specifically on football are discussed and are given more importance, since they are closely related to this analysis. Finally, papers that analyse sports datasets using a frequentist approach, which is the opposite of the Bayesian approach, are also considered. This helps to highlight some improvements of the Bayesian approach

[1] https://www.kaggle.com/datasets/swaptr/fifa-world-cup-2022-player-data

over the frequentist approach, as well as some limitations.

Chapter 3 discusses the theoretical aspects of Bayesian hierarchical models and introduces several models that are crucial to this study. The chapter begins by presenting the statistical formulation of hierarchical models and discusses their theoretical foundations, including exchangeability and de Finetti's general representation theorem. It proceeds with a description of Markov Chain Monte Carlo (MCMC) algorithms, focusing mainly on Gibbs sampling, which is then applied through the software JAGS. A more detailed discussion of Gibbs sampling within the context of Bayesian hierarchical modelling is also provided. Convergence diagnostics are then introduced, with particular attention given to the most important diagnostics that will later be used in the application to assess whether the Markov chains have converged and to ensure reliable posterior estimates. It then proceeds to the subject at hand in this dissertation, namely sports analytics, and reviews important papers that use similar modelling approaches. In particular, Whitaker's (2020) paper is discussed, as its linear predictor is of inspiration for the model used in this study. The model used in this dissertation is then presented, including both the Poisson and negative binomial specifications. Finally, predictive error measures are described, which will be used to compare the performance of the Poisson and negative binomial models.

Chapter 4 is initiated with an overview of the dataset, describing the important metrics that are used in this study. The chapter then proceeds with an exploratory data analysis section, which examines the data to determine whether it is suitable for fitting the proposed models. Consequently, the results of the model are outputted. Both the Poisson and negative binomial specifications are discussed for each football performance metric considered, focusing on convergence and the reliability of the results. Each pair of models is then compared using the predictive error measures that were mentioned and described in Chapter 3.

Chapter 5 presents the main conclusions drawn from the model and discusses both aspects that worked well and the difficulties encountered. It also provides some suggestions for future studies and improvements that could be considered.

Finally, the appendix contains supplementary tables, including convergence diagnostics for each effect considered in the model. It also contains tables reporting team effects for the football performance metrics considered, together with the corresponding posterior means, posterior standard deviations and lower and upper HPD bounds. It also contains tables showing the top and bottom 10 results based on the model estimates.

Chapter 2

Literature Review

This chapter reviews available studies that apply Bayesian hierarchical models to sports data. The aim of this review is to highlight key developments and identify gaps that motivate this dissertation. The Bayesian framework is particularly suitable for such analyses, as sports data usually contain multi-level structures, such as players nested within teams or seasons, which can be naturally modelled within a hierarchical framework.

Several studies have applied this approach to football data. Premier League data, for instance, have been used by Whitaker et al. (2021) to estimate a player's ability in individual matches. Similarly, Egidi and Gabry (2018) estimated football players' abilities using data from the Italian fantasy game Fantacalcio.

One of the earliest and most influential contributions in the application of Bayesian hierarchical modelling to football data was introduced by Baio and Blangiardo (2010), who successfully applied a Bayesian hierarchical model to football. Their study showcased the potential of hierarchical modelling for sports applications, showing how outcomes can be affected by factors operating at different levels. Since the publication of this paper, a growing number of papers have built upon this approach.

Beyond football, Attard et al. (2023) also applied the Bayesian hierarchical framework to basketball. For example, their study analysed NBA data from the 2008–2009 season and fitted both Poisson and negative binomial models to account for the higher scoring frequency relative to football.

2.1 Frequentist Approaches in Football

In frequentist statistics, probability has a different interpretation and meaning compared to the Bayesian approach. Under the frequentist framework, probabilities are not assigned to parameters, but rather, parameters are fixed but unknown, while the data are random variables before observation. In contrast, Bayesian statistics assign probability distributions to parameters, treating them as random variables, and update prior beliefs using the observed data to obtain posterior distributions.

In this section, several studies that adopt a frequentist approach to modelling football are reviewed in order to emphasize the differences between frequentist and Bayesian methods in the context of sports analytics, with a particular focus on football.

Maher (1982) proposed one of the earliest statistical models for football scores with the aim of predicting match outcomes and quantifying team strength. The author assumed that the number of goals scored by the home and away team both follow a Poisson distribution and that these were independent from each other. The parameters of the model were estimated using maximum likelihood estimation and no priors were introduced, making the approach purely frequentist. Once the model was fitted, it was used to predict win, draw and loss probabilities, as well as rank teams based on attacking and defensive strength.

Dixon and Coles (1997) extended Maher's (1982) model by accounting for low-scoring football matches, which were not handled well by the original model. They retained the same modelling framework but introduced a correction factor to the joint score distribution for small scorelines. In particular, they modified the probabilities of matches ending 0-1, 0-0, 1-0, and 1-1 to better reflect tactical aspects. Moreover, the authors gave more importance to recent matches, allowing team strengths to change over time to reflect changes in form. After fitting the model, they computed win, draw, and loss probabilities and compared them with bookmakers' odds, arguing that the betting market was not hugely efficient.

Another frequentist paper in the literature is Lee, A.J. (1997), which addressed the question of whether Manchester United deserved to win the 1995-1996 Premier League. The statistical model used in the study is a frequentist Poisson regression model and the results show that Manchester United had the strongest attack and Arsenal exhibited the strongest defensive strength. The paper used Monte Carlo to simulate the 380-match season 1000 times, and results showed that Manchester United had the highest probability of finishing top of the table. However, the analysis suggested that Liverpool was estimated to be slightly stronger than Newcastle, despite Newcastle finishing second in the actual season. Therefore, Manchester United were somewhat lucky because the differences between teams were extremely small, implying that chance still played a role. Nevertheless, United were still judged to have been the best team overall.

Karlis and Ntzoufras (2003) note that standard football score models assume independence between goals scored by each team, but in reality these scores tend to be somewhat correlated, and ignoring this leads to poor model fit, particularly for draws. To address this issue, the authors propose a bivariate Poisson distribution to account for score dependence, thereby improving model fit and enhancing the prediction of match draws, which tend to be underestimated by independent models. The goals scored by teams X and Y are modelled jointly: $(X, Y) \sim BP(\lambda_1, \lambda_2, \lambda_3)$. In this model, $Cov(X, Y) = \lambda_3$ refers to the covariance between scores and when $\lambda_3 = 0$, the model reduces to independent Poisson. Accounting for covariance is important because even small positive covariances can significantly increase the predicted number of draws and affect the estimation of team strengths.

Dutta et al. (2024) is a paper that investigates the timing of the first goal in the second half of a football match from both a Bayesian and a frequentist perspective. The authors use data from the Indian Super League (ISL) from the 2022-2023 season. According to the frequentist approach, the number of completed passes in the first half is identified as the significant predictor of scoring early in the second half. Under the Bayesian Cox modelling approach, corner kicks and goalkeeper saves emerge as the most important indicators based on posterior summaries. Dutta et al. (2024) use concordance (C-index) and the Brier score to compare predictive performance of the two approaches. They determine that the Bayesian model performs better based on the two metrics. A crucial outcome from this comparison is that the two approaches led to different conclusions. The frequentist approach highlights completed passes as the main predictor, whereas the Bayesian approach identifies corners and goalkeeper saves as crucial indicators.

2.2 Applications of Bayesian Hierarchical Models in Football

The paper by Baio and Blangiardo (2010) is considered as a seminal paper in the context of Bayesian hierarchical modelling applied to football, since it was one of the first to successfully use such a model for the analysis of football match outcomes. Match scores are modelled using Poisson goal models for home and away teams where scoring intensity is affected by the home advantage and on team-specific attacking and defensive abilities. Team effects are treated hierarchically as they are assumed to come from the same prior distributions, allowing information to be shared across teams, and producing more stable estimates of performance. The study also extends the model using a mixture structure because of the presence of overshrinkage, improving fit and predictive accuracy.

Egidi and Gabry (2018) propose a method for predicting individual football performance based on data from the Italian fantasy game Fantacalcio. The main goal of this paper is to compare some small Bayesian hierarchical models for predicting player ratings, even if players miss a lot of matches due to injuries or suspensions. Player data is made up of two components: a raw score and a fantasy point score, where raw score is a general rating from one to ten reflecting individual performance, and fantasy points assign fixed values to crucial match events like goals, with additional deductions for negative actions. The authors estimate three hierarchical models to predict player performance. First, they fit a hierarchical autoregressive model in which missing ratings are treated as zero, therefore assuming that a player who does not play contributes no performance rating. Second, they propose a mixture model that explicitly accounts for the probability of a player not appearing in a match, jointly modelling both participation and performance. Finally, they re-estimate the hierarchical autoregressive model but treat missing observations as unobserved rather than zero, allowing the model to predict the rating a player would have received if they had played.

An important contribution to this dissertation is the paper by Whitaker et al. (2021) who use data from the 2013/2014 and 2014/2015 English Premier League seasons to apply an interpretable Bayesian model that estimates an individual footballer's ability for any given event in a match. Whitaker et al. (2021) is more important to this dissertation than other papers because it builds directly on the same modelling approach and underlying principles, while also extending them. A crucial difference between their model and the model presented in this dissertation is the type of data used. They use match-by-match data, whereas this dissertation uses aggregated data for the whole tournament. While their model uses a Poisson-based likelihood for event counts, this dissertation extends the approach by also considering a negative binomial model to account for potential overdispersion. Despite the differences in data structure, the modelling framework adopted in this dissertation is based on the work of Whitaker et al. (2021). For this reason, their paper is discussed in more detail than the other studies reviewed in this chapter.

2.3 Applications of Bayesian Hierarchical Models in Other Sports

Ingram (2019) proposes a Bayesian hierarchical model to predict the outcome of tennis matches. The model estimates the probability that a player wins the point on serve and each match is divided into two parts: Player A serving against player B and player B serving against player A. The model is fitted using Hamiltonian Monte Carlo based on ATP 2014 data. A Bayesian hierarchical structure is utilised, with defined prior distributions and automatic partial pooling. The proposed model outperforms other point-based models with a 68.8% accuracy. This paper is closely related to the present work and motivates the use of a Bayesian hierarchical framework in this dissertation.

Another interesting paper in this context is Gabrio (2020) volleyball paper, whose main goal was to predict both match results and league rankings using data from the Italian women's Serie A1 2017-2018 season. The author proposes a three-module framework for important sport-specific reasons. In volleyball, total points scored in a match does not always identify the winner, since a volleyball team can score fewer points but win more sets and therefore wins the match. As a result, modeling only total points can lead to incorrect predictions of match outcomes.

The modeling framework in Gabrio (2020) is structured into three modules. The first module models the number of points scored by home and away teams using a Poisson distribution which accounts for team attack and defence strengths, home advantage and uses weakly informative priors. The second module models the probability that the sum of sets won by the home and away team is equal to 5. This module is important because league points in volleyball depend on both the match outcome and the number of sets played. Therefore, predicting the winner alone is not enough, as it is also necessary to determine whether the match went to five sets. The third module models the probability that the home team wins the match, depending on whether the match went to five sets and on the number of points scored. Together, these modules jointly predict match outcomes and league rankings while also respecting the sport-specific rules of volleyball.

Attard et al., (2023) propose and compare two Bayesian hierarchical modelling approaches for basketball using individual match data from the 2008/2009 NBA regular season. The first approach models scoring by scoring type: 1-pointers, 2-pointers, and 3-pointers. They fit Poisson and negative binomial models for each type of goal and combine the three models to obtain predicted total points in a single match, incorporating team-specific attack and defence effects, a home advantage term, and an intercept. After analyzing RMSE, they found that the negative binomial models perform better than the Poisson in predictive accuracy, implying the presence of overdispersion. The second approach assesses whether the home team wins using a Bernoulli hierarchical model. However, they had some MCMC convergence issues, so they simplified team effects to a single strength parameter per team instead of having separate parameters for attack and defence. The second model's predictions for match outcomes were worse than the first model's but performed well for predicting final standings. Both models have success probabilities modelled using a log-linear hierarchical structure, and parameter estimation for both scoring and winning probability models was performed using Gibbs sampling.

Van Kesteren and Bergkamp (2023) proposed a Bayesian multilevel model to address one of the most persistent questions in Formula One: how much success is due to the driver's ability and how much is due to the car, mechanics, and engineers. They modelled the full finishing order of each race using a rank-ordered logit likelihood, allowing every finishing position to contribute information. Latent ability was decomposed into four additive components, enabling partial pooling across drivers, teams, and seasons and allowing a direct comparison of driver and constructor effects. They also considered whether driver skill varies across circuit types and wet races, but ultimately chose the simpler model for greater clarity, following the principle that complexity should only be added when it improves insight or results. Their variance-component analysis found that 88% of the variance was attributed to constructor-related effects and the rest to drivers, suggesting that the car is much more important than the driver. owever, this does not imply that driver skill is unimportant; rather, car differences are often substantial, while driver skill remains crucial when cars are closely matched.

The review of Bayesian hierarchical modeling in various sports and contexts highlights the flexibility of this modelling framework across different settings. Despite data structures being different across papers, these studies share common theoretical foundations to the approach developed in this dissertation. The following section moves from extensive literature to the underlying theory used in these papers, with particular focus on the works of Baio and Blangiardo (2010), Whitaker et al. (2021) and Attard et al. (2023), whose work is extremely important to this dissertation.

Chapter 3

Bayesian Hierarchical Models in Sports

The main aim of this dissertation is to develop a Bayesian hierarchical model for individual football players. It is thus crucial to introduce the general theoretical framework of Bayesian hierarchical models. This will be done to establish the foundational concepts, explain their hierarchical structure, and highlight why they are particularly well-suited for modeling grouped or nested data, for instance, players within a football team. This chapter begins with an overview of Bayesian hierarchical modelling and its motivation, followed by a review of the fundamental principles of Bayesian inference, including exchangeability and de Finetti's theorem. MCMC algorithms are then implemented, with particular emphasis given to the Gibbs Sampling algorithm, which serves as the main computational method for estimating the models developed in this dissertation. After this, the different MCMC diagnostics are introduced, since these are important for checking convergence and assessing whether the estimates are reliable. Some important papers that are particularly relevant to this dissertation are then discussed. Following this, the model developed for this study is presented, which is designed specifically for this type of data. Finally, the theory behind comparing the Poisson and negative binomial models using predictive error measures is presented.

3.1 Introduction

In many real-world applications, data exhibit a group structure in which individuals belong to units that influence their behaviour and performance. Players within teams in sports, students within schools, and patients within hospitals are common examples of such settings. In football, individuals belong to different teams of varying quality, and these teams operate under different conditions, such as tactical systems and match conditions. Individuals themselves differ in qualities such as experience, ability, and roles on the pitch, and their performance is also affected by broader external factors such as opponent strength. Any statistical model that aims to model individual performance should therefore account for this nested structure. A hierarchical framework allows these dependencies to be expressed through a joint probability model, capturing how parameters at one level influence information with those at another. Bayesian hierarchical models account for this structured way of linking parameters together, which helps explain their effectiveness in modeling complex and structured data.

3.2 Theoretical Framework for Bayesian Hierarchical Models

In Bayesian theory we assume the existence of a joint distribution with random variable Y and a realised vector of model parameters $\boldsymbol{\theta}$, denoted by $f(y, \boldsymbol{\theta})$, where y is the observed value of Y. This is never directly formulated but we use Bayes' theorem instead. Bayes' theorem was developed in the 18th century by Reverend Thomas Bayes. The theorem provides an equation for updating prior beliefs about unknown parameters based on some data. In particular, given a sample of size n, for $\boldsymbol{y} = (y_1, y_2, ..., y_n)'$, Bayes' equation is defined as:

$$f(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{f(\mathbf{y} \mid \boldsymbol{\theta})\, f(\boldsymbol{\theta})}{f(\mathbf{y})}.$$

Here:

- $f(\boldsymbol{\theta})$ is the prior distribution of $\boldsymbol{\Theta}$,
- $f(\mathbf{y} \mid \boldsymbol{\theta})$ is the likelihood,
- $f(\boldsymbol{\theta} \mid \mathbf{y})$ is the posterior distribution of $\boldsymbol{\Theta}$,
- $f(\mathbf{y})$ is the marginal distribution of the random variable Y.

This is the basic Bayes' equation, where the prior distribution $f(\boldsymbol{\theta})$ expresses prior beliefs about the parameters before observing the data, and the posterior distribution $f(\boldsymbol{\theta} \mid \mathbf{y})$ gives updated beliefs about the parameters after observing the data $\mathbf{y}$. Moreover, in the likelihood function $f(\mathbf{y} \mid \boldsymbol{\theta})$, the observed data $\mathbf{y}$ are assumed to depend on a single realisation of model parameters $\boldsymbol{\theta}$.

Since a Bayesian hierarchical model consists of a minimum of three levels, Bayes' theorem is extended to account for this structure. Let Y be a random variable with probability density function conditional on the random parameter vector $\boldsymbol{\Theta}$. The data level of the model is represented by the likelihood function $f(\mathbf{y} \mid \boldsymbol{\theta})$. Since $\boldsymbol{\Theta}$ is a random parameter vector, it is assigned a distribution that depends on a second random parameter vector, say $\boldsymbol{\Phi}$. This vector of parameters is known as the vector of hyperparameters. The elements in $\boldsymbol{\Phi}$ describe the population-level structure governing the parameters $\boldsymbol{\Theta}$ and determine the distribution from which the individual-level parameters arise. Therefore, in the Bayesian hierarchical framework, the observed data $\mathbf{y}$ depend on the parameters in $\boldsymbol{\Theta}$, while the parameters in $\boldsymbol{\Theta}$ themselves depend on the hyperparameters in $\boldsymbol{\Phi}$.

This gives rise to the second level of the model represented by the conditional prior distribution $f(\boldsymbol{\theta} \mid \boldsymbol{\phi})$. This formulation assumes that all observations arise from the same underlying process, which may be unrealistic, as observed data can exhibit variation that cannot be adequately captured by a single realisation of parameters of $\boldsymbol{\Theta}$. Observations may arise from different groups or units, each of which may have different characteristics.

Therefore, in a Bayesian hierarchical framework, three types of probability distributions are specified: the likelihood, the prior and the hyperprior. The likelihood describes the distribution of the observed data $\mathbf{y}$ given the parameters $\boldsymbol{\theta}$. The prior distribution specifies prior beliefs about the parameters $\boldsymbol{\Theta}$ before data is observed. The hyperprior is the distribution placed on the hyperparameters $\boldsymbol{\Phi}$. Combining the likelihood, prior, and hyperprior distributions, Bayes' equation can be extended to the hierarchical form by considering the joint posterior distribution of both the parameters and hyperparameters given the observed data:

$$f(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \mathbf{y}) = \frac{f(\boldsymbol{\theta}, \boldsymbol{\phi}, \mathbf{y})}{f(\mathbf{y})} = \frac{f(\mathbf{y} \mid \boldsymbol{\theta})\, f(\boldsymbol{\theta} \mid \boldsymbol{\phi})\, f(\boldsymbol{\phi})}{f(\mathbf{y})}. \tag{3.2.1}$$

where:

- $f(\mathbf{y} \mid \boldsymbol{\theta})$ is the likelihood, describing the distribution of the observed data given the parameters,
- $f(\boldsymbol{\theta} \mid \boldsymbol{\phi})$ is the prior distribution of the parameters conditional on the hyperparameters,
- $f(\boldsymbol{\phi})$ is the hyperprior distribution describing prior beliefs about the hyperparameters,
- $f(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \mathbf{y})$ is the joint posterior distribution of the parameters and hyperparameters given the observed data,
- $f(\mathbf{y})$ is the marginal distribution of the observed data.

Moreover,

$$f(\mathbf{y}) = \int\int f(y \mid \boldsymbol{\theta})\, f(\boldsymbol{\theta} \mid \boldsymbol{\phi})\, f(\boldsymbol{\phi})\, d\boldsymbol{\theta}\, d\boldsymbol{\phi} \quad \text{if both } \boldsymbol{\Theta} \text{ and } \boldsymbol{\Phi} \text{ are continuous,}$$

In this case, the integration is taken over all possible values of both $\boldsymbol{\Theta}$ and $\boldsymbol{\Phi}$ when evaluating the marginal likelihood. If $\boldsymbol{\Phi}$, $\boldsymbol{\Theta}$ or both are discrete, the corresponding integral is replaced by a summation. By inspecting Equation (3.2.1), it can be observed that the marginal likelihood $f(\mathbf{y})$ does not depend on any of the parameters and serves only as a normalizing constant. Its sole purpose is to ensure that the posterior distribution integrates to one, thereby it always satisfies the requirements of a probability density function. When this normalizing constant is removed, the following proportional relationship is obtained:

$$f(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \mathbf{y}) \propto f(\mathbf{y} \mid \boldsymbol{\theta})\, f(\boldsymbol{\theta} \mid \boldsymbol{\phi})\, f(\boldsymbol{\phi}). \tag{3.2.2}$$

The right side of this equation is therefore known as the unnormalized posterior density, since it excludes the normalizing constant $f(\mathbf{y})$.

Considered together, these equations illustrate how Bayesian inference works by combining the prior distribution with the likelihood to form the posterior distribution. A similar logic is followed when deriving the *prior predictive distribution*. Before the data $\mathbf{y}$ are observed, predictions about the unknown but observable $\mathbf{y}$ are made by integrating the lower-level $\boldsymbol{\theta}$ (lower-level) and hyperparameters $\boldsymbol{\phi}$ (hyperparameters).

$$f(\mathbf{y}) = \iint f(\mathbf{y}, \boldsymbol{\theta}, \boldsymbol{\phi})\, d\boldsymbol{\theta}\, d\boldsymbol{\phi} = \iint f(\mathbf{y}|\boldsymbol{\theta})\, f(\boldsymbol{\theta}|\boldsymbol{\phi})\, f(\boldsymbol{\phi})\, d\boldsymbol{\theta}\, d\boldsymbol{\phi}.$$

The prior predictive distribution is also often called the marginal distribution of $\mathbf{y}$. It is described as prior because it does not depend on a previous observation of the process, and predictive because it is the distribution for a quantity which is still unknown but observable.

This multi-level structure forms the basis of Bayesian hierarchical modelling. Figure 3.2.1 illustrates the hierarchical structure of the model. The direction of the arrows indicates how the hyperparameters at the population level influence the individual-level parameters, which in turn are associated with the observed data $y_1, y_2, ..., y_n$ obtained from a sample of size n.

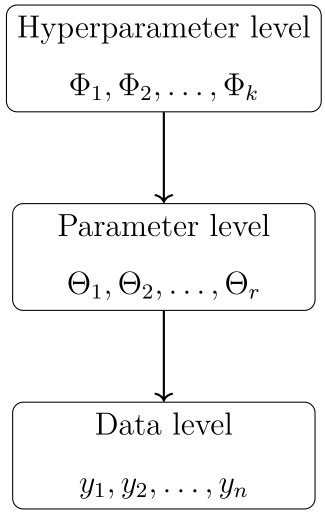

Figure 3.2.1: *Schematic representation of a Bayesian hierarchical model with three levels.*

The data level forms the base of the hierarchy, as it is the stage at which the observed data enter the model. The parameter level occupies the middle layer of the hierarchy, while the hyperparameter level represents the top layer and specifies prior distributions on global parameters. These priors, often referred to as hyperpriors represent a key difference between a standard Bayesian model and a Bayesian hierarchical model. A standard Bayesian model consists of a prior, a likelihood, and a posterior, whereas a Bayesian hierarchical model includes these components together with an additional hyperprior level.

This level introduces uncertainty about the global parameters, which describe the average behaviour across all individuals in the model. Additional hyperpriors can be introduced as needed, in which case the model is extended beyond the typical three levels. In reality, the hierarchy can be expanded infinitely, as there is no theoretical upper limit on the number of layers. However, in this dissertation, the hierarchy is restricted to three levels.

These levels create a system such that the model learns about both individuals and groups. Hierarchical models are especially useful when the data are grouped, for example, students within schools. Instead of fitting separate models for students and schools, hierarchical models partially pool the available information. Thanks to partial pooling, group-level estimates borrow strength from each other. This is the case also for teams within divisions and players within teams, as visualized in Figure 3.2.2. The hierarchical structure in Figure 3.2.2 relies on the assumption that individual units such as players or teams, are related through a common distribution.

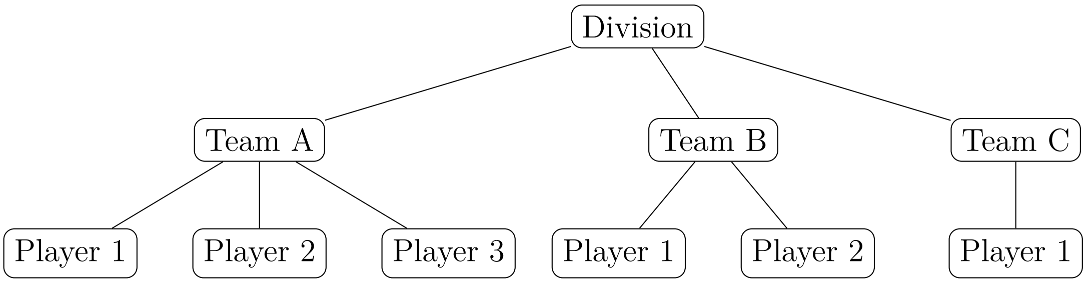

Figure 3.2.2: *Hierarchical structure of players nested within teams and teams within a division.*

Partial pooling provides a compromise between assuming that all θ_i are identical and allowing them to differ freely across individuals. Although θ_i are modelled as arising from a common distribution, individual-level differences are still considered. By combining population-level and group-specific information, hierarchical models yield more stable estimates, even when data is limited. In the context of our dataset, each θ_i represents a player-specific or team-specific effect. Here $i = 1, ..., I$ where I represents the total number of players or teams.

For partial pooling to be appropriate, these units must be comparable so that information can be shared across them. This idea leads to the concept of exchangeability, which assumes that, in the absence of additional information, the ordering of units does not affect our beliefs. It therefore serves as a more flexible alternative to independent and identically distributed assumptions. Exchangeability provides an important theoretical justification for modelling group-specific parameters as arising from a common population distribution. Its connection to hierarchical modelling through the important de Finetti representation theorem is discussed in more detail in Section 3.3.

3.3 Exchangeability and De Finetti's General Representation Theorem

Exchangeability provides an important theoretical justification for modelling group-specific parameters as arising from a common population distribution. Before data is observed, all orderings of observations are considered to be equally plausible. This idea can be formalized through the concepts of finite and infinite exchangeability, which distinguish between exchangeability in a fixed sample and exchangeability in an infinitely extendable sequence of observations.

A finite sequence of random variables $Y_1, Y_2, \ldots, Y_b$ is said to be *finitely exchangeable* if its joint distribution $f(y_1, \ldots, y_b)$ remains unchanged under any reordering of the variables. That is, for every permutation π of the index set $\{1, \ldots, b\}$,

$$f(y_1, \ldots, y_b) = f(y_{\pi(1)}, \ldots, y_{\pi(b)}).$$

Moreover, an infinite sequence of random variables $Y_1, Y_2, \ldots$ is infinitely exchangeable if every finite subsequence $Y_1, Y_2, \ldots, Y_{i_b}$ is finitely exchangeable for all $b \in \mathbb{N}$. Infinite exchangeability has a crucial implication in statistical modelling. Particularly, it must be assumed in order to apply de Finetti's general representation theorem. The general representation theorem, first established by Bruno de Finetti in 1937, is stated in Theorem 1.

Theorem 1 (*The General Representation Theorem*)**.** *Let $Y_1, Y_2, \ldots$ be an infinitely exchangeable sequence of random variables. For this sequence, there exists a unique probability measure μ on the space of distribution functions $\mathscr{F}$ on $\mathbb{R}$ such that the joint distribution of any b variables is*

$$F(y_1, \ldots, y_b) = \int_{\mathscr{F}} \prod_{i=1}^{b} F(y_i)\, d\mu(F)$$

where

$$\mu(F) = \lim_{q \to \infty} P(F_q).$$

and F_q *is the empirical distribution function defined by* $Y_1, Y_2, ...$

The original proof of the theorem was given by de Finetti (1937). However, as this work is written in French, a clearer statement, discussion, and interpretation can be found in Bernardo and Smith (1994). The reason why this theorem is important in Bayesian hierarchical modelling is that it provides justification for the introduction of a prior distribution. By showing that each variable in an infinitely exchangeable sequence can be represented as arising from a common underlying distribution, de Finetti's theorem suggests that the observations are governed by this shared distribution.

The general representation theorem guarantees the existence of a prior distribution for any infinitely exchangeable sequence. Therefore, when the data points are exchangeable, the Bayesian framework provides a natural approach for modelling data. Moreover, in a Bayesian framework, it is typically assumed that the data collected are exchangeable. When these observations are exchangeable, de Finetti's theorem suggests that there must exist a distribution governing them. In Bayesian terminology, this distribution is referred to as a hyperprior, and its introduction creates a third level within the hierarchical model, as discussed in Section 3.2.

In summary, de Finetti's theorem provides a theoretical foundation for Bayesian hierarchical modelling since through exchangeability of Y_i's, it implies that their joint distribution can be represented as a mixture distribution with a prior. Therefore, information can be shared across groups through partial pooling, leading to more reliable estimation of player- and team-specific effects.

However, in hierarchical models, the posterior distribution cannot usually be derived directly, since there are many parameters involved, and these parameters depend on other parameters through the hierarchical structure. Therefore, inference requires computational methods to approximate the posterior distribution, such as Markov Chain Monte Carlo algorithms, and in particular Gibbs sampling, which are discussed in Section 3.4.

3.4 Markov Chain Monte Carlo Algorithms

MCMC methods were first introduced in the context of statistical physics in 1953, when Metropolis proposed the Metropolis algorithm as a solution to computationally intensive integration problems. Later in the 1970s, the idea of the Metropolis Algorithm was generalized to allow asymmetric distributions by W.K. Hastings, leading to the development of the Metropolis–Hastings algorithm. His student Peskun also contributed by providing theoretical foundations for comparing the efficiency of different Markov chains (Peskun, 1973), an idea that remains crucial to MCMC research today. They aimed to overcome the curse of dimensionality, which was already highlighted and emphasized by Metropolis et al. (1953).

At the same time (circa 1980), Gibbs sampling was being described by Stuart Geman and Donald Geman in a paper titled 'Stochastic Relaxation, Gibbs Distributions, and the Bayesian Restoration of Images'. They were investigating methods to recover an image corrupted by random noise. The model they were trying to use had an immensely complicated probability distribution, making it impractical to sample from one joint distribution over all the pixels. They discovered that it was easier to sample each pixel sequentially from its conditional distribution, repeating the process for all pixels. This approach forms the basis of Gibbs sampling, which proved to be highly effective, not only in image analysis but also across a wide range of Bayesian models. This realisation led to its adoption within the Bayesian community during the 1990s.

Gibbs sampling and the Metropolis-Hastings algorithms are the two foundational MCMC algorithms. While more advanced MCMC algorithms like Hamiltonian Monte Carlo (Neal, 2011) have been developed for more complex or high-dimensional models, this dissertation focuses mainly on Gibbs sampling. This choice is motivated by its close connection to Bayesian hierarchical models, as Gibbs sampling arises naturally from full conditional distributions, which are available for the models considered in this dissertation. Moreover, since Bayesian hierarchical models are typically too complex to be solved manually, software such as JAGS (Just Another Gibbs Sampler) is used to perform the computations. JAGS mainly uses Gibbs sampling to estimate parameters of Bayesian hierarchical models, and it provides a computational framework for obtaining posterior distributions. The following section introduces Gibbs sampling in more detail and explains its role in estimating the parameters of Bayesian hierarchical models.

3.4.1 Gibbs Sampling

Among the MCMC algorithms, the Gibbs Sampler is one of the most widely used due to its simplicity, generality and operational efficiency across low and high-dimensional spaces. The name 'Gibbs' originates from 'Josiah Willard Gibbs', whose contributions to statistical mechanics gave birth to the concept of the Gibbs distribution. When Geman and Geman (1984) developed their sampling algorithm for Gibbs distributions, they named it in his honour.

The Gibbs sampler, also called alternating conditional sampling, has proven useful in many multidimensional problems. The method begins by specifying the posterior distribution from which we want to sample. In many practical problems, sampling from the full joint distribution is infeasible. Gibbs sampling overcomes this issue by focusing on one component at a time.

A crucial point regarding the Gibbs sampler is that it can be viewed as a special case of the Metropolis-Hastings algorithm. When the full conditional distribution cannot be sampled directly because its form is too complex for normal closed-form sampling, a Metropolis-Hastings step can be inserted for that component, which leads to a *Metropolis-within-Gibbs* algorithm.

An important reason for using Gibbs Sampling is that it generates samples from the full conditional posterior distribution given the current values of all the other components of $\boldsymbol{\theta}$ and the observed data $\mathbf{y}$, which are often simpler than sampling from the full joint posterior. Instead of sampling from $f(\boldsymbol{\theta}|\mathbf{y})$, we sample from $f(\theta_i|\boldsymbol{\theta_{-i}}, \mathbf{y})$. Here, θ_i is the parameter currently being updated and $\boldsymbol{\theta}_{-i} = (\theta_1, ..., \theta_{i-1}, \theta_{i+1}, ..., \theta_p)'$. Mathematically, at iteration t of the Gibbs sampler, a new value for θ_i is drawn from its conditional distribution given the data $\mathbf{y}$ and the current values of all the other parameters. Thus, $\theta_i^{(t)} \sim f(\theta_i \mid \boldsymbol{\theta}_{-i}^{(t-1)}, \mathbf{y})$ indicates that $\theta_i^{(t)}$ is sampled conditional on $\boldsymbol{\theta}_{-i}^{(t-1)}$, where $\boldsymbol{\theta}_{-i}^{(t-1)}$ refers to the vector of all parameters except for θ_i, using their values from the previous iteration $t-1$.

The general Gibbs Sampling algorithm is executed as follows:

Algorithm 1: General Gibbs Sampling Algorithm.

1: **Initialization:** Choose initial values for all p unknown parameters in the model. These initial values are selected based on prior information or reasonable estimates, as these can affect the rate of convergence of the Markov chain. Mathematically, let the initial parameter vector be:

$$\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, \theta_2^{(0)}, \dots, \theta_p^{(0)})'$$

where p denotes the total number of unknown parameters in the model.

2: **Iterative Sampling:** At each iteration the following steps are followed: Given the current values of all parameters and the observed data, each component of the parameter vector is updated in turn by sampling from its full conditional posterior distribution. Mathematically, for $t = 1, ..., N$, produce:

- $\theta_1^{(t+1)} \sim f(\theta_1 \mid \theta_2^{(t)}, \theta_3^{(t)}, \dots, \theta_p^{(t)}, \mathbf{y})$
- $\theta_2^{(t+1)} \sim f(\theta_2 \mid \theta_1^{(t+1)}, \theta_3^{(t)}, \dots, \theta_p^{(t)}, \mathbf{y})$
- $\dots$
- $\theta_p^{(t+1)} \sim f\Big(\theta_p \mid \theta_1^{(t+1)}, \dots, \theta_{p-1}^{(t+1)}, \mathbf{y}\Big)$

where N denotes the total number of Gibbs sampling iterations. Once all parameters have been updated, one full iteration of the algorithm is completed, yielding a new parameter vector $\boldsymbol{\theta}^{(t+1)}$.

3: Repeating this procedure for N iterations produces a sequence of samples:

$$\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)}, \dots, \boldsymbol{\theta}^{(N)},$$

which form a Markov chain whose stationary distribution is the joint posterior distribution. At each iteration, the parameter vector is therefore modified by replacing the old values with the updated ones.

In methods where the proposal distribution must be chosen, selecting an inappropriate distribution can lead to slow convergence and poor mixing of the Markov chain. Gibbs sampling avoids this kind of issue since the full conditional posterior distribution acts as the proposal distribution. Therefore, every draw is automatically accepted, and no tuning of the proposal distribution is required. Moreover, despite Gibbs sampling updating one parameter at a time, which can lead to slow convergence, especially when the model has many parameters, each step occurs in a low-dimensional setting. This makes it easier to apply than other Monte Carlo methods such as importance sampling and rejection sampling. Despite its advantages, Gibbs sampling can perform poorly if model parameters are highly correlated, as the Markov chain may move slowly (Gelman et al., 2013). In such cases, successive samples can become highly similar, leading to strong autocorrelation and a smaller effective sample size.

After the Gibbs updates are run, the early samples may still be influenced by the initial values. Therefore, the first B iterations are typically discarded as the burn-in period. The remaining samples are then used to approximate the posterior distribution, provided that convergence has been achieved. This is assessed using diagnostics such as traceplots, autocorrelation analysis and the Gelman-Rubin statistic, which are discussed later in Section 3.6.

3.4.2 Gibbs Sampling for Bayesian Hierarchical Models

After providing a general overview of Gibbs sampling, the technique can be extended to Bayesian hierarchical models with some modifications. It is widely used for performing inference in such models. Since Bayesian hierarchical models involve the vector of hyperparameters $\mathbf{\Phi}$, in addition to the vector of parameters $\mathbf{\Theta}$ and the observed data $\mathbf{y}$ they typically lead to complex joint posterior distributions, which lack analytical solutions. Therefore, direct sampling from the posterior is generally impractical, making MCMC algorithms, in particular Gibbs sampling, more useful for posterior inference. This is often the case because Bayesian hierarchical models are typically constructed so that the full conditional posterior distribution of each parameter (given the other parameters) belongs to a standard distribution family, such as the normal or gamma distributions, from which sampling is straightforward. Moreover, Bayesian hierarchical models typically involve a high-dimensional parameter space, whereas the Gibbs sampler iteratively samples from these low-dimensional conditional distributions, which is computationally efficient. The main aim here is sampling from the joint posterior distribution $f(\boldsymbol{\theta}, \boldsymbol{\phi}|\mathbf{y})$.

As shown in Algorithm 2, sampling alternates between the conditional posterior distributions of the parameters and hyperparameters. The Gibbs sampler is run until the Markov chain appears to have reached stationarity and the samples provide a reliable representation of the joint posterior distribution.

The Bayesian Gibbs sampling algorithm proceeds as follows:

Algorithm 2: Gibbs sampling for a Bayesian hierarchical model.

1: **Initialization:** Choose initial values for all unknown quantities in the model. These initial values may be selected based on prior information or reasonable estimates, as they can affect the rate of convergence of the Markov chain. Mathematically,

$$(\boldsymbol{\theta}^{(0)'}, \boldsymbol{\phi}^{(0)'})'.$$

2: **Iterative Sampling:** For $t = 1, \ldots, N$, generate the parameter vector from its conditional posterior distribution

$$\boldsymbol{\theta}^{(t+1)} \sim f(\boldsymbol{\theta} \mid \boldsymbol{\phi}^{(t)}, \mathbf{y}).$$

3: Given the updated value of $\boldsymbol{\theta}^{(t+1)}$, generate the hyperparameter vector from its conditional posterior distribution

$$\boldsymbol{\phi}^{(t+1)} \sim f(\boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t+1)}, \mathbf{y}).$$

4: Repeating this procedure for N iterations produces the sequence of samples

$$(\boldsymbol{\theta}^{(1)'}, \boldsymbol{\phi}^{(1)'})', (\boldsymbol{\theta}^{(2)'}, \boldsymbol{\phi}^{(2)'})', \ldots, (\boldsymbol{\theta}^{(N)'}, \boldsymbol{\phi}^{(N)'})',$$

which, after convergence of the Markov chain, can be regarded as draws from the joint posterior distribution $f(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \mathbf{y})$.

Furthermore, the convergence of the Markov chain to the posterior distribution is confirmed using MCMC diagnostics such as traceplots and Geweke's convergence diagnostic, among others. Therefore, five important convergence diagnostics are explained and presented in Section 3.5.

3.5 MCMC Diagnostics

MCMC diagnostics are crucial statistical tools and graphical techniques used to check whether a Markov chain Monte Carlo has converged to the target posterior distribution and whether it exhibits adequate mixing. MCMC diagnostics also evaluate the presence of autocorrelation and determine whether a sufficient sample size was produced to ensure reliable posterior inference. These diagnostics provide evidence that the posterior summaries obtained are trustworthy and reliable. Without loss of generality, MCMC diagnostics are being defined strictly for the standard Bayesian model. In this section, we use θ_i to refer to the value of a single parameter, rather than to the vector of parameters $\boldsymbol{\theta}$ used previously.

The first thing in assessing convergence is to establish an appropriate burn-in period, which refers to the initial part of the chain which is discarded to remove the influence of starting values. This makes sure that the retained samples are drawn from the equilibrium (stationary) distribution of the chain. To determine an appropriate burn-in period, multiple chains are run from different starting points, and their traceplots are shown together. The burn-in period is then chosen as the iteration after which the chains appear to mix well and exhibit stable, stationary behaviour. If such behaviour is not clearly observed, the number of iterations is increased.

Some common MCMC diagnostic methods which will be utilised in this dissertation include traceplots, Gelman-Rubin diagnostic, effective sample size (ESS), ACF plots, Geweke diagnostic and Heidelberger-Welch stationarity.

3.5.1 Traceplots

The traceplot is a visual inspection of the values obtained from the sample against the iteration number. Good mixing and convergence occur when the traceplot results in a stable horizontal band with the chains fluctuating around at a constant level, as can be seen in Figure 3.5.1.

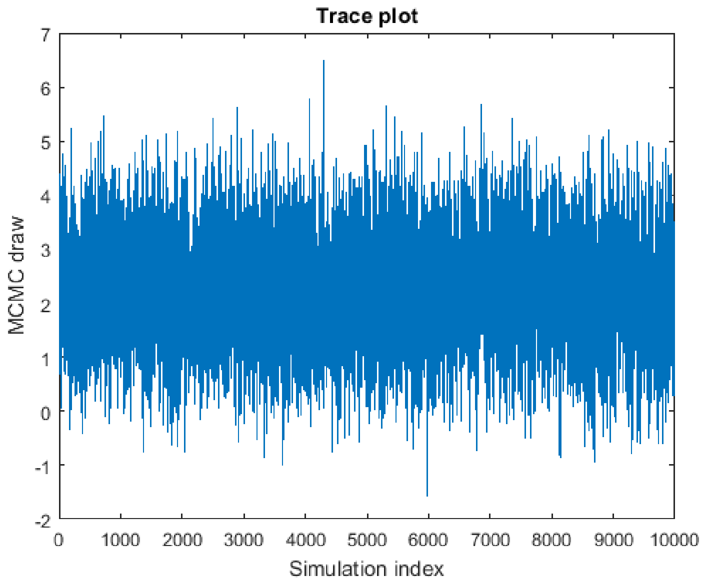

Figure 3.5.1: *Example traceplot illustrating good mixing and convergence.*

3.5.2 Autocorrelation Function

The autocorrelation function (ACF) assesses convergence by measuring the correlation between successive samples in a Markov chain at different lags. If autocorrelation is high, then successive samples are similar to each other, meaning that the chain explores the target distribution slowly, which reduces independence between samples making the algorithm less efficient and meaningful, hence more iterations are needed. A low value of autocorrelation, indicates good mixing and an efficient exploration of the posterior distribution, reflecting desirable MCMC sampling.

3.5.3 Effective Sample Size

The effective sample size (ESS) is a key diagnostic in MCMC analysis. It measures the number of independent samples that the MCMC chain is equivalent to after accounting for high autocorrelation between successive draws. The ESS quantifies the true amount of information contained in the chain. Since MCMC draws are typically correlated, the total number of iterations tends to overestimate the actual amount of information of the chain. If the chain mixes well and autocorrelation is low, the ESS will be similar to the number of iterations. Nonetheless, if the chain is highly autocorrelated (successive draws are particularly similar), the ESS will be smaller than the total number of draws. Mathematically, the ESS is approximated as:

$$ESS = \frac{N}{1 + 2\sum_{k=1}^{\infty} \rho_k} \tag{3.5.1}$$

where:

- N is the total number of MCMC samples,
- ρ_k is the autocorrelation at lag k.

A higher autocorrelation will increase the denominator, therefore reducing the ESS.

A high effective sample size, typically greater than 400, indicates good mixing of the Markov Chain and low autocorrelation between successive draws. In the case of high ESS, the chain explores the posterior distribution efficiently, and the resulting parameter estimates are considered reliable and may be used for further analysis. A moderate effective sample size, usually between 100-400, is acceptable but may require further attention. While current estimates may be utilised for inference, more iterations may help improve convergence, reduce autocorrelation and enhance mixing of the chains, leading to more accurate estimates. A low effective sample size, typically less than 100, exhibits a sticky chain with high autocorrelation, in which case posterior estimates may be unreliable and more iterations are usually needed. In some cases, model reparameterisation or tuning of the sampling algorithm may be required to achieve better mixing and convergence and less autocorrelation between successive draws.

For instance, if 1,000 iterations are run and autocorrelation is high, the ESS could drop significantly from 1,000 to 200-300, meaning that the chain contains much less true information than we expected. In summary, the effective sample size helps determine whether an MCMC chain has run long enough and mixed well enough to produce reliable results. In some cases, increasing the number of iterations is sufficient, but modifying the model specification may sometimes be needed to improve convergence.

3.5.4 Gelman-Rubin Statistic

The Gelman-Rubin diagnostic, also known as the potential scale reduction factor can also be used to assess convergence of the Markov chain. This diagnostic compares multiple chains to determine if they have converged to the same posterior distribution. If convergence has been achieved, the Gelman–Rubin statistic $\hat{R}$ which is based on the ratio of between-chain variance to within-chain variance, should be close to 1. Mathematically, suppose that m chains are being run, each of length n, and let θ_{ij} be the i'th draw from chain j of a single parameter, where $i = 1, ..., n$ and $j = 1, ..., m$. Here, scalar θ is used because this statistic is computed separately for each parameter. The sample mean of a single chain j is

$$\bar{\theta}_{\cdot j} = \frac{1}{n} \sum_{i=1}^{n} \theta_{ij},$$

and the overall mean across all m chains is

$$\bar{\theta}_{\cdot\cdot} = \frac{1}{m} \sum_{j=1}^{m} \bar{\theta}_{\cdot j}.$$

The between-chain variance is defined as

$$B = \frac{n}{m-1} \sum_{j=1}^{m} \left(\bar{\theta}_{\cdot j} - \bar{\theta}_{\cdot\cdot} \right)^2,$$

and the within-chain variance is

$$W = \frac{1}{m} \sum_{j=1}^{m} s_j^2,$$

where

$$s_j^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(\theta_{ij} - \bar{\theta}_{\cdot j} \right)^2$$

is the sample variance of chain j.

An estimate of the marginal posterior variance of some parameter θ is then given by

$$\widehat{\text{Var}}(\theta) = \frac{n-1}{n}W + \frac{1}{n}B.$$

The Gelman-Rubin statistic is defined as

$$\hat{R} = \sqrt{\frac{\widehat{\text{Var}}(\theta)}{W}}.$$

$\hat{R}$ measures how similar the sampled values from different chains are and is therefore used to assess convergence. More specifically, it measures the potential scale reduction, meaning how much the spread of the chains might still decrease if they were run for longer. This statistic is typically computed for each parameter separately, since some parameters may converge faster than others.

Values of $\hat{R}$ being considerably greater than 1 indicate that the chains are still sampling different regions of the parameter space and convergence has not yet been achieved. In practice, values of $\hat{R}$ below 1.05 are evidence of satisfactory convergence, while values below 1.01 indicate excellent convergence.

3.5.5 Heidelberger-Welch Test

The Heidelberger-Welch test is a statistical diagnostic test used to assess whether an MCMC chain is sampling from the target posterior distribution or is still being influenced by the initial values. Although this test is considered somewhat outdated, it is still included since it remains in use in the literature. If the test concludes that the early part of the chain differs significantly from the later part, this indicates that the chain is still being influenced by initial values and stationarity has not yet been reached. In such cases, a longer burn-in period is required before reliable inference can be made.

The test is performed in two main stages: a stationarity test and a half-width test. The first part of the Heidelberger-Welch diagnostic is where it evaluates whether the Markov chain has reached stationarity, meaning that the distribution is no longer being affected by the initial values.

This test uses a Cramér-von Mises statistic to check whether the sampled values come from a stationary distribution. Specifically, the Cramér-von Mises statistic measures how far the standardized cumulative sum process is from what is expected under stationarity (Anderson and Darling, 1952). Therefore, a small statistic value implies that the behaviour is consistent with stationarity, and a large statistic value implies that the process is still far from what is expected under stationarity. If the test fails, a portion of the initial samples is discarded and the test is repeated on the remaining chain. This process continues until either the test passes the stationarity test or a big part of the chain has been discarded, in which case convergence has not been achieved. If the stationarity test is passed, it is concluded that the chain has converged and the remaining samples can be considered as draws from the target posterior distribution.

When stationarity is achieved in the first test, the second part of the Heidelberger-Welch diagnostic evaluates the accuracy of the estimated posterior mean using the half-width confidence interval test. This test develops a confidence interval around the sample mean to check if the Monte Carlo error is sufficiently small. The test checks whether:

$$\frac{\text{Half-width of confidence interval}}{\left|\bar{\theta}\right|} < \epsilon \tag{3.5.2}$$

- $\bar{\theta}$ is the sample mean,
- ϵ is a chosen tolerance level (commonly 0.1).

The commonly used threshold value $\epsilon = 0.1$ in the Heidelberger-Welch diagnostic is not a fixed requirement. Alternative values may be selected according to the original Heidelberger-Welch diagnostic paper. The original method does not fix 0.1 as a universal rule, but later implementations introduced 0.1 as a default value (Heidelberg and Welch 1983). In this dissertation, $\epsilon = 0.2$ is used as the threshold value. This is a more relaxed choice than the default value of 0.1. Since the diagnostic is relatively outdated as already discussed, and the choice of threshold is flexible, this value is considered appropriate here.

If the condition in Equation (3.5.2) is satisfied, the estimate is considered sufficiently precise. If not, the chain would require more iterations to reduce Monte Carlo error and improve accuracy in estimation.

In conclusion, the Heidelberger–Welch diagnostic is a useful tool for evaluating convergence and reliability in MCMC simulations. It helps ensure the validity of Bayesian estimation results.

3.5.6 Geweke Statistic

Finally, the Geweke diagnostic is a statistical test which assesses convergence by comparing the early part of the chain to the later part. This test works on a single chain, and if convergence was reached, then the early and later part of the chain should look like they were drawn from the same distribution. Convergence is further confirmed through the Geweke statistic by testing the hypothesis:

$$H_0 : \text{The early and late segments of the chain have the same mean}$$
$$\mathbb{E}[\theta_{\text{early}}] = \mathbb{E}[\theta_{\text{late}}]$$

$$H_1 : \text{The early and late segments of the chain have different means}$$
$$\mathbb{E}[\theta_{\text{early}}] \neq \mathbb{E}[\theta_{\text{late}}]$$

The Geweke statistic (Z) is a Z-score calculated as follows:

$$Z = \frac{\bar{\theta}_1 - \bar{\theta}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{3.5.3}$$

where:

- N is the total number of retained MCMC samples excluding the burn-in period.
- $\bar{\theta}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} \theta^{(i)}$ denotes the sample mean of the early segment of the Markov

chain,

- $\bar{\theta}_2 = \frac{1}{n_2}\sum_{i=N-n_2+1}^{N}\theta^{(i)}$ denotes the sample mean of the late segment of the Markov chain,
- $\theta^{(i)}$ denotes the i-th draw from the Markov chain.
- s_1^2 is the estimated variance of the early segment after adjusting for autocorrelation,
- s_2^2 is the estimated variance of the late segment after adjusting for autocorrelation,
- n_1 is the number of observations in the early segment,
- n_2 is the number of observations in the late segment.

The null hypothesis is not rejected if the Z-score lies in the interval $-1.96 < z < 1.96$. In that case, there is no evidence against stationarity, suggesting that convergence has been reached. If the Z-score does not lie in that interval, the null hypothesis is rejected, indicating that the chain may not have converged and modifications to the sampling process may be required. The Geweke diagnostic can also be analysed graphically through a Geweke plot.

The dotted lines in Figure 3.5.2 represent the -1.96 and $+1.96$ limits used to assess whether the Geweke statistics fall within the acceptable range. The top plot is an example of a chain that shows no evidence against stationarity, indicating that convergence has likely been achieved. In contrast, the bottom plot shows some points below the -1.96 threshold, suggesting that the chain may not have converged.

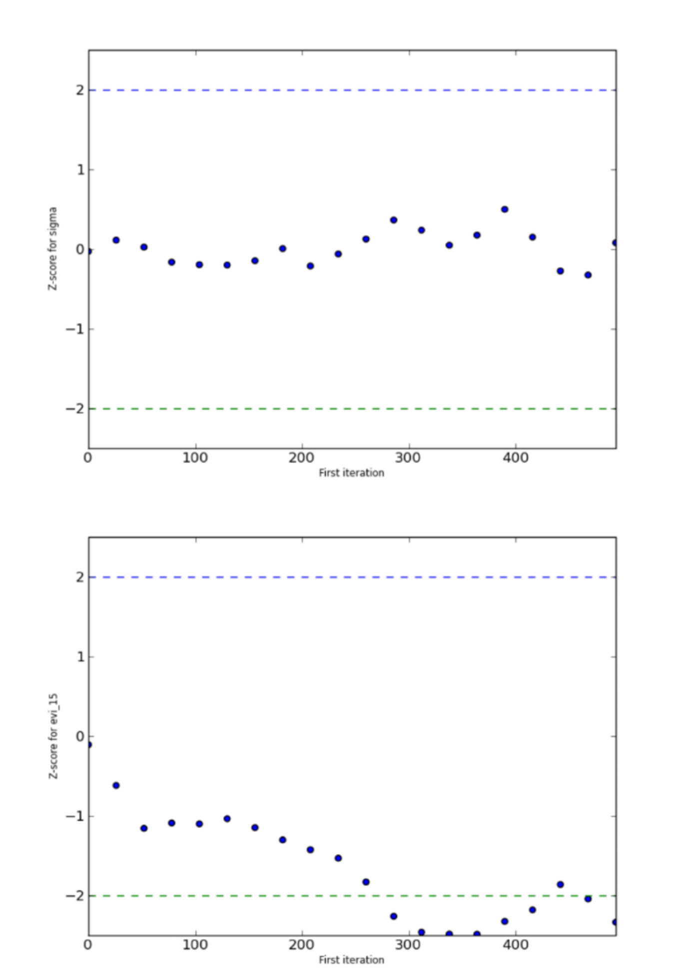

Figure 3.5.2: *Geweke diagnostic plot for assessing convergence obtained from Patil et al. (2010) .*

Having discussed the MCMC diagnostics used to assess convergence and reliability, the next section reviews relevant literature on Bayesian hierarchical models in sports analytics. It focuses on key papers in this area, highlighting how these models have been used and developed in practice.

3.6 Bayesian Hierarchical Modelling in Sports Analytics

This section reviews existing literature in which the theoretical principles discussed earlier are applied to sports performance data. Bayesian hierarchical models have become a crucial tool in sports analytics, which account for player and team performance while also capturing variation in performance across players, teams and matches. Three important papers are reviewed in this section. First, Baio and Blangiardo (2010) present a Bayesian hierarchical model for football match results, using latent ability parameters and partial pooling to model team performance across a season. Secondly, Attard et al. (2023) developed and compared two Bayesian hierarchical models using basketball data, employing Poisson and negative binomial specifications similar to the models that will be presented in this dissertation in section 3.7. Finally, Whitaker et al. (2021) apply an interpretable Bayesian model that estimates an individual footballer's ability for any given event in a match. The paper by Whitaker et al. (2021) is of particular importance to this study because a variant of this model is later proposed using aggregate data, rather than match-by-match data.

3.6.1 Team and Player Based Poisson Models

In this section, the models proposed by Baio and Blangiardo (2010) and Whitaker et al. (2021) are presented. Both studies fit Bayesian hierarchical models using a Poisson specification and are therefore crucial to the model developed in this dissertation. Since these studies are closely related to the models proposed in this dissertation, they are analysed and described in technical detail.

Baio and Blangiardo (2010)

In their paper, Baio and Blangiardo first fit a simple Bayesian hierarchical model using the 1991-1992 Serie A season. There were a total of $m = 18$ teams, which play against each other twice, bringing a total of $G = 306$ matches. The number of goals scored by the home and away team in match g were denoted by Y_{g1} and Y_{g2}. Consequently, each pair $\mathbf{Y}_g = (Y_{g1}, Y_{g2})'$ is modelled independently using a Poisson distribution:

$$Y_{gj} \mid H_{gj} = \eta_{gj} \sim \text{Poisson}(\eta_{gj})$$

where H_{g1} and H_{g2} represent the scoring intensities in the g'th match for the home ($j = 1$) and away ($j = 2$) team. The parameters H_{g1} and H_{g2} were modelled using a log-linear random effect model:

$$\log(H_{g1}) = \text{home} + \text{att}_{h(g)} + \text{def}_{a(g)} \tag{3.6.1}$$

$$\log(H_{g2}) = \text{att}_{a(g)} + \text{def}_{h(g)}. \tag{3.6.2}$$

Here, *home* represents the home-advantage effect. The functions *h(g)* and *a(g)* denote the indices of the home and away teams, respectively, in the g^{th} match of the season, so that $h(g), a(g) \in \{1, ..., m\}$, where m is the total number of teams. In particular, $att_{h(g)}$ and $def_{a(g)}$ represent the attack and defence effects associated with those teams in match g.

Baio and Blangiardo (2010) assigned weakly informative priors to let the data dominate inference and avoid extremes:

$$\text{home} \sim \text{Normal}(0,\ 0.0001).$$

Then, for each $j = 1, ..., m$ the team-specific effects are assumed to follow normal distributions with means μ_{att} and μ_{def} and precisions τ_{att} and τ_{def}, respectively:

$$att_j \sim \text{Normal}(\mu_{att}, \tau_{att}), \quad def_j \sim \text{Normal}(\mu_{def}, \tau_{def}).$$

Here, att_j and def_j represent the attack and defence effects for team j, respectively. In the match-level model, these are indexed through $h(g)$ and $a(g)$, so that the home and away teams are distinguished.

To ensure identifiability, sum-to-zero constraints are enforced on the team-specific effects:

$$\sum_{j=1}^{m} att_j = 0, \qquad \sum_{j=1}^{m} def_j = 0.$$

The hyperpriors of the attack and defence effects are chosen to be weakly informative. In particular, the location parameters are assigned normal priors, while the precision parameters are given Gamma priors:

$$\mu_{\text{att}},\ \mu_{\text{def}} \sim \text{Normal}(0,\ 0.0001),$$
$$\tau_{\text{att}},\ \tau_{\text{def}} \sim \text{Gamma}(0.1,\ 0.1).$$

Baio and Blangiardo (2010) aimed to estimate the characteristics which lead a team to lose or win a football match and to predict the score of a particular match. The authors also observed that the model specified above contained a lot of overshrinkage, hence they proceeded to fit a mixture model to that same dataset, which was somewhat more complex but helped avoid overshrinkage.

The structure of the Bayesian hierarchical model proposed by Baio and Blangiardo (2010), in terms of how the parameters influence one another, is clearly illustrated using a directed acyclic graph (DAG), shown in Figure 3.6.1. The graphical representation demonstrates the relationships between parameters and highlights the hierarchical nature of the model. The top level of the hierarchy represents the hyperparameters ($\mu_{att}, \tau_{att}, \mu_{def}, \tau_{def}$) which govern the distributions of the attack and defence effects for both the home and away teams. Here, μ represents population average for attack and defence effects, whereas τ represents precision for attack and defence effects.

The middle layer of the hierarchy represents the home advantage effect together with the attacking and defensive abilities of both the home and away teams. The DAG demonstrates how the attacking hyperparameters influence the attacking abilities of the home and away teams, and how the defensive hyperparameters influence the defensive abilities of the home and away teams. These team-specific parameters then determine the scoring intensities for each match for both teams, which then determine the predicted number of goals scored through the Poisson likelihood. This part shows how the hyperparameters (higher level parameters) affect team-specific parameters (middle level parameters).

The elements in the middle layer are combined to predict the scoring intensities H_{g1} and H_{g2}, which represent the expected number of goals scored by home (1) and away (2) teams in match g. Hence, at the third level, the predicted scoring ability of each team in any given match is determined.

Finally, the last level contains the observed number of goals for home and away teams, denoted by Y_{g1} and Y_{g2}, respectively. These are modelled as arising from Poisson distributions with means given by the scoring intensities. Therefore, H determines the distribution of goals scored, while Y represents the observed goal outcomes generated from these distributions.

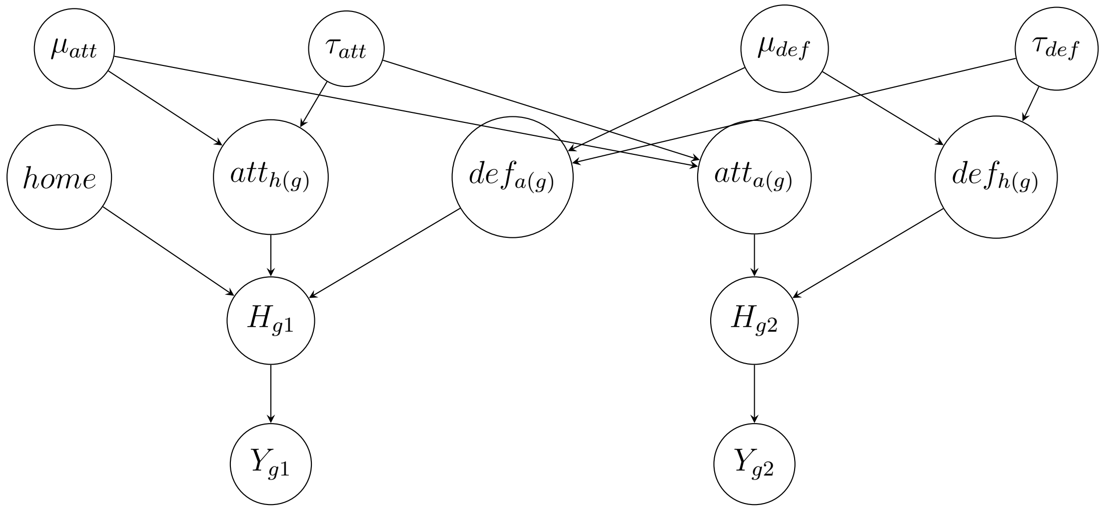

Figure 3.6.1: *Directed acyclic graph representing the Bayesian hierarchical model proposed by Baio and Blangiardo (2010).*

Whitaker et al. (2021)

A related, but more player-oriented approach was developed by Whitaker et al. (2021), whose work focuses on individual players rather than teams. In this study, event types were split into four categories: Stop (stoppage in play, for example, a substitution), control (team is in control of the ball and trying to attack), disruption (defensive events, interceptions, etc.), and miscellanea (those that do not fit in the other 3 categories). In the model, $Y^e_{i,g}$; denotes number of occurrences of event type e, by player i (who currently plays for team j) in match g. Event type e refers to an in-match event that can occur in the match, such as a goal or an assist. The quantity $Y^e_{i,g}$ is assumed to follow a Poisson distribution:

$$Y^e_{i,g} \mid H^e_{i,g} = \eta^e_{i,g}, \tau_{i,g} \sim \text{Poisson}(\eta^e_{i,g}\tau_{i,g}) \tag{3.6.3}$$

where:

$$H_{i,g}^{e} = \exp\left\{ \Delta_i^e + \tau_{i,g} \left(\Lambda_1^e \sum_{i' \in P_g^j} \Delta_{i'}^e - \Lambda_2^e \sum_{i' \in P_g^{T_g \setminus j}} \Delta_{i'}^{E \setminus e} \right) + \left(\delta_{T_g^h, j} \right) \gamma^e \right\}, \tag{3.6.4}$$

and:

- j denotes the team of player i in match g, and j' denotes the opposing team,
- i' denotes other players in the team of player i,
- e and $E \setminus e$ denote two event types that are assumed to interact, for instance, shots taken and interceptions,
- $\delta_{r,s}$ is the Kronecker delta, taking value of 1 when $r = s$ and 0 otherwise,
- $\tau_{i,g}$ measures the proper fraction of match g that player i spends on the pitch from the full 90 minutes while playing for team j. For instance, playing 70 minutes in a normal 90-minute match corresponds to $\tau_{i,g} = 7/9$,
- γ^e represents the home advantage for the home team associated with event type e, reflecting the benefit of playing at home,
- Δ_i^e describes player i's latent ability for event type e,
- $\Delta_{i'}^{E \setminus e}$ is the latent ability of the opposing player i' associated with the interacting event type $E \setminus e$.
- Λ_1^e reflects the quality of the player's own team of event type e,
- Λ_2^e represents how well the opposing team is able to restrict event type e,
- P is the set of all players who appear in the dataset whereas P_g^j and $P_g^{j'}$ denote the sets of players appearing for teams j and j' in match g, respectively,
- T_g^h denotes the home team in match g and T_g^a denotes the away team, hence $T_g = \{T_g^h, T_g^a\}$,

- $H^e_{i,g}$ is the random Poisson intensity (rate) parameter for player i in match g and event type e.

- $\eta^e_{i,g}$ is the specific realisation Poisson intensity (rate) parameter for player i in match g and event type e.

Independent Gaussian priors are put over all player abilities, and unknown parameters are treated as hyperparameters to be fitted by the marginal likelihood function. Initially, the model proposed by Whitaker et al. (2021) is not formulated as a Bayesian hierarchical model. However, later in the paper, this structure is introduced by following the framework used by Baio and Blangiardo (2010).

Using Equations (3.6.3) and (3.6.4) and assuming for simplicity that each team consists only of the starting eleven players while temporarily ignoring the effect of playing time $\tau_{i,g}$, the log-likelihood function is given by:

$$\ell = \sum_{e\in E}\sum_{g=1}^{G}\sum_{j\in T_g}\sum_{i\in P_g^j}\left[Y^e_{i,g}\log\Big(\eta^e_{i,g}\tau_{i,g}\Big) - \eta^e_{i,g}\tau_{i,g} - \log\Big(Y^e_{i,g}!\Big)\right]. \tag{3.6.5}$$

The log-likelihood function is built by adding contributions over all competitions, games, players and events. Here, $Y^e_{i,g}$ denotes the observed count for player i in match g, for event type e while $\eta^e_{i,g}\tau_{i,g}$ is the corresponding Poisson mean. The notation G denotes the total number of matches.

Figure 3.6.2 provides a graphical representation of the model for a single player in a single match, showing how player abilities, team effects, opponent strength, and home advantage combine to determine the observed event count. In the figure, it is assumed for simplicity that only 11 players play for each team. This implies that substitutions and injuries are not accounted for.

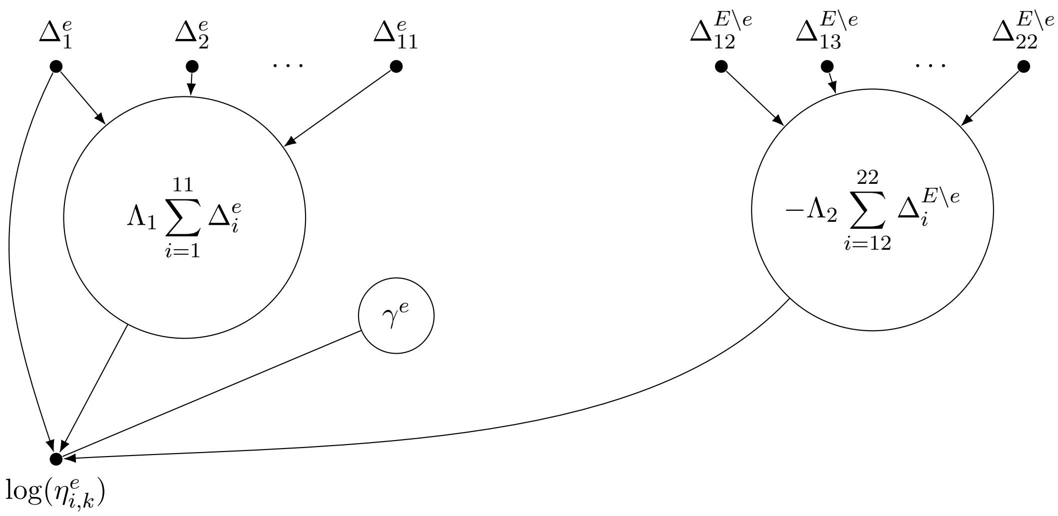

Figure 3.6.2: *Graphical representation of the Whitaker et al. (2021) model for a single player for a single match.*

Firstly, Δ_i^e is the latent player ability for player i in event type e. Furthermore, $\Delta_1^e, \Delta_2^e, \dots, \Delta_{11}^e$ refer to the 11 players on the same team as the player whose event rate is being modelled, representing their respective abilities for event e. On the other hand, $\Delta_{12}^{E\setminus e}, \Delta_{13}^{E\setminus e}, \dots, \Delta_{22}^{E\setminus e}$ refer to the 11 players on the opponent team and their abilities for event type $E \setminus e$. The event type $E \setminus e$ is the opposite of the event type e (for example, passes (e) and interceptions ($E \setminus e$) or shots (e) and blocked shots ($E \setminus e$)). Each large circle represents a team-level summary, which eventually feeds into the player's event rate.

The large circle on the left represents the contribution of the team with whom the player plays. This emphasizes the idea that a player's performance also depends on their teammates. The parameter Λ_1 controls how much the team influences the player's event rate. A similar interpretation applies to the large circle on the right, where Λ_2 controls how strongly the opposing team affects the player's rate. The negative sign indicates that a stronger opposition ability reduces the player's event rate. In addition, the small circle labelled γ^e represents a home advantage parameter for event type e. This parameter contributes when the player's team is playing at home but does not contribute when the team is playing away. All arrows ultimately point to $\log(\eta_{i,g}^e)$, which represents the log event rate for player i for event type e in match g. The log-likelihood function in Equation (3.4.6) extends this idea to all players and all matches by summing over the relevant indices.

3.6.2 Extending Event Modelling Using a Negative Binomial Distribution

Attard et al. (2023) developed and compared two Bayesian hierarchical model approaches using basketball data of the 2008/2009 NBA season. This paper will also be discussed in greater detail, as it fits both Poisson and negative binomial models using a strategy similar to that of this dissertation. Furthermore, the authors perform posterior inference using Gibbs sampling, which is the main computational approach employed in this dissertation and has already been discussed in detail in Sections 3.4.1 and 3.4.2, both in general terms and in the context of Bayesian hierarchical models.

The first model discussed in Attard et al. (2023) aimed to estimate all match outcomes by considering the offensive and defensive ability for each team. The authors attempted to fit both a Poisson and a negative binomial distribution. The negative binomial distribution, which provided a better fit, was subsequently used to estimate each team's attacking and defending abilities. These estimated abilities were then used to create a final score for each match and also final standings at the end of the season. Attard et al. (2023) model basketball data using a negative binomial distribution in order to account for overdispersion, which regularly appears in sports data and cannot be captured by a Poisson model. This allows for greater variability between matches and teams, increasing model flexibility, as some matches differ greatly from each other, while still retaining the hierarchical structure for teams and players. This idea is applicable to the work presented in this dissertation, and for the same reasons, a negative binomial model is also fitted in the proposed Bayesian hierarchical model in order to better capture variability.

For notation purposes, FT, $TwoPT$, and $ThreePT$ denote free throws, two-point shots, and three-point shots, respectively. Three Poisson models were considered, one for each type of shot; the free throw (one point), the two point shots (two points) and the three point shots (three points), as illustrated below:

$$
\begin{aligned}
FT_{gj} \mid H_{gj}^{FT} = \eta_{gj}^{FT} &\sim \text{Poisson}(\eta_{gj}^{FT}) \\
TwoPT_{gj} \mid H_{gj}^{2PT} = \eta_{gj}^{2PT} &\sim \text{Poisson}(\eta_{gj}^{2PT}) \\
ThreePT_{gj} \mid H_{gj}^{3PT} = \eta_{gj}^{3PT} &\sim \text{Poisson}(\eta_{gj}^{3PT})
\end{aligned}
\tag{3.6.6}
$$

where j is the team index (with $j = 1$ for the home team and $j = 2$ for the away team) and g is the match index. The variables FT_{gj}, $TwoPT_{gj}$ and $ThreePT_{gj}$ represent the observed count for the three types of goals and H_{gj}^{FT}, H_{gj}^{2PT} and H_{gj}^{3PT} denote the scoring intensity in the g^{th} match. The home advantage effect is considered for the home team in each match, and a common intercept is introduced to account for the typically high scoring nature of basketball games.

In the log-linear formulation found in Equation (3.6.7), the logarithm of a team's scoring intensity is expressed as the sum of several components. These are the team's attacking intensity, the defensive intensity of the opposing team, a common intercept and, when the team plays at home, an additional home-advantage term:

$$
\begin{aligned}
\log(H_{g1}^{M}) &= att_{h(g)}^{M} + def_{a(g)}^{M} + c^{M} + home^{M} \\
\log(H_{g2}^{M}) &= att_{a(g)}^{M} + def_{h(g)}^{M} + c^{M}
\end{aligned}
\tag{3.6.7}
$$

where $h(g)$ denotes the home team in match g, $a(g)$ denotes the away team in match g, and M represents one of the 3 models (FT, $TwoPT$ or $ThreePT$). Moreover, $att_{h(g)}^{M}$ and $att_{a(g)}^{M}$ denote the attack intensity for the home and away teams, respectively, under model M (which can be FT, $TwoPT$ or $ThreePT$), whereas $def_{h(g)}^{M}$ and $def_{a(g)}^{M}$ denote the defence ability for the home and away teams, respectively, under model M. $Home^{M}$ indicates the home advantage effect for the home team and c^{M} represents the common intercept under model M. As said earlier, H_{gj}^{M} represents the scoring intensity in the g^{th} match for the home ($j = 1$) and away ($j = 2$) teams.

Similar to Baio and Blangiardo (2010), weakly informative prior distributions are assigned to these parameters:

$$
\begin{aligned}
home^M &\sim \mathrm{Norm}(0, 0.0001) \\
c^M &\sim \mathrm{Norm}(0, 0.0001).
\end{aligned}
\tag{3.6.8}
$$

Following this hierarchical structure, the parameters att_j^M and def_j^M, where $j = h(g)$ or $j = a(g)$ are assigned two hyperparameters each. These parameters are assumed to follow normal distributions with means μ_{att} and μ_{def}, respectively and precisions τ_{att} and τ_{def}:

$$att_j^M \sim \mathrm{Norm}(\mu_{att}, \tau_{att}), \quad def_j^M \sim \mathrm{Norm}(\mu_{def}, \tau_{def}).$$

The location and precision parameters are also modelled using flat prior distributions, similar to Baio and Blangiardo (2010):

$$
\begin{aligned}
\mu_{\text{att}},\ \mu_{\text{def}} &\sim \text{Normal}(0,\ 0.0001), \\
\tau_{\text{att}},\ \tau_{\text{def}} &\sim \text{Gamma}(0.1,\ 0.1).
\end{aligned}
$$

Sum-to-zero constraints were also imposed for identifiability purposes:

$$\sum_{j=1}^{m} att_j = 0, \qquad \sum_{j=1}^{m} def_j = 0.$$

Here, j represents teams and m denotes the total number of teams.

Figure 3.5.3 offers a graphical representation of the hierarchical structure used by Attard et al. (2021). The four nodes at the top are the hyperparameters, while the distributions assigned to them, discussed earlier, are the hyperpriors. Specifically, μ_{att}^M and τ_{att}^M control the attack effects whereas μ_{def}^M and τ_{def}^M control the defence effects. The middle layer is the latent-effect level, where $att_{h(g)}^M$, $def_{h(g)}^M$, $att_{a(g)}^M$ and $def_{a(g)}^M$ denote the latent team effects, each of which depends on the four hyperparameters in the layer above. Moreover, $home^M$ represents the home advantage parameter and c^M is the common effect, and these are assigned their own prior distributions. Finally, $Y_{g_1}^M$ and $Y_{g_2}^M$ denote the observed scores for the home and away teams in game g, respectively, under scoring model M. These are directly affected by the latent variables H_{g1}^M and H_{g2}^M.

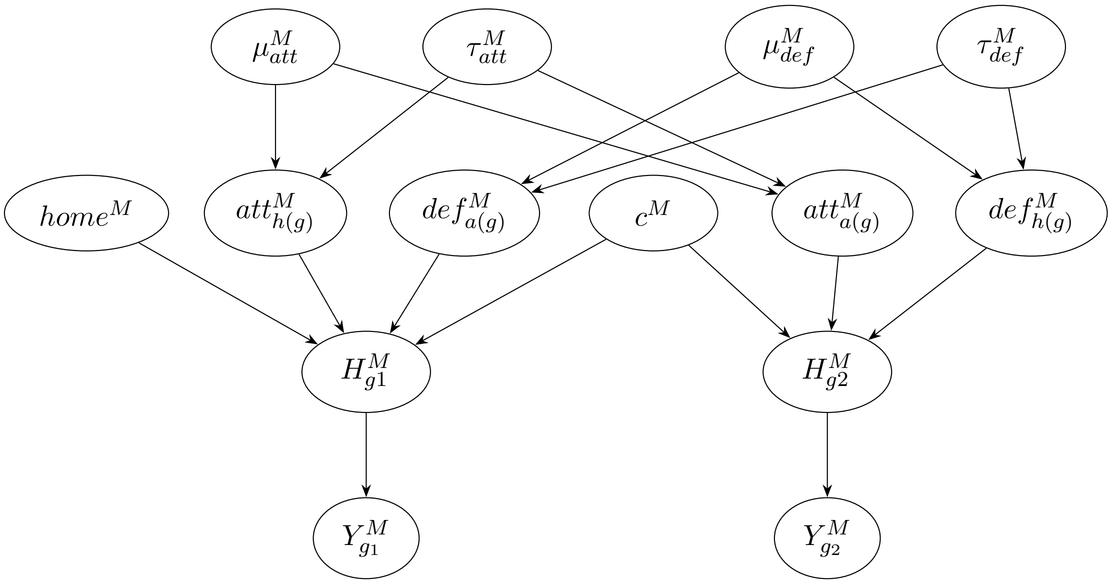

Figure 3.6.3: *DAG for the scoring intensity model using the Poisson distribution.*

Negative binomial models were also fitted since their second parameter allows them to accommodate the consistently high scores typical of basketball matches, and thus the authors do not solely rely on the Poisson model. This model helps account for the typically high scores observed in basketball games, with scores far from zero. The models considered are presented below:

$$\begin{aligned} FT_{gj} \mid L_{gj}^{FT} = l_{gj}^{FT}, R_{gj}^{FT} = r_{gj}^{FT} &\sim \text{NegBin}(l_{gj}^{FT}, r_{gj}^{FT}) \\ TwoPT_{gj} \mid L_{gj}^{2PT} = l_{gj}^{2PT}, R_{gj}^{2PT} = r_{gj}^{2PT} &\sim \text{NegBin}(l_{gj}^{2PT}, r_{gj}^{2PT}) \\ ThreePT_{gj} \mid L_{gj}^{3PT} = l_{gj}^{3PT}, R_{gj}^{3PT} = r_{gj}^{3PT} &\sim \text{NegBin}(l_{gj}^{3PT}, r_{gj}^{3PT}) \end{aligned} \tag{3.6.9}$$

In Equation (3.6.9), R_{gj}^{FT}, R_{gj}^{2PT} and R_{gj}^{3PT} are the overdispersion parameters in the g^{th} match for team j with respect each of the three types of shots. A key difference compared to the Poisson model, is that L_{gj}^{FT}, L_{gj}^{2PT} and L_{gj}^{3PT} denote the parameters measuring success probability for team j in the g^{th} match with respect to free throws, two point shots and three point shots, respectively. These parameters are once again modelled using a similar log-linear random-effects specification used in the Poisson model:

$$\begin{aligned}\log\left(R^M_{g1}\right) &= att^M_{h(g)} + def^M_{a(g)} + c^M + home^M,\\ \log\left(R^M_{g2}\right) &= att^M_{a(g)} + def^M_{h(g)} + c^M.\end{aligned} \tag{3.6.10}$$

The elements in Equation (3.6.10) have already been defined for the log-linear specification in the Poisson model. Since the parameters L^M_{gj} represent a success probability, their values need to range between 0 and 1, hence they are assumed to follow a uniform distribution: $L^M_{gj} \sim \text{Unif}(0,1)$. In the negative binomial model, similar to the Poisson model, the attack and defence effects are assumed to be normally distributed:

$$att^M_j \sim \text{Norm}(\mu_{att}, \tau_{att}), \quad def^M_j \sim \text{Norm}(\mu_{def}, \tau_{def}).$$

The location and precision parameters are also modelled using flat prior distributions:

$$\begin{aligned}\mu_{\text{att}},\, \mu_{\text{def}} &\sim \text{Normal}(0,\ 0.0001),\\ \tau_{\text{att}},\, \tau_{\text{def}} &\sim \text{Gamma}(0.1,\ 0.1).\end{aligned}$$

Sum-to-zero constraints were also imposed for identifiability purposes:

$$\sum_{j=1}^{m} att_j = 0, \qquad \sum_{j=1}^{m} def_j = 0.$$

Figure 3.6.4 shows the directed acyclic graph for the negative binomial model. The difference here is that there are two linear predictors, R^M_{gj} and L^M_{g1}. R^M_{gj} represents the negative binomial overdispersion parameters with respect to the three types of shots by team j in the g^{th} match, while L^M_{gj} represents the success probability parameters with respect to the three types of shots by team j in the g^{th} match. The stopping parameters are influenced by the home advantage parameter when the team is playing at home, the team's attacking ability, the opposing team's defensive ability, and the common intercept. Mathematically, this is presented in Equation (3.6.10). Furthermore, since L^M_{g1} represent a success probability, they must take a value between 0 and 1, therefore, $L^M_{gj} \sim Unif(0,1)$.

Figure 3.6.4 shows the directed acyclic graph for the negative binomial model, which is similar to the DAG for the Poisson model in Figure 3.6.3, with the addition of the overdispersion parameter.

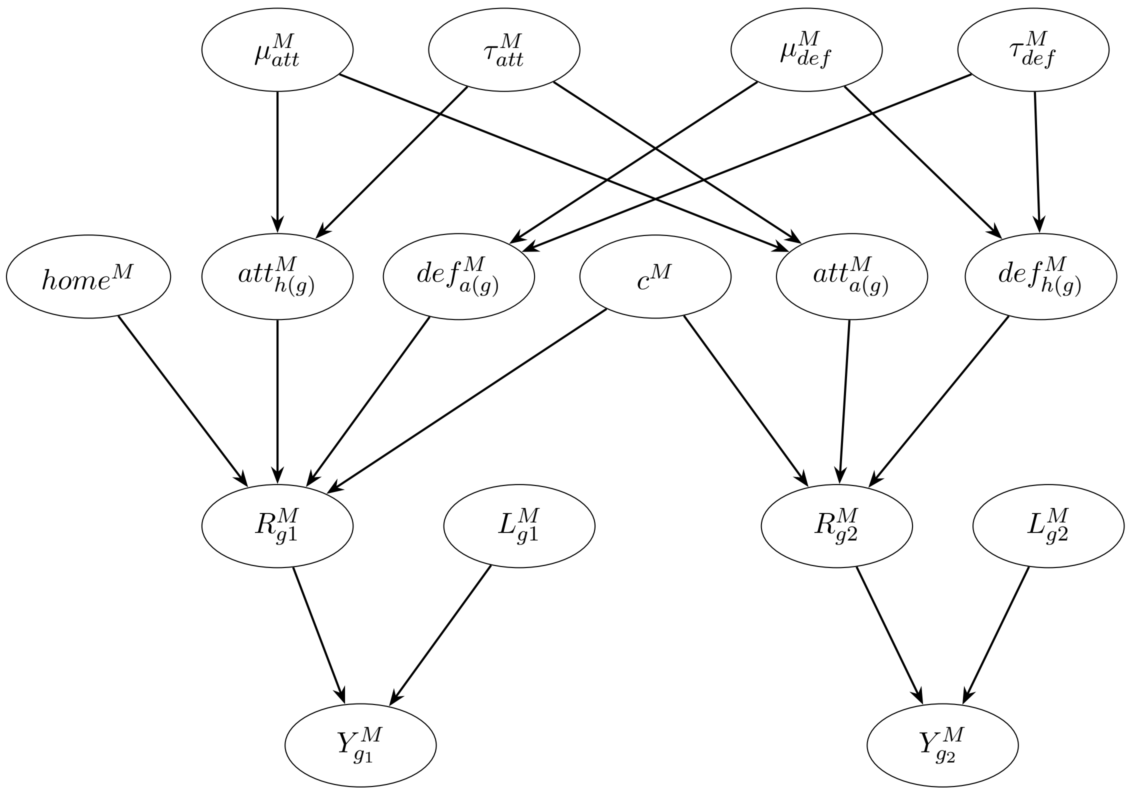

Figure 3.6.4: *DAG for the scoring intensity model using the negative binomial distribution.*

Posterior inference was performed using Markov Chain Monte Carlo (MCMC) methods. In particular, Gibbs sampling was used to generate samples from the posterior distribution using the JAGS software through the 'rjags' package in R.

So as seen in the papers in this section, a player's performance is influenced not only by their individual ability but also by their own team's ability, the ability of the opposing team, and match conditions such as home advantage. Combining all of these components results in a rich and realistic model but also leads to a large number of parameters that are linked in complicated ways. In Section 3.7, a model similar to that proposed by Whittaker et al. (2021), together with some features from Baio and Blangiardo (2010), is used in the model presented in this dissertation.

3.7 Proposed model for Predicting Individual Player Ability Using Match Events

Recall that the aim of this dissertation is to estimate individual player performance based on selected football performance metrics from the 2022 World Cup, while separating individual player ability from team quality. Throughout this dissertation, Θ refers to a random parameter, while θ refers to a single realisation of parameters, and Φ refers to a random hyperparameter, whereas ϕ denotes a single realisation of hyperparameters. Both Poisson and negative binomial specifications are considered. The Poisson model is standard for count data, while the negative binomial model allows for overdispersion. In both cases, the same linear predictor is used, depending on player ability, team ability, opposing team ability, and minutes played. Partial pooling is achieved by modelling player and team effects using common normal prior distributions with shared hyperparameters k and s, allowing information to be shared across players and teams.

3.7.1 The Poisson Model

Football performance metrics are quantitative measures to analyse different aspects of a player's contribution in a match, such as blocked shots or passes attempted. The model estimates player tendencies to generate events. In order to analyse the data, a Bayesian hierarchical model is employed. This section introduces the general statistical formulation of the model for the three performance metrics considered in this dissertation, namely interceptions, shots and passes. The prior distributions are chosen to be as uninformative as possible in order to minimise their influence on the posterior inference. However, if convergence is not reached, more informative priors are considered in order to help enforce convergence.

Initially, the number of successful events is modelled using a Poisson distribution, since this is suitable for count data, where events transpire in a fixed period of time. The model is later extended to a negative binomial specification to account for overdispersion.

In the Poisson specification, the quantity Y_i^e is assumed to follow a Poisson distribution:

$$Y_i^e | H_i^e = \eta_i^e, \tau_i \sim \text{Poisson}(\eta_i^e \tau_i) \tag{3.7.1}$$

Furthermore, the linear predictor for events in general is defined mathematically in Equation (3.7.2):

$$H_i^e = \exp\left\{\Delta_i^e + \tau_i \Lambda_i^e - \tau_i^* \sum_{j=1}^{m} \Lambda_j^{E\setminus e} \delta_{ij}\right\} \tag{3.7.2}$$

where i represents one observation for each player, j represents opponent's teams, m is the number of opponent teams faced by player i, e is the type of event being modelled such as passes or shots, and $E \setminus e$ is the interacting event that affects or stops the event e. For instance, if e is passing, then $E \setminus e$ is interceptions. Moreover, Δ_i^e denotes latent player ability for player i for event type e, τ_i represents the minutes played by player i, Λ_i^e denotes the team skill based on event e of player i and $\tau_i^* = \left(\frac{\tau_i}{\sum_j \delta_{ij}}\right)$ is the number of minutes played per match for player i. Furthermore, $\Lambda_j^{E\setminus e}$ is the skill of the opposing team, team j, based on an interacting event ability $E \setminus e$, δ_{ij} denotes the number of matches played against team j in observation i and $\sum_{j=1}^{m} \Lambda_j^{E\setminus e} \delta_{ij}$ refers to the combined team strength of the opponents faced.

The general modelling framework remains the same across all football performance metrics, and the same form of linear predictor is used throughout. However, the interpretation of the player and team effects depends on the football metric under consideration. For example, in the passing model, the player effect reflects individual passing ability, the team effect reflects overall team passing ability and the opponent effect reflects the intercepting ability of the opposing team as a whole. In τ_i, one unit corresponds to 90 minutes. So if a player played 270 minutes in the tournament, then $\tau_i = 3$. Moreover, the player abilities and the team effects are summed to zero so that the average player and the average team have zero effect. The player and team effects specified below were inspired by the code provided in Baio and Blangiardo (2010):

$$\Delta_i \sim \text{Normal}(k_\Delta, s_\Delta),$$

$$\Lambda_i^e \sim \text{Normal}(k_{\Lambda^e}, s_{\Lambda^e}),$$

$$\Lambda_j^{E\setminus e} \sim \text{Normal}(k_{\Lambda^{E\setminus e}}, s_{\Lambda^{E\setminus e}}),$$

where s_Δ, s_{Λ^e}, and $s_{\Lambda^{E\setminus e}}$ denote precision parameters. Moreover, sum-to-zero constraints were imposed on the player effects and on both team effects to improve identifiability, as suggested by multiple studies such as Baio and Blangiardo (2010) and Whitaker et al. (2021):

$$\sum_{j=1}^{m} \Lambda_j^e = 0, \qquad \sum_{j=1}^{m} \Lambda_j^{E\setminus\mathrm{e}} = 0, \qquad \sum_{i=1}^{n} \Delta_i = 0.$$

Therefore, Δ_i represents the difference of player i's event ability from the average player on his team. In addition, Λ_j^e represents the difference of team j's event ability from the average across teams and $\Lambda_j^{E\setminus\mathrm{e}}$ represents the difference of the opponent team j's event ability from the average across teams. Positive values imply players and teams who tend to complete more events than average, while negative values indicate players and teams who tend to complete fewer events. Here, i is the index of the player and j is the index of the team, while n refers to the total number of players and m refers to the total number of teams.

The hyperpriors for the team and player effects were chosen to be as weakly informative as possible so that the data, rather than the priors, would determine the parameter values:

$$
\begin{aligned}
k_{\Lambda^e} &\sim \text{Normal}(0, 0.0001) & s_{\Lambda^e} &\sim \text{Gamma}(0.01, 0.01),\\
k_{\Lambda^{E\setminus e}} &\sim \text{Normal}(0, 0.0001) & s_{\Lambda^{E\setminus e}} &\sim \text{Gamma}(0.01, 0.01),\\
k_{\Delta} &\sim \text{Normal}(0, 0.0001) & s_{\Delta} &\sim \text{Gamma}(0.1, 0.01).
\end{aligned}
$$

The prior means are assigned zero-centred normal distributions. Under the JAGS parametrisation, the second value denotes the precision, so a precision of 0.0001 corresponds to a large variance of 10,000. This gives a diffuse prior, allowing teams to deviate positively or negatively from the league average while expressing minimal prior information. The normal prior is standard in Bayesian hierarchical models because it is symmetric, flexible, and discourages extremely large parameter values unless supported by the data. Following Baio and Blangiardo (2010), the team-level precision parameters are assigned Gamma distributions with shape and rate parameters equal to 0.01, while the player-level precision parameter is assigned a Gamma distribution with shape 0.1 and rate 0.01, giving weakly informative priors.

Figure 3.7.1 presents the model described in this section in graphical form. The hyperparameters in the top layer, each of which is assigned a hyperprior distribution, govern the team and player ability parameters in the layer below. In turn, the player and team abilities combine to determine the linear predictor for each player, through which H_i is defined. Finally, H_i determines the distribution of the observed results Y_i.

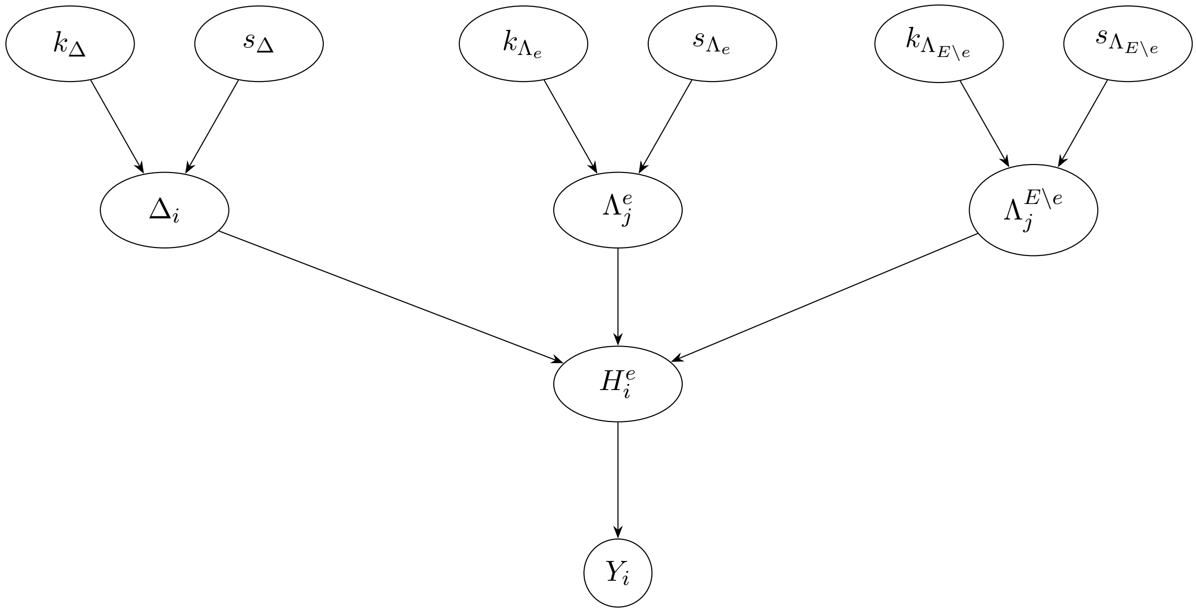

Figure 3.7.1: *Graphical representation of the Bayesian hierarchical Poisson model.*

The Poisson distribution is typically used for count data because it is suitable for modelling the number of times an event occurs within a fixed period of time. An important assumption of the Poisson model is equidispersion, because it assumes equality between the mean and variance (Mathematically, $\mathbb{E}[Y_i] = \text{Var}(Y_i)$). As already mentioned, this may be true for some datasets, but it is not always a logical assumption. The negative binomial model is used to cater for cases when the mean and variance are not equal.

3.7.2 The Negative Binomial Model

To allow for possible overdispersion in the data, a negative binomial model was also considered. This was specified using the mean-dispersion parameterisation, with mean μ_i and dispersion parameter φ, which differs from the parameterisation used by Attard et al. (2023). Hence, under this specification, the number of events Y_i^e is assumed to follow a negative binomial distribution:

$$Y_i^e \sim \text{NegBin}(\mu_i, \varphi)$$

where

$$\mu_i = \eta_i \tau_i.$$

Moreover,

$$\mathbb{E}[Y_i] = \mu_i, \qquad \mathrm{Var}(Y_i) = \mu_i + \varphi\mu_i^2.$$

The mean of Y_i is μ_i and this gives the expected number of events for player i. The variance exceeds the mean whenever $\varphi > 0$. Therefore, φ controls the degree of overdispersion. As φ increases, the variance increases, and when $\varphi = 0$, the variance becomes equal to μ_i which corresponds to the Poisson case. The variance specification introduces flexibility when the data exhibit overdispersion, which is not accommodated by the Poisson model.

For implementation in JAGS, using the 'dnegbin' function, the negative binomial model is written in terms of the probability parameter p_i and the size parameter r. We thus introduce the following:

$$p_i = \frac{r}{r + \mu_i}, \qquad r = \frac{1}{\varphi}.$$

This allows the model to be coded in JAGS while also retaining a mean-dispersion parametrisation. In this model, r is common across all players i. The prior distribution assigned to r is:

$$r \sim \mathrm{Gamma}(0.01, 0.01).$$

This prior distribution was chosen because the size parameter r must be positive and governs the degree of overdispersion in the model. The Gamma distribution has support on the positive real line and gives a flexible prior for parameters that must be positive. Since $r = 1/\varphi$, finite values of r correspond to positive overdispersion relative to the Poisson model. As $r \to \infty$ (that is, as $\varphi \to 0$), the negative binomial distribution approaches the Poisson distribution.

The same linear predictor used in the Poisson model is adopted here

$$H_i^e = \exp\left\{ \Delta_i^e + \tau_i \Lambda_i^e - \tau_i^* \sum_{j=1}^{m} \Lambda_j^{E\backslash e} \delta_{ij} \right\}. \tag{3.7.3}$$

The player and team effects specified below were inspired by the code provided in

Baio and Blangiardo (2010). These are identical to those used in the Poisson model (Section 3.7.1):

$$\Delta_i \sim \text{Normal}(k_\Delta, s_\Delta),$$

$$\Lambda_j^e \sim \text{Normal}(k_{\Lambda^e}, s_{\Lambda^e}),$$

$$\Lambda_j^{E\backslash e} \sim \text{Normal}(k_{\Lambda^{E\backslash e}}, s_{\Lambda^{E\backslash e}}).$$

Moreover, sum-to-zero constraints were again imposed on the player effects and on both team effects to ensure identifiability:

$$\sum_{j=1}^{m} \Lambda_j^e = 0, \qquad \sum_{j=1}^{m} \Lambda_j^{E\backslash \mathrm{e}} = 0, \qquad \sum_{i=1}^{n} \Delta_i = 0.$$

The hyperpriors for the team and player effects were again chosen to be as weakly informative as possible, so that the parameter values would be driven mainly by the data rather than by prior assumptions:

$$k_{\Lambda^e} \sim \text{Normal}(0, 0.0001) \qquad s_{\Lambda^e} \sim \text{Gamma}(0.01, 0.01),$$

$$k_{\Lambda^{E\backslash e}} \sim \text{Normal}(0, 0.0001) \quad s_{\Lambda^{E\backslash e}} \sim \text{Gamma}(0.01, 0.01),$$

$$k_\Delta \sim \text{Normal}(0, 0.0001) \qquad s_\Delta \sim \text{Gamma}(0.1, 0.01).$$

The hyperpriors above are used as the default choice. However, if convergence problems are encountered, tighter hyperpriors may be used to represent a more informative prior.

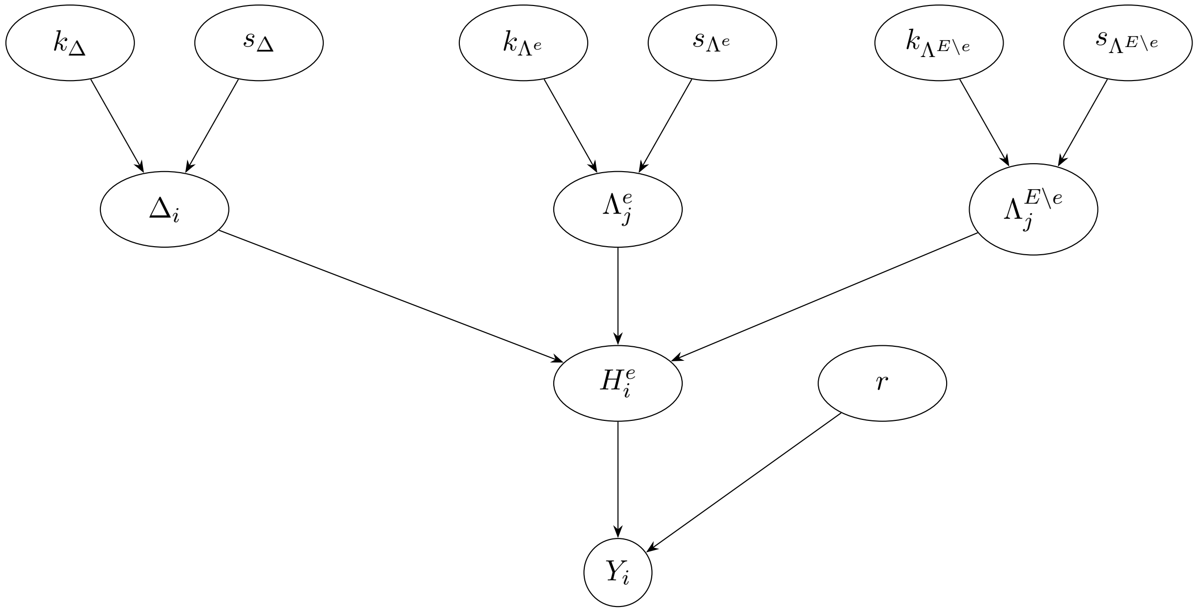

Figure 3.7.2: *Graphical representation of the Bayesian hierarchical negative binomial model.*

Similar to the Poisson model, Figure 3.7.2 shows how the hyperparameters influence the player and team abilities, which subsequently affect the linear predictor. The observed data vector $\boldsymbol{y}$ is then determined by the linear predictor together with the overdispersion parameter.

3.8 Predictive Error Measures

In predictive modelling, model performance is evaluated by calculating the difference between the actual observed values and the corresponding predicted values. In the Bayesian framework, predictions are based on the posterior predictive distribution. Predictive error measures are typically considered in model development, because a model may fit the observed data adequately yet perform poorly in terms of predictive accuracy. In this study, the predictive performance will be assessed for both the Poisson and the negative binomial models. Predictive error measures are used to assess whether the extra parameter utilised in the negative binomial model, which allows for overdispersion in the response variable, results in improved predictive accuracy.

Let y_i denote the actual value and let $\hat{y}_i(\boldsymbol{\theta}, \boldsymbol{\phi})$ denote the Bayesian predicted value for the i^{th} player, $i = 1, ..., n$, in the sample, under parameterisation $(\boldsymbol{\theta}, \boldsymbol{\phi})$. Then, the residual is defined as

$$e_i(\boldsymbol{\theta}, \boldsymbol{\phi}) = y_i - \hat{y}_i(\boldsymbol{\theta}, \boldsymbol{\phi}).$$

To assess predictive accuracy, all residuals are combined into a single numerical value that measures prediction error. Important predictive error measures used for continuous response variables include Bayesian mean absolute error (MAE) and the Bayesian root mean squared error (RMSE).

3.8.1 Bayesian Root Mean Squared Error

The root mean squared error (RMSE) measures the average magnitude of prediction errors. The errors are squared before being averaged to ensure that negative and positive values do not cancel each other out. The Bayesian RMSE corresponds to the quadratic loss function, meaning that squaring the errors gives more weight to larger deviations and therefore makes the measure sensitive to outliers. For this reason, Bayesian MAE is also considered in this dissertation, since it is less sensitive to outliers and therefore provides a fairer assessment when large errors are present.

The Bayesian version of the RMSE has been utilised in this dissertation and is obtained as follows:

$$RMSE_{Bayes} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\frac{1}{L}\sum_{k=1}^{L} e_k(\boldsymbol{\theta}_k, \boldsymbol{\phi}_k)^2} \tag{3.8.1}$$

where:

- $(\boldsymbol{\theta}_k, \boldsymbol{\phi}_k) \sim p(\boldsymbol{\theta}, \boldsymbol{\phi} \mid y)$
- L is the post burn-in sample size of the chain
- n: total number of observations/data points
- i: index of each observation $(i = 1, \ldots, n)$

- k: index for each MCMC sample.

The RMSE is computed for each player and team effect across the three MCMC chains used in Section 4 of this dissertation. The model with the lowest RMSE is considered to have better predictive accuracy, as it indicates smaller deviations between predicted and observed values. For each iteration of the posterior chains, predicted values are generated and the RMSE is computed for all observations.

3.8.2 Bayesian Mean Absolute Error

The mean absolute error (MAE) is an important statistical metric that measures the average magnitude of errors between predicted and observed values. The Bayesian mean absolute error is defined in the same way as the Bayesian root mean squared error, except that the squared error loss is replaced by the absolute error loss. It is defined as:

$$\frac{1}{n}\sum_{i=1}^{n}\frac{1}{L}\sum_{k=1}^{L}|e_k(\boldsymbol{\theta}_k, \boldsymbol{\phi}_k)| \tag{3.8.2}$$

After establishing the theoretical framework of Bayesian hierarchical models in the context of sports, the following section turns to the analysis of results.

Chapter 4

Modelling FIFA World Cup 2022 Data

4.1 Description of Dataset

The dataset utilised in this study is derived from the publicly available FIFA 2022 World Cup dataset on Kaggle [1]. The dataset originally consists of eleven Excel files, and three of these are considered as the most relevant to this analysis: *Players' minutes, Players' defence* and *Players' passing*. In all three Excel files, each row corresponds to one player and their nation. The original dataset contained 680 players, but the final sample was restricted to outfield players who played at least 50% of their nation's total minutes to reduce noise from limited playing time. Goalkeepers were excluded, and Ismaila Mohamad was removed because his statistical information was missing across all files in the original Kaggle dataset. As a result, the final sample was made up of 310 players.

The *Players' Minutes* file contains information about each player regarding their total playing time as well as their participation against every opponent in the World Cup. This file also includes various variables of information, which will not all be

[1] https://www.kaggle.com/datasets/swaptr/fifa-world-cup-2022-player-data

mentioned here, but the ones that will be used are described below:

- **player**: Name of the player.
- **position**: The player's primary position on the pitch, classified as:
  - **DF** – Defender
  - **MD** – Midfielder
  - **FW** – Forward
- **team**: The national team that the player represents during the tournament.
- **games**: Total number of matches the player participated in.
- **minutes**: Total number of minutes accumulated by the player during the tournament.
- **Opponent Columns**: There are 32 different columns, each corresponding to a single nation participating in the World Cup. The value in each column indicates how many matches the player played against that nation in the World Cup (0, 1 or 2). This could occur in a World Cup if two teams met in the group stage and then faced each other again at a later stage. In particular, in this dataset, '2' appears only for players who participated in both the Morocco–Croatia Group F match and the third-place playoff.

Furthermore, the *Players' defence* file includes defensive statistics for every player throughout the whole tournament. Similar to the file mentioned before, each row corresponds to an individual player, and the columns capture several important defensive actions such as tackles won in the attacking third, tackles won in the midfield third, interceptions, blocked shots and other indicators of defensive contribution. This information allows for a deeper analysis of a player's contribution when the team is out of possession. The variables that are used in this analysis found in the *Players' defence* file are outlined below:

- **minutes 90s**: Total playing time expressed in 90-minute intervals to reflect full-match duration.

- **interceptions**: The number of instances the player intercepted an opponent's pass by reading the play and stepping in to cut it out.

Finally, the *Players' passing* file contains information regarding each individual players' general passing statistics during the course of the tournament. As with the other excel files, each row corresponds to an individual player, and each column reports a measure describing how frequently the player passed the ball, if the pass was accurate and the types of passes attempted in various different areas of the pitch. Together, these metrics help show how much the player contributes to their team when in possession of the ball, both in terms of creating chances and in terms of their role in retaining possession of the ball through accurate passing. This file also includes different variables of information, which will not all be mentioned here, but the ones that will be used are described below:

- **passes completed**: Number of passes which successfully reached a teammate.

- **shots**: The number of attempts made with the intention of scoring, regardless of whether the shot was on target.

4.2 Exploratory Data Analysis

Before building the model, it is crucial to first understand the dataset and the variables involved, and to verify that the information is suitable for the analysis. In this section, some relevant summary statistics and a histogram for each football performance metric included in the analysis are presented to illustrate the type of data we are working with.

Table 4.2.1 yields the summary statistics for the number of interceptions made by midfielders and defenders in the full dataset. Evidently, interceptions are generally low-count events in this dataset. A minimum of zero shows that even though defensive players are being recorded, a player may not have any interceptions throughout the tournament. The first quartile (Q1) is 1, which means that 25% of the observations have one or fewer interceptions. The median is 3, which indicates that an average performance throughout the tournament involves around three interceptions. The mean of 3.178 is slightly higher than the median, indicating a small right-skew in the distribution. The third quartile (Q3) is 4, meaning that 75% of observations involve four or fewer interceptions. The maximum value of 14 indicates that, although interceptions are generally a low-count event, higher values can still occur depending on the player over the course of the tournament.

Statistic	Value
Minimum	0
1st Quartile (Q1)	1
Median	3
Mean	3.178
3rd Quartile (Q3)	4
Maximum	14

Table 4.2.1: *Summary statistics for interceptions among selected defenders and midfielders throughout the tournament.*

The histogram in Figure 4.2.1 illustrates the distribution of interceptions by defenders and midfielders across the dataset. Clearly, players typically record a low number of interceptions throughout the tournament, around 0-4 interceptions. The frequency declines as the number of interceptions completed increases, which creates a right-skewed distribution. Here, most players recorded low numbers of interceptions across the tournament, but a subset recorded higher totals. A small number of players reached 10 or more interceptions, with a maximum of 14, likely reflecting particularly high defensive activity over the course of the tournament.

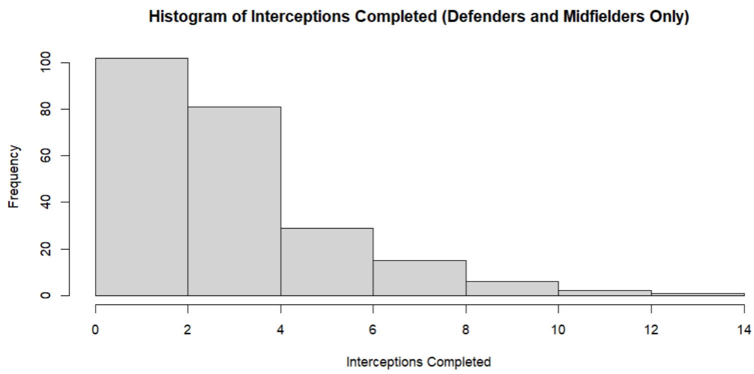

Figure 4.2.1: *Histogram showing the distribution of interceptions for selected defenders and midfielders throughout the tournament.*

Table 4.2.2 reports summary statistics for shots by midfielders and forwards, and the minimum value is 0. This indicates that some players attempted no shots throughout the tournament, which is realistic, as not every midfielder or forward is expected to attempt a shot over the course of the entire tournament. The first quartile (Q1) is 2, meaning that 25% of players attempted two or fewer shots. The median is 3, indicating that a typical player attempts around 3 shots over the course of a tournament. The mean is 4.703, which contributes to a right-skewed distribution since the mean is larger than the median. This suggests that a small number of observations with high shot counts increase the average. The third quartile (Q3) is 7, meaning that 75% of observations attempted seven or fewer shots. Finally, the maximum value of 29 is noticeably higher than the third quartile value of 7, indicating that although most players take 7 or fewer shots, there are some observations with remarkably high shot counts, likely reflecting players with particularly strong offensive performances.

Statistic	Value
Minimum	0
1st Quartile (Q1)	2
Median	3
Mean	4.703
3rd Quartile (Q3)	7
Maximum	29

Table 4.2.2: *Summary statistics for shots attempted among selected midfielders and forwards throughout the tournament.*

The histogram in Figure 4.2.2 shows the distribution of shots taken by attackers and midfielders. Most observations are concentrated between 0 and 10 shots taken. After this range, there is a noticeable decrease in frequency, meaning there are fewer observations with high shot count totals. This also results in a right-skewed distribution, where some players attempt a large number of shots throughout the tournament. The distribution extends to 29 shots, which suggests that while most players attempt a few shots, a small number of players have exceptionally high attacking involvement.

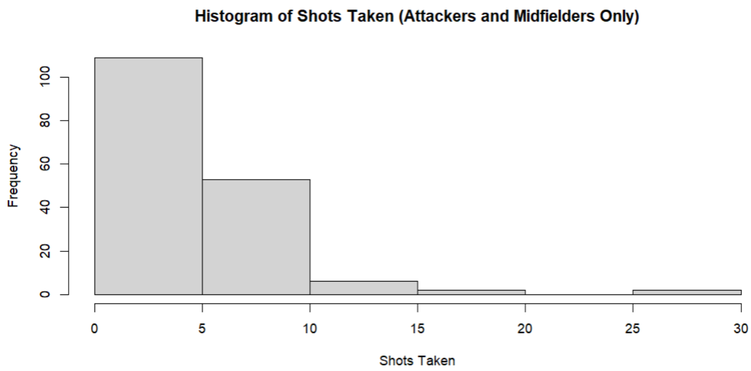

Figure 4.2.2: *Histogram showing the distribution of shots attempted by selected midfielders and forwards throughout the tournament.*

The summary statistics in Table 4.2.3 show a wide range of values. The player with the lowest number of completed passes recorded only 8, while the highest recorded total was 642, indicating considerable variation in successful passing involvement among the players. The median is 107, which is a realistic central value, and it lies below the mean (138). This indicates a right-skewed distribution, where most players record moderate passing totals while only a smaller group achieves exceptionally high counts. This observation is consistent with the histogram in Figure 4.2.3.

Statistic	Value
Minimum	8
1st Quartile (Q1)	67
Median	107.5
Mean	138.0
3rd Quartile (Q3)	172.8
Maximum	642

Table 4.2.3: *Summary statistics for passes completed among selected defenders, midfielders and forwards throughout the tournament.*

In Figure 4.2.3, the bars exhibit the highest frequency between 50 and 150 passes, which aligns well with the median value of 107 and indicates that the majority of players fall within this central range. In contrast, relatively few players complete more than 200 passes and an even smaller number exceed 400 passes. The high-passing players generate the long right tail of the distribution, and their notably large values increase the overall mean, causing it to rise above the median, which is a common occurrence in right-skewed data. Moreover, the considerable spread of the bars reflects the substantial variation in passing completion across players, illustrating the wide range observed in the summary statistics.

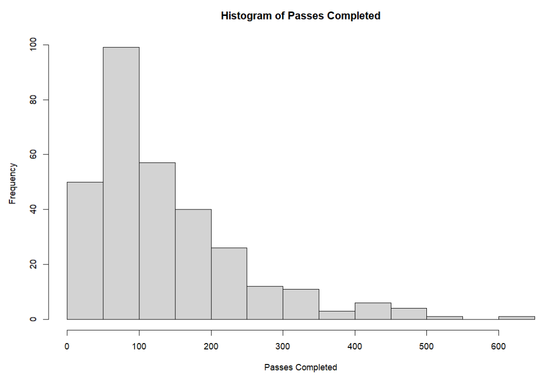

Figure 4.2.3: *Histogram showing the distribution of passes completed among selected defenders, midfielders and forwards throughout the tournament.*

The graphs discussed in this section help to show what type of data we are working with. In particular, the passing data is different when compared to interception and shot data because it is measured on a higher scale.

4.3 Player Analysis Based on Various Football Performance Metrics

Recall that the aim of this dissertation is to estimate individual player performance in the 2022 FIFA World Cup using football performance metrics. In this section, the response variable is the specific football performance metric, and covariates such as minutes played and player position are included.

In this section, statistical models are presented to evaluate players based on their latent abilities. The aim is to compare players across different positions, which tends to be challenging since players in different roles are tasked with different responsibilities within their teams. To address this, Bayesian hierarchical Poisson and negative binomial regression models are employed, where the distributions refer to the likelihood of the response variable. In particular, the number of recorded events is modelled as a count variable and is initially assumed to follow a Poisson distribution. We then assume that the number of events follows a negative binomial distribution. This change can be an improvement to the Poisson model since it does not assume equality between the variance and the mean, which may be inappropriate for count data. The Poisson model does not properly account for uncertainty in the case of overdispersion, whereas the negative binomial model introduces a dispersion parameter that allows the variance to exceed the mean. However, the Poisson model has fewer parameters than the negative binomial and is therefore more parsimonious. In cases where overdispersion is small and adequately explained by the hierarchical structure, the additional parameter included in the negative binomial specification may be unnecessary.

Since the Poisson model has fewer parameters than the negative binomial model, it generally requires less computational time to run. All models were run on an ASUS laptop, using R software (version 4.4.3), specifically using the 'rjags' package.

Computations were performed on a laptop equipped with an Intel Core i5-1035G1 processor (4 cores, 1 GHz) and 8 GB of RAM, utilizing Windows 11. Table 4.3.1 presents the computational time necessary for the MCMC estimation of the Poisson and negative binomial models, based on 10,000 iterations and 3 chains. The computational times reported in Table 4.3.1 were obtained using the laptop specifications described above.

In the model for interceptions, attackers are omitted since they rarely make interceptions, and including them would lead to an unfair representation of their performances. Similarly, defenders are omitted from models for attempted shots because they rarely attempt shots.

Table 4.3.1: *Computational time for MCMC estimation by response variable and player group, using 3 chains and 10,000 iterations.*

Response Variable	Distribution	Player Group	Time (minutes)
Interceptions	Poisson	Defenders + Midfielders	45
Interceptions	Negative binomial	Defenders + Midfielders	104
Shots	Poisson	Midfielders + Attackers	25
Shots	Negative binomial	Midfielders + Attackers	58
Passes	Poisson	Whole Dataset	61
Passes	Negative binomial	Whole Dataset	160

The two Poisson and negative binomial models are compared using Bayesian RMSE and MAE in order to determine which provides a better fit for modelling the football performance metric. In each model, the number of observed events recorded by player i (y_i) is assumed to follow a Poisson or a negative binomial distribution with mean μ_i, where the mean depends on the player's latent ability, the strength of their team and of the opponent, and the number of minutes played. These are all relevant factors affecting a player's performance.

To demonstrate convergence across the different player groups, one representative player was selected from each position group: Nicolas Tagliafico, Sofyan Amrabat, and Kylian Mbappé. Where relevant, their traceplots, empirical density function

(ECDF) plots, and MCMC diagnostic statistics are examined throughout this chapter. However, not every player is included in every model. For example, Kylian Mbappé is not analysed in the interceptions model, since attackers were not included in that dataset, while Tagliafico is not analysed in the shots model.

4.3.1 Analysing Defenders and Midfielders Based on Interceptions Made

In this subsection, only defenders and midfielders are analysed, since interceptions are primarily a defensive metric. Attackers are far less likely to record a high number of interceptions during a match. A Poisson and a negative binomial statistical model were employed and eventually compared. In this model, interception counts are modelled as a rate that depends on individual player interception ability, team intercepting ability, opponent passing quality, and minutes played. Opponent passing quality is included in the linear predictor because interceptions may be influenced by the passing ability of the opposing team. In this case, $E = \{\text{passes}, \text{interceptions}\}$, $e = \{\text{interceptions}\}$, and $E \backslash e = \{\text{passes}\}$. The linear predictor in Equation (3.7.2) is used for analysis here and in the models presented in Sections 4.3.2 and 4.3.3. The analysis is then extended to the separate Poisson and negative binomial models, which are later compared using Bayesian RMSE and MAE. Finally, the necessary tables containing the team and opponent effects, the top and bottom ten performing players based on the η_i's, and the convergence diagnostics for all players and teams are provided in Appendix Section A. The full set of player effects, including all players rather than only the top and bottom ten, is available on GitHub[2].

4.3.1.1 Poisson Model

The number of interceptions made by player i is assumed to follow a Poisson distribution. Three independent Markov chains were run in parallel to analyse convergence. Prior to the burn-in phase, an adaptation phase of 2,000 iterations was carried out to help the Markov chains move efficiently in the parameter space before actually being used for analysis. The samples generated here are not used for inference, only

[2] https://github.com/andreaspsaila9/A-Bayesian-Hierarchical-Model-of-Individual-Player-Performance-in-the-2022-FIFA-World-Cup

done for stability and better mixing of the chains. Following this, a burn-in period of 20,000 iterations was implemented to allow the Markov chains to move away from their initial values and approach the target posterior distribution. In addition, 80,000 iterations from each chain were retained for analysis. Therefore, in total we have 240,000 posterior draws across all 3 chains.

The traceplot in Figure 4.3.1 demonstrates good mixing between chains. The chains fluctuate around a mean level of around 0.2 and evidently, there is no visible trend. Moreover, the empirical density function (ECDF) plot is extremely stable and shows no irregularities. The bell-shaped curve reaches a peak at around 0.2, which shows that Tagliafico's estimated average interception rate is slightly higher than that of the average player on his team.

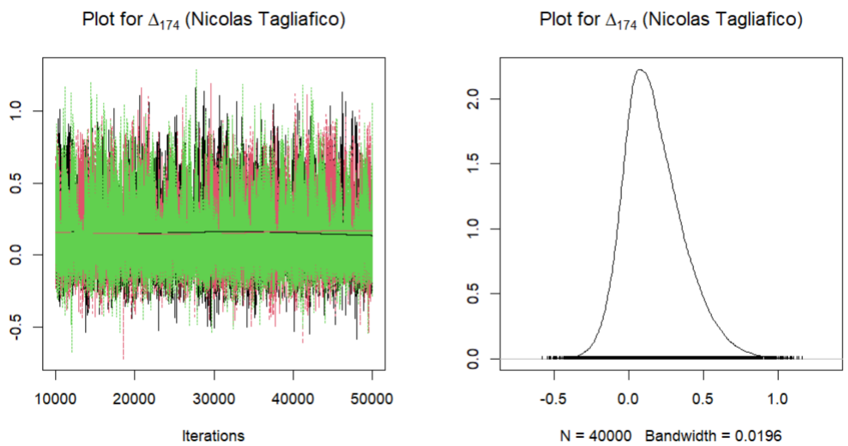

Figure 4.3.1: *Traceplot and ECDF plot for Δ_{174} (Nicolas Tagliafico) under the Poisson model for interceptions.*

The Gelman-Rubin $\hat{R}$ statistic for Tagliafico's parameter is equal to 1, which indicates excellent convergence, since it is less than 1.01. The Geweke diagnostic yields Z-scores of 1.25, -1.06, and -0.36 for the three chains, all of which are within the $(-1.96, 1.96)$ interval, indicating that convergence has likely been reached. An effective sample size of 3598.02 implies low autocorrelation between successive draws. Finally, the Heidelberger-Welch test provides p-values of 0.58, 0.13, and 0.07, which are all above the 0.05 threshold; hence, there is no evidence against convergence. Moreover, the posterior means of 0.17, with half-width values of 0.01 for the three chains respectively, are satisfactory since the ratio of the half-width to the posterior mean is less than 0.2. This implies convergence and reliable posterior estimates.

The traceplot for Sofyan Amrabat in Figure 4.3.2 also seems to show good mixing and convergence. Once again, there are no trends or drifts over iterations. The chains fluctuate around a mean level of around 0. The empirical density function (ECDF) plot seems stable and centered around 0, implying that Sofyan's estimated interception rate is extremely close to the rate of the average player on his team.

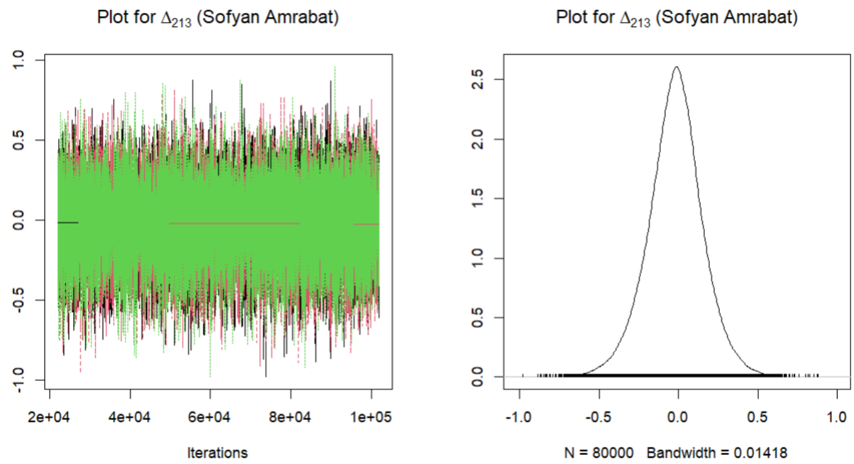

Figure 4.3.2: *Traceplot and ECDF plot for Δ_{213} (Sofyan Amrabat) under the Poisson model for interceptions.*

The Heidelberger-Welch test provides p-values of 0.71, 0.05 and 0.76. The p-values for all 3 chains are all equal to or above the 0.05 threshold, which shows that no evidence was found against convergence. The posterior means of -0.02 have corresponding half-width values of 0.002. The ratio of the half-width relative to the posterior means is less than 0.2, indicating that estimates are reliable. The Gelman-Rubin $\hat{R}$ statistic is equal to 1, from which we can conclude excellent convergence for this parameter. The Geweke diagnostic yields Z-scores of 0.56, -0.01 and -0.67. These are all within, the $(-1.96, 1.96)$ threshold; suggesting convergence for all three chains. Finally, the effective sample size of 123415.30 indicates good mixing and a large number of effectively independent posterior samples for reliable inference.

Tables 4.3.2 and 4.3.3 present the top and bottom 10 players ranked by posterior mean Δ_i estimates, where Δ_i represents the individual player effect in the Poisson model for interceptions. In the model, each player's interception count is explained by three components: an individual player effect, a team effect, and an opponent passing-related effect. The values reported in these tables correspond to our Δ_i terms, that is, the individual player effects alone, and therefore represent each player's own estimated contribution to the final predicted number of interceptions, separate from the team and opponent components.

Throughout this chapter, the top 10 tables are presented in descending order, whereas the bottom 10 tables are presented in ascending order. In addition to the posterior mean and posterior standard deviation, the lower and upper HPD bounds are included to reflect the uncertainty surrounding each estimate. This is important because relying only on the posterior mean may give an incomplete impression of a player's effect.

In Table 4.3.2, the posterior means range from 0.13 to 0.20, suggesting that these player effects are likely to be positive. This interpretation is also supported by the relatively large positive upper HPD bounds. However, since the lower HPD bounds are negative, the intervals still include zero. Therefore, although the estimated effects are probably positive, there remains uncertainty, and this should be taken into consideration when interpreting the rankings.

Table 4.3.2: *Top 10 player effects for the Poisson model for interceptions, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Declan Rice	0.20	0.21	-0.14	0.67
2	Nicolas Tagliafico	0.18	0.20	-0.15	0.65
3	Ro-Ro	0.18	0.21	-0.17	0.65
4	David Raum	0.17	0.21	-0.20	0.65
5	Jurien Timber	0.15	0.19	-0.18	0.59
6	Rodrigo Bentancur	0.15	0.20	-0.21	0.60
7	Ellyes Skhiri	0.14	0.20	-0.21	0.60
8	Jean-Charles Castelletto	0.13	0.20	-0.22	0.59
9	Josko Gvardiol	0.13	0.19	-0.20	0.54
10	Selim Amallah	0.13	0.19	-0.21	0.56

It is also worth noting that some of the players in the top ten, such as Declan

Rice (England), Nicolas Tagliafico (Argentina), Jurrien Timber (Netherlands), Josko Gvardiol (Croatia), and Selim Amallah (Morocco), played for teams that reached at least the quarter-finals, suggesting that strong individual interception effects are partly associated with teams that progressed further in the tournament. However, other players in the table, such as Ro-Ro (Qatar), David Raum (Germany), Rodrigo Bentancur (Uruguay), Ellyes Skhiri (Tunisia), and Jean-Charles Castelletto (Cameroon), represented teams that were eliminated earlier, indicating above-average individual contributions in interceptions despite playing for teams that did not progress as far in the tournament. This indicates that their rankings may reflect good individual defensive ability, despite playing for teams that did not go far in the tournament.

In Table 4.3.3, the posterior means range from −0.12 to −0.09, indicating that these player effects are more likely to be negative. However, the HPD intervals have positive upper bounds, meaning that zero is included in each interval. Therefore, the effects are likely to be negative overall, but this interpretation should be made with caution because the intervals do not rule out small positive effects.

Table 4.3.3: *Bottom 10 player effects, ranked in ascending order by posterior mean, with posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	John Stones	-0.12	0.20	-0.56	0.22
2	Yahya Attiat Allah	-0.11	0.20	-0.55	0.24
3	Matty Cash	-0.11	0.20	-0.55	0.24
4	Ricardo Rodriguez	-0.10	0.20	-0.55	0.25
5	Weston McKennie	-0.10	0.20	-0.54	0.26
6	Toby Alderweireld	-0.10	0.20	-0.54	0.27
7	Aaron Ramsey	-0.10	0.20	-0.54	0.26
8	Noussair Mazraoui	-0.09	0.19	-0.52	0.25
9	Thomas Partey	-0.09	0.20	-0.54	0.27
10	Niklas Sule	-0.09	0.20	-0.54	0.26

The team and opponent effects under the Poisson model for interceptions that contribute to the linear predictor are provided in Appendix Sections A.1.1 and A.1.2.

Appendix Sections A.1.3 and A.1.4 present the top and bottom ten players according to the η_i's, including the lower and upper HPD bounds. These tables differ from those in Appendix Sections A.1.1 and A.1.2 because they are not based only on the individual player effect. Instead, they are based on the full linear predictor from the fitted model, which combines the player effect, team effect, and opponent effect. Therefore, these values represent the model-predicted number of interceptions for each player, rather than only the player's contribution to the linear predictor. The HPD bounds are included to show the uncertainty around these predicted values and to allow comparison between the model-predicted totals and the observed totals in the data. Moreover, the convergence diagnostics for all players and team effects are provided in Appendix Sections A.1.5–A.1.13. This appendix structure is repeated for all fitted models, and the same interpretation applies in each case.

4.3.1.2 Negative Binomial Model

The number of interceptions by player i is now assumed to follow a negative binomial distribution with mean μ_i and dispersion parameter φ. This model was introduced in Section 3.7.2, and its linear predictor, presented in Equation (3.7.2), is the same as that used for the Poisson model. Three Markov chains were run to assess convergence. An initial 2,000 iterations were run to let the sampler adapt to the posterior distribution of the model parameters and improve sampling efficiency. Next, a further 20,000 iterations were run as the burn-in phase. These iterations were also discarded and not used for inference, as their purpose was to allow the Markov chain to move away from initial values, and hence the final samples will not be affected by initial values and are more representative of the posterior distribution.

The model was run for 80,000 iterations in each chain, giving a total of 240,000 posterior draws. This provided sufficient time for the player and team effects to converge, leading to reliable estimates. Since interception counts are not too large, as seen in Section 4.2, this was considered sufficient for the type of data analysed here.

The traceplot in Figure 4.3.3 shows reasonably good mixing between chains. The chains fluctuate around a stable mean of around 0.1, while the ECDF plot reaches a peak at around 0.1. Together, these results suggest that Nicolas Tagliafico makes slightly more interceptions than the average player on his team. The ECDF plot seems to be unimodal, and the absence of drifts or trends in the traceplot indicates that the chains are exploring the target distribution effectively.

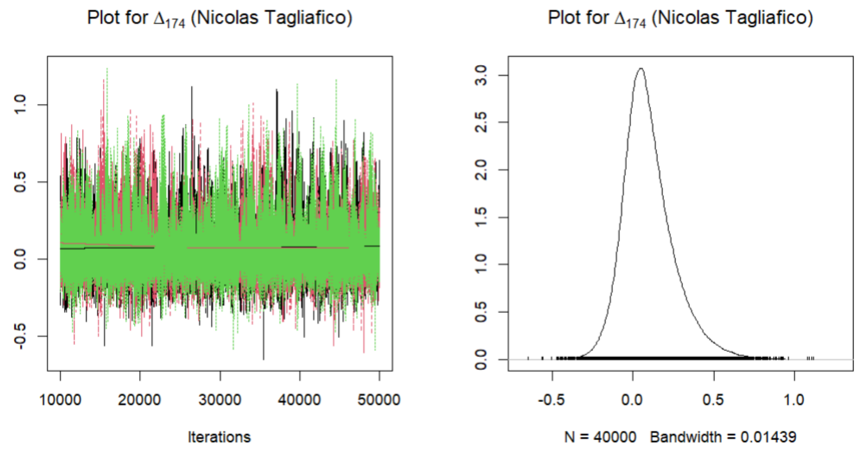

Figure 4.3.3: *Traceplot and ECDF plot for Δ_{174} (Nicolas Tagliafico) under the negative binomial model for interceptions.*

The Heidelberger-Welch test provides a p-value for each chain and these are 0.41, 0.16 and 0.88, therefore convergence is supported since these are all greater than 0.05. The posterior means of 0.09, 0.09 and 0.10 together with half-width values of 0.01 indicate that the estimates are reliable, since the ratio of each half-width value to the respective posterior mean values is less than 0.2. A Gelman-Rubin $\hat{R}$ statistic of 1 indicates excellent convergence. Furthermore, the three Z-scores obtained from the Geweke diagnostic are 0.00, 2.39 and -0.64. Although the second value lies slightly outside the $(-1.96, 1.96)$ threshold, it is not far from the cutoff. This suggests that convergence for the second chain may be slightly weaker, but not particularly poor. Finally, an effective sample size of 3932.57 is large enough to support the reliability of estimates.

The traceplot in Figure 4.3.4 shows acceptable mixing, which is a good indication of convergence. There are no visible drifts or trends, and the chains fluctuate around a mean of approximately 0. The ECDF plot supports this conclusion, as it is smooth and peaks at around 0. This suggests that Sofyan Amrabat makes about the same number of interceptions as the average player on his team. Overall, the smoothness of the ECDF plot and the good mixing of the chains in the traceplot indicate that the chains are successfully exploring the target distribution.

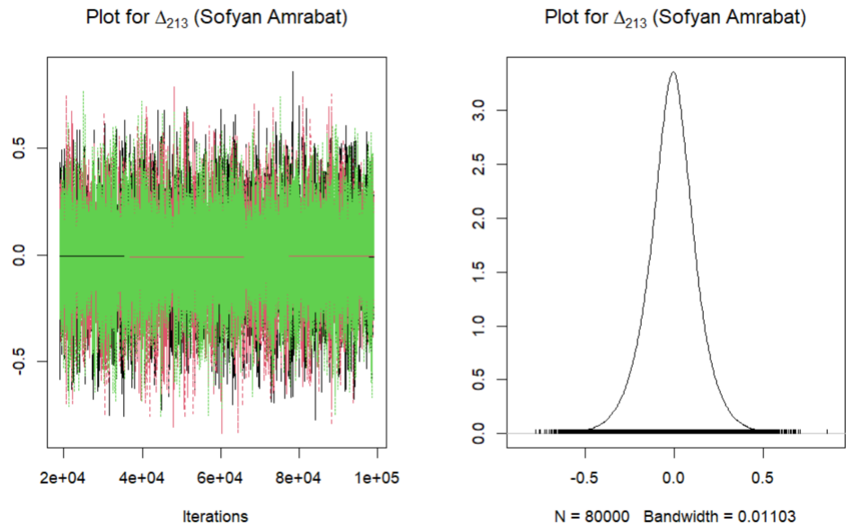

Figure 4.3.4: *Traceplot and ECDF plot for Δ_{212} (Sofyan Amrabat) under the negative binomial model for interceptions.*

The Heidelberger-Welch test provides three p-values, one for each chain, all of which are above the 0.05 threshold (0.61, 0.18 and 0.25). Therefore, convergence is supported according to this test. The posterior means for the three chains are -0.01, with corresponding half-width values of 0.001. The ratio of the half-width values to the posterior means is less than 0.2, which implies that estimates are considered to be reliable according to this test. The Gelman-Rubin $\hat{R}$ statistic is 1 which indicates excellent convergence. The Geweke diagnostic produces Z-scores of 1.61, 0.14 and 0.01. Since all values lie within the interval $(-1.96, 1.96)$, all chains satisfy the threshold, and convergence is supported. Finally, the effective sample size is 122967.30, which is sufficiently large since it implies that the amount of information obtained is equal to approximately 122967 independent samples. All convergence diagnostics for this model, including those for all player and team effects, are provided in Appendix Sections A.2.5–A.2.13.

Tables 4.3.4 and 4.3.5 present the top and bottom 10 players ranked by posterior mean Δ_i estimates under the negative binomial model for interceptions. In Table 4.3.4, the posterior means range from 0.07 to 0.11, suggesting that these player effects are likely to be positive. This interpretation is also supported by the positive upper

HPD bounds, which range from 0.41 to 0.52. However, since the lower HPD bounds are negative, the intervals still include zero. Therefore, although the estimated effects are probably positive, there remains uncertainty, and this should be taken into consideration when interpreting the rankings.

Table 4.3.4: *Top 10 player effects for the negative binomial model for interceptions, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Declan Rice	0.11	0.17	-0.16	0.52
2	Nicolas Tagliafico	0.10	0.16	-0.17	0.49
3	Ro-Ro	0.10	0.17	-0.17	0.50
4	David Raum	0.10	0.17	-0.18	0.49
5	Rodrigo Bentancur	0.08	0.16	-0.19	0.45
6	Jurrien Timber	0.08	0.16	-0.19	0.45
7	Ellyes Skhiri	0.08	0.16	-0.20	0.45
8	Jean-Charles Castelletto	0.08	0.16	-0.21	0.45
9	Nikola Milenković	0.07	0.16	-0.21	0.44
10	Selim Amallah	0.07	0.15	-0.20	0.41

Tables 4.3.2 and 4.3.4 are extremely similar in terms of the players included. The main differences are that Jurrien Timber and Rodrigo Bentancur switch rankings, although their estimated effects remain similar and are equal when rounded to two decimal places. In Table 4.3.4, Nikola Milenković appears in the top ten, while Joško Gvardiol drops out.

The same general group of players appears in both bottom ten tables (Tables 4.3.3 and 4.3.5), and the posterior mean values of Δ_i are also extremely similar. The main differences are that Ricardo Rodriguez and Matty Cash switch rankings, although their estimates are equal when rounded to two decimal places. Aaron Ramsey now has a higher Δ_i value than Weston McKennie and Toby Alderweireld. In addition, Steven Vitoria and Antonio Rudiger enter the bottom ten, replacing Noussair Mazraoui and Niklas Sule.

Table 4.3.5: *Bottom 10 player effects for the negative binomial model for interceptions, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	John Stones	-0.07	0.16	-0.43	0.21
2	Yahya Attiat Allah	-0.06	0.16	-0.42	0.22
3	Ricardo Rodriguez	-0.06	0.16	-0.41	0.23
4	Matty Cash	-0.06	0.16	-0.42	0.23
5	Aaron Ramsey	-0.06	0.16	-0.41	0.23
6	Weston McKennie	-0.05	0.16	-0.41	0.23
7	Toby Alderweireld	-0.05	0.16	-0.41	0.23
8	Thomas Partey	-0.05	0.16	-0.41	0.24
9	Steven Vitoria	-0.05	0.16	-0.40	0.23
10	Antonio Rudiger	-0.05	0.16	-0.41	0.24

The team and opponent effects under the negative binomial model for interceptions, which contribute to the linear predictor, are provided in Appendix Sections A.2.1 and A.2.2. Appendix Sections A.2.3 and A.2.4 present the top and bottom ten players according to the η_i's, including the lower and upper HPD bounds. These tables differ from those in Appendix Sections A.2.1 and A.2.2, since they are not based only on the individual player effect.

4.3.1.3 Comparison Of The Poisson And Negative Binomial Models

The Bayesian RMSE for the Poisson model for interceptions, computed from the posterior predictive distribution, is 1.68, while the corresponding Bayesian MAE is 1.37. For the negative binomial model for interceptions, the Bayesian RMSE is 1.78 and the Bayesian MAE is 1.45. The Bayesian RMSE and MAE were introduced theoretically in Sections 3.8.1 and 3.8.2, respectively. These results show that both models provide reasonably accurate predictions, since both RMSE and MAE are low. In other words, the predicted number of interceptions is close to the observed number of interceptions.

The Poisson model seems to perform slightly better than the negative binomial model, since it has lower values of both RMSE and MAE. However, the difference is small, with the RMSE increasing by only 0.10 and the MAE increasing by only 0.08 under the negative binomial model. This suggests that, although the Poisson model provides a slightly better fit, the two models perform extremely similarly in practice.

A possible reason why the Poisson model performs slightly better is that the interceptions data may contain little overdispersion. In that case, the dispersion parameter in the negative binomial model does not provide any major benefit, and the simpler Poisson model may be better suited to the data, offering better estimates and therefore slightly more accurate predictive performance.

4.3.2 Analysing Midfielders and Forwards Based on Shots Completed

In this section, a subset of the dataset consisting of midfielders and forwards is analysed using an attacking football performance metric, namely shots. Since the aim of this dissertation is to evaluate player quality through selected football performance metrics, this analysis extends the earlier examination of interceptions by focusing on a more attacking contribution. By considering shots as an offensive player statistic, the study attempts to determine how effectively the proposed modelling framework captures differences in attacking contribution among midfielders and forwards. This study adopts the perspective that a higher number of shots indicates a player's greater ability to position themselves in opportunities to strike the ball. Midfielders, who also contribute to the attacking contribution of their team, are therefore included in the analysis of shots. In this case, $E = \{\text{shots}, \text{blocked shots}\}$, $e = \{\text{shots}\}$, and $E \setminus e = \{\text{blocked shots}\}$. The linear predictor given in Equation (3.7.2) is used here again. Finally, the tables containing the team and opponent effects, the top and bottom ten performing players according to the η_i's, and the convergence diagnostics for all players and teams are provided in Appendix Section B.

4.3.2.1 Poisson Model

Firstly, the number of shots by player i is assumed to follow a Poisson distribution. Three independent Markov chains were run to assess convergence of the posterior distributions and an adaptation phase of 2,000 iterations was chosen. A burn-in phase of 17,000 iterations is chosen, which gives enough time for the chains to converge to the stationary posterior distribution. In the actual sampling phase, 70,000 iterations per chain are drawn from the posterior distribution. Therefore, there are 210,000 posterior samples across the three chains.

Figure 4.3.5 displays the traceplot and the ECDF plot for Sofyan Amrabat. The traceplot indicates good convergence, since the three chains seem to overlap and mix well with no patterns, trends, or separation between chains. The samples in the traceplot fluctuate around a stable mean level of -0.9 which suggests that Sofyan Amrabat's estimated shot rate is below the average player's on his team.. The empirical distribution function (ECDF) plot is smooth and centered around a value of -0.9.

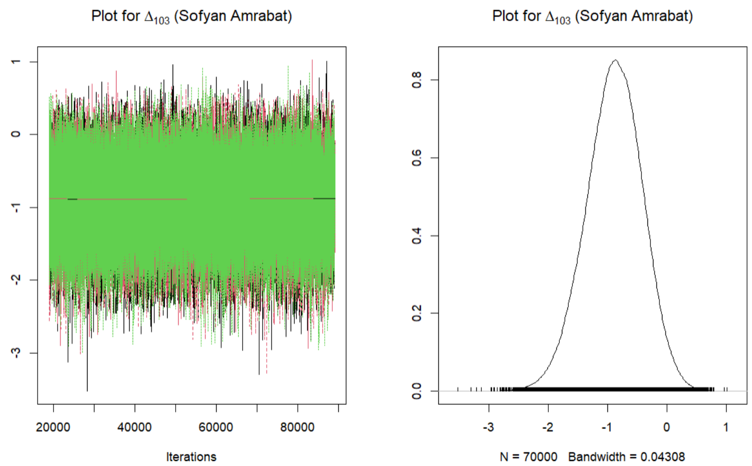

Figure 4.3.5: *Traceplot and ECDF plot for Δ_{103} (Sofyan Amrabat) under the Poisson model for shots attempted.*

The Heidelberger–Welch test further supports convergence, as the p-values of 0.72, 0.68, and 0.50 are all above the 0.05 threshold. In addition, the half-width test is satisfied, with posterior means of -0.89 corresponding to half-width values of approximately 0.01. The ratio of the half-width to the posterior mean is therefore less than 0.2 for all three chains. This suggests that convergence has been achieved throughout and the posterior estimates are reliable. The Gelman–Rubin $\hat{R}$ statistic is equal to 1, which is less than 1.01, indicating excellent convergence for this parameter. Furthermore, the Geweke diagnostic reported Z-scores of -0.27, 0.11 and 0.19. The three chains have values within the $(-1.96, 1.96)$ threshold, and hence convergence is also supported through the Geweke diagnostic. Finally, an effective sample size of 69487.64 is extremely large, which implies low autocorrelation and precise estimation. All convergence diagnostics for this model, including those for all player and team effects, are provided in Appendix Sections B.1.5–B.1.11.

Similarly to Figure 4.3.5, the traceplot for the parameter of Kylian Mbappé in Figure 4.3.6 shows good mixing of chains without any separation or patterns. This suggests that the stationary distribution has likely been reached. Furthermore, the samples seem to fluctuate around a stable level, roughly around 0.8. This suggests that Mbappé attempts more shots than the average player on his team.

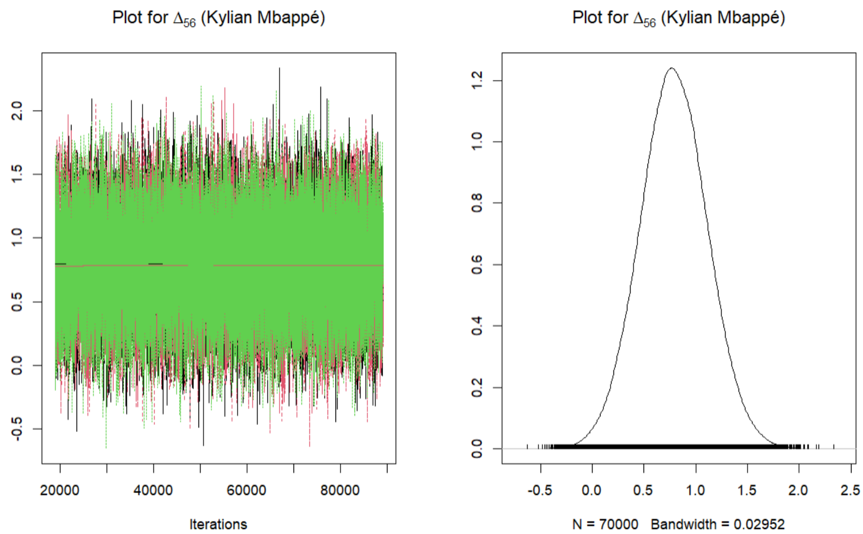

Figure 4.3.6: *Traceplot and ECDF plot for Δ_{56} (Kylian Mbappe) under the Poisson model for shots completed.*

The Empirical distribution function plot (ECDF) is unimodal which indicates stable sampling. The distribution is centred around a positive value, which indicates that Kylian Mbappé's estimated shot rate is above the average player on his team.

The Heidelberger-Welch test yields adequate p-values across all three chains, with p-values of 0.53, 0.81 and 0.63. The half-width values are approximately 0.01 for all chains, with corresponding posterior means of 0.8. The resulting ratio of the half-width to the posterior mean is therefore less than 0.2, suggesting that posterior estimates are reliable and convergence has likely been reached for all three chains. The Gelman-Rubin $\hat{R}$ statistic is equal to 1, indicating excellent convergence for this parameter. Furthermore, an effective sample size of 19733.00 suggests excellent mixing of the Markov chains and provides a large number of effectively independent posterior samples. Finally, the Geweke diagnostic provides Z-scores of 0.13, 0.16, and 0.58, all of which lie within the interval $(-1.96, 1.96)$, further supporting convergence of chains.

Tables 4.3.6 and 4.3.7 present the top and bottom ten players ranked by their posterior mean Δ_i estimates under the Poisson model for shots. In Table 4.3.6, the posterior means range from 0.76 to 1.07, suggesting that these player effects are highly likely to be positive. This interpretation is further supported by the fact that most of the lower HPD bounds are also positive, indicating that the model provides strong evidence that these players have positive effects on the expected number of shots.

It is also useful to consider the teams represented by the players in Table 4.3.6. Some of the players, such as Marco Asensio (Spain), Memphis (Netherlands), Alvaro Morata (Spain), and Kylian Mbappe (France), played for teams that reached at least the round of 16. This suggests that high individual shot effects are partly associated with players from stronger teams. However, other players in the table, such as Youssef Msakni (Tunisia), Aleksandar Mitrovic (Serbia), Alexis Vega (Mexico), Mohammed Kudus (Ghana), Vincent Aboubakar (Cameroon), and Jonathan David (Canada), represented teams that were eliminated in the group stage. Their inclusion in the top ten is therefore important, as they were still estimated to have strong positive individual effects despite playing for teams that did not progress beyond the group stage. This indicates that their rankings may reflect strong individual attacking ability, rather than simply reflecting the advantage of playing in stronger teams.

Table 4.3.6: *Top 10 player effects for the Poisson model for shots, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Youssef Msakni	1.07	0.38	0.31	1.81
2	Marco Asensio	0.94	0.35	0.24	1.63
3	Aleksandar Mitrovic	0.92	0.38	0.16	1.66
4	Alexis Vega	0.85	0.37	0.12	1.57
5	Memphis	0.85	0.37	0.14	1.57
6	Alvaro Morata	0.84	0.37	0.11	1.57
7	Kylian Mbappe	0.79	0.32	0.16	1.44
8	Kudus Mohammed	0.79	0.39	0.02	1.55
9	Vincent Aboubakar	0.78	0.42	-0.05	1.59
10	Jonathan David	0.76	0.38	0.02	1.50

In Table 4.3.7, all posterior means are negative, suggesting that the model estimates lower shot rates for these players, even after accounting for team and opponent effects. Although the upper HPD bounds are positive, meaning that there is still some uncertainty, the lower HPD bounds are strongly negative, ranging approximately from -1.92 to -1.51. Therefore, these player effects are likely to be negative overall, but this interpretation should still be made with some caution since the HPD intervals do not completely rule out small positive effects.

It is also important to interpret these effects in relation to the players' roles and teams. Many of the players in Table 4.3.7 are midfielders whose main responsibilities are not necessarily related to shooting. Therefore, their negative shot effects may reflect their tactical roles rather than poor overall performance. This is supported by the fact that several of them played for teams that progressed beyond the group stage, including England, Morocco, Portugal, Croatia, Spain, Japan, the United States, and South Korea. On the other hand, players such as Hector Herrera (Mexico) and Celso Borges (Costa Rica) both were eliminated in the group stage, so weaker team performance may also partly explain their lower shot effects. However, the main interpretation remains that, for midfielders, shooting is usually not the main responsibility, and a negative shot effect should therefore be understood mainly as lower involvement in shooting rather than as a negative overall contribution.

Table 4.3.7: *Bottom 10 player effects for the Poisson model for shots, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Pedri	-0.92	0.50	-1.92	0.05
2	Declan Rice	-0.91	0.50	-1.92	0.05
3	Sofyan Amrabat	-0.89	0.47	-1.82	0.03
4	Bernardo Silva	-0.75	0.47	-1.70	0.16
5	Yuto Nagatomo	-0.66	0.52	-1.69	0.34
6	Tyler Adams	-0.66	0.48	-1.62	0.28
7	Mateo Kovacic	-0.65	0.43	-1.51	0.19
8	Hector Herrera	-0.64	0.52	-1.71	0.34
9	Celso Borges	-0.62	0.54	-1.68	0.42
10	Jung Woo-young	-0.62	0.48	-1.59	0.30

The team and opponent effects under the Poisson model for shots, which contribute to the linear predictor, are provided in Appendix Sections B.1.1 and B.1.2. Appendix Sections B.1.3 and B.1.4 present the top and bottom ten players according to the η_i's, including the lower and upper HPD bounds.

4.3.2.2 Negative Binomial Model

The negative binomial model for the number of shots by player i assumes that player i follows a negative binomial distribution with mean μ_i and dispersion parameter φ. The first 2,000 iterations are used as the adaptation phase and the following 17,000 iterations are used for the burn-in phase. After discarding the burn-in phase, we proceed to the sampling stage, where 70,000 iterations are retained for analysis. Across the three chains, 210,000 posterior draws are obtained, and these are sufficient for reliable results.

In Figure 4.3.7, the traceplot and ECDF plot for Sofyan Amrabat are visualized. The three chains in the traceplot mix well and fluctuate around a stable mean of -0.9, with no visible trends or drifts, suggesting no evidence of non-convergence.

The smooth unimodal ECDF plot supports this conclusion and peaks around -0.9, indicating that Amrabat attempts fewer shots than his average teammate.

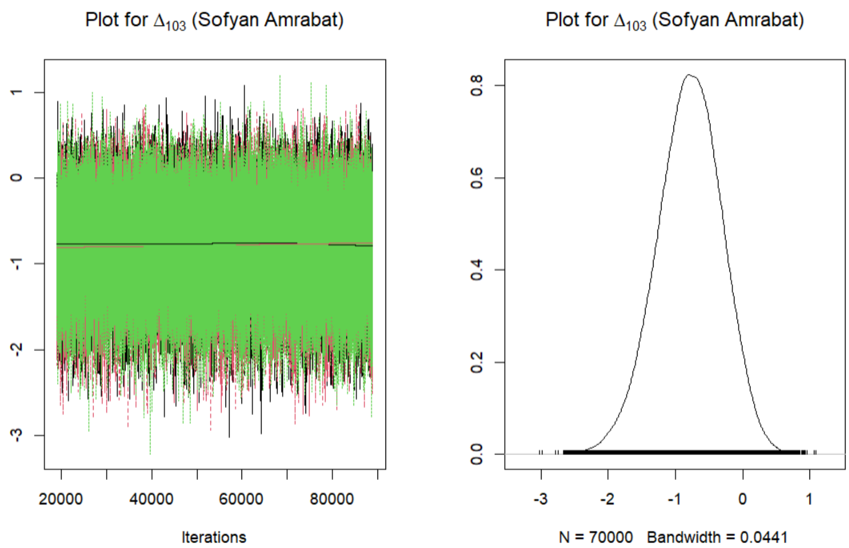

Figure 4.3.7: *Traceplot and ECDF plot for Δ_{103} (Sofyan Amrabat) under the negative binomial model for shots.*

According to the Heidelberger-Welch test, convergence is supported since the p-values of 0.59, 0.14, and 0.09 are all above the 0.05 threshold. Moreover, the posterior means of approximately -0.8 for all three chains with corresponding half-width values of 0.01 for all three chains show that we have reliable estimates. Since the ratio of the half-width value to the posterior means is less than 0.2 for all three chains, estimates are sufficiently precise. The Gelman-Rubin $\hat{R}$ is equal to 1, which indicates excellent convergence for this player effect. The Geweke diagnostic values for the three chains are -1.15, -3.83, and -0.06. Since the second value lies outside the $(-1.96, 1.96)$ interval, convergence is supported for the first and third chains but not for the second. Finally, the effective sample size of 30722.32 shows that the amount of information obtained is equal to about 30722 independent samples, which is a large value for reliable inference. All convergence diagnostics for this model, including those for all player and team effects, are provided in Appendix Sections B.2.5–B.2.11.

Figure 4.3.8 displays the traceplot and ECDF plot for Kylian Mbappé under the negative binomial model for shots. The traceplot shows that the three chains are mixing well and fluctuate around a mean value of around 0.7. There are no drifts, hence there is no evidence of non-convergence. The ECDF plot also appears consistent and smooth, with the distribution reaching a peak at around 0.7, which is consistent with the traceplot. We may therefore conclude that, according to the model, Mbappé takes shots more often than the average player on his team.

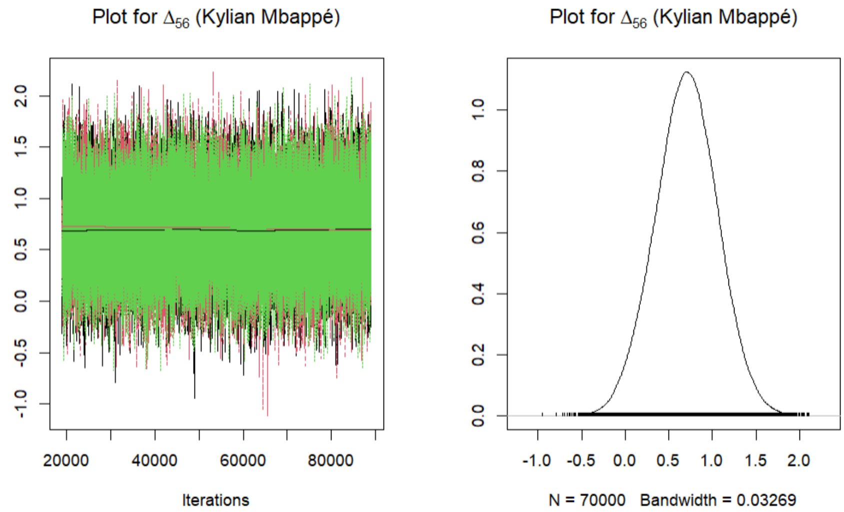

Figure 4.3.8: *Traceplot and ECDF plot for Δ_{56} (Kylian Mbappé) under the negative binomial model for shots.*

The Gelman-Rubin $\hat{R}$ statistic is 1, which is evidence of excellent convergence. The p-values of the Heidelberger-Welch test are 0.58, 0.09 and 0.32, respectively, for the three chains, each of which is bigger than the 0.05 threshold, which is further evidence of convergence. The posterior means are all approximately equal to 0.7, and the respective half-width values are all approximately equal to 0.01. The ratio of the half-width to the posterior mean is less than 0.2, which suggests that convergence has been reached and estimates are reliable. Furthermore, the Geweke diagnostic values for the three chains are 0.35, 2.35, and 0.77. Since the second value lies just outside the $(-1.96, 1.96)$ interval, this may indicate mild convergence issues, so the result should be interpreted with caution. The effective sample size of 18104.2 implies that the amount of real information achieved from the model is approximately equal to 18104 independent samples, which is large enough for reliable inference.

Table 4.3.8 is extremely similar to Table 4.3.6. The same players appear in both tables and in the same order, except for Kylian Mbappe and Mohammed Kudus, who switch rankings. However, their posterior mean estimates remain extremely similar across both models.

Table 4.3.8: *Top 10 player effects for the negative binomial model for shots, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Youssef Msakni	0.96	0.41	0.14	1.76
2	Marco Asensio	0.84	0.38	0.07	1.58
3	Aleksandar Mitrovic	0.82	0.41	0.02	1.60
4	Alexis Vega	0.76	0.39	-0.03	1.51
5	Memphis	0.76	0.39	-0.02	1.51
6	Alvaro Morata	0.75	0.40	-0.04	1.52
7	Kudus Mohammed	0.71	0.41	-0.10	1.51
8	Kylian Mbappe	0.70	0.36	-0.01	1.40
9	Vincent Aboubakar	0.70	0.43	-0.13	1.55
10	Jonathan David	0.69	0.40	-0.10	1.46

The players listed in Table 4.3.7 also appear in Table 4.3.9. The only differences in the ordering are that Tyler Adams and Mateo Kovacic switch rankings, while Jung Woo-young moves ahead of Celso Borges. However, when rounded to two decimal places, their posterior mean estimates are the same.

Table 4.3.9: *Bottom 10 player effects for the negative binomial model for shots, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Pedri	-0.83	0.50	-1.82	0.15
2	Declan Rice	-0.81	0.51	-1.81	0.17
3	Sofyan Amrabat	-0.79	0.48	-1.73	0.16
4	Bernardo Silva	-0.67	0.48	-1.62	0.24
5	Yuto Nagatomo	-0.60	0.52	-1.62	0.39
6	Mateo Kovacic	-0.59	0.44	-1.41	0.26
7	Tyler Adams	-0.58	0.48	-1.53	0.35
8	Hector Herrera	-0.58	0.51	-1.59	0.41
9	Jung Woo-young	-0.55	0.48	-1.51	0.36
10	Celso Borges	-0.55	0.52	-1.60	0.44

The team and opponent effects under the negative binomial model for shots, which contribute to the linear predictor, are provided in Appendix Sections B.2.1 and B.2.2. Appendix Sections B.2.3 and B.2.4 present the top and bottom ten players according to the η_i's, including the lower and upper HPD bounds.

4.3.2.3 Comparison Of The Poisson And Negative Binomial Model

The Bayesian RMSE for the negative binomial model for shots is 1.41, whereas the Bayesian MAE for the same model is 1.12. On the other hand, the Bayesian RMSE for the Poisson model for shots is 1.18, whereas the Bayesian MAE is 0.96. These values suggest that both models give reasonably accurate predictions, since the squared differences between observed values and predicted values are particularly similar.

However, since the Poisson model for shots has lower Bayesian RMSE and MAE values, it is considered the better model in this case. This model shows smaller overall prediction errors, as indicated by the Bayesian RMSE, and a smaller average absolute prediction error, as reflected by the Bayesian MAE.

4.3.3 Analysing Players Based on Passes Completed

After examining two football performance metrics focusing on defenders and attackers, the analysis now turns to the full dataset by considering passing, a football performance metric that is strongly associated with midfielders, but nevertheless important for all playing positions. This analysis examines all players in terms of their passing quality. Passing is a fundamental ability in football, as players in every position are expected to have the quality to retain possession for their team. As in the previous analyses, both a Poisson and a negative binomial model are fitted. In this section, $E = \{\text{passes}, \text{interceptions}\}$, $e = \{\text{passes}\}$, and $E \setminus e = \{\text{interceptions}\}$.

In this section, the negative binomial model is discussed before the Poisson model, since convergence was achieved for almost all parameters, with only a small number being close to convergence. In contrast, the Poisson model showed poor convergence, possibly due to the presence of overdispersion in the data. The linear predictor given in Equation (3.7.2) is used here again. Finally, Appendix Section C includes all relevant results for this model.

4.3.3.1 Main Model - Negative Binomial

Here, the number of completed passes is assumed to follow a negative binomial distribution with mean $\mu_i = \eta_i \tau_i$ and dispersion parameter φ. Three Markov chains were run to assess convergence. An adaptation phase of 2,000 iterations was used, while a burn-in period of 20,000 iterations allowed the chain sufficient time to move from its initial values to the stationary posterior distribution.

Using the same weakly informative priors as in the interception and shot models, convergence was not achieved in this model, even with a large number of iterations. A likely reason for this, as shown in Section 4.2, is that passes have an average of 138 per match per player, whereas interceptions have an average of 3.2 and shots have an average of 4.7. Moreover, the passing data involves more parameters and a larger number of players to estimate than the interception and shot data. Another possible reason for lack of convergence is that the passing model may struggle to distinguish between player and team skill. While the model is able to separate these effects more successfully for interceptions and shots, this distinction can be much more complex for passes since players in high-possession teams tend to pass more and players in low-possession teams tend to pass less. This does not mean that the model is misspecified, but the player-within-team hierarchical structure is much less suitable for passing data, because passing is much more team-dependent than interceptions and shots. In this case, the lack of convergence may not necessarily be resolved by adding more iterations but may instead reflect a limitation of the modelling approach for passing data.

To address this issue, the priors were tightened to reduce excessive movement in the chains. This increases the influence of the priors on the posterior distribution; however, the model may yield more reliable estimates if convergence is improved. Under this updated specification, 50,000 iterations were initially retained, giving a total of 150,000 posterior draws. Convergence had still not been achieved for all parameters, so the chains were extended by a further 30,000 iterations per chain. Since convergence was still not fully reached, the chains were extended once more by an additional 40,000 iterations. Even after these extensions, full convergence was not achieved.

The posterior distribution was not fully explored, and this problem persisted even after tightening the priors. This suggests that the data may not be sufficient for this type of model, as increasing the number of iterations and using tighter priors did not resolve the issue. In this dataset, as is typical in football passing event data, the observed variability in event counts is greater than the mean, indicating the presence of overdispersion. The negative binomial model gives more reliable results since it accounts for overdispersion through the parameter φ. This parameter reduces the need for player and team effects to inflate in order to explain the extra variability.

Convergence diagnostics indicate that not all player and team effects have converged in a satisfactory manner. Starting from the Gelman-Rubin statistic, the worst $\hat{R}$ value across all player and team effects is 1.08, indicating that convergence had not yet been reached for all parameters. Although the value of 1.08 is not far above the threshold of 1.05, the resulting estimates should still be interpreted with some caution since their reliability may be compromised. This conclusion is further supported by the effective sample size (ESS) diagnostics. The worst mixed parameter has an effective sample size of 79, indicating an extremely small number of effectively independent draws. Overall, these results point to low MCMC efficiency and provide evidence that convergence has not been fully achieved for all parameters. Despite this, we now focus on the three representative players and analyse their individual convergence diagnostics in more detail.

The Gelman-Rubin statistic for Nicolas Tagliafico results to 1 which indicates excellent convergence. The effective sample size for Nicolas Tagliafico results to 16471.54 which indicates low autocorrelation amd efficient sampling for this parameter. The Heidelberger–Welch stationarity test returned p-values of 0.02, 0.00 and 0.04, all less than 0.05, so there is evidence against stationarity. The Geweke diagnostic for the three chains yields Z-scores of -0.58, -1.5 and 3. The first 2 chains have Z-scores within the $(-1.96, 1.96)$ interval, indicating no evidence against stationarity. The Z-score for the third chain falls outside the interval, suggesting evidence against stationarity. The traceplot and empirical posterior density function shown in Figure 4.3.6 display good mixing and no visible trends. Some diagnostics suggest that convergence may not have been fully achieved.

This model yields a Bayesian RMSE of 115443.6 and an MAE of 7264. These values indicate an extremely poor model fit and suggest that the resulting estimates are highly unreliable and unstable. This is not unexpected, given that convergence was not fully achieved. Both the Bayesian RMSE and MAE are large, showing that the model predictions are far from the observed values. Therefore, this model does not describe the passing data well, and the results should be interpreted with great caution.

However, the plots in figures 4.3.9, 4.3.10 and 4.3.11 show good mixing across the chains and no visible trends or drift, suggesting that convergence has been achieved for these parameters on the basis of visual diagnostics.

The traceplot in Figure 4.3.9 appears to fluctuate around a stable mean of approximately -0.6, while the ECDF plot is also centered around this value. This suggests that, according to this analysis, Tagliafico makes fewer successful passes than expected within the context of his team.

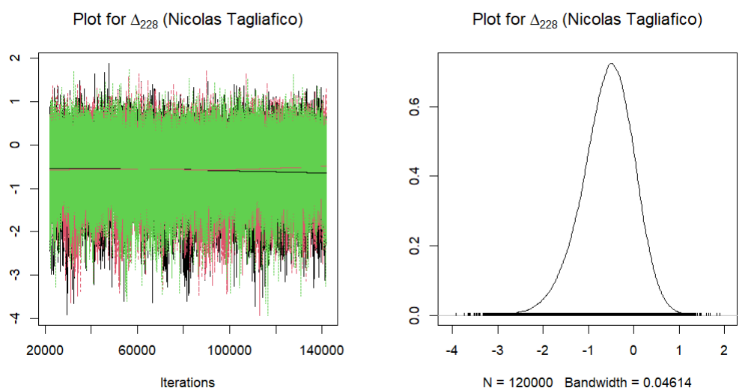

Figure 4.3.9: *Traceplot and ECDF plot for Δ_{228} (Nicolas Tagliafico) under the negative binomial model for passing.*

An $\hat{R}$ value of 1 indicates excellent convergence for Sofyan Amrabat's player effect. The effective sample size of 183539.3 is a huge value, implying low autocorrelation and precise posterior estimation for this parameter. Results from the Heidelberger–Welch diagnostic further support convergence, with p-values of 0.18, 0.92 and 0.12 for the three chains, all comfortably above the 0.05 threshold. The corresponding posterior means of -0.09, -0.08 and -0.08 together with the small half-width values of 0.004 suggest low Monte Carlo error and accurate posterior estimates. In addition, the Geweke diagnostic produced Z-scores of 1.13, -0.18 and 0.07 all of which lie within the interval $(-1.96, 1.96)$. Visual inspection of the traceplot and empirical posterior density function in Figure 4.3.12 shows good mixing and no trends, emphasizing the conclusion that the Markov chains have converged and are sampling effectively from the posterior distribution.

The traceplot in Figure 4.3.10 appears to vary around a stable mean of approximately 0, while the ECDF plot is likewise concentrated around this value. This indicates that, according to this analysis, Amrabat's passing ability is roughly in line with what would be expected for a player in his team.

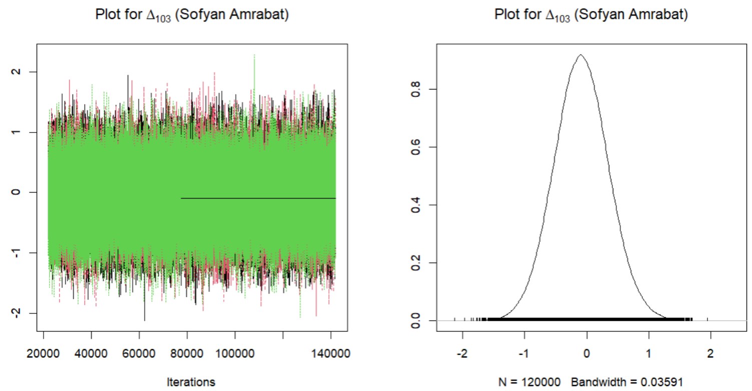

Figure 4.3.10: *Traceplot and ECDF plot for Δ_{280} (Sofyan Amrabat) under the negative binomial model for passing.*

Convergence diagnostics for Kylian Mbappé's player effect also indicate reliable posterior sampling. The Gelman–Rubin statistic equals 1, highlighting strong agreement between the three chains. The effective sample size of 74057.2 is large, suggesting remarkably weak autocorrelation and a high degree of precision in the posterior estimates. The Heidelberger–Welch test supports convergence for the first and third chains, as the stationarity p-values (0.08 and 0.07) both exceed the 5% significance level. The second chain has a p-value of 0.01, which is smaller than the 0.05 threshold; hence, this parameter has not yet converged. The estimated posterior means are both approximately -0.12 with corresponding half-width values of 0.01. The ratios of the half-width value relative to the mean are less than 0.2, indicating reliable estimates. Similarly, the Geweke Z-scores (-0.73, -1.95 and 1.96) fall within the acceptable range of (-1.96, 1.96), providing no indication of non-stationarity. The traceplot and empirical posterior density displayed in Figure 4.3.8 show stable chains without visible patterns, confirming that the sampling procedure has successfully reached the target posterior distribution.

The traceplot in Figure 4.3.9 appears to vary around a stable mean of approximately -0.1, while the ECDF plot is likewise concentrated around this value. This indicates that, according to this analysis, Mbappe's passing ability is slightly below what would be expected for a player in his team.

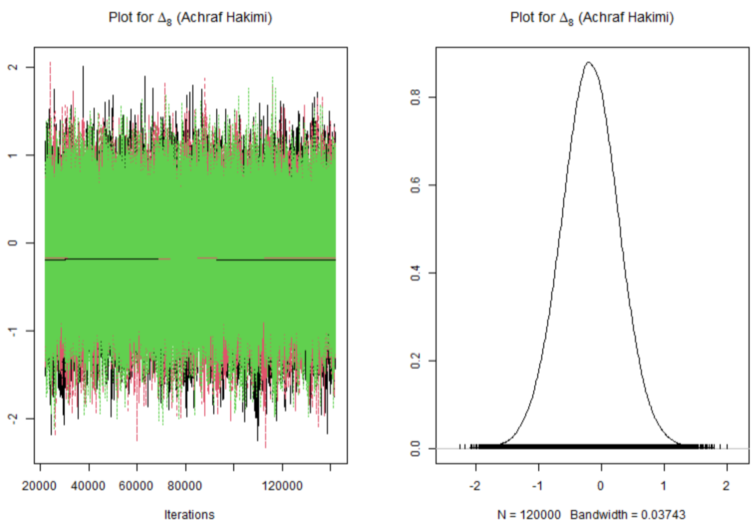

Figure 4.3.11: *Traceplot and ECDF plot for Δ_{181} (Kylian Mbappé) under the negative binomial model for passing.*

The top and bottom ten player effects are presented in Tables 4.3.10 and 4.3.11. These player effects are combined with the set of team and opponent effects, which are provided in Appendix Sections C.1.1 and C.1.2, to obtain the predicted number of passes. The corresponding top and bottom ten predicted values according to the η_i's are presented in Appendix Sections C.1.3 and C.1.4.

The results in Tables 4.3.10 and 4.3.11 should be interpreted with caution, since the model did not consistently achieve convergence and produced a large RMSE. All players in Table 4.3.10 played no more than four matches, as their teams (Mexico, South Korea, Australia, Tunisia, Serbia, Canada, and Japan) did not progress beyond the round of 16. However, Morocco is an exception, as they advanced further in the tournament. If the model were reliable, this would suggest that passing ability is not closely linked to overall team quality, since most of these players came from teams that were eliminated relatively early, while Morocco provides an exception. However, because the model did not converge, this should not be taken as strong evidence.

Table 4.3.10: *Top 10 player effects under the negative binomial model for passing, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Edson Alvarez	0.87	0.48	-0.05	1.82
2	Javad El Yamiq	0.87	0.43	0.04	1.72
3	Hector Herrera	0.81	0.51	-0.17	1.83
4	Jorge Eduardo Sanchez	0.75	0.47	-0.14	1.70
5	Lee Jae-sung	0.72	0.47	-0.19	1.65
6	Milos Degenek	0.62	0.39	-0.09	1.45
7	Dylan Bronn	0.63	0.48	-0.28	1.61
8	Filip Kostic	0.63	0.48	-0.29	1.59
9	Atiba Hutchinson	0.63	0.40	-0.14	1.43
10	Yuto Nagatomo	0.63	0.41	-0.15	1.44

In Table 4.3.11, Nicolas Tagliafico, Marcos Acuña, and Lisandro Martínez were part of the Argentina team that won the tournament. Jawad El Yamiq (Morocco) represented Morocco, who finished fourth, while Lee Jae-sung (South Korea), Milos Degenek (Australia), and Yuto Nagatomo (Japan) represented teams that reached the round of 16. Therefore, the players in this table do not all come from poorly performing teams. If the model were reliable, this would again suggest that passing ability does not simply reflect team success, since some players from successful or relatively successful teams still appear among the lowest-ranked passers.

Table 4.3.11: *Bottom 10 player effects under the negative binomial model for passing, ranked in ascending order by posterior mean, with posterior standard deviations, and lower and upper HPD bounds.*

Rank	Player	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Assim Madibo	-0.59	0.59	-1.80	0.52
2	Nicolas Tagliafico	-0.57	0.58	-1.76	0.52
3	Phil Foden	-0.53	0.55	-1.65	0.52
4	Jhegson Mendez	-0.53	0.55	-1.63	0.54
5	Shogo Taniguchi	-0.52	0.56	-1.66	0.53
6	Jordan Henderson	-0.52	0.55	-1.61	0.54
7	Marcos Acuna	-0.49	0.54	-1.57	0.55
8	Lisandro Martinez	-0.49	0.54	-1.56	0.54
9	Hirving Lozano	-0.48	0.51	-1.43	0.58
10	Abdullelah Al-Malki	-0.41	0.53	-1.48	0.61

The full list of the convergence diagnostics can be found in Appendix C.1.5–C.1.15.

4.3.3.2 Poisson Model

For the Poisson model, the same informative priors as in the negative binomial model in Section 4.3.3.1 were used, since the negative binomial model showed that non-informative priors did not give good convergence. However, despite using a large number of iterations, convergence was still not achieved. The largest $\hat{R}$ value was 3.16, which is well above the 1.05 threshold. Therefore, the Poisson model is not used as the main model for the passing data and is included only for comparison. All convergence diagnostics for this model, including those for all player and team effects, are provided in Appendix Sections C.2.5–C.2.15. One likely reason for this poor performance is overdispersion. As shown in Table 4.2.3, the mean number of passes is 138, while the variance is 10444.7. Since the variance is much larger than the mean, the Poisson assumption is not suitable for these data. The model cannot account for this extra variability directly, which may lead to inflated player and team effects, poor model fit, and strong posterior dependencies between parameters. As a result, the MCMC algorithm struggles to explore the posterior distribution efficiently. The diagnostics showed that several player, team, and opponent-team effects did not converge, based on the diagnostics mentioned in Section 3.5. The posterior summaries for the team and opponent-team effects are reported in Appendix Tables C.2.1 and C.2.2, while the top and bottom ten players according to the η_i's are presented in Appendix Sections C.2.3 and C.2.4. The full convergence diagnostics for the Δ, λ^e, and $\lambda^{E\setminus e}$ parameters are provided in Appendix Tables C.2.5–C.2.15.

Chapter 5

Conclusion

In this dissertation, a Bayesian hierarchical framework was applied to a dataset containing collective performance statistics for the 2022 FIFA World Cup on intercepting, shooting and passing abilities. Players who participated in at least 50% of their team's total playing minutes were all included in the analysis, while goalkeepers were excluded. Typically, count data is modelled using the Poisson distribution. Since datasets which exhibit overdispersion are typically better modelled using a negative binomial distribution, we assessed whether the negative binomial specification provided an improved fit over the Poisson model. Hence, player ability was modelled using both Poisson and negative binomial Bayesian hierarchical models, which were subsequently compared. Having presented the modelling framework and discussed the results in detail, it is useful to bring these elements together. The following section provides a summary of the modelling framework and the key findings obtained throughout the analysis.

5.1 A Summary of the Modelling Framework and Results

Since no readily available Bayesian hierarchical model was found for a World Cup dataset of this kind, a Bayesian hierarchical modelling framework had to be developed, drawing inspiration from Baio and Blangiardo (2010), Attard et al. (2023), and Whitaker et al. (2021). A model suitable for non-granular data, which is the type of data used in this dissertation, was proposed in Section 3.7.

The newly formulated Poisson and negative binomials were first fitted to interceptions and shots. For each of these two models, subsets of the data were used. Defenders and midfielders data was considered for interceptions, while midfielders and forwards data was considered for shots. Overall, these models produced a reasonably good fit to the data. Finally, Poisson and negative binomial models were also fitted on the full data to model passing ability. However, this proved to be significantly more challenging than expected, since players typically complete a large number of passes across a relatively small number of matches and this is largely dependent on the player's team's ball possession. This could have potentially led to difficulties for the Poisson and negative binomial models to achieve convergence. Convergence was still not achieved even after using 170000 iterations. As a result, an attempt was made to achieve convergence for both models by tightening the prior distributions, meaning that the priors were no longer weakly informative. Even after this adjustment, while improvement was noticed, convergence was still not achieved. The Poisson model is preferred for both interceptions and shots, since it gives a smaller RMSE than the negative binomial model. This suggests that overdispersion was limited, and that the Poisson model provided an adequate fit. For the passing model, the negative binomial model is preferred despite its considerably large RMSE, because overdispersion was present and the Poisson model did not converge satisfactorily.

A Bayesian hierarchical structure was appropriate because it accounts for the player and team structure in the data. It estimates player abilities while controlling for team effects, opponent abilities, and minutes played, with partial pooling allowing players with limited minutes to still be assigned stable estimates. The Bayesian framework also quantifies uncertainty rather than producing only single estimates. The credible interval offers a measure of uncertainty around the final estimate. This structure emphasizes that different data can require different modelling considerations, since the passing dataset required more informative priors than the intercepting and shooting datasets, and yet convergence was still not achieved. This analysis separates individual player ability from team and opponent effects, which is important because a player who plays for an extremely strong team may appear to be much better than a player who plays for an average team, simply because the team creates more chances and is more effective. Consequently, it may make sense in the case of passing ability, to consider a similar model but with a different nesting structure than the team/player one considered here, or to subsets of the data, as tightening the priors did not solve the convergence issue.

5.1.1 Limitations and Potential Future Research

One limitation of this dissertation is that the analysis is restricted to the 2022 FIFA World Cup. Since some players had fewer minutes played, their abilities were estimated using less direct information. Moreover, other factors, such as tactics, team form, and weather conditions, were not considered. These factors may also influence player performance and could be included in future work. The tournament has a small number of matches played, especially when compared to the datasets used by Baio and Blangiardo (2010) and Whitaker et al. (2021), which limits the amount of data available per player.

In terms of possible improvements, splitting the data by position and fitting three different models for passing, one for defenders, one for midfielders and one for forwards might have possibly led to a better fitted model for passing data as passing behaviour may vary a lot across positions so modelling all players together may have made convergence harder. Moreover, for a more comprehensive study, additional football performance metrics could have been analysed, rather than only the three considered here. Metrics such as goals or successful tackles could also be included so that player contribution can be evaluated more thoroughly.

This analysis can be extended to multiple tournaments or to a domestic league to assess whether the results are similar to those obtained here. Future models can also account for contextual factors such as player role, making the inference more realistic. Moreover, player ability could be allowed to vary over time, whereas in this dissertation it is treated as constant.

Appendix A

Interceptions

A.1 Poisson Model

Table A.1.1: *Team effects for each national team under the Poisson model for interceptions, showing each team's ability to intercept opponent passing, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-0.01	0.03	-0.08	0.06
2	Australia	-0.01	0.05	-0.10	0.09
3	Belgium	-0.00	0.06	-0.11	0.10
4	Brazil	0.02	0.05	-0.11	0.07
5	Cameroon	0.10	0.06	-0.02	0.22
6	Canada	0.02	0.06	-0.13	0.09
7	Costa Rica	0.04	0.05	-0.06	0.15
8	Croatia	0.01	0.03	-0.07	0.06
9	Denmark	-0.04	0.06	-0.16	0.07
10	Ecuador	0.01	0.05	-0.09	0.15
11	England	0.00	0.04	-0.09	0.08
12	France	0.05	0.03	-0.02	0.12
13	Germany	0.03	0.06	-0.14	0.09
14	Ghana	-0.03	0.06	-0.14	0.08
15	IR Iran	0.06	0.06	-0.05	0.18
16	Japan	-0.01	0.05	-0.11	0.09
17	Korea Republic	0.01	0.05	-0.11	0.09
18	Mexico	-0.02	0.06	-0.13	0.10
19	Morocco	0.01	0.04	-0.06	0.09
20	Netherlands	0.03	0.04	-0.05	0.11
21	Poland	-0.03	0.05	-0.13	0.07
22	Portugal	0.02	0.05	-0.11	0.08
23	Qatar	0.01	0.05	-0.10	0.12
24	Saudi Arabia	0.00	0.05	-0.11	0.11
25	Senegal	-0.01	0.05	-0.11	0.08
26	Serbia	-0.02	0.06	-0.14	0.08
27	Spain	-0.04	0.05	-0.14	0.06
28	Switzerland	-0.06	0.05	-0.17	0.04
29	Tunisia	0.02	0.06	-0.09	0.13
30	United States	-0.01	0.05	-0.10	0.09
31	Uruguay	0.07	0.06	-0.04	0.18
32	Wales	-0.02	0.06	-0.13	0.08

Table A.1.2: *Opponent team effects for each national team under the Poisson model for interceptions, showing each team's passing ability against opponent interceptions, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	0.05	0.08	-0.11	0.22
2	Australia	0.01	0.09	-0.17	0.18
3	Belgium	0.01	0.09	-0.17	0.19
4	Brazil	0.00	0.08	-0.16	0.17
5	Cameroon	0.05	0.09	-0.12	0.25
6	Canada	-0.01	0.09	-0.20	0.17
7	Costa Rica	0.02	0.09	-0.16	0.19
8	Croatia	0.05	0.08	-0.11	0.22
9	Denmark	-0.03	0.09	-0.22	0.15
10	Ecuador	-0.03	0.09	-0.22	0.14
11	England	0.02	0.09	-0.15	0.19
12	France	0.06	0.08	-0.11	0.23
13	Germany	-0.02	0.08	-0.20	0.16
14	Ghana	-0.03	0.09	-0.21	0.14
15	IR Iran	0.00	0.09	-0.18	0.18
16	Japan	0.01	0.09	-0.16	0.18
17	Korea Republic	0.03	0.09	-0.15	0.21
18	Mexico	0.01	0.08	-0.15	0.17
19	Morocco	0.02	0.09	-0.15	0.19
20	Netherlands	0.00	0.09	-0.18	0.17
21	Poland	0.01	0.08	-0.16	0.18
22	Portugal	-0.03	0.09	-0.22	0.14
23	Qatar	-0.01	0.09	-0.18	0.17
24	Saudi Arabia	0.00	0.09	-0.17	0.18
25	Senegal	-0.05	0.09	-0.23	0.13
26	Serbia	-0.02	0.08	-0.19	0.15
27	Spain	0.01	0.09	-0.17	0.18
28	Switzerland	-0.02	0.08	-0.18	0.16
29	Tunisia	-0.03	0.09	-0.21	0.14
30	United States	-0.04	0.08	-0.22	0.13
31	Uruguay	0.02	0.09	-0.15	0.21
32	Wales	-0.05	0.09	-0.24	0.12

Table A.1.3: *Top 10 players under the Poisson model for interceptions, ranked by posterior predicted total interceptions, with lower and upper HPD bounds.*

Rank	Player	Predicted Interceptions	Lower HPD Bound	Upper HPD Bound	Observed Interceptions	90-minute units
1	Aurelien Tchouameni	11.00	6.76	15.54	14	6.9
2	Antoine Griezmann	8.22	5.10	11.56	8	6.0
3	Josko Gvardiol	7.80	4.55	11.35	11	7.7
4	Adrien Rabiot	7.61	4.87	10.86	10	5.4
5	Raphael Varane	7.33	4.45	10.33	6	5.8
6	Luka Modric	7.31	4.43	10.57	10	7.3
7	Achraf Hakimi	7.21	4.19	10.53	9	7.0
8	Nathan Ake	7.08	4.24	10.16	8	5.3
9	Denzel Dumfries	6.90	4.15	9.86	7	5.3
10	Sofyan Amrabat	6.85	3.87	10.06	6	7.3

Table A.1.4: *Bottom 10 players under the Poisson model for interceptions, ranked by posterior predicted total interceptions, with lower and upper HPD bounds.*

Rank	Player	Predicted Interceptions	Lower HPD Bound	Upper HPD Bound	Observed Interceptions	90-minute units
1	Alidu Seidu	1.54	0.89	2.22	0	1.7
2	Mikkel Damsgaard	1.62	0.99	2.33	2	1.7
3	Richie Laryea	1.63	0.96	2.33	0	1.8
4	Anis Ben Slimane	1.64	0.98	2.36	1	1.6
5	Atiba Hutchinson	1.69	1.01	2.41	1	1.8
6	Leander Dendoncker	1.69	0.99	2.46	0	1.8
7	Leon Goretzka	1.69	1.02	2.42	1	1.8
8	Harry Wilson	1.74	1.05	2.50	1	1.8
9	Assim Madibo	1.76	1.06	2.55	2	1.7
10	Filip Kostic	1.78	1.02	2.60	0	2.0

Table A.1.5: *Convergence diagnostic results for the first 35 parameters, under the Poisson model for interceptions. The parameter column reports Δ_i for player effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Aaron Mooy	Δ_1	1.00	132302	-0.01	-0.17	0.51	TRUE
2	Aaron Ramsey	Δ_2	1.00	26010	0.28	-1.51	-1.92	TRUE
3	Abdelkarim Hassan	Δ_3	1.00	42544	-0.92	-1.94	-2.43	TRUE
4	Abdou Diallo	Δ_4	1.00	32757	-0.14	-1.38	-2.18	TRUE
5	Abdulaziz Hatem	Δ_5	1.00	100634	-0.66	-1.45	-0.71	TRUE
6	Abdullelah Al-Amri	Δ_6	1.00	31057	0.13	1.57	2.01	TRUE
7	Abdullelah Al-Malki	Δ_7	1.00	132712	-0.24	0.29	2.37	TRUE
8	Achraf Hakimi	Δ_8	1.00	33872	-0.40	0.39	0.35	TRUE
9	Adrien Rabiot	Δ_9	1.00	23788	-0.12	1.69	1.50	TRUE
10	Ahmad Nourollahi	Δ_{10}	1.00	20237	0.36	0.48	1.73	TRUE
11	Aissa Laidouni	Δ_{11}	1.00	70279	1.17	2.60	0.72	TRUE
12	Alexis Mac Allister	Δ_{12}	1.00	63227	-0.87	-0.45	-2.28	TRUE
13	Ali Abdi	Δ_{13}	1.00	37250	0.65	1.47	1.66	TRUE
14	Ali Al-Bulaihi	Δ_{14}	1.00	102915	-0.97	0.33	-0.45	TRUE
15	Ali Gholizadeh	Δ_{15}	1.00	125659	1.11	-0.37	0.51	TRUE
16	Alidu Seidu	Δ_{16}	1.00	71320	0.68	-1.48	-0.78	TRUE
17	Alistair Johnston	Δ_{17}	1.00	133611	-0.25	-0.83	-3.22	TRUE
18	Alphonso Davies	Δ_{18}	1.00	127752	0.05	0.17	0.70	TRUE
19	Andre-Frank Zambo Anguissa	Δ_{19}	1.00	83117	0.52	1.44	2.01	TRUE
20	Andre Ayew	Δ_{20}	1.00	45328	0.05	-0.36	-1.53	TRUE
21	Andreas Christensen	Δ_{21}	1.00	59504	1.44	1.16	1.81	TRUE
22	Andrija Zivkovic	Δ_{22}	1.00	87937	1.46	1.03	1.17	TRUE
23	Angelo Preciado	Δ_{23}	1.00	20116	0.17	0.88	1.78	TRUE
24	Anis Ben Slimane	Δ_{24}	1.00	130449	-0.92	-0.54	-0.93	TRUE
25	Antoine Griezmann	Δ_{25}	1.00	123502	1.29	1.38	1.04	TRUE
26	Antonee Robinson	Δ_{26}	1.00	22030	-0.43	2.19	1.93	TRUE
27	Antonio Rudiger	Δ_{27}	1.00	26698	-0.47	-0.61	-1.53	TRUE
28	Assim Madibo	Δ_{28}	1.00	127450	0.40	-0.57	0.35	TRUE
29	Atiba Hutchinson	Δ_{29}	1.00	130318	-0.95	0.42	1.17	TRUE
30	Aurelien Tchouameni	Δ_{30}	1.00	17467	0.29	1.09	1.94	TRUE
31	Axel Witsel	Δ_{31}	1.00	75419	-0.14	2.69	2.49	TRUE
32	Aymeric Laporte	Δ_{32}	1.00	125316	0.67	0.45	-0.86	TRUE
33	Aziz Behich	Δ_{33}	1.00	126302	-0.02	-1.07	-0.87	TRUE
34	Azzedine Ounahi	Δ_{34}	1.00	79228	0.45	-1.45	2.96	TRUE
35	Baba Rahman	Δ_{35}	1.00	127535	-1.27	-2.05	-1.37	TRUE

Table A.1.6: *Convergence diagnostic results for parameters 36–70, under the Poisson model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
36	Bartosz Bereszynski	Δ_{36}	1.00	115742	-0.91	0.10	0.03	TRUE
37	Ben Davies	Δ_{37}	1.00	77933	0.69	-2.82	-1.55	TRUE
38	Bernardo Silva	Δ_{38}	1.00	55850	0.22	-0.90	-0.14	TRUE
39	Borna Sosa	Δ_{39}	1.00	43983	-0.06	-1.26	-1.42	TRUE
40	Boualem Khoukhi	Δ_{40}	1.00	132125	-0.55	0.48	-1.61	TRUE
41	Bryan Oviedo	Δ_{41}	1.00	137813	0.81	-0.91	2.76	TRUE
42	Casemiro	Δ_{42}	1.00	26491	0.54	1.29	3.10	TRUE
43	Celso Borges	Δ_{43}	1.00	99571	-0.24	0.92	0.97	TRUE
44	Cesar Montes	Δ_{44}	1.00	131462	-0.15	0.08	-2.64	TRUE
45	Chris Mepham	Δ_{45}	1.00	55096	0.97	-0.37	-2.35	TRUE
46	Christian Eriksen	Δ_{46}	1.00	32087	-0.90	-0.96	-1.56	TRUE
47	Collins Fai	Δ_{47}	1.00	122971	-0.49	-1.05	0.40	TRUE
48	Connor Roberts	Δ_{48}	1.00	118032	0.62	-0.48	-0.87	TRUE
49	Craig Goodwin	Δ_{49}	1.00	33462	0.48	-1.44	-0.77	TRUE
50	Cristian Romero	Δ_{50}	1.00	117872	0.36	0.69	2.29	TRUE
51	Daley Blind	Δ_{51}	1.00	121943	2.50	-0.51	-0.94	TRUE
52	Daniel Amartey	Δ_{52}	1.00	119815	-0.15	1.11	0.47	TRUE
53	Danilo	Δ_{53}	1.00	131384	-0.46	0.32	2.68	TRUE
54	David Raum	Δ_{54}	1.00	8707	-0.64	0.84	1.16	TRUE
55	Dayot Upamecano	Δ_{55}	1.00	122810	-0.80	-1.49	0.56	TRUE
56	Declan Rice	Δ_{56}	1.00	63799	-0.06	1.25	1.28	TRUE
57	Dejan Lovren	Δ_{57}	1.00	127950	-0.27	-0.61	-0.92	TRUE
58	Denzel Dumfries	Δ_{58}	1.00	125146	-0.16	0.29	1.77	TRUE
59	Diego Godin	Δ_{59}	1.00	119003	0.12	-1.56	-0.90	TRUE
60	Diogo Dalot	Δ_{60}	1.00	118804	-0.66	-1.88	-1.39	TRUE
61	Djibril Sow	Δ_{61}	1.00	135295	-0.16	-0.59	-0.36	TRUE
62	Dusan Tadic	Δ_{62}	1.00	66090	1.79	-0.29	0.49	TRUE
63	Dylan Bronn	Δ_{63}	1.00	52417	-0.15	-0.52	-1.98	TRUE
64	Eder Militao	Δ_{64}	1.00	126432	-1.44	-1.14	-0.11	TRUE
65	Edson Alvarez	Δ_{65}	1.00	32003	0.07	1.90	1.39	TRUE
66	Ehsan Hajsafi	Δ_{66}	1.00	88819	-0.25	-0.40	-2.84	TRUE
67	Ellyes Skhiri	Δ_{67}	1.00	12568	-0.18	1.47	1.44	TRUE
68	Enzo Fernandez	Δ_{68}	1.00	49453	0.19	-0.36	-1.24	TRUE
69	Ethan Ampadu	Δ_{69}	1.00	62989	-0.56	2.13	1.98	TRUE
70	Facundo Pellistri	Δ_{70}	1.00	35911	-0.05	-0.39	-1.47	TRUE

Table A.1.7: *Convergence diagnostic results for parameters 71–105, under the Poisson model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
71	Federico Valverde	Δ_{71}	1.00	74220	-0.41	1.90	1.44	TRUE
72	Felix Torres Caicedo	Δ_{72}	1.00	102611	-0.11	1.36	0.66	TRUE
73	Filip Kostic	Δ_{73}	1.00	52970	0.57	0.06	-0.65	TRUE
74	Firas Al-Buraikan	Δ_{74}	1.00	126820	-1.27	-2.44	1.73	TRUE
75	Francisco Calvo	Δ_{75}	1.00	134532	0.25	-0.30	0.32	TRUE
76	Frenkie de Jong	Δ_{76}	1.00	125619	-1.33	-0.22	0.03	TRUE
77	Gavi	Δ_{77}	1.00	62214	0.51	-2.46	-2.26	TRUE
78	Gonzalo Plata	Δ_{78}	1.00	97623	0.84	-2.39	-2.30	TRUE
79	Granit Xhaka	Δ_{79}	1.00	101500	0.38	0.54	-3.15	TRUE
80	Grzegorz Krychowiak	Δ_{80}	1.00	39961	0.41	1.47	2.39	TRUE
81	Guillermo Varela	Δ_{81}	1.00	105999	-1.46	0.09	-0.47	TRUE
82	Harry Maguire	Δ_{82}	1.00	125983	-0.32	-0.43	-1.08	TRUE
83	Harry Souttar	Δ_{83}	1.00	33722	0.32	0.80	1.24	TRUE
84	Harry Wilson	Δ_{84}	1.00	130014	0.37	-0.34	-2.00	TRUE
85	Hassan Al-Haydos	Δ_{85}	1.00	131892	0.63	-0.77	-1.20	TRUE
86	Hassan Al Tambakti	Δ_{86}	1.00	65730	0.04	1.48	-0.39	TRUE
87	Hector Herrera	Δ_{87}	1.00	56491	-0.83	0.84	2.14	TRUE
88	Hector Moreno	Δ_{88}	1.00	58696	0.25	-1.14	-0.59	TRUE
89	Hidemasa Morita	Δ_{89}	1.00	130883	0.61	1.30	-0.10	TRUE
90	Homam Ahmed	Δ_{90}	1.00	118172	0.03	-2.28	-0.58	TRUE
91	Hwang In-beom	Δ_{91}	1.00	132983	0.28	0.53	-0.32	TRUE
92	Ibrahima Konate	Δ_{92}	1.00	18577	0.07	1.18	1.50	TRUE
93	Idrissa Gana Gueye	Δ_{93}	1.00	38549	-0.33	1.83	0.95	TRUE
94	Ilkay Gundogan	Δ_{94}	1.00	111412	1.15	-0.82	-1.39	TRUE
95	Ismail Jakobs	Δ_{95}	1.00	92656	-0.44	0.27	0.96	TRUE
96	Jackson Irvine	Δ_{96}	1.00	129478	0.09	-0.28	-0.02	TRUE
97	Jakub Kaminski	Δ_{97}	1.00	93981	-0.25	-0.09	-0.52	TRUE
98	Jakub Kiwior	Δ_{98}	1.00	99618	-0.95	2.65	1.20	TRUE
99	Jamal Musiala	Δ_{99}	1.00	61462	1.58	-1.33	-0.72	TRUE
100	Jan Vertonghen	Δ_{100}	1.00	37635	0.25	1.46	2.21	TRUE
101	Javad El Yamiq	Δ_{101}	1.00	52625	-0.31	1.45	0.02	TRUE
102	Jean-Charles Castelletto	Δ_{102}	1.00	13103	-0.38	0.87	1.34	TRUE
103	Jesper Lindstrom	Δ_{103}	1.00	123129	-0.27	-0.66	-0.16	TRUE
104	Jesus Gallardo	Δ_{104}	1.00	6076	-0.27	-0.50	-1.45	TRUE
105	Jhegson Mendez	Δ_{105}	1.00	106531	0.27	-0.33	-1.31	TRUE

Table A.1.8: *Convergence diagnostic results for parameters 106–140, under the Poisson model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
106	Joachim Andersen	Δ_{106}	1.00	31079	0.26	-0.65	-1.01	TRUE
107	Joakim Maehle	Δ_{107}	1.00	132510	-0.25	-1.09	-0.15	TRUE
108	Joao Cancelo	Δ_{108}	1.00	112245	1.44	-1.01	1.10	TRUE
109	Joe Rodon	Δ_{109}	1.00	76226	0.15	3.07	0.67	TRUE
110	John Stones	Δ_{110}	1.00	16845	0.33	-1.61	-1.12	TRUE
111	Jordan Henderson	Δ_{111}	1.00	47976	0.39	-1.24	-0.77	TRUE
112	Jordi Alba	Δ_{112}	1.00	131828	1.02	-0.07	0.94	TRUE
113	Jorge Eduardo Sanchez	Δ_{113}	1.00	7100	1.09	0.03	1.09	TRUE
114	Jose Maria Gimenez	Δ_{114}	1.00	18286	-0.24	1.14	1.17	TRUE
115	Joshua Kimmich	Δ_{115}	1.00	12897	0.05	1.26	0.35	TRUE
116	Josip Juranovic	Δ_{116}	1.00	33414	2.05	-2.60	-2.36	TRUE
117	Josko Gvardiol	Δ_{117}	1.00	14404	-0.44	1.43	0.91	TRUE
118	Jude Bellingham	Δ_{118}	1.00	128034	2.20	0.30	0.89	TRUE
119	Jules Kounde	Δ_{119}	1.00	81908	1.07	0.18	-0.49	TRUE
120	Jung Woo-young	Δ_{120}	1.00	133155	-0.45	1.36	-0.44	TRUE
121	Junya Ito	Δ_{121}	1.00	47652	0.23	1.34	1.40	TRUE
122	Jurien Timber	Δ_{122}	1.00	10620	0.09	1.93	1.93	TRUE
123	Kalidou Koulibaly	Δ_{123}	1.00	54871	0.31	-0.81	-1.22	TRUE
124	Kamal Miller	Δ_{124}	1.00	64324	0.14	0.60	0.90	TRUE
125	Kamil Glik	Δ_{125}	1.00	126518	0.79	-0.65	1.12	TRUE
126	Karim Boudiaf	Δ_{126}	1.00	44780	0.62	-0.70	-1.96	TRUE
127	Kendall Waston	Δ_{127}	1.00	31801	0.48	1.72	2.15	TRUE
128	Keysher Fuller	Δ_{128}	1.00	129687	-2.22	-0.61	1.85	TRUE
129	Kim Jin-su	Δ_{129}	1.00	132473	-0.80	0.85	-0.78	TRUE
130	Kim Min-jae	Δ_{130}	1.00	123915	1.67	-0.16	-0.58	TRUE
131	Kim Moon-hwan	Δ_{131}	1.00	130951	-0.57	-0.40	0.13	TRUE
132	Kim Young-gwon	Δ_{132}	1.00	131719	-1.41	-1.35	-1.44	TRUE
133	Ko Itakura	Δ_{133}	1.00	75255	1.66	0.74	0.73	TRUE
134	Krystian Bielik	Δ_{134}	1.00	18983	-0.43	1.40	1.17	TRUE
135	Kye Rowles	Δ_{135}	1.00	18012	-0.06	0.63	1.69	TRUE
136	Kyle Walker	Δ_{136}	1.00	131487	-0.73	0.09	1.12	TRUE
137	Leander Dendoncker	Δ_{137}	1.00	52990	-0.15	-1.48	-1.59	TRUE
138	Lee Jae-sung	Δ_{138}	1.00	10692	0.50	0.77	-0.20	TRUE
139	Leon Goretzka	Δ_{139}	1.00	135431	-1.92	-2.21	-2.10	TRUE
140	Lisandro Martinez	Δ_{140}	1.00	132378	-0.93	1.53	-0.44	TRUE

Table A.1.9: *Convergence diagnostic results for parameters 141–175, under the Poisson model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
141	Lucas Paqueta	Δ_{141}	1.00	80740	0.49	-0.75	-1.15	TRUE
142	Luis Chavez	Δ_{142}	1.00	53299	0.25	-0.97	-1.99	TRUE
143	Luka Modric	Δ_{143}	1.00	18569	-0.23	1.41	1.48	TRUE
144	Luke Shaw	Δ_{144}	1.00	10209	0.86	-0.47	-0.61	TRUE
145	Majid Hosseini	Δ_{145}	1.00	35051	-0.07	2.29	1.51	TRUE
146	Manuel Akanji	Δ_{146}	1.00	12038	0.15	0.95	0.48	TRUE
147	Marcelo Brozovic	Δ_{147}	1.00	94676	0.58	-1.38	-1.76	TRUE
148	Marcos Acuna	Δ_{148}	1.00	71577	0.32	0.32	-1.53	TRUE
149	Marquinhos	Δ_{149}	1.00	45995	1.72	-1.08	-1.02	TRUE
150	Mateo Kovacic	Δ_{150}	1.00	55064	-0.46	-2.69	-0.05	TRUE
151	Mathew Leckie	Δ_{151}	1.00	36072	-0.36	-1.36	-1.33	TRUE
152	Mathias Olivera	Δ_{152}	1.00	77609	1.69	-1.86	-0.76	TRUE
153	Matias Vecino	Δ_{153}	1.00	54935	0.22	0.81	1.81	TRUE
154	Matty Cash	Δ_{154}	1.00	20662	0.61	-1.06	-2.22	TRUE
155	Maya Yoshida	Δ_{155}	1.00	116272	-2.09	-0.04	-0.08	TRUE
156	Mikkel Damsgaard	Δ_{156}	1.00	116766	-0.72	-0.79	0.14	TRUE
157	Milad Mohammadi	Δ_{157}	1.00	67567	-0.03	-2.71	-0.23	TRUE
158	Milos Degenek	Δ_{158}	1.00	120257	-0.90	-1.54	-1.07	TRUE
159	Milos Veljkovic	Δ_{159}	1.00	37689	-0.06	-0.82	-1.48	TRUE
160	Mohamed Kanno	Δ_{160}	1.00	127640	-0.98	-1.00	1.82	TRUE
161	Mohammed Salisu	Δ_{161}	1.00	129947	0.49	-0.28	2.14	TRUE
162	Moises Caicedo	Δ_{162}	1.00	97950	1.07	0.16	-1.60	TRUE
163	Montassar Talbi	Δ_{163}	1.00	113505	-0.63	-0.86	-0.31	TRUE
164	Morteza Pouraliganji	Δ_{164}	1.00	36820	-0.63	-2.08	1.33	TRUE
165	Nahuel Molina	Δ_{165}	1.00	50065	1.16	-0.94	-1.80	TRUE
166	Nampalys Mendy	Δ_{166}	1.00	129447	-0.55	0.71	-1.39	TRUE
167	Nathan Ake	Δ_{167}	1.00	84010	-0.37	0.57	0.82	TRUE
168	Nayef Aguerd	Δ_{168}	1.00	131471	-0.26	-1.51	-1.44	TRUE
169	Nico Williams	Δ_{169}	1.00	31901	0.00	1.16	1.59	TRUE
170	Neymar	Δ_{170}	1.00	46814	0.47	-0.35	-3.13	TRUE
171	Nico Elvedi	Δ_{171}	1.00	12735	0.33	-2.32	0.00	TRUE
172	Nicolas Nkoulou	Δ_{172}	1.00	25686	0.30	1.36	1.56	TRUE
173	Nicolas Otamendi	Δ_{173}	1.00	28373	-1.03	1.54	2.11	TRUE
174	Nicolas Tagliafico	Δ_{174}	1.00	7037	-0.63	1.25	1.12	TRUE
175	Niklas Sule	Δ_{175}	1.00	28612	0.25	-0.29	-1.96	TRUE

Table A.1.10: *Convergence diagnostic results for parameters 176–210, under the Poisson model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
176	Nikola Milenkovic	Δ_{176}	1.00	16126	-0.13	1.85	1.52	TRUE
177	Nouhou Tolo	Δ_{177}	1.00	47282	-0.16	2.24	2.36	TRUE
178	Noussair Mazraoui	Δ_{178}	1.00	28307	-0.15	-1.70	-1.73	TRUE
179	Oscar Duarte	Δ_{179}	1.00	26771	-0.44	0.52	1.35	TRUE
180	Pedri	Δ_{180}	1.00	125725	-0.85	1.71	0.87	TRUE
181	Pepe	Δ_{181}	1.00	37506	0.56	1.39	2.57	TRUE
182	Pervis Estupinan	Δ_{182}	1.00	131556	0.10	0.02	-1.56	TRUE
183	Piero Hincapie	Δ_{183}	1.00	111314	0.35	-1.89	-0.05	TRUE
184	Pierre Hojbjerg	Δ_{184}	1.00	130498	0.21	0.61	-0.81	TRUE
185	Piotr Zielinski	Δ_{185}	1.00	53414	1.07	-0.33	-1.17	TRUE
186	Przemyslaw Frankowski	Δ_{186}	1.00	97855	-0.82	2.84	-1.07	TRUE
187	Ramin Rezaeian	Δ_{187}	1.00	121677	0.37	1.62	0.79	TRUE
188	Raphael Guerreiro	Δ_{188}	1.00	12731	-1.77	1.18	0.66	TRUE
189	Raphael Varane	Δ_{189}	1.00	116462	0.42	1.62	-1.07	TRUE
190	Rasmus Nissen	Δ_{190}	1.00	10225	0.03	-0.23	-2.64	TRUE
191	Remo Freuler	Δ_{191}	1.00	130413	1.27	0.15	-0.68	TRUE
192	Ricardo Rodriguez	Δ_{192}	1.00	21177	0.43	-1.28	-1.72	TRUE
193	Richie Laryea	Δ_{193}	1.00	58988	0.89	-1.67	-1.69	TRUE
194	Ro-Ro	Δ_{194}	1.00	7677	-0.23	1.59	1.36	TRUE
195	Rodri	Δ_{195}	1.00	12285	-0.90	-0.68	-2.88	TRUE
196	Rodrigo Bentancur	Δ_{196}	1.00	10703	-0.17	0.95	1.35	TRUE
197	Rodrigo De Paul	Δ_{197}	1.00	63071	-0.39	2.03	2.72	TRUE
198	Romain Saiss	Δ_{198}	1.00	56567	0.28	-1.61	-1.05	TRUE
199	Ruben Dias	Δ_{199}	1.00	118672	-0.31	0.61	-0.30	TRUE
200	Ruben Neves	Δ_{200}	1.00	134063	0.79	-0.56	-0.20	TRUE
201	Saeid Ezatolahi	Δ_{201}	1.00	121052	-0.51	-0.71	-0.00	TRUE
202	Salem Al-Dawsari	Δ_{202}	1.00	50432	0.74	-1.67	-1.14	TRUE
203	Sali Abdul Samed	Δ_{203}	1.00	61338	-0.80	1.47	1.39	TRUE
204	Sasa Lucic	Δ_{204}	1.00	73842	-0.41	-0.75	-1.39	TRUE
205	Saud Abdulhamid	Δ_{205}	1.00	127294	-1.86	-0.48	-0.07	TRUE
206	Sebastian Coates	Δ_{206}	1.00	90620	0.92	-2.14	-2.45	TRUE
207	Selim Amallah	Δ_{207}	1.00	14484	-0.70	0.52	1.43	TRUE
208	Sergej Milinkovic-Savic	Δ_{208}	1.00	135677	-1.38	0.81	0.40	TRUE
209	Sergio Dest	Δ_{209}	1.00	89278	-0.51	-1.03	-1.95	TRUE
210	Sergio Busquets	Δ_{210}	1.00	35658	-0.63	-2.41	-1.89	TRUE

Table A.1.11: *Convergence diagnostic results for parameters 211–236, under the Poisson model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
211	Shogo Taniguchi	Δ_{211}	1.00	108571	0.88	-0.37	1.02	TRUE
212	Silvan Widmer	Δ_{212}	1.00	131865	-0.17	0.46	-0.94	TRUE
213	Sofyan Amrabat	Δ_{213}	1.00	123415	0.56	-0.01	-0.67	TRUE
214	Steven Vitoria	Δ_{214}	1.00	30292	0.04	-1.33	-1.52	TRUE
215	Strahinja Pavlovic	Δ_{215}	1.00	92938	-1.06	-0.53	-0.60	TRUE
216	Tajon Buchanan	Δ_{216}	1.00	12959	-0.62	0.06	0.07	TRUE
217	Takehiro Tomiyasu	Δ_{217}	1.00	47329	0.32	-1.21	-1.10	TRUE
218	Theo Hernandez	Δ_{218}	1.00	34229	0.39	-2.23	-1.14	TRUE
219	Thiago Silva	Δ_{219}	1.00	110973	-1.23	0.41	0.53	TRUE
220	Thomas Meunier	Δ_{220}	1.00	13480	-1.40	-1.47	-1.40	TRUE
221	Thomas Partey	Δ_{221}	1.00	27434	0.90	-1.34	-1.34	TRUE
222	Tim Ream	Δ_{222}	1.00	134583	-0.32	0.49	1.19	TRUE
223	Timothy Castagne	Δ_{223}	1.00	124494	0.01	-0.79	-0.50	TRUE
224	Toby Alderweireld	Δ_{224}	1.00	24602	0.21	-2.52	-1.09	TRUE
225	Tyler Adams	Δ_{225}	1.00	14377	0.15	1.40	0.60	TRUE
226	Virgil van Dijk	Δ_{226}	1.00	42283	1.48	-2.11	-1.49	TRUE
227	Walker Zimmerman	Δ_{227}	1.00	87742	-0.72	2.25	-0.04	TRUE
228	Wataru Endo	Δ_{228}	1.00	132321	-0.31	0.88	-1.38	TRUE
229	Weston McKennie	Δ_{229}	1.00	24490	-0.32	-1.88	-1.66	TRUE
230	Yahya Attiat Allah	Δ_{230}	1.00	20250	0.62	-1.20	-2.04	TRUE
231	Yassine Meriah	Δ_{231}	1.00	47243	0.29	-3.13	-2.29	TRUE
232	Yeltsin Tejeda	Δ_{232}	1.00	39138	-0.11	-0.58	-2.05	TRUE
233	Youssef Msakni	Δ_{233}	1.00	44135	0.74	-1.84	-1.24	TRUE
234	Youssouf Sabaly	Δ_{234}	1.00	13121	0.58	-0.09	-0.02	TRUE
235	Yunus Musah	Δ_{235}	1.00	65252	0.97	-1.92	-1.19	TRUE
236	Yuto Nagatomo	Δ_{236}	1.00	41590	0.04	-0.90	-1.26	TRUE

Table A.1.12: *Convergence diagnostic results for the team-effect parameters under the Poisson model for interceptions. The parameter column reports λ_i^e for team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	λ_1^e	1.00	45897	-1.55	-1.74	0.50	TRUE
2	Australia	λ_2^e	1.00	85193	-0.41	-0.90	-1.03	TRUE
3	Belgium	λ_3^e	1.00	110472	-0.35	1.17	-1.11	TRUE
4	Brazil	λ_4^e	1.00	8163	0.67	0.21	-0.10	TRUE
5	Cameroon	λ_5^e	1.00	55810	-0.42	0.55	-2.32	TRUE
6	Canada	λ_6^e	1.00	106434	0.82	0.81	2.13	TRUE
7	Costa Rica	λ_7^e	1.00	87427	-1.67	1.68	-2.41	TRUE
8	Croatia	λ_8^e	1.00	40682	-0.36	0.36	3.62	TRUE
9	Denmark	λ_9^e	1.00	89832	-1.05	-1.39	1.65	TRUE
10	Ecuador	λ_{10}^e	1.00	97566	-0.14	0.90	0.57	TRUE
11	England	λ_{11}^e	1.00	70934	2.19	0.16	-1.34	TRUE
12	France	λ_{12}^e	1.00	42511	-1.42	0.35	-2.34	TRUE
13	Germany	λ_{13}^e	1.00	101591	0.49	-0.15	0.89	TRUE
14	Ghana	λ_{14}^e	1.00	103663	0.68	-0.59	0.94	TRUE
15	IR Iran	λ_{15}^e	1.00	68134	1.28	0.99	-3.23	TRUE
16	Japan	λ_{16}^e	1.00	95825	0.93	-0.04	-2.13	TRUE
17	Korea Republic	λ_{17}^e	1.00	93642	1.19	-1.03	0.26	TRUE
18	Mexico	λ_{18}^e	1.00	107192	-0.84	0.60	1.25	TRUE
19	Morocco	λ_{19}^e	1.00	43298	-0.07	0.77	-0.00	TRUE
20	Netherlands	λ_{20}^e	1.00	52543	0.80	0.66	-1.03	TRUE
21	Poland	λ_{21}^e	1.00	79208	-0.74	-2.89	2.62	TRUE
22	Portugal	λ_{22}^e	1.00	91551	-2.86	-1.08	0.17	TRUE
23	Qatar	λ_{23}^e	1.00	104758	-1.17	0.45	-0.11	TRUE
24	Saudi Arabia	λ_{24}^e	1.00	108169	-1.02	0.75	2.07	TRUE
25	Senegal	λ_{25}^e	1.00	92219	1.37	-0.37	0.60	TRUE
26	Serbia	λ_{26}^e	1.00	103835	0.30	-0.60	1.04	TRUE
27	Spain	λ_{27}^e	1.00	81399	-0.63	1.31	2.36	TRUE
28	Switzerland	λ_{28}^e	1.00	73759	-0.92	0.07	1.95	TRUE
29	Tunisia	λ_{29}^e	1.00	104193	0.64	0.11	-2.94	TRUE
30	United States	λ_{30}^e	1.00	86023	0.71	-0.44	-1.41	TRUE
31	Uruguay	λ_{31}^e	1.00	68113	0.17	-0.31	-1.41	TRUE
32	Wales	λ_{32}^e	1.00	105456	1.20	-0.86	1.70	TRUE

Table A.1.13: *Convergence diagnostic results for the opponent team-effect parameters under the Poisson model for interceptions. The parameter column reports $\lambda_i^{E\backslash e}$ for opponent team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	$\lambda_{1}^{E\backslash e}$	1.00	39471	-1.91	-0.04	1.32	TRUE
2	Australia	$\lambda_{2}^{E\backslash e}$	1.00	64294	-1.16	-0.05	-0.59	TRUE
3	Belgium	$\lambda_{3}^{E\backslash e}$	1.00	67346	1.56	-0.29	-0.49	TRUE
4	Brazil	$\lambda_{4}^{E\backslash e}$	1.00	70420	0.55	-0.04	1.36	TRUE
5	Cameroon	$\lambda_{5}^{E\backslash e}$	1.00	49982	-1.83	1.16	1.67	TRUE
6	Canada	$\lambda_{6}^{E\backslash e}$	1.00	64065	0.70	-0.87	0.86	TRUE
7	Costa Rica	$\lambda_{7}^{E\backslash e}$	1.00	81060	0.45	2.40	-1.73	TRUE
8	Croatia	$\lambda_{8}^{E\backslash e}$	1.00	41358	-0.69	1.03	0.32	TRUE
9	Denmark	$\lambda_{9}^{E\backslash e}$	1.00	56048	1.93	-0.15	-3.11	TRUE
10	Ecuador	$\lambda_{10}^{E\backslash e}$	1.00	60085	0.50	-0.32	-0.81	TRUE
11	England	$\lambda_{11}^{E\backslash e}$	1.00	55335	0.65	0.60	-0.14	TRUE
12	France	$\lambda_{12}^{E\backslash e}$	1.00	42947	0.40	-0.74	1.90	TRUE
13	Germany	$\lambda_{13}^{E\backslash e}$	1.00	81225	0.90	-0.55	-0.62	TRUE
14	Ghana	$\lambda_{14}^{E\backslash e}$	1.00	7619	1.97	-0.26	-1.71	TRUE
15	IR Iran	$\lambda_{15}^{E\backslash e}$	1.00	77342	-0.67	-0.38	0.38	TRUE
16	Japan	$\lambda_{16}^{E\backslash e}$	1.00	73212	-1.24	-0.01	3.14	TRUE
17	Korea Republic	$\lambda_{17}^{E\backslash e}$	1.00	76863	0.98	-1.14	2.46	TRUE
18	Mexico	$\lambda_{18}^{E\backslash e}$	1.00	49668	-1.87	1.82	1.56	TRUE
19	Morocco	$\lambda_{19}^{E\backslash e}$	1.00	59567	-2.98	-1.05	-0.68	TRUE
20	Netherlands	$\lambda_{20}^{E\backslash e}$	1.00	71974	-1.59	1.47	-0.81	TRUE
21	Poland	$\lambda_{21}^{E\backslash e}$	1.00	77952	0.34	1.42	0.15	TRUE
22	Portugal	$\lambda_{22}^{E\backslash e}$	1.00	60601	2.12	-0.80	0.39	TRUE
23	Qatar	$\lambda_{23}^{E\backslash e}$	1.00	79125	1.93	-0.94	-0.78	TRUE
24	Saudi Arabia	$\lambda_{24}^{E\backslash e}$	1.00	71587	-0.60	1.19	0.80	TRUE
25	Senegal	$\lambda_{25}^{E\backslash e}$	1.00	61667	1.72	-0.05	-1.08	TRUE
26	Serbia	$\lambda_{26}^{E\backslash e}$	1.00	84074	0.27	-1.52	-0.78	TRUE
27	Spain	$\lambda_{27}^{E\backslash e}$	1.00	73703	-2.15	-0.07	-1.74	TRUE
28	Switzerland	$\lambda_{28}^{E\backslash e}$	1.00	89411	-0.47	-0.32	0.32	TRUE
29	Tunisia	$\lambda_{29}^{E\backslash e}$	1.00	70055	1.06	-0.95	-0.88	TRUE
30	United States	$\lambda_{30}^{E\backslash e}$	1.00	59662	1.77	0.37	-1.19	TRUE
31	Uruguay	$\lambda_{31}^{E\backslash e}$	1.00	66785	-2.71	-0.60	1.26	TRUE
32	Wales	$\lambda_{32}^{E\backslash e}$	1.00	51968	2.83	-0.58	-0.83	TRUE

A.2 Negative Binomial Model

Table A.2.1: *Team effects for each national team under the negative binomial model for interceptions, showing each team's ability to intercept opponent passing, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-0.01	0.04	-0.08	0.06
2	Australia	-0.01	0.05	-0.10	0.09
3	Belgium	-0.00	0.06	-0.11	0.11
4	Brazil	-0.02	0.05	-0.11	0.07
5	Cameroon	0.09	0.06	-0.03	0.21
6	Canada	-0.02	0.06	-0.13	0.09
7	Costa Rica	0.04	0.05	-0.07	0.15
8	Croatia	-0.01	0.04	-0.08	0.06
9	Denmark	-0.04	0.06	-0.16	0.07
10	Ecuador	0.01	0.05	-0.10	0.12
11	England	0.00	0.04	-0.09	0.08
12	France	0.05	0.04	-0.02	0.12
13	Germany	-0.02	0.06	-0.14	0.09
14	Ghana	-0.03	0.06	-0.14	0.08
15	IR Iran	0.06	0.06	-0.05	0.17
16	Japan	-0.01	0.05	-0.11	0.09
17	Korea Republic	-0.01	0.05	-0.11	0.09
18	Mexico	-0.01	0.06	-0.13	0.10
19	Morocco	0.01	0.04	-0.06	0.09
20	Netherlands	0.03	0.04	-0.05	0.11
21	Poland	-0.03	0.05	-0.13	0.07
22	Portugal	-0.01	0.05	-0.11	0.08
23	Qatar	0.01	0.06	-0.10	0.11
24	Saudi Arabia	0.00	0.06	-0.11	0.10
25	Senegal	-0.01	0.05	-0.11	0.08
26	Serbia	-0.03	0.06	-0.14	0.09
27	Spain	-0.04	0.05	-0.14	0.06
28	Switzerland	-0.06	0.05	-0.16	0.05
29	Tunisia	0.02	0.05	-0.09	0.13
30	United States	-0.01	0.05	-0.10	0.09
31	Uruguay	0.06	0.06	-0.05	0.17
32	Wales	-0.02	0.06	-0.13	0.09

Table A.2.2: *Opponent team effects for each national team under the negative binomial model for interceptions, showing each team's passing ability against opponent interceptions, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	0.04	0.08	-0.12	0.21
2	Australia	0.01	0.09	-0.17	0.18
3	Belgium	0.01	0.09	-0.17	0.19
4	Brazil	0.00	0.08	-0.16	0.17
5	Cameroon	0.05	0.09	-0.12	0.25
6	Canada	-0.02	0.09	-0.19	0.16
7	Costa Rica	0.02	0.08	-0.16	0.19
8	Croatia	0.04	0.08	-0.11	0.21
9	Denmark	-0.03	0.09	-0.21	0.15
10	Ecuador	-0.03	0.09	-0.21	0.14
11	England	0.01	0.08	-0.15	0.19
12	France	0.05	0.08	-0.12	0.22
13	Germany	-0.01	0.09	-0.19	0.16
14	Ghana	-0.03	0.09	-0.21	0.14
15	IR Iran	0.00	0.09	-0.17	0.18
16	Japan	0.00	0.09	-0.17	0.18
17	Korea Republic	0.02	0.09	-0.15	0.21
18	Mexico	0.01	0.08	-0.15	0.17
19	Morocco	0.01	0.09	-0.15	0.18
20	Netherlands	-0.00	0.08	-0.17	0.17
21	Poland	0.01	0.08	-0.16	0.18
22	Portugal	-0.02	0.09	-0.21	0.14
23	Qatar	-0.00	0.09	-0.18	0.17
24	Saudi Arabia	0.00	0.09	-0.17	0.17
25	Senegal	-0.04	0.09	-0.22	0.13
26	Serbia	-0.02	0.08	-0.20	0.14
27	Spain	0.01	0.08	-0.16	0.18
28	Switzerland	-0.01	0.08	-0.19	0.15
29	Tunisia	-0.02	0.08	-0.20	0.15
30	United States	-0.04	0.09	-0.22	0.13
31	Uruguay	0.02	0.09	-0.15	0.20
32	Wales	-0.04	0.09	-0.23	0.13

Table A.2.3: *Top 10 players under the negative binomial model for interceptions, ranked by posterior predicted total interceptions, with lower and upper HPD bounds.*

Rank	Player	Predicted Interceptions	Lower HPD Bound	Upper HPD Bound	Observed Interceptions	90-minute units
1	Aurelien Tchouameni	10.45	6.60	15.77	14	6.9
2	Antoine Griezmann	8.19	5.32	12.05	8	6.0
3	Josko Gvardiol	7.50	4.65	11.52	11	7.7
4	Raphael Varane	7.47	4.81	10.84	6	5.8
5	Adrien Rabiot	7.37	4.88	10.84	10	5.4
6	Luka Modric	7.09	4.44	10.74	10	7.3
7	Achraf Hakimi	7.06	4.32	10.86	9	7.0
8	Nathan Ake	7.02	4.46	10.54	8	5.3
9	Sofyan Amrabat	7.00	4.18	10.83	6	7.3
10	Theo Hernandez	6.97	4.42	10.18	4	5.6

Table A.2.4: *Bottom 10 players under the negative binomial model for interceptions, ranked by posterior predicted total interceptions, with lower and upper HPD bounds.*

Rank	Player	Predicted Interceptions	Lower HPD Bound	Upper HPD Bound	Observed Interceptions	90-minute units
1	Alidu Seidu	1.57	1.03	2.25	0	1.7
2	Mikkel Damsgaard	1.61	1.10	2.31	2	1.7
3	Anis Ben Slimane	1.63	1.08	2.35	1	1.6
4	Richie Laryea	1.66	1.10	2.34	0	1.8
5	Atiba Hutchinson	1.70	1.14	2.41	1	1.8
6	Leon Goretzka	1.70	1.14	2.40	1	1.8
7	Leander Dendoncker	1.71	1.11	2.45	0	1.8
8	Assim Madibo	1.74	1.17	2.51	2	1.7
9	Harry Wilson	1.76	1.18	2.51	1	1.8
10	Hector Herrera	1.79	1.18	2.63	3	1.8

Table A.2.5: *Convergence diagnostic results for parameters 1–35, under the negative binomial model for interceptions. The parameter column reports Δ_i for player effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Aaron Mooy	Δ_1	1.00	13048	0.50	0.28	1.09	TRUE
2	Aaron Ramsey	Δ_2	1.00	29652	3.27	-0.15	-0.59	TRUE
3	Abdelkarim Hassan	Δ_3	1.00	46763	1.18	0.06	-0.74	TRUE
4	Abdou Diallo	Δ_4	1.00	35156	1.88	-0.46	-0.49	TRUE
5	Abdulaziz Hatem	Δ_5	1.00	92932	1.37	-0.55	-1.62	TRUE
6	Abdullelah Al-Amri	Δ_6	1.00	32966	-2.45	-0.21	0.53	TRUE
7	Abdullelah Al-Malki	Δ_7	1.00	126783	-0.71	-2.68	0.18	TRUE
8	Achraf Hakimi	Δ_8	1.00	41452	-2.30	0.02	-0.74	TRUE
9	Adrien Rabiot	Δ_9	1.00	28523	-2.61	0.62	0.26	TRUE
10	Ahmad Nourollahi	Δ_{10}	1.00	23935	-1.86	-0.00	0.52	TRUE
11	Aissa Laidouni	Δ_{11}	1.00	69478	-1.88	0.37	-0.04	TRUE
12	Alexis Mac Allister	Δ_{12}	1.00	63227	2.54	-0.17	-1.03	TRUE
13	Ali Abdi	Δ_{13}	1.00	43122	-2.29	0.27	0.47	TRUE
14	Ali Al-Bulaihi	Δ_{14}	1.00	107910	-1.82	-1.24	-0.06	TRUE
15	Ali Gholizadeh	Δ_{15}	1.00	126038	0.13	1.28	0.63	TRUE
16	Alidu Seidu	Δ_{16}	1.00	66419	2.38	0.07	-1.43	TRUE
17	Alistair Johnston	Δ_{17}	1.00	12077	0.73	0.50	0.69	TRUE
18	Alphonso Davies	Δ_{18}	1.00	120771	-1.11	-1.37	-1.44	TRUE
19	Andre-Frank Zambo Anguissa	Δ_{19}	1.00	83355	-1.99	-0.23	0.37	TRUE
20	Andre Ayew	Δ_{20}	1.00	51245	3.11	-1.26	0.09	TRUE
21	Andreas Christensen	Δ_{21}	1.00	59924	-2.24	1.06	0.49	TRUE
22	Andrija Zivkovic	Δ_{22}	1.00	91536	0.11	0.21	-0.31	TRUE
23	Angelo Preciado	Δ_{23}	1.00	21410	-1.98	0.31	0.57	TRUE
24	Anis Ben Slimane	Δ_{24}	1.00	12205	2.39	1.15	0.58	TRUE
25	Antoine Griezmann	Δ_{25}	1.00	119054	0.23	-0.37	0.08	TRUE
26	Antonee Robinson	Δ_{26}	1.00	24457	-1.90	0.92	0.44	TRUE
27	Antonio Rudiger	Δ_{27}	1.00	29406	2.61	-0.22	-0.10	TRUE
28	Assim Madibo	Δ_{28}	1.00	122600	-2.03	0.63	-0.26	TRUE
29	Atiba Hutchinson	Δ_{29}	1.00	12689	0.76	-0.83	-0.65	TRUE
30	Aurelien Tchouameni	Δ_{30}	1.00	20111	-2.74	-0.26	0.35	TRUE
31	Axel Witsel	Δ_{31}	1.00	81262	-1.43	1.07	0.66	TRUE
32	Aymeric Laporte	Δ_{32}	1.00	124060	-2.48	-1.11	0.23	TRUE
33	Aziz Behich	Δ_{33}	1.00	12851	3.15	-0.05	1.40	TRUE
34	Azzedine Ounahi	Δ_{34}	1.00	9465	-2.78	0.51	-0.56	TRUE
35	Baba Rahman	Δ_{35}	1.00	12047	1.15	0.38	-1.88	TRUE

Table A.2.6: *Convergence diagnostic results for parameters 36–70, under the negative binomial model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
36	Bartosz Bereszynski	Δ_{36}	1.00	11279	0.88	-0.46	-1.40	TRUE
37	Ben Davies	Δ_{37}	1.00	73465	2.60	-0.56	0.14	TRUE
38	Bernardo Silva	Δ_{38}	1.00	56598	2.91	0.29	1.16	TRUE
39	Borna Sosa	Δ_{39}	1.00	51070	2.80	1.31	-0.89	TRUE
40	Boualem Khoukhi	Δ_{40}	1.00	128672	-0.18	0.33	0.34	TRUE
41	Bryan Oviedo	Δ_{41}	1.00	123271	0.13	0.14	-0.01	TRUE
42	Casemiro	Δ_{42}	1.00	25757	-2.58	-0.22	0.17	TRUE
43	Celso Borges	Δ_{43}	1.00	98876	-2.00	0.30	1.40	TRUE
44	Cesar Montes	Δ_{44}	1.00	121449	-0.14	0.03	-0.39	TRUE
45	Chris Mepham	Δ_{45}	1.00	64531	1.68	-0.01	-0.35	TRUE
46	Christian Eriksen	Δ_{46}	1.00	36214	2.36	-1.10	-0.54	TRUE
47	Collins Fai	Δ_{47}	1.00	120745	-1.01	1.01	1.38	TRUE
48	Connor Roberts	Δ_{48}	1.00	113867	2.41	0.71	0.41	TRUE
49	Craig Goodwin	Δ_{49}	1.00	34010	0.93	-0.69	0.29	TRUE
50	Cristian Romero	Δ_{50}	1.00	114345	4.30	-0.89	-1.18	TRUE
51	Daley Blind	Δ_{51}	1.00	127121	-0.94	-0.57	-1.01	TRUE
52	Daniel Amartey	Δ_{52}	1.00	112490	-1.52	-0.38	-0.06	TRUE
53	Danilo	Δ_{53}	1.00	129313	0.11	0.35	0.14	TRUE
54	David Raum	Δ_{54}	1.00	9403	-2.68	0.16	0.60	TRUE
55	Dayot Upamecano	Δ_{55}	1.00	127116	1.80	0.37	0.88	TRUE
56	Declan Rice	Δ_{56}	1.00	6954	-2.06	0.43	-0.02	TRUE
57	Dejan Lovren	Δ_{57}	1.00	12295	-0.48	-0.41	0.55	TRUE
58	Denzel Dumfries	Δ_{58}	1.00	122408	0.50	0.73	1.96	TRUE
59	Diego Godin	Δ_{59}	1.00	109514	1.80	-1.20	0.01	TRUE
60	Diogo Dalot	Δ_{60}	1.00	120427	1.03	0.63	-0.04	TRUE
61	Djibril Sow	Δ_{61}	1.00	126330	0.93	-2.54	1.73	TRUE
62	Dusan Tadic	Δ_{62}	1.00	65798	1.63	-0.25	-0.51	TRUE
63	Dylan Bronn	Δ_{63}	1.00	48338	3.35	1.28	0.01	TRUE
64	Eder Militao	Δ_{64}	1.00	123633	0.51	-1.36	0.12	TRUE
65	Edson Alvarez	Δ_{65}	1.00	32164	-2.63	-0.16	0.60	TRUE
66	Ehsan Hajsafi	Δ_{66}	1.00	96106	1.29	0.92	-0.48	TRUE
67	Ellyes Skhiri	Δ_{67}	1.00	13753	-2.01	0.21	0.19	TRUE
68	Enzo Fernandez	Δ_{68}	1.00	48296	1.80	0.28	0.85	TRUE
69	Ethan Ampadu	Δ_{69}	1.00	67896	-2.66	-0.16	0.30	TRUE
70	Facundo Pellistri	Δ_{70}	1.00	38776	2.32	-0.73	-0.50	TRUE

Table A.2.7: *Convergence diagnostic results for parameters 71–105, under the negative binomial model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
71	Federico Valverde	Δ_{71}	1.00	78986	-0.99	0.37	-0.33	TRUE
72	Felix Torres Caicedo	Δ_{72}	1.00	93603	-1.63	-0.70	0.83	TRUE
73	Filip Kostic	Δ_{73}	1.00	56206	3.12	0.41	-0.02	TRUE
74	Firas Al-Buraikan	Δ_{74}	1.00	112439	0.29	-0.13	-0.60	TRUE
75	Francisco Calvo	Δ_{75}	1.00	128173	-0.98	0.36	0.78	TRUE
76	Frenkie de Jong	Δ_{76}	1.00	118679	0.96	-1.84	1.40	TRUE
77	Gavi	Δ_{77}	1.00	61543	2.61	1.32	-1.38	TRUE
78	Gonzalo Plata	Δ_{78}	1.00	104246	0.67	-0.27	-0.70	TRUE
79	Granit Xhaka	Δ_{79}	1.00	101661	1.92	1.37	-1.72	TRUE
80	Grzegorz Krychowiak	Δ_{80}	1.00	44017	-1.93	-1.14	0.91	TRUE
81	Guillermo Varela	Δ_{81}	1.00	124235	-1.24	0.17	-0.57	TRUE
82	Harry Maguire	Δ_{82}	1.00	121799	2.31	0.95	0.25	TRUE
83	Harry Souttar	Δ_{83}	1.00	38287	-4.37	-0.29	0.30	TRUE
84	Harry Wilson	Δ_{84}	1.00	120427	1.75	0.80	0.06	TRUE
85	Hassan Al-Haydos	Δ_{85}	1.00	12772	0.91	1.28	-0.11	TRUE
86	Hassan Al Tambakti	Δ_{86}	1.00	67112	-3.09	-0.54	0.28	TRUE
87	Hector Herrera	Δ_{87}	1.00	62754	-0.91	0.09	0.45	TRUE
88	Hector Moreno	Δ_{88}	1.00	60834	2.27	0.54	0.78	TRUE
89	Hidemasa Morita	Δ_{89}	1.00	126333	0.40	0.13	0.28	TRUE
90	Homam Ahmed	Δ_{90}	1.00	110145	2.14	-0.93	-2.31	TRUE
91	Hwang In-beom	Δ_{91}	1.00	123002	0.38	0.33	-0.39	TRUE
92	Ibrahima Konate	Δ_{92}	1.00	21647	-2.35	0.00	-0.11	TRUE
93	Idrissa Gana Gueye	Δ_{93}	1.00	43234	-3.54	-0.30	-0.17	TRUE
94	Ilkay Gundogan	Δ_{94}	1.00	102714	-1.05	-0.06	2.22	TRUE
95	Ismail Jakobs	Δ_{95}	1.00	94057	-2.62	0.07	1.45	TRUE
96	Jackson Irvine	Δ_{96}	1.00	119664	0.02	2.28	-0.31	TRUE
97	Jakub Kaminski	Δ_{97}	1.00	43645	2.50	0.05	0.18	TRUE
98	Jakub Kiwior	Δ_{98}	1.00	99593	-0.76	1.79	-0.61	TRUE
99	Jamal Musiala	Δ_{99}	1.00	64302	1.07	-0.05	0.85	TRUE
100	Jan Vertonghen	Δ_{100}	1.00	34961	-2.09	0.33	0.96	TRUE
101	Javad El Yamiq	Δ_{101}	1.00	51665	-0.91	0.23	-0.13	TRUE
102	Jean-Charles Castelletto	Δ_{102}	1.00	13631	-2.39	0.57	0.72	TRUE
103	Jesper Lindstrom	Δ_{103}	1.00	114098	1.79	-1.86	1.16	TRUE
104	Jesus Gallardo	Δ_{104}	1.00	61169	2.01	0.33	-0.58	TRUE
105	Jhegson Mendez	Δ_{105}	1.00	117423	1.10	-1.84	-1.61	TRUE

Table A.2.8: *Convergence diagnostic results for parameters 106–140, under the negative binomial model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
106	Joachim Andersen	Δ_{106}	1.00	33159	2.03	0.07	-0.29	TRUE
107	Joakim Maehle	Δ_{107}	1.00	120977	-0.47	-0.26	1.38	TRUE
108	Joao Cancelo	Δ_{108}	1.00	91060	1.85	0.79	0.01	TRUE
109	Joe Rodon	Δ_{109}	1.00	80003	-1.86	-1.13	0.41	TRUE
110	John Stones	Δ_{110}	1.00	19001	2.56	0.62	0.27	TRUE
111	Jordan Henderson	Δ_{111}	1.00	51522	1.70	-0.47	-0.12	TRUE
112	Jordi Alba	Δ_{112}	1.00	119078	-1.16	1.99	0.69	TRUE
113	Jorge Eduardo Sanchez	Δ_{113}	1.00	67844	-4.67	-0.12	0.13	TRUE
114	Jose Maria Gimenez	Δ_{114}	1.00	22036	-3.08	0.33	0.85	TRUE
115	Joshua Kimmich	Δ_{115}	1.00	12395	-0.59	1.13	0.30	TRUE
116	Josip Juranovic	Δ_{116}	1.00	37903	2.69	-2.01	-0.50	TRUE
117	Josko Gvardiol	Δ_{117}	1.00	18028	-2.21	0.26	0.07	TRUE
118	Jude Bellingham	Δ_{118}	1.00	118546	-2.57	-0.04	0.64	TRUE
119	Jules Kounde	Δ_{119}	1.00	95536	1.03	-1.56	-0.39	TRUE
120	Jung Woo-young	Δ_{120}	1.00	130910	-0.59	-0.05	-0.67	TRUE
121	Junya Ito	Δ_{121}	1.00	52359	-2.92	0.86	0.40	TRUE
122	Jurien Timber	Δ_{122}	1.00	12251	-1.62	0.58	0.01	TRUE
123	Kalidou Koulibaly	Δ_{123}	1.00	53997	3.04	-0.18	-0.59	TRUE
124	Kamal Miller	Δ_{124}	1.00	65045	-1.19	-0.68	1.35	TRUE
125	Kamil Glik	Δ_{125}	1.00	122947	1.68	-0.50	-1.28	TRUE
126	Karim Boudiaf	Δ_{126}	1.00	45805	2.49	-1.47	-0.00	TRUE
127	Kendall Waston	Δ_{127}	1.00	35972	-2.30	0.18	0.33	TRUE
128	Keysher Fuller	Δ_{128}	1.00	127891	0.80	-1.11	-0.42	TRUE
129	Kim Jin-su	Δ_{129}	1.00	127263	1.41	-1.52	-2.43	TRUE
130	Kim Min-jae	Δ_{130}	1.00	120259	1.89	-1.24	-2.48	TRUE
131	Kim Moon-hwan	Δ_{131}	1.00	124065	-1.73	1.26	-1.05	TRUE
132	Kim Young-gwon	Δ_{132}	1.00	117966	2.24	0.83	-0.30	TRUE
133	Ko Itakura	Δ_{133}	1.00	78485	-1.05	-0.12	1.41	TRUE
134	Krystian Bielik	Δ_{134}	1.00	19901	-3.77	-0.78	-0.12	TRUE
135	Kye Rowles	Δ_{135}	1.00	19618	-2.15	-0.60	-0.43	TRUE
136	Kyle Walker	Δ_{136}	1.00	12065	-0.48	1.28	-0.12	TRUE
137	Leander Dendoncker	Δ_{137}	1.00	60634	2.13	0.17	-0.47	TRUE
138	Lee Jae-sung	Δ_{138}	1.00	100463	-0.74	0.02	-1.03	TRUE
139	Leon Goretzka	Δ_{139}	1.00	113993	-0.65	0.41	-0.40	TRUE
140	Lisandro Martinez	Δ_{140}	1.00	122770	-1.70	0.59	-0.05	TRUE

Table A.2.9: *Convergence diagnostic results for parameters 141–175, under the negative binomial model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
141	Lucas Paqueta	Δ_{141}	1.00	83234	2.01	0.00	-0.84	TRUE
142	Luis Chavez	Δ_{142}	1.00	59347	2.67	-0.29	-0.38	TRUE
143	Luka Modric	Δ_{143}	1.00	22785	-1.99	0.23	0.15	TRUE
144	Luke Shaw	Δ_{144}	1.00	12307	1.35	1.00	-1.05	TRUE
145	Majid Hosseini	Δ_{145}	1.00	38262	-1.11	0.34	0.23	TRUE
146	Manuel Akanji	Δ_{146}	1.00	124376	-0.46	0.74	1.86	TRUE
147	Marcelo Brozovic	Δ_{147}	1.00	86306	1.86	-0.49	-0.55	TRUE
148	Marcos Acuna	Δ_{148}	1.00	73735	3.52	-0.95	-0.56	TRUE
149	Marquinhos	Δ_{149}	1.00	47738	2.25	0.24	0.24	TRUE
150	Mateo Kovacic	Δ_{150}	1.00	62712	1.63	0.11	0.94	TRUE
151	Mathew Leckie	Δ_{151}	1.00	44228	1.40	-0.10	0.57	TRUE
152	Mathias Olivera	Δ_{152}	1.00	80286	4.12	-0.40	-0.54	TRUE
153	Matias Vecino	Δ_{153}	1.00	60650	-1.44	0.79	-0.31	TRUE
154	Matty Cash	Δ_{154}	1.00	23853	2.71	-1.26	-0.05	TRUE
155	Maya Yoshida	Δ_{155}	1.00	121497	0.69	1.25	-0.63	TRUE
156	Mikkel Damsgaard	Δ_{156}	1.00	118535	-1.72	-0.29	-1.75	TRUE
157	Milad Mohammadi	Δ_{157}	1.00	7066	2.84	0.34	-1.95	TRUE
158	Milos Degenek	Δ_{158}	1.00	118235	2.05	-0.92	-0.20	TRUE
159	Milos Veljkovic	Δ_{159}	1.00	41368	1.05	0.82	-0.44	TRUE
160	Mohamed Kanno	Δ_{160}	1.00	12325	0.67	0.24	0.29	TRUE
161	Mohammed Salisu	Δ_{161}	1.00	119420	-1.08	-1.24	-0.02	TRUE
162	Moises Caicedo	Δ_{162}	1.00	96235	2.64	1.18	0.42	TRUE
163	Montassar Talbi	Δ_{163}	1.00	96902	0.13	-0.22	-1.40	TRUE
164	Morteza Pouraliganji	Δ_{164}	1.00	39045	-2.31	-0.52	0.34	TRUE
165	Nahuel Molina	Δ_{165}	1.00	53251	2.37	0.65	-0.94	TRUE
166	Nampalys Mendy	Δ_{166}	1.00	124969	0.19	0.72	0.85	TRUE
167	Nathan Ake	Δ_{167}	1.00	73644	-1.33	0.27	0.32	TRUE
168	Nayef Aguerd	Δ_{168}	1.00	123446	-1.40	-2.45	0.87	TRUE
169	Nico Williams	Δ_{169}	1.00	45834	-2.65	1.30	0.02	TRUE
170	Neymar	Δ_{170}	1.00	59268	2.95	0.02	-0.15	TRUE
171	Nico Elvedi	Δ_{171}	1.00	11373	1.59	-1.21	1.52	TRUE
172	Nicolas Nkoulou	Δ_{172}	1.00	25185	-1.99	0.92	-0.32	TRUE
173	Nicolas Otamendi	Δ_{173}	1.00	36161	-4.32	0.09	-0.52	TRUE
174	Nicolas Tagliafico	Δ_{174}	1.00	8261	-3.03	0.31	0.35	TRUE
175	Niklas Sule	Δ_{175}	1.00	31698	3.16	-0.22	-1.07	TRUE

Table A.2.10: *Convergence diagnostic results for parameters 176–210, under the negative binomial model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
176	Nikola Milenkovic	Δ_{176}	1.00	17960	-3.42	-0.13	0.36	TRUE
177	Nouhou Tolo	Δ_{177}	1.00	47194	-2.94	0.68	0.76	TRUE
178	Noussair Mazraoui	Δ_{178}	1.00	29933	2.05	-0.71	-0.47	TRUE
179	Oscar Duarte	Δ_{179}	1.00	29114	-1.08	0.88	0.10	TRUE
180	Pedri	Δ_{180}	1.00	111859	-0.07	0.03	-0.37	TRUE
181	Pepe	Δ_{181}	1.00	38841	-2.45	0.07	0.17	TRUE
182	Pervis Estupinan	Δ_{182}	1.00	122968	1.57	-0.63	-0.10	TRUE
183	Piero Hincapie	Δ_{183}	1.00	91244	1.76	0.48	0.95	TRUE
184	Pierre Hojbjerg	Δ_{184}	1.00	123066	-3.32	-0.57	-1.53	TRUE
185	Piotr Zielinski	Δ_{185}	1.00	56556	3.52	0.00	-0.62	TRUE
186	Przemyslaw Frankowski	Δ_{186}	1.00	109593	-2.87	1.28	-0.04	TRUE
187	Ramin Rezaeian	Δ_{187}	1.00	114118	-0.90	-0.87	1.37	TRUE
188	Raphael Guerreiro	Δ_{188}	1.00	120320	1.17	0.67	-1.00	TRUE
189	Raphael Varane	Δ_{189}	1.00	110574	0.20	1.24	-0.90	TRUE
190	Rasmus Nissen	Δ_{190}	1.00	101174	1.74	1.21	0.89	TRUE
191	Remo Freuler	Δ_{191}	1.00	124662	-1.54	0.43	-0.17	TRUE
192	Ricardo Rodriguez	Δ_{192}	1.00	23219	1.71	-0.22	0.25	TRUE
193	Richie Laryea	Δ_{193}	1.00	59842	1.20	-0.76	0.49	TRUE
194	Ro-Ro	Δ_{194}	1.00	8333	-2.65	-0.03	0.36	TRUE
195	Rodri	Δ_{195}	1.00	12535	1.43	1.71	-0.08	TRUE
196	Rodrigo Bentancur	Δ_{196}	1.00	11811	-2.35	0.14	0.13	TRUE
197	Rodrigo De Paul	Δ_{197}	1.00	80328	-1.16	0.99	-0.57	TRUE
198	Romain Saiss	Δ_{198}	1.00	64195	1.33	-0.37	-2.07	TRUE
199	Ruben Dias	Δ_{199}	1.00	12357	1.00	0.51	-0.63	TRUE
200	Ruben Neves	Δ_{200}	1.00	123119	-2.75	0.86	0.85	TRUE
201	Saeid Ezatolahi	Δ_{201}	1.00	115173	0.47	-0.28	1.11	TRUE
202	Salem Al-Dawsari	Δ_{202}	1.00	61007	1.72	-1.21	-1.49	TRUE
203	Sali Abdul Samed	Δ_{203}	1.00	57135	-1.01	-0.29	-0.20	TRUE
204	Sasa Lucic	Δ_{204}	1.00	85256	1.76	0.57	-1.18	TRUE
205	Saud Abdulhamid	Δ_{205}	1.00	108943	-1.01	2.08	-0.41	TRUE
206	Sebastian Coates	Δ_{206}	1.00	80064	2.92	-0.20	-0.05	TRUE
207	Selim Amallah	Δ_{207}	1.00	17423	-3.02	0.35	-0.34	TRUE
208	Sergej Milinkovic-Savic	Δ_{208}	1.00	123417	-0.37	0.18	0.09	TRUE
209	Sergio Dest	Δ_{209}	1.00	94839	0.64	1.84	1.19	TRUE
210	Sergio Busquets	Δ_{210}	1.00	41691	3.11	-1.06	-0.40	TRUE

Table A.2.11: *Convergence diagnostic results for parameters 211–236, under the negative binomial model for interceptions.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
211	Shogo Taniguchi	Δ_{211}	1.00	101883	1.66	-1.34	-1.90	TRUE
212	Silvan Widmer	Δ_{212}	1.00	127194	0.05	0.22	0.16	TRUE
213	Sofyan Amrabat	Δ_{213}	1.00	122968	1.61	0.14	0.00	TRUE
214	Steven Vitoria	Δ_{214}	1.00	31719	2.46	0.21	-0.36	TRUE
215	Strahinja Pavlovic	Δ_{215}	1.00	8534	2.09	-0.38	0.95	TRUE
216	Tajon Buchanan	Δ_{216}	1.00	116046	-2.07	-2.25	1.09	TRUE
217	Takehiro Tomiyasu	Δ_{217}	1.00	50434	3.08	-0.56	-0.76	TRUE
218	Theo Hernandez	Δ_{218}	1.00	37296	1.67	0.13	-0.41	TRUE
219	Thiago Silva	Δ_{219}	1.00	96350	-0.03	-0.58	1.95	TRUE
220	Thomas Meunier	Δ_{220}	1.00	131053	-0.30	-2.14	1.42	TRUE
221	Thomas Partey	Δ_{221}	1.00	29416	2.42	1.16	-0.44	TRUE
222	Tim Ream	Δ_{222}	1.00	122866	-1.91	0.51	-0.23	TRUE
223	Timothy Castagne	Δ_{223}	1.00	117876	0.32	-0.50	2.92	TRUE
224	Toby Alderweireld	Δ_{224}	1.00	28509	2.87	-0.40	0.02	TRUE
225	Tyler Adams	Δ_{225}	1.00	120143	-0.98	-0.48	0.41	TRUE
226	Virgil van Dijk	Δ_{226}	1.00	47490	3.21	0.07	-0.57	TRUE
227	Walker Zimmerman	Δ_{227}	1.00	95922	-1.36	1.45	0.48	TRUE
228	Wataru Endo	Δ_{228}	1.00	129334	-0.56	-1.41	-1.95	TRUE
229	Weston McKennie	Δ_{229}	1.00	26550	2.85	-0.51	-0.78	TRUE
230	Yahya Attiat Allah	Δ_{230}	1.00	23113	2.30	-0.88	-0.10	TRUE
231	Yassine Meriah	Δ_{231}	1.00	51717	2.40	0.25	-0.37	TRUE
232	Yeltsin Tejeda	Δ_{232}	1.00	44268	2.22	-0.78	-1.25	TRUE
233	Youssef Msakni	Δ_{233}	1.00	46551	2.25	-0.06	-0.76	TRUE
234	Youssouf Sabaly	Δ_{234}	1.00	130197	-0.44	-1.14	2.12	TRUE
235	Yunus Musah	Δ_{235}	1.00	69634	0.95	0.38	0.98	TRUE
236	Yuto Nagatomo	Δ_{236}	1.00	41560	1.75	0.83	0.93	TRUE

Table A.2.12: *Convergence diagnostic results for the team-effect parameters under the negative binomial model for interceptions. The parameter column reports λ_i^e for team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	λ_1^e	1.00	58785	-0.24	-1.47	1.24	TRUE
2	Australia	λ_2^e	1.00	94921	-1.48	-0.25	-0.30	TRUE
3	Belgium	λ_3^e	1.00	117115	-0.50	0.58	0.99	TRUE
4	Brazil	λ_4^e	1.00	95249	-0.28	0.79	-1.04	TRUE
5	Cameroon	λ_5^e	1.00	65717	-0.27	0.84	-0.87	TRUE
6	Canada	λ_6^e	1.00	111698	1.22	-0.33	-1.24	TRUE
7	Costa Rica	λ_7^e	1.00	96246	-1.43	0.95	-0.50	TRUE
8	Croatia	λ_8^e	1.00	52866	1.77	0.01	-0.20	TRUE
9	Denmark	λ_9^e	1.00	99286	1.34	-0.19	-0.54	TRUE
10	Ecuador	λ_{10}^e	1.00	110671	-0.22	0.02	0.58	TRUE
11	England	λ_{11}^e	1.00	85522	0.09	0.64	1.11	TRUE
12	France	λ_{12}^e	1.00	54673	-0.59	0.45	-0.74	TRUE
13	Germany	λ_{13}^e	1.00	109871	0.01	-0.16	0.44	TRUE
14	Ghana	λ_{14}^e	1.00	111387	-0.38	0.37	-0.41	TRUE
15	IR Iran	λ_{15}^e	1.00	77404	-1.14	0.65	-0.14	TRUE
16	Japan	λ_{16}^e	1.00	107506	-1.46	-0.51	-0.25	TRUE
17	Korea Republic	λ_{17}^e	1.00	105749	-0.70	0.45	0.98	TRUE
18	Mexico	λ_{18}^e	1.00	114099	-0.56	-1.15	1.45	TRUE
19	Morocco	λ_{19}^e	1.00	58645	-0.04	-0.49	0.91	TRUE
20	Netherlands	λ_{20}^e	1.00	66072	-0.35	0.60	-0.83	TRUE
21	Poland	λ_{21}^e	1.00	87680	0.76	-0.98	2.00	TRUE
22	Portugal	λ_{22}^e	1.00	109968	0.36	-0.42	-1.68	TRUE
23	Qatar	λ_{23}^e	1.00	116016	-0.40	0.27	0.41	TRUE
24	Saudi Arabia	λ_{24}^e	1.00	116618	0.39	-1.36	-0.44	TRUE
25	Senegal	λ_{25}^e	1.00	106584	0.13	0.18	-0.75	TRUE
26	Serbia	λ_{26}^e	1.00	108241	0.56	0.32	-0.45	TRUE
27	Spain	λ_{27}^e	1.00	93176	0.12	-1.07	0.69	TRUE
28	Switzerland	λ_{28}^e	1.00	84320	2.10	-0.44	0.71	TRUE
29	Tunisia	λ_{29}^e	1.00	113506	0.21	0.92	-0.81	TRUE
30	United States	λ_{30}^e	1.00	97397	-0.39	-0.68	0.97	TRUE
31	Uruguay	λ_{31}^e	1.00	79058	-0.10	0.91	-1.88	TRUE
32	Wales	λ_{32}^e	1.00	111244	1.41	0.69	0.46	TRUE

Table A.2.13: *Convergence diagnostic results for the opponent team-effect parameters under the negative binomial model for interceptions. The parameter column reports $\lambda_i^{E\backslash e}$ for opponent team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	$\lambda_1^{E\backslash e}$	1.00	51051	-1.28	0.77	-1.01	TRUE
2	Australia	$\lambda_2^{E\backslash e}$	1.00	75613	0.46	1.63	-0.27	TRUE
3	Belgium	$\lambda_3^{E\backslash e}$	1.00	75084	1.66	-0.41	1.18	TRUE
4	Brazil	$\lambda_4^{E\backslash e}$	1.00	80115	0.57	0.68	0.43	TRUE
5	Cameroon	$\lambda_5^{E\backslash e}$	1.00	56190	0.15	-1.29	0.27	TRUE
6	Canada	$\lambda_6^{E\backslash e}$	1.00	74847	0.28	1.23	0.35	TRUE
7	Costa Rica	$\lambda_7^{E\backslash e}$	1.00	88293	1.78	-0.59	-1.24	TRUE
8	Croatia	$\lambda_8^{E\backslash e}$	1.00	50282	-2.02	0.23	0.44	TRUE
9	Denmark	$\lambda_9^{E\backslash e}$	1.00	67354	-2.13	0.24	-0.23	TRUE
10	Ecuador	$\lambda_{10}^{E\backslash e}$	1.00	72377	-0.03	0.02	-1.81	TRUE
11	England	$\lambda_{11}^{E\backslash e}$	1.00	69902	0.09	0.09	1.19	TRUE
12	France	$\lambda_{12}^{E\backslash e}$	1.00	53530	0.74	-1.06	0.21	TRUE
13	Germany	$\lambda_{13}^{E\backslash e}$	1.00	87431	0.32	0.84	-0.02	TRUE
14	Ghana	$\lambda_{14}^{E\backslash e}$	1.00	82012	-0.25	0.58	-0.56	TRUE
15	IR Iran	$\lambda_{15}^{E\backslash e}$	1.00	89217	0.14	0.88	0.41	TRUE
16	Japan	$\lambda_{16}^{E\backslash e}$	1.00	83892	-0.13	-0.76	1.01	TRUE
17	Korea Republic	$\lambda_{17}^{E\backslash e}$	1.00	78069	0.66	-1.13	0.29	TRUE
18	Mexico	$\lambda_{18}^{E\backslash e}$	1.00	60607	1.08	-1.23	-1.24	TRUE
19	Morocco	$\lambda_{19}^{E\backslash e}$	1.00	75654	-1.60	-0.46	0.74	TRUE
20	Netherlands	$\lambda_{20}^{E\backslash e}$	1.00	83732	-0.11	2.88	-0.48	TRUE
21	Poland	$\lambda_{21}^{E\backslash e}$	1.00	84162	0.90	0.57	-0.90	TRUE
22	Portugal	$\lambda_{22}^{E\backslash e}$	1.00	75747	0.33	0.30	0.77	TRUE
23	Qatar	$\lambda_{23}^{E\backslash e}$	1.00	88246	0.79	-0.83	2.10	TRUE
24	Saudi Arabia	$\lambda_{24}^{E\backslash e}$	1.00	82072	-0.31	0.23	0.37	TRUE
25	Senegal	$\lambda_{25}^{E\backslash e}$	1.00	69710	-0.39	1.43	-1.00	TRUE
26	Serbia	$\lambda_{26}^{E\backslash e}$	1.00	86491	0.51	0.82	-0.88	TRUE
27	Spain	$\lambda_{27}^{E\backslash e}$	1.00	81504	-1.08	-0.63	-0.92	TRUE
28	Switzerland	$\lambda_{28}^{E\backslash e}$	1.00	94173	0.24	0.00	-1.03	TRUE
29	Tunisia	$\lambda_{29}^{E\backslash e}$	1.00	91953	0.92	0.20	0.91	TRUE
30	United States	$\lambda_{30}^{E\backslash e}$	1.00	67162	-0.10	-0.18	0.31	TRUE
31	Uruguay	$\lambda_{31}^{E\backslash e}$	1.00	79598	-0.81	-0.55	0.67	TRUE
32	Wales	$\lambda_{32}^{E\backslash e}$	1.00	60249	-0.44	1.06	0.63	TRUE

Appendix B

Shots

B.1 Poisson Model

Table B.1.1: *Team effects for each national team under the Poisson model for shots, showing each team's ability to create shooting chances, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	0.05	0.05	-0.06	0.15
2	Australia	-0.05	0.07	-0.20	0.09
3	Belgium	-0.01	0.08	-0.17	0.15
4	Brazil	0.09	0.07	-0.05	0.24
5	Cameroon	-0.03	0.08	-0.20	0.13
6	Canada	0.03	0.08	-0.13	0.19
7	Costa Rica	-0.09	0.09	-0.29	0.08
8	Croatia	-0.02	0.05	-0.13	0.09
9	Denmark	-0.04	0.08	-0.21	0.13
10	Ecuador	-0.01	0.08	-0.17	0.15
11	England	0.02	0.07	-0.11	0.16
12	France	0.06	0.05	-0.05	0.17
13	Germany	0.12	0.09	-0.04	0.30
14	Ghana	-0.04	0.08	-0.20	0.12
15	IR Iran	-0.01	0.08	-0.17	0.15
16	Japan	-0.02	0.07	-0.16	0.13
17	Korea Republic	0.03	0.07	-0.11	0.18
18	Mexico	0.05	0.08	-0.11	0.22
19	Morocco	-0.03	0.05	-0.13	0.08
20	Netherlands	-0.01	0.07	-0.15	0.13
21	Poland	-0.02	0.07	-0.17	0.12
22	Portugal	0.03	0.07	-0.11	0.17
23	Qatar	-0.09	0.09	-0.27	0.08
24	Saudi Arabia	-0.00	0.08	-0.16	0.16
25	Senegal	0.05	0.08	-0.09	0.20
26	Serbia	-0.01	0.08	-0.17	0.15
27	Spain	0.00	0.07	-0.14	0.14
28	Switzerland	0.01	0.07	-0.13	0.15
29	Tunisia	-0.01	0.08	-0.17	0.16
30	United States	0.01	0.07	-0.13	0.15
31	Uruguay	-0.01	0.08	-0.17	0.15
32	Wales	-0.06	0.08	-0.24	0.10

Table B.1.2: *Opponent team effects for each national team under the Poisson model for shots, showing how strongly each team limits their opponents from shooting, with posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-0.05	0.12	-0.31	0.19
2	Australia	0.03	0.12	-0.22	0.29
3	Belgium	-0.09	0.13	-0.36	0.15
4	Brazil	-0.00	0.13	-0.26	0.26
5	Cameroon	-0.03	0.13	-0.30	0.25
6	Canada	-0.06	0.13	-0.34	0.20
7	Costa Rica	-0.00	0.13	-0.28	0.26
8	Croatia	0.00	0.13	-0.27	0.28
9	Denmark	-0.02	0.13	-0.28	0.24
10	Ecuador	0.03	0.13	-0.23	0.30
11	England	-0.00	0.13	-0.26	0.26
12	France	-0.02	0.13	-0.29	0.24
13	Germany	0.03	0.14	-0.23	0.31
14	Ghana	0.05	0.14	-0.22	0.33
15	IR Iran	-0.01	0.13	-0.29	0.26
16	Japan	0.02	0.13	-0.26	0.29
17	Korea Republic	0.05	0.14	-0.21	0.34
18	Mexico	0.06	0.14	-0.21	0.36
19	Morocco	-0.01	0.13	-0.27	0.25
20	Netherlands	0.04	0.14	-0.22	0.34
21	Poland	0.08	0.14	-0.19	0.38
22	Portugal	0.03	0.13	-0.24	0.30
23	Qatar	0.02	0.13	-0.25	0.29
24	Saudi Arabia	0.10	0.15	-0.17	0.43
25	Senegal	-0.05	0.13	-0.32	0.22
26	Serbia	-0.07	0.14	-0.37	0.20
27	Spain	-0.03	0.13	-0.23	0.31
28	Switzerland	0.01	0.13	-0.26	0.29
29	Tunisia	0.02	0.13	-0.24	0.30
30	United States	0.10	0.15	-0.17	0.43
31	Uruguay	-0.05	0.13	-0.32	0.22
32	Wales	-0.07	0.14	-0.37	0.20

Table B.1.3: *Top 10 players under the Poisson model for shots, ranked by posterior predicted total shots, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Shots	Lower HPD Bound	Upper HPD Bound	Observed Shots	90-minute units
1	Kylian Mbappe	26.67	17.29	36.56	29	6.6
2	Lionel Messi	25.09	16.13	34.84	27	7.7
3	Ivan Perisic	14.32	7.71	21.32	16	7.4
4	Olivier Giroud	14.23	7.83	21.26	16	4.7
5	Julian Alvarez	10.41	4.78	15.73	11	5.2
6	Dani Olmo	9.87	4.62	15.53	12	4.1
7	Jamal Musiala	9.81	4.67	15.48	12	2.9
8	Serge Gnabry	9.80	4.66	15.56	12	2.9
9	Neymar	9.12	4.30	14.62	11	3.1
10	Joao Felix	9.06	4.11	14.46	11	3.7

Table B.1.4: *Bottom 10 players under the Poisson model for shots, ranked by posterior predicted total shots, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Shots	Lower HPD Bound	Upper HPD Bound	Observed Shots	90-minute units
1	Assim Madibo	0.98	0.17	2.10	0	1.7
2	Daniel James	1.02	0.16	2.11	0	1.7
3	Mikkel Damsgaard	1.04	0.18	2.19	0	1.7
4	Karim Boudiaf	1.09	0.19	2.31	0	2.1
5	Jhegson Mendez	1.23	0.21	2.54	0	2.0
6	Celso Borges	1.25	0.22	2.64	0	2.8
7	Jordan Ayew	1.28	0.25	2.62	1	1.5
8	Hector Herrera	1.33	0.25	2.75	0	1.8
9	Anis Ben Slimane	1.35	0.28	2.78	1	1.6
10	Yuto Nagatomo	1.36	0.26	2.78	0	2.3

Table B.1.5: *Convergence diagnostic results for the first 35 parameters, under the Poisson model for shots. The parameter column reports Δ_i for player effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Julian Alvarez	Δ_1	1.00	38243	0.33	0.96	-0.18	TRUE
2	Lionel Messi	Δ_2	1.00	18180	-0.25	0.73	0.01	TRUE
3	Enzo Fernandez	Δ_3	1.00	35678	-0.26	0.96	0.18	TRUE
4	Rodrigo De Paul	Δ_4	1.00	36156	-0.37	0.81	0.15	TRUE
5	Alexis Mac Allister	Δ_5	1.00	40680	-0.71	0.89	-0.29	TRUE
6	Mitchel Duke	Δ_6	1.00	83445	-0.23	0.08	1.84	TRUE
7	Riley McGree	Δ_7	1.00	104289	0.65	0.87	-0.25	TRUE
8	Aaron Mooy	Δ_8	1.00	91117	-0.26	0.72	-1.70	TRUE
9	Craig Goodwin	Δ_9	1.00	115491	-0.70	0.01	-0.91	TRUE
10	Jackson Irvine	Δ_{10}	1.00	92484	0.53	1.51	-2.14	TRUE
11	Mathew Leckie	Δ_{11}	1.00	97599	-0.73	1.47	-0.58	TRUE
12	Kevin De Bruyne	Δ_{12}	1.00	60849	-0.39	-1.95	-0.11	TRUE
13	Michy Batshuayi	Δ_{13}	1.00	92765	0.45	-0.66	1.67	TRUE
14	Axel Witsel	Δ_{14}	1.00	96499	-1.50	-0.64	-1.62	TRUE
15	Raphinha	Δ_{15}	1.00	56301	0.28	0.92	0.35	TRUE
16	Richarlison	Δ_{16}	1.00	54201	-0.22	1.13	0.80	TRUE
17	Vinicius Junior	Δ_{17}	1.00	53465	0.33	0.69	-1.16	TRUE
18	Neymar	Δ_{18}	1.00	45294	0.51	0.38	-0.06	TRUE
19	Casemiro	Δ_{19}	1.00	40652	0.63	0.36	-0.42	TRUE
20	Lucas Paqueta	Δ_{20}	1.00	76305	-1.00	-0.12	0.27	TRUE
21	Bryan Mbeumo	Δ_{21}	1.00	105995	0.95	-0.03	0.54	TRUE
22	Eric Maxim Choupo-Moting	Δ_{22}	1.00	101192	-0.22	-1.64	0.29	TRUE
23	Karl Toko Ekambi	Δ_{23}	1.00	121031	-0.08	-2.00	0.50	TRUE
24	Vincent Aboubakar	Δ_{24}	1.00	103324	-0.44	-0.32	0.19	TRUE
25	Andre-Frank Zambo Anguissa	Δ_{25}	1.00	104211	1.19	-0.18	-0.25	TRUE
26	Jonathan David	Δ_{26}	1.00	77037	-0.14	1.57	0.38	TRUE
27	Cyle Larin	Δ_{27}	1.00	118503	-1.73	0.51	0.17	TRUE
28	Junior Hoilett	Δ_{28}	1.00	108385	1.07	-0.53	-1.17	TRUE
29	Alphonso Davies	Δ_{29}	1.00	90116	2.51	0.12	-0.01	TRUE
30	Tajon Buchanan	Δ_{30}	1.00	8002	0.39	-1.31	0.67	TRUE
31	Atiba Hutchinson	Δ_{31}	1.00	118161	-1.72	-0.67	0.80	TRUE
32	Joel Campbell	Δ_{32}	1.00	9704	-1.33	0.13	0.63	TRUE
33	Celso Borges	Δ_{33}	1.00	95951	-1.58	0.28	0.37	TRUE
34	Yeltsin Tejeda	Δ_{34}	1.00	92258	-0.35	1.27	-0.58	TRUE
35	Andrej Kramaric	Δ_{35}	1.00	50588	1.96	1.82	-0.93	TRUE

Table B.1.6: *Convergence diagnostic results for parameters 36–70, under the Poisson model for shots.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
36	Ivan Perisic	Δ_{36}	1.00	29337	1.01	1.23	-0.16	TRUE
37	Mario Pasalic	Δ_{37}	1.00	110524	-0.98	2.38	0.08	TRUE
38	Luka Modric	Δ_{38}	1.00	45919	0.77	1.51	1.16	TRUE
39	Mateo Kovacic	Δ_{39}	1.00	57733	0.70	1.00	1.29	TRUE
40	Marcelo Brozovic	Δ_{40}	1.00	59739	-0.16	1.20	-0.82	TRUE
41	Christian Eriksen	Δ_{41}	1.00	48170	-1.05	1.54	-0.49	TRUE
42	Jesper Lindstrom	Δ_{42}	1.00	109023	2.20	1.43	-0.63	TRUE
43	Pierre Hojbjerg	Δ_{43}	1.00	95804	-2.18	-0.81	-0.72	TRUE
44	Mikkel Damsgaard	Δ_{44}	1.00	110482	-1.95	0.19	0.83	TRUE
45	Enner Valencia	Δ_{45}	1.00	77682	-0.57	1.08	-0.79	TRUE
46	Michael Estrada	Δ_{46}	1.00	109557	0.08	1.54	0.18	TRUE
47	Gonzalo Plata	Δ_{47}	1.00	103008	-0.81	-0.49	-0.63	TRUE
48	Moises Caicedo	Δ_{48}	1.00	100065	-0.04	1.26	-0.86	TRUE
49	Jhegson Mendez	Δ_{49}	1.00	106721	-0.24	1.25	-0.63	TRUE
50	Bukayo Saka	Δ_{50}	1.00	76440	-0.01	1.36	1.54	TRUE
51	Phil Foden	Δ_{51}	1.00	87616	-0.45	-0.34	-1.76	TRUE
52	Harry Kane	Δ_{52}	1.00	52463	-0.01	-1.36	0.78	TRUE
53	Jordan Henderson	Δ_{53}	1.00	104334	-0.30	-1.32	-0.03	TRUE
54	Declan Rice	Δ_{54}	1.00	82421	-1.23	0.17	-0.81	TRUE
55	Jude Bellingham	Δ_{55}	1.00	70428	-0.61	0.29	1.32	TRUE
56	Kylian Mbappe	Δ_{56}	1.00	19733	0.13	0.16	0.58	TRUE
57	Ousmane Dembele	Δ_{57}	1.00	4904	0.04	-0.03	0.82	TRUE
58	Olivier Giroud	Δ_{58}	1.00	31166	0.03	-0.07	1.67	TRUE
59	Randal Kolo Muani	Δ_{59}	1.00	108797	-0.11	1.07	-1.57	TRUE
60	Antoine Griezmann	Δ_{60}	1.00	44733	-0.20	1.22	0.52	TRUE
61	Aurelien Tchouameni	Δ_{61}	1.00	35999	-0.13	0.52	0.81	TRUE
62	Adrien Rabiot	Δ_{62}	1.00	36632	-0.44	-0.68	0.79	TRUE
63	Serge Gnabry	Δ_{63}	1.00	36222	2.10	-0.21	0.05	TRUE
64	Thomas Muller	Δ_{64}	1.00	9682	2.56	0.50	0.50	TRUE
65	Jamal Musiala	Δ_{65}	1.00	34887	2.05	-0.44	0.02	TRUE
66	Joshua Kimmich	Δ_{66}	1.00	49014	2.61	-2.08	0.02	TRUE
67	Ilkay Gundogan	Δ_{67}	1.00	84472	3.50	-0.71	-0.24	TRUE
68	Leon Goretzka	Δ_{68}	1.00	93719	2.13	-1.54	-1.16	TRUE
69	Inaki Williams	Δ_{69}	1.00	103822	-0.27	1.53	-1.97	TRUE
70	Kudus Mohammed	Δ_{70}	1.00	76313	0.77	0.48	-0.66	TRUE

Table B.1.7: *Convergence diagnostic results for parameters 71–105, under the Poisson model for shots.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
71	Jordan Ayew	Δ_{71}	1.00	12324	0.25	-1.32	-1.46	TRUE
72	Andre Ayew	Δ_{72}	1.00	108434	-0.69	1.46	-0.39	TRUE
73	Salis Abdul Samed	Δ_{73}	1.00	101879	-0.42	-0.86	1.02	TRUE
74	Thomas Partey	Δ_{74}	1.00	101021	-1.31	-0.45	0.84	TRUE
75	Mehdi Taremi	Δ_{75}	1.00	88493	-1.42	-0.71	0.06	TRUE
76	Ahmad Nourollahi	Δ_{76}	1.00	94492	-1.59	1.13	0.63	TRUE
77	Ali Gholizadeh	Δ_{77}	1.00	17123	-0.28	-0.50	0.35	TRUE
78	Ehsan Hajsafi	Δ_{78}	1.00	107739	-1.15	-1.12	-0.83	TRUE
79	Saeid Ezatolahi	Δ_{79}	1.00	111032	-1.63	0.08	-2.32	TRUE
80	Ritsu Doan	Δ_{80}	1.00	108968	0.62	0.87	0.97	TRUE
81	Daizen Maeda	Δ_{81}	1.00	116544	0.26	0.39	1.91	TRUE
82	Daichi Kamada	Δ_{82}	1.00	79555	0.72	0.06	0.61	TRUE
83	Yuto Nagatomo	Δ_{83}	1.00	107003	-0.21	2.59	-1.25	TRUE
84	Junya Ito	Δ_{84}	1.00	97912	1.09	0.08	0.40	FALSE
85	Wataru Endo	Δ_{85}	1.00	80999	0.96	1.03	0.22	TRUE
86	Hidemasa Morita	Δ_{86}	1.00	86154	-0.19	0.36	-0.23	FALSE
87	Cho Gue-sung	Δ_{87}	1.00	63813	1.87	-1.06	0.08	TRUE
88	Son Heung-min	Δ_{88}	1.00	54664	1.64	-0.52	-0.41	TRUE
89	Lee Jae-sung	Δ_{89}	1.00	116055	-0.09	-0.92	0.31	TRUE
90	Hwang In-beom	Δ_{90}	1.00	69577	1.03	-1.96	0.35	TRUE
91	Jung Woo-young	Δ_{91}	1.00	9996	0.32	0.97	0.81	TRUE
92	Alex Vega	Δ_{92}	1.00	65157	0.03	-0.40	0.09	TRUE
93	Hirving Lozano	Δ_{93}	1.00	68376	1.00	0.42	-0.21	TRUE
94	Henry Martin	Δ_{94}	1.00	102951	1.66	-0.52	-0.09	TRUE
95	Luis Chavez	Δ_{95}	1.00	45714	-0.96	-0.93	1.40	TRUE
96	Edson Alvarez	Δ_{96}	1.00	115819	0.18	0.66	1.31	TRUE
97	Hector Herrera	Δ_{97}	1.00	114803	1.28	-0.48	-0.52	TRUE
98	Hakim Ziyech	Δ_{98}	1.00	42948	-0.93	-0.94	0.08	TRUE
99	Sofiane Boufal	Δ_{99}	1.00	66264	-0.30	-1.30	0.42	TRUE
100	Youssef En-Nesyri	Δ_{100}	1.00	53756	-0.00	1.22	1.11	TRUE
101	Azzedine Ounahi	Δ_{101}	1.00	60774	-0.42	-1.40	-0.09	TRUE
102	Selim Amallah	Δ_{102}	1.00	9268	-0.84	0.29	1.76	TRUE
103	Sofyan Amrabat	Δ_{103}	1.00	69488	-0.27	0.11	0.19	TRUE
104	Cody Gakpo	Δ_{104}	1.00	63590	-0.44	0.67	1.31	TRUE
105	Memphis	Δ_{105}	1.00	68037	-0.30	-0.30	1.01	TRUE

Table B.1.8: *Convergence diagnostic results for parameters 106–140, under the Poisson model for shots.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
106	Frenkie de Jong	Δ_{106}	1.00	76257	-0.64	-0.43	0.47	TRUE
107	Robert Lewandowski	Δ_{107}	1.00	58017	-0.98	-0.06	0.91	TRUE
108	Jakub Kaminski	Δ_{108}	1.00	106316	-0.75	-0.57	-1.45	TRUE
109	Krystian Bielik	Δ_{109}	1.00	111787	-1.05	0.81	0.44	TRUE
110	Grzegorz Krychowiak	Δ_{110}	1.00	94027	-0.62	0.47	-0.50	TRUE
111	Piotr Zielinski	Δ_{111}	1.00	92425	-0.86	-0.84	0.05	TRUE
112	Przemyslaw Frankowski	Δ_{112}	1.00	113257	-1.18	0.13	1.26	TRUE
113	Cristiano Ronaldo	Δ_{113}	1.00	61383	1.91	0.40	-1.82	TRUE
114	Bruno Fernandes	Δ_{114}	1.00	57318	3.30	-0.69	-1.69	TRUE
115	Joao Felix	Δ_{115}	1.00	49488	1.72	0.15	-0.48	TRUE
116	Ruben Neves	Δ_{116}	1.00	10402	1.76	-0.98	0.27	TRUE
117	Bernardo Silva	Δ_{117}	1.00	88031	1.15	0.12	0.64	TRUE
118	Akram Afif	Δ_{118}	1.00	10420	-1.04	-0.62	0.00	TRUE
119	Almoez Ali	Δ_{119}	1.00	105014	-1.50	1.38	-1.42	TRUE
120	Abdulaziz Hatem	Δ_{120}	1.00	11320	-1.04	0.14	-1.29	TRUE
121	Hassan Al-Haydos	Δ_{121}	1.00	104381	-1.08	0.81	-0.09	TRUE
122	Karim Boudiaf	Δ_{122}	1.00	105377	-1.39	0.24	-0.16	TRUE
123	Assim Madibo	Δ_{123}	1.00	100573	-0.06	0.36	-1.23	TRUE
124	Saleh Al-Shehri	Δ_{124}	1.00	11267	-0.87	-0.95	-0.45	TRUE
125	Firas Al-Buraikan	Δ_{125}	1.00	98358	0.02	-2.04	0.39	TRUE
126	Mohamed Kanno	Δ_{126}	1.00	82105	0.93	0.44	0.14	TRUE
127	Salem Al-Dawsari	Δ_{127}	1.00	8053	1.46	-0.95	0.46	TRUE
128	Abdulelah Al-Malki	Δ_{128}	1.00	114316	-1.45	0.44	0.95	TRUE
129	Krepin Diatta	Δ_{129}	1.00	115270	0.17	1.62	-0.39	TRUE
130	Boulaye Dia	Δ_{130}	1.00	60690	0.63	2.49	-0.57	TRUE
131	Ismaila Sarr	Δ_{131}	1.00	50916	1.32	0.94	-0.38	TRUE
132	Nampalys Mendy	Δ_{132}	1.00	101906	-0.64	-0.51	-1.20	TRUE
133	Idrissa Gana Gueye	Δ_{133}	1.00	87196	0.49	1.05	-0.38	TRUE
134	Aleksandar Mitrovic	Δ_{134}	1.00	66390	0.21	-0.77	1.64	TRUE
135	Dusan Tadic	Δ_{135}	1.00	102546	0.70	1.46	1.54	TRUE
136	Sasa Lukic	Δ_{136}	1.00	105713	-0.60	-0.39	0.98	TRUE
137	Sergej Milinkovic-Savic	Δ_{137}	1.00	103513	-2.24	-1.80	1.25	TRUE
138	Ferran Torres	Δ_{138}	1.00	112853	0.50	-1.24	1.29	TRUE
139	Marco Asensio	Δ_{139}	1.00	72746	0.85	-0.34	0.57	TRUE
140	Alvaro Morata	Δ_{140}	1.00	89395	-1.29	0.36	2.04	TRUE

Table B.1.9: *Convergence diagnostic results for parameters 141–172, under the Poisson model for shots.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
141	Dani Olmo	Δ_{141}	1.00	46934	1.31	-1.33	0.45	TRUE
142	Gavi	Δ_{142}	1.00	101282	0.05	-0.62	0.25	TRUE
143	Pedri	Δ_{143}	1.00	88921	0.78	1.22	1.02	TRUE
144	Sergio Busquets	Δ_{144}	1.00	91470	0.83	-0.98	3.27	TRUE
145	Xherdan Shaqiri	Δ_{145}	1.00	101182	-0.39	-0.96	0.78	TRUE
146	Breel Embolo	Δ_{146}	1.00	80244	-0.28	0.02	0.80	TRUE
147	Ruben Vargas	Δ_{147}	1.00	98901	1.28	-2.20	-1.03	TRUE
148	Djibril Sow	Δ_{148}	1.00	114975	0.65	0.16	0.47	TRUE
149	Granit Xhaka	Δ_{149}	1.00	75040	0.36	-0.51	0.32	TRUE
150	Remo Freuler	Δ_{150}	1.00	93386	-0.51	-1.21	-0.17	TRUE
151	Issam Jebali	Δ_{151}	1.00	112484	-0.61	-1.90	-0.96	TRUE
152	Aissa Laidouni	Δ_{152}	1.00	106813	-0.20	-1.08	2.07	TRUE
153	Elyes Skhiri	Δ_{153}	1.00	100140	2.56	0.73	0.16	TRUE
154	Anis Ben Slimane	Δ_{154}	1.00	119927	1.44	-1.87	-1.83	TRUE
155	Youssef Msakni	Δ_{155}	1.00	73788	1.03	-0.95	-1.08	TRUE
156	Christian Pulisic	Δ_{156}	1.00	67496	1.02	0.33	-0.82	TRUE
157	Timothy Weah	Δ_{157}	1.00	84855	-2.21	0.42	-1.42	TRUE
158	Tyler Adams	Δ_{158}	1.00	93451	1.80	-0.28	1.09	TRUE
159	Weston McKennie	Δ_{159}	1.00	93306	-1.77	-0.49	0.20	FALSE
160	Yunus Musah	Δ_{160}	1.00	84593	-1.99	0.02	-0.67	TRUE
161	Darwin Nunez	Δ_{161}	1.00	102958	2.02	-0.29	0.54	TRUE
162	Luis Suarez	Δ_{162}	1.00	122931	-1.10	-0.88	-1.08	TRUE
163	Facundo Pellistri	Δ_{163}	1.00	111791	-0.13	0.40	-1.09	TRUE
164	Federico Valverde	Δ_{164}	1.00	77479	1.32	0.28	-0.39	TRUE
165	Matias Vecino	Δ_{165}	1.00	119428	0.48	0.62	2.42	TRUE
166	Rodrigo Bentancur	Δ_{166}	1.00	106763	0.98	0.18	0.19	TRUE
167	Gareth Bale	Δ_{167}	1.00	105404	-1.15	1.49	-1.68	TRUE
168	Kieffer Moore	Δ_{168}	1.00	87188	0.02	1.17	-0.70	TRUE
169	Daniel James	Δ_{169}	1.00	110898	-1.22	-1.52	-0.61	TRUE
170	Aaron Ramsey	Δ_{170}	1.00	100937	-1.69	-0.83	-0.75	TRUE
171	Ethan Ampadu	Δ_{171}	1.00	109649	1.65	1.43	-0.72	TRUE
172	Harry Wilson	Δ_{172}	1.00	119637	-0.84	-0.31	-0.46	TRUE

Table B.1.10: *Convergence diagnostic results for the team-effect parameters under the Poisson model for shots. The parameter column reports λ_i^e for team effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	λ_{1}^e	1.00	12216	-0.27	-0.42	0.53	TRUE
2	Australia	λ_{2}^e	1.00	41675	-0.86	-0.65	-0.45	TRUE
3	Belgium	λ_{3}^e	1.00	66983	0.95	-0.37	-0.67	TRUE
4	Brazil	λ_{4}^e	1.00	19755	-0.74	-1.57	0.58	TRUE
5	Cameroon	λ_{5}^e	1.00	66710	0.11	1.29	0.62	TRUE
6	Canada	λ_{6}^e	1.00	51918	0.56	0.32	-0.23	TRUE
7	Costa Rica	λ_{7}^e	1.00	35924	2.01	1.08	-0.75	TRUE
8	Croatia	λ_{8}^e	1.00	16479	-0.14	-2.37	-0.08	TRUE
9	Denmark	λ_{9}^e	1.00	59730	0.92	0.29	0.21	TRUE
10	Ecuador	λ_{10}^e	1.00	5988	0.18	-0.22	1.39	TRUE
11	England	λ_{11}^e	1.00	32991	-0.49	0.14	-0.84	TRUE
12	France	λ_{12}^e	1.00	11702	0.80	-0.73	-0.69	TRUE
13	Germany	λ_{13}^e	1.00	21045	-1.86	0.64	0.20	TRUE
14	Ghana	λ_{14}^e	1.00	57696	0.86	-1.31	0.71	TRUE
15	IR Iran	λ_{15}^e	1.00	6427	0.95	-0.06	-0.46	TRUE
16	Japan	λ_{16}^e	1.00	47444	-0.27	0.72	-0.26	TRUE
17	Korea Republic	λ_{17}^e	1.00	38130	-1.04	1.44	-0.51	TRUE
18	Mexico	λ_{18}^e	1.00	34949	-0.72	-0.53	0.08	TRUE
19	Morocco	λ_{19}^e	1.00	20412	-1.55	0.88	-0.84	TRUE
20	Netherlands	λ_{20}^e	1.00	42364	1.18	0.59	-1.79	TRUE
21	Poland	λ_{21}^e	1.00	44931	0.49	-0.65	-0.13	TRUE
22	Portugal	λ_{22}^e	1.00	34091	-0.74	-0.85	1.15	TRUE
23	Qatar	λ_{23}^e	1.00	39753	0.96	0.48	0.49	TRUE
24	Saudi Arabia	λ_{24}^e	1.00	56235	0.03	0.08	1.67	TRUE
25	Senegal	λ_{25}^e	1.00	33448	-1.65	-0.62	-0.28	TRUE
26	Serbia	λ_{26}^e	1.00	58735	0.36	1.11	-0.35	TRUE
27	Spain	λ_{27}^e	1.00	35933	0.17	0.77	-0.68	TRUE
28	Switzerland	λ_{28}^e	1.00	44881	-0.01	0.44	0.30	TRUE
29	Tunisia	λ_{29}^e	1.00	6936	-2.78	2.63	-0.32	TRUE
30	United States	λ_{30}^e	1.00	43169	1.08	-1.07	0.53	TRUE
31	Uruguay	λ_{31}^e	1.00	6292	-0.62	-1.67	-0.70	TRUE
32	Wales	λ_{32}^e	1.00	49357	2.07	-0.29	1.48	TRUE

Table B.1.11: *Convergence diagnostic results for the opponent team-effect parameters under the Poisson model for shots. The parameter column reports $\lambda_i^{E\setminus e}$ for opponent team effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	$\lambda_{1}^{E\setminus e}$	1.00	26385	0.81	-0.81	-0.50	TRUE
2	Australia	$\lambda_{2}^{E\setminus e}$	1.00	25092	0.56	0.33	0.61	TRUE
3	Belgium	$\lambda_{3}^{E\setminus e}$	1.00	25293	-0.32	0.74	0.30	TRUE
4	Brazil	$\lambda_{4}^{E\setminus e}$	1.00	36293	1.21	-0.18	0.26	TRUE
5	Cameroon	$\lambda_{5}^{E\setminus e}$	1.00	50109	-0.53	1.45	0.84	TRUE
6	Canada	$\lambda_{6}^{E\setminus e}$	1.00	24453	-1.84	-1.01	-2.09	TRUE
7	Costa Rica	$\lambda_{7}^{E\setminus e}$	1.00	2437	1.27	-0.68	0.08	TRUE
8	Croatia	$\lambda_{8}^{E\setminus e}$	1.00	14166	-0.36	-0.03	-0.36	TRUE
9	Denmark	$\lambda_{9}^{E\setminus e}$	1.00	32727	-0.68	-0.29	-1.58	TRUE
10	Ecuador	$\lambda_{10}^{E\setminus e}$	1.00	4103	-1.13	1.30	0.27	TRUE
11	England	$\lambda_{11}^{E\setminus e}$	1.00	31935	0.20	-1.49	-0.17	TRUE
12	France	$\lambda_{12}^{E\setminus e}$	1.00	29828	-1.04	0.89	-0.52	TRUE
13	Germany	$\lambda_{13}^{E\setminus e}$	1.00	14385	-0.77	2.29	-0.07	TRUE
14	Ghana	$\lambda_{14}^{E\setminus e}$	1.00	42709	1.25	-0.45	-1.13	TRUE
15	IR Iran	$\lambda_{15}^{E\setminus e}$	1.00	3796	0.19	0.11	0.98	TRUE
16	Japan	$\lambda_{16}^{E\setminus e}$	1.00	32498	-0.95	-0.72	-0.10	TRUE
17	Korea Republic	$\lambda_{17}^{E\setminus e}$	1.00	33539	-0.48	-0.50	0.42	TRUE
18	Mexico	$\lambda_{18}^{E\setminus e}$	1.00	11553	1.97	-1.27	0.72	TRUE
19	Morocco	$\lambda_{19}^{E\setminus e}$	1.00	33818	-1.36	1.35	-0.83	TRUE
20	Netherlands	$\lambda_{20}^{E\setminus e}$	1.00	23351	0.97	-0.70	1.77	TRUE
21	Poland	$\lambda_{21}^{E\setminus e}$	1.00	36267	1.04	-0.42	-0.31	TRUE
22	Portugal	$\lambda_{22}^{E\setminus e}$	1.00	47412	1.30	1.37	-1.09	TRUE
23	Qatar	$\lambda_{23}^{E\setminus e}$	1.00	28095	0.71	-0.66	1.17	TRUE
24	Saudi Arabia	$\lambda_{24}^{E\setminus e}$	1.00	44639	-0.69	-0.40	1.30	TRUE
25	Senegal	$\lambda_{25}^{E\setminus e}$	1.00	36686	0.38	0.02	1.14	TRUE
26	Serbia	$\lambda_{26}^{E\setminus e}$	1.00	4039	-0.89	-1.87	1.21	TRUE
27	Spain	$\lambda_{27}^{E\setminus e}$	1.00	37038	1.26	0.56	1.27	TRUE
28	Switzerland	$\lambda_{28}^{E\setminus e}$	1.00	39319	0.63	-1.35	0.37	TRUE
29	Tunisia	$\lambda_{29}^{E\setminus e}$	1.00	22731	-0.95	1.07	-1.63	TRUE
30	United States	$\lambda_{30}^{E\setminus e}$	1.00	45469	0.99	0.99	-1.46	TRUE
31	Uruguay	$\lambda_{31}^{E\setminus e}$	1.00	43291	0.61	1.56	-0.07	TRUE
32	Wales	$\lambda_{32}^{E\setminus e}$	1.00	47634	-1.67	-1.09	-0.56	TRUE

B.2 Negative Binomial Model

Table B.2.1: *Team effects for each national team under the negative binomial model for shots, showing each team's ability to create shooting chances, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	0.05	0.05	-0.06	0.15
2	Australia	-0.05	0.07	-0.20	0.10
3	Belgium	-0.01	0.08	-0.17	0.16
4	Brazil	0.09	0.07	-0.05	0.24
5	Cameroon	-0.03	0.08	-0.21	0.13
6	Canada	0.03	0.08	-0.13	0.19
7	Costa Rica	-0.09	0.09	-0.29	0.08
8	Croatia	-0.02	0.05	-0.13	0.09
9	Denmark	-0.04	0.08	-0.20	0.13
10	Ecuador	-0.01	0.08	-0.17	0.15
11	England	0.02	0.07	-0.11	0.16
12	France	0.06	0.05	-0.05	0.17
13	Germany	0.12	0.09	-0.04	0.30
14	Ghana	-0.04	0.08	-0.20	0.12
15	IR Iran	-0.01	0.08	-0.18	0.14
16	Japan	-0.02	0.07	-0.16	0.13
17	Korea Republic	0.03	0.07	-0.11	0.18
18	Mexico	0.06	0.08	-0.10	0.23
19	Morocco	-0.03	0.06	-0.14	0.08
20	Netherlands	-0.01	0.07	-0.15	0.13
21	Poland	-0.02	0.07	-0.17	0.12
22	Portugal	0.03	0.07	-0.11	0.17
23	Qatar	-0.09	0.09	-0.27	0.08
24	Saudi Arabia	-0.00	0.08	-0.16	0.15
25	Senegal	0.05	0.08	-0.09	0.21
26	Serbia	-0.01	0.08	-0.18	0.15
27	Spain	0.01	0.07	-0.13	0.15
28	Switzerland	0.01	0.07	-0.13	0.15
29	Tunisia	-0.01	0.08	-0.17	0.16
30	United States	0.01	0.07	-0.13	0.16
31	Uruguay	-0.01	0.08	-0.17	0.15
32	Wales	-0.06	0.09	-0.24	0.10

Table B.2.2: *Opponent team effects for each national team under the negative binomial model for shots, showing how strongly each team limits their opponents from shooting, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-0.06	0.13	-0.32	0.19
2	Australia	-0.04	0.14	-0.32	0.24
3	Belgium	0.04	0.14	-0.23	0.34
4	Brazil	-0.01	0.13	-0.27	0.26
5	Cameroon	0.03	0.14	-0.24	0.32
6	Canada	0.06	0.15	-0.22	0.37
7	Costa Rica	-0.07	0.14	-0.37	0.20
8	Croatia	-0.10	0.13	-0.37	0.14
9	Denmark	0.01	0.14	-0.26	0.29
10	Ecuador	0.05	0.14	-0.22	0.35
11	England	-0.03	0.13	-0.31	0.22
12	France	0.02	0.13	-0.23	0.28
13	Germany	0.10	0.16	-0.17	0.43
14	Ghana	-0.01	0.13	-0.29	0.27
15	IR Iran	0.06	0.14	-0.21	0.34
16	Japan	-0.02	0.14	-0.31	0.25
17	Korea Republic	0.03	0.14	-0.23	0.33
18	Mexico	-0.12	0.13	-0.40	0.13
19	Morocco	-0.00	0.13	-0.27	0.25
20	Netherlands	-0.06	0.14	-0.34	0.20
21	Poland	-0.01	0.13	-0.27	0.26
22	Portugal	0.00	0.14	-0.28	0.28
23	Qatar	-0.04	0.14	-0.34	0.22
24	Saudi Arabia	0.04	0.13	-0.23	0.31
25	Senegal	-0.03	0.14	-0.31	0.24
26	Serbia	-0.01	0.13	-0.27	0.26
27	Spain	-0.00	0.13	-0.28	0.26
28	Switzerland	-0.02	0.13	-0.29	0.25
29	Tunisia	0.09	0.14	-0.18	0.39
30	United States	0.03	0.13	-0.23	0.31
31	Uruguay	0.02	0.14	-0.25	0.30
32	Wales	0.03	0.14	-0.25	0.30

Table B.2.3: *Top 10 players under the negative binomial model for shots, ranked by posterior predicted total shots, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Shots	Lower HPD Bound	Upper HPD Bound	Observed Shots	90-minute units
1	Kylian Mbappe	25.08	12.92	38.14	29	6.6
2	Lionel Messi	24.07	12.51	36.77	27	7.7
3	Ivan Perisic	13.80	6.22	21.89	16	7.4
4	Olivier Giroud	13.67	6.43	21.38	16	4.7
5	Julian Alvarez	9.93	4.35	16.32	11	5.2
6	Dani Olmo	9.34	3.82	15.03	12	4.1
7	Adrien Rabiot	9.30	3.90	15.58	9	5.4
8	Serge Gnabry	9.24	3.79	15.25	12	2.9
9	Jamal Musiala	9.22	3.78	15.27	12	2.9
10	Hakim Ziyech	8.77	3.59	14.65	10	7.1

Table B.2.4: *Bottom 10 players under the negative binomial model for shots, ranked by posterior predicted total shots, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Shots	Lower HPD Bound	Upper HPD Bound	Observed Shots	90-minute units
1	Assim Madibo	1.04	0.19	2.17	0	1.7
2	Daniel James	1.06	0.20	2.19	0	1.7
3	Mikkel Damsgaard	1.09	0.21	2.25	0	1.7
4	Karim Boudiaf	1.15	0.22	2.40	0	2.1
5	Jordan Ayew	1.30	0.29	2.65	1	1.5
6	Jhegson Mendez	1.31	0.25	2.74	0	2.0
7	Celso Borges	1.32	0.22	2.77	0	2.8
8	Anis Ben Slimane	1.38	0.28	2.79	1	1.6
9	Hector Herrera	1.43	0.27	2.95	0	1.8
10	Yuto Nagatomo	1.45	0.28	2.97	0	3.3

Table B.2.5: *Convergence diagnostic results for the first 35 parameters, under the negative binomial model for shots. The parameter column reports Δ_i for player effects, λ_i^e for team effects, and $\lambda_i^{E\setminus e}$ for opponent team effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Aaron Mooy	Δ_{8}	1.00	58961	-1.54	-2.71	-0.71	TRUE
2	Aaron Ramsey	Δ_{170}	1.00	82103	0.42	-3.08	-0.52	TRUE
3	Abdulaziz Hatem	Δ_{120}	1.00	97855	-0.17	-0.93	-1.14	TRUE
4	Abdullelah Al-Malki	Δ_{128}	1.00	118489	1.36	0.06	-0.57	TRUE
5	Adrien Rabiot	Δ_{62}	1.00	53185	0.44	-0.02	-0.16	TRUE
6	Ahmad Nourollahi	Δ_{76}	1.00	76585	-0.86	0.17	0.10	TRUE
7	Aissa Laidouni	Δ_{152}	1.00	103729	-0.54	-2.07	1.29	FALSE
8	Akram Afif	Δ_{118}	1.00	99291	0.02	0.41	-0.37	TRUE
9	Aleksandar Mitrovic	Δ_{134}	1.00	28434	1.60	1.98	0.03	TRUE
10	Alexis Mac Allister	Δ_{5}	1.00	50812	0.24	-0.47	-0.27	TRUE
11	Alex Vega	Δ_{92}	1.00	25075	0.61	2.53	2.11	TRUE
12	Ali Gholizadeh	Δ_{77}	1.00	94178	-1.50	-0.99	0.89	TRUE
13	Almoez Ali	Δ_{119}	1.00	110880	-0.50	0.53	0.62	TRUE
14	Alphonso Davies	Δ_{29}	1.00	92444	-1.40	0.05	0.67	TRUE
15	Alvaro Morata	Δ_{140}	1.00	29909	0.24	2.47	-0.30	TRUE
16	Andre-Frank Zambo Anguissa	Δ_{25}	1.00	73019	-0.27	-2.59	-0.12	TRUE
17	Andre Ayew	Δ_{72}	1.00	95797	-0.77	-1.19	-0.46	TRUE
18	Andrej Kramaric	Δ_{35}	1.00	49436	0.41	0.67	1.96	TRUE
19	Anis Ben Slimane	Δ_{154}	1.00	112122	-1.70	-0.36	-1.71	TRUE
20	Antoine Griezmann	Δ_{60}	1.00	47699	-0.52	-0.68	0.12	FALSE
21	Assim Madibo	Δ_{123}	1.00	65191	-0.53	-2.45	-0.36	TRUE
22	Atiba Hutchinson	Δ_{31}	1.00	119809	0.23	2.17	-1.28	TRUE
23	Aurelien Tchouameni	Δ_{61}	1.00	39223	-0.15	-0.76	-0.39	TRUE
24	Axel Witsel	Δ_{14}	1.00	60246	-0.59	-0.86	-0.94	TRUE
25	Azzedine Ounahi	Δ_{101}	1.00	71493	-0.75	-2.14	1.49	TRUE
26	Bernardo Silva	Δ_{117}	1.00	42259	-0.35	-2.46	-1.19	TRUE
27	Boulaye Dia	Δ_{130}	1.00	52122	0.10	2.83	-0.90	TRUE
28	Breel Embolo	Δ_{146}	1.00	83009	-0.46	-0.35	1.03	TRUE
29	Bruno Fernandes	Δ_{114}	1.00	64370	0.20	0.81	0.46	TRUE
30	Bryan Mbeumo	Δ_{21}	1.00	102912	-2.02	-1.86	1.09	TRUE
31	Bukayo Saka	Δ_{50}	1.00	50448	-0.43	2.68	1.62	TRUE
32	Casemiro	Δ_{19}	1.00	53286	-0.09	0.98	1.04	TRUE
33	Celso Borges	Δ_{33}	1.00	50724	-0.07	-2.22	-1.14	TRUE
34	Cho Guesung	Δ_{87}	1.00	36793	1.33	1.32	0.05	TRUE
35	Christian Eriksen	Δ_{41}	1.00	92132	-1.05	1.10	0.38	TRUE

Table B.2.6: *Convergence diagnostic results for parameters 36–70, under the negative binomial model for shots. The parameter column reports Δ_i for player effects, λ_i^e for team effects, and $\lambda_i^{E\backslash e}$ for opponent team effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
36	Christian Pulisic	Δ_{156}	1.00	39645	0.56	1.74	2.40	TRUE
37	Cody Gakpo	Δ_{104}	1.00	70835	1.07	0.23	1.21	TRUE
38	Craig Goodwin	Δ_{9}	1.00	115617	0.03	0.55	-0.77	TRUE
39	Cristiano Ronaldo	Δ_{113}	1.00	32141	1.00	1.97	0.80	TRUE
40	Cyle Larin	Δ_{27}	1.00	120573	-0.46	2.00	0.58	FALSE
41	Daichi Kamada	Δ_{82}	1.00	80390	2.57	-0.80	0.45	TRUE
42	Daizen Maeda	Δ_{81}	1.00	119824	-0.30	0.84	-0.80	TRUE
43	Dani Olmo	Δ_{141}	1.00	29923	0.66	2.87	0.76	TRUE
44	Daniel James	Δ_{169}	1.00	73143	-1.37	2.55	-2.30	FALSE
45	Darwin Nunez	Δ_{161}	1.00	111900	-1.04	0.80	2.00	TRUE
46	Declan Rice	Δ_{54}	1.00	33255	-1.20	-0.81	-0.15	TRUE
47	Djibril Sow	Δ_{148}	1.00	102123	-0.03	-0.01	2.13	TRUE
48	Dusan Tadic	Δ_{135}	1.00	71537	-0.34	-2.12	-1.00	TRUE
49	Edson Alvarez	Δ_{96}	1.00	93257	-1.85	-1.98	-2.30	TRUE
50	Ehsan Hajsafi	Δ_{78}	1.00	77178	-0.39	-1.01	-1.00	TRUE
51	Ellyes Skhiri	Δ_{153}	1.00	94887	-1.30	0.26	-0.03	TRUE
52	Enner Valencia	Δ_{45}	1.00	38006	0.99	1.81	-0.34	TRUE
53	Enzo Fernandez	Δ_{3}	1.00	48582	-0.86	0.84	0.12	TRUE
54	Eric Maxim Choupo-Moting	Δ_{22}	1.00	105865	0.62	-1.98	0.38	TRUE
55	Ethan Ampadu	Δ_{171}	1.00	81257	1.97	-0.51	-1.87	TRUE
56	Facundo Pellistri	Δ_{163}	1.00	90029	-1.53	-0.89	0.83	TRUE
57	Federico Valverde	Δ_{164}	1.00	52649	-0.44	2.89	2.84	FALSE
58	Ferran Torres	Δ_{138}	1.00	117332	-1.76	1.82	-2.10	TRUE
59	Firas Al-Buraikan	Δ_{125}	1.00	88727	0.08	-1.41	0.46	TRUE
60	Frenkie de Jong	Δ_{106}	1.00	58029	1.23	0.88	0.16	TRUE
61	Gareth Bale	Δ_{167}	1.00	88171	0.10	-0.91	0.67	TRUE
62	Gavi	Δ_{142}	1.00	60367	-1.68	-1.35	-0.48	FALSE
63	Gonzalo Plata	Δ_{47}	1.00	99634	1.04	-1.56	-3.00	TRUE
64	Granit Xhaka	Δ_{149}	1.00	80962	-0.12	-0.43	-0.33	TRUE
65	Grzegorz Krychowiak	Δ_{110}	1.00	53958	-0.65	-1.01	-1.96	TRUE
66	Hakim Ziyech	Δ_{98}	1.00	49965	-0.22	-1.81	-0.06	TRUE
67	Harry Kane	Δ_{52}	1.00	37225	-0.08	2.43	1.07	TRUE
68	Harry Wilson	Δ_{172}	1.00	121089	0.17	-0.94	-1.69	TRUE
69	Hassan Al-Haydos	Δ_{121}	1.00	90154	-0.91	-0.86	0.10	TRUE
70	Hector Herrera	Δ_{97}	1.00	61162	-0.84	-2.53	-0.06	TRUE

Table B.2.7: *Convergence diagnostic results for parameters 71–105, under the negative binomial model for shots.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
71	Henry Martin	Δ_{94}	1.00	81291	-0.40	1.02	0.10	TRUE
72	Hidemasa Morita	Δ_{86}	1.00	87734	1.50	-0.77	-1.83	TRUE
73	Hirving Lozano	Δ_{93}	1.00	66355	0.78	1.25	0.01	TRUE
74	Hwang In-beom	Δ_{90}	1.00	78193	1.29	-1.79	-0.22	TRUE
75	Idrissa Gana Gueye	Δ_{133}	1.00	93146	-1.27	1.74	-1.38	TRUE
76	Ilkay Gundogan	Δ_{67}	1.00	73714	-0.63	-0.72	0.25	TRUE
77	Inaki Williams	Δ_{69}	1.00	103574	1.05	0.39	0.96	TRUE
78	Ismaila Sarr	Δ_{131}	1.00	37907	-0.17	2.60	-0.14	TRUE
79	Issam Jebali	Δ_{151}	1.00	116613	-0.21	-0.33	0.84	TRUE
80	Ivan Perisic	Δ_{36}	1.00	30302	1.08	0.58	1.92	TRUE
81	Jackson Irvine	Δ_{10}	1.00	93472	1.93	-1.11	-1.18	TRUE
82	Jakub Kaminski	Δ_{108}	1.00	106784	-0.57	-0.30	0.06	FALSE
83	Jamal Musiala	Δ_{65}	1.00	24301	0.92	1.84	0.35	FALSE
84	Jesper Lindstrom	Δ_{42}	1.00	114946	1.63	0.61	-0.36	TRUE
85	Jhegson Mendez	Δ_{49}	1.00	63677	-0.87	-2.65	-1.26	TRUE
86	Joao Felix	Δ_{115}	1.00	32504	0.61	1.76	0.38	TRUE
87	Joel Campbell	Δ_{32}	1.00	79961	0.03	-1.58	-0.79	TRUE
88	Jonathan David	Δ_{26}	1.00	32347	2.18	2.41	0.17	TRUE
89	Jordan Ayew	Δ_{71}	1.00	121577	-0.32	0.10	-0.08	TRUE
90	Jordan Henderson	Δ_{53}	1.00	93883	-0.34	0.69	-0.52	TRUE
91	Joshua Kimmich	Δ_{66}	1.00	56762	0.76	0.93	0.12	TRUE
92	Jude Bellingham	Δ_{55}	1.00	78699	-0.74	0.92	-1.05	TRUE
93	Julian Alvarez	Δ_{1}	1.00	48583	-0.14	1.61	1.45	TRUE
94	Jung Woo-young	Δ_{91}	1.00	53177	-0.09	-2.60	-1.46	TRUE
95	Junior Hoilett	Δ_{28}	1.00	95060	-0.16	1.91	0.39	TRUE
96	Junya Ito	Δ_{84}	1.00	76031	0.50	-1.76	-2.01	TRUE
97	Karim Boudiaf	Δ_{122}	1.00	64831	-0.19	-0.44	0.09	TRUE
98	Karl Toko Ekambi	Δ_{23}	1.00	121152	1.05	-1.91	0.39	FALSE
99	Kevin De Bruyne	Δ_{12}	1.00	48686	1.27	1.71	0.64	TRUE
100	Kieffer Moore	Δ_{168}	1.00	50658	1.45	2.26	0.87	FALSE
101	Krepin Diatta	Δ_{129}	1.00	117329	1.06	0.86	-2.04	TRUE
102	Krystian Bielik	Δ_{109}	1.00	108955	-0.18	-1.85	-1.23	FALSE
103	Kudus Mohammed	Δ_{70}	1.00	35942	0.58	0.11	1.89	TRUE
104	Kylian Mbappe	Δ_{56}	1.00	18104	0.35	2.35	0.77	TRUE
105	Lee Jae-sung	Δ_{89}	1.00	83827	-2.06	-1.45	0.62	TRUE

Table B.2.8: *Convergence diagnostic results for parameters 106–140, under the negative binomial model for shots.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
106	Leon Goretzka	Δ_{68}	1.00	102451	-0.07	0.93	0.83	FALSE
107	Lionel Messi	Δ_{2}	1.00	21998	0.03	1.64	1.71	TRUE
108	Lucas Paqueta	Δ_{20}	1.00	77979	-2.92	-1.48	1.04	TRUE
109	Luis Chavez	Δ_{95}	1.00	30750	0.35	1.85	1.27	FALSE
110	Luis Suarez	Δ_{162}	1.00	124499	1.30	1.13	1.90	TRUE
111	Luka Modric	Δ_{38}	1.00	57831	-0.38	-1.98	0.71	TRUE
112	Marcelo Brozovic	Δ_{40}	1.00	60729	-0.10	-0.55	-0.30	TRUE
113	Marco Asensio	Δ_{139}	1.00	25263	0.06	1.82	1.29	TRUE
114	Mario Pasalic	Δ_{37}	1.00	115677	1.29	-0.21	0.59	TRUE
115	Mateo Kovacic	Δ_{39}	1.00	44957	-0.46	-2.11	0.21	TRUE
116	Mathew Leckie	Δ_{11}	1.00	63346	0.16	-2.78	-0.04	FALSE
117	Matias Vecino	Δ_{165}	1.00	90779	0.52	-0.47	0.09	TRUE
118	Mehdi Taremi	Δ_{75}	1.00	89429	-0.53	2.21	1.56	TRUE
119	Memphis	Δ_{105}	1.00	28643	1.11	3.03	1.57	TRUE
120	Michael Estrada	Δ_{46}	1.00	110049	1.04	-1.15	0.36	TRUE
121	Michy Batshuayi	Δ_{13}	1.00	41948	-0.03	2.59	3.40	TRUE
122	Mikkel Damsgaard	Δ_{44}	1.00	68207	-0.41	-3.57	-1.15	FALSE
123	Mitchell Duke	Δ_{6}	1.00	45210	0.85	0.84	-0.19	FALSE
124	Mohamed Kanno	Δ_{126}	1.00	75473	0.71	1.75	0.70	TRUE
125	Moises Caicedo	Δ_{48}	1.00	104836	-0.72	-0.52	-1.39	TRUE
126	Nampalys Mendy	Δ_{132}	1.00	82528	-0.99	-0.60	-0.56	TRUE
127	Neymar	Δ_{18}	1.00	33994	0.26	1.87	-0.31	TRUE
128	Olivier Giroud	Δ_{58}	1.00	26975	0.61	1.83	1.08	TRUE
129	Ousmane Dembele	Δ_{57}	1.00	68443	-0.15	0.07	-0.55	TRUE
130	Pedri	Δ_{143}	1.00	34086	-1.59	-1.75	-2.34	TRUE
131	Phil Foden	Δ_{51}	1.00	71753	0.42	2.10	-0.25	TRUE
132	Pierre Hojbjerg	Δ_{43}	1.00	97408	-0.72	-1.00	-0.65	TRUE
133	Piotr Zielinski	Δ_{111}	1.00	94795	-0.40	0.23	-1.66	TRUE
134	Randal Kolo Muani	Δ_{59}	1.00	100584	-0.39	1.09	0.07	TRUE
135	Przemyslaw Frankowski	Δ_{112}	1.00	68734	0.69	-1.88	-0.62	TRUE
136	Raphinha	Δ_{15}	1.00	57423	-0.21	1.77	0.82	TRUE
137	Remo Freuler	Δ_{150}	1.00	95970	0.45	-2.59	-0.08	FALSE
138	Richarlison	Δ_{16}	1.00	60557	-0.20	1.60	0.91	TRUE
139	Riley McGree	Δ_{7}	1.00	110943	0.63	0.01	-2.27	TRUE
140	Ritsu Doan	Δ_{80}	1.00	108990	0.76	0.09	0.40	TRUE

Table B.2.9: *Convergence diagnostic results for parameters 141–172, under the negative binomial model for shots.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
141	Robert Lewandowski	Δ_{107}	1.00	36418	1.10	0.75	1.00	TRUE
142	Rodrigo Bentancur	Δ_{166}	1.00	115689	-0.58	0.07	0.88	TRUE
143	Rodrigo De Paul	Δ_{4}	1.00	44663	-0.66	-0.53	-0.22	TRUE
144	Ruben Neves	Δ_{116}	1.00	88449	-0.43	-0.93	-1.56	TRUE
145	Ruben Vargas	Δ_{147}	1.00	104849	-1.32	-1.40	-0.14	TRUE
146	Saeid Ezatolahi	Δ_{79}	1.00	109684	-1.88	-0.03	-1.35	TRUE
147	Saleh Al-Shehri	Δ_{124}	1.00	101717	-1.00	-0.92	-0.96	TRUE
148	Salem Al-Dawsari	Δ_{127}	1.00	73990	0.32	2.39	-0.49	TRUE
149	Salis Abdul Samed	Δ_{73}	1.00	80082	-1.05	-1.81	0.31	TRUE
150	Sasa Lukic	Δ_{136}	1.00	76313	0.76	-1.53	-0.30	TRUE
151	Selim Amallah	Δ_{102}	1.00	64298	-0.50	-1.27	0.34	TRUE
152	Serge Gnabry	Δ_{63}	1.00	23064	1.12	1.96	0.41	FALSE
153	Sergej Milinkovic-Savic	Δ_{137}	1.00	109455	1.20	-0.58	-0.45	TRUE
154	Sergio Busquets	Δ_{144}	1.00	65263	-0.39	-1.96	-1.28	FALSE
155	Sofiane Boufal	Δ_{99}	1.00	75319	-0.15	-1.29	1.23	TRUE
156	Sofyan Amrabat	Δ_{103}	1.00	30722	-1.15	-3.83	-0.06	TRUE
157	Son Heung-min	Δ_{88}	1.00	49738	1.35	1.72	-0.45	FALSE
158	Tajon Buchanan	Δ_{30}	1.00	85521	-0.24	-0.07	-0.37	TRUE
159	Thomas Muller	Δ_{64}	1.00	105788	-1.34	2.05	-1.17	TRUE
160	Thomas Partey	Δ_{74}	1.00	96549	0.40	-3.05	-0.19	TRUE
161	Timothy Weah	Δ_{157}	1.00	91081	-0.57	0.71	-0.36	TRUE
162	Tyler Adams	Δ_{158}	1.00	48838	-1.07	-1.20	-0.76	FALSE
163	Vincent Aboubakar	Δ_{24}	1.00	40367	0.77	1.34	1.35	TRUE
164	Vinicius Junior	Δ_{17}	1.00	50531	-0.51	2.63	1.60	TRUE
165	Wataru Endo	Δ_{85}	1.00	84209	0.45	1.03	-1.07	TRUE
166	Weston McKennie	Δ_{159}	1.00	89308	-0.67	2.01	2.23	FALSE
167	Xherdan Shaqiri	Δ_{145}	1.00	106388	0.10	0.46	-1.09	TRUE
168	Yeltsin Tejeda	Δ_{34}	1.00	97626	0.64	-0.76	-2.71	TRUE
169	Youssef En-Nesyri	Δ_{100}	1.00	58444	1.67	-1.94	1.06	TRUE
170	Youssef Msakni	Δ_{155}	1.00	20631	0.54	2.08	0.66	TRUE
171	Yunus Musah	Δ_{160}	1.00	92241	-0.34	-0.59	-0.69	TRUE
172	Yuto Nagatomo	Δ_{83}	1.00	52650	1.36	-1.48	-1.33	TRUE

Table B.2.10: *Convergence diagnostic results for the team-effect parameters under the negative binomial model for shots. The parameter column reports λ_i^e for team effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	λ_1^e	1.00	17601	0.83	-0.46	-1.56	TRUE
2	Australia	λ_2^e	1.00	48178	-2.99	0.86	-0.28	TRUE
3	Belgium	λ_3^e	1.00	72307	-2.09	-1.09	-0.92	TRUE
4	Brazil	λ_4^e	1.00	23350	0.68	-0.86	0.33	TRUE
5	Cameroon	λ_5^e	1.00	69124	1.10	2.18	-0.30	TRUE
6	Canada	λ_6^e	1.00	58718	0.15	-1.24	0.25	TRUE
7	Costa Rica	λ_7^e	1.00	37091	-0.74	1.97	0.37	TRUE
8	Croatia	λ_8^e	1.00	21147	0.55	0.06	0.82	TRUE
9	Denmark	λ_9^e	1.00	66894	-0.66	0.35	0.86	TRUE
10	Ecuador	λ_{10}^e	1.00	66212	1.04	0.10	0.39	TRUE
11	England	λ_{11}^e	1.00	38373	-1.27	-0.92	0.90	TRUE
12	France	λ_{12}^e	1.00	16608	1.17	-1.56	0.58	TRUE
13	Germany	λ_{13}^e	1.00	24106	0.82	-0.95	-0.55	TRUE
14	Ghana	λ_{14}^e	1.00	62627	-0.70	2.34	-0.34	TRUE
15	IR Iran	λ_{15}^e	1.00	71750	0.74	0.20	1.23	TRUE
16	Japan	λ_{16}^e	1.00	55126	-0.53	0.24	-0.65	TRUE
17	Korea Republic	λ_{17}^e	1.00	45204	-0.46	-0.25	0.75	TRUE
18	Mexico	λ_{18}^e	1.00	38533	0.72	-1.21	-1.25	TRUE
19	Morocco	λ_{19}^e	1.00	24398	-0.28	1.58	-1.28	TRUE
20	Netherlands	λ_{20}^e	1.00	50170	-1.40	-1.25	-0.69	TRUE
21	Poland	λ_{21}^e	1.00	49377	-0.78	1.70	0.98	TRUE
22	Portugal	λ_{22}^e	1.00	39152	0.37	0.10	1.98	TRUE
23	Qatar	λ_{23}^e	1.00	41024	-0.30	1.15	1.17	TRUE
24	Saudi Arabia	λ_{24}^e	1.00	62917	1.48	0.64	-0.27	TRUE
25	Senegal	λ_{25}^e	1.00	39493	1.26	-1.67	-0.26	TRUE
26	Serbia	λ_{26}^e	1.00	64720	-0.71	0.50	1.21	TRUE
27	Spain	λ_{27}^e	1.00	42944	0.78	-1.08	0.87	TRUE
28	Switzerland	λ_{28}^e	1.00	52060	0.64	0.56	0.04	TRUE
29	Tunisia	λ_{29}^e	1.00	76191	-0.29	0.42	-0.46	TRUE
30	United States	λ_{30}^e	1.00	48535	0.93	-0.48	-1.77	TRUE
31	Uruguay	λ_{31}^e	1.00	68798	0.70	-1.10	-0.93	TRUE
32	Wales	λ_{32}^e	1.00	51711	-1.91	0.34	-1.22	TRUE

Table B.2.11: *Convergence diagnostic results for the opponent team-effect parameters under the negative binomial model for shots. The parameter column reports $\lambda_i^{E\setminus e}$ for opponent team effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	$\lambda_{1}^{E\setminus e}$	1.00	29312	0.95	0.50	1.12	TRUE
2	Australia	$\lambda_{2}^{E\setminus e}$	1.00	32346	2.28	-0.87	0.48	TRUE
3	Belgium	$\lambda_{3}^{E\setminus e}$	1.00	31223	-0.16	-1.41	0.72	FALSE
4	Brazil	$\lambda_{4}^{E\setminus e}$	1.00	42921	1.14	-0.11	-0.20	TRUE
5	Cameroon	$\lambda_{5}^{E\setminus e}$	1.00	54043	-0.26	0.27	-1.89	TRUE
6	Canada	$\lambda_{6}^{E\setminus e}$	1.00	26423	-0.79	0.25	-1.62	TRUE
7	Costa Rica	$\lambda_{7}^{E\setminus e}$	1.00	25888	2.57	-0.96	0.06	TRUE
8	Croatia	$\lambda_{8}^{E\setminus e}$	1.00	17600	-0.18	0.54	-1.22	TRUE
9	Denmark	$\lambda_{9}^{E\setminus e}$	1.00	44931	-0.83	-1.26	-1.84	TRUE
10	Ecuador	$\lambda_{10}^{E\setminus e}$	1.00	46728	-2.08	2.52	0.36	TRUE
11	England	$\lambda_{11}^{E\setminus e}$	1.00	35502	2.43	-0.80	0.54	TRUE
12	France	$\lambda_{12}^{E\setminus e}$	1.00	36792	-1.02	0.40	0.93	TRUE
13	Germany	$\lambda_{13}^{E\setminus e}$	1.00	17238	-1.63	0.56	-0.56	TRUE
14	Ghana	$\lambda_{14}^{E\setminus e}$	1.00	52120	-0.94	-1.14	-0.61	TRUE
15	IR Iran	$\lambda_{15}^{E\setminus e}$	1.00	42036	-1.74	1.26	-2.01	TRUE
16	Japan	$\lambda_{16}^{E\setminus e}$	1.00	39191	-0.47	1.60	-0.76	TRUE
17	Korea Republic	$\lambda_{17}^{E\setminus e}$	1.00	36373	-0.12	0.47	-1.19	TRUE
18	Mexico	$\lambda_{18}^{E\setminus e}$	1.00	13438	1.04	-0.34	2.30	TRUE
19	Morocco	$\lambda_{19}^{E\setminus e}$	1.00	41226	0.87	-0.36	-1.59	TRUE
20	Netherlands	$\lambda_{20}^{E\setminus e}$	1.00	29397	1.01	-0.44	-0.45	TRUE
21	Poland	$\lambda_{21}^{E\setminus e}$	1.00	42007	1.09	0.25	0.96	TRUE
22	Portugal	$\lambda_{22}^{E\setminus e}$	1.00	52503	1.60	-0.25	-1.67	TRUE
23	Qatar	$\lambda_{23}^{E\setminus e}$	1.00	35241	-0.27	-0.01	0.54	TRUE
24	Saudi Arabia	$\lambda_{24}^{E\setminus e}$	1.00	48582	-0.67	-1.11	0.89	TRUE
25	Senegal	$\lambda_{25}^{E\setminus e}$	1.00	43083	0.68	-1.59	1.68	TRUE
26	Serbia	$\lambda_{26}^{E\setminus e}$	1.00	46571	0.17	0.97	1.67	TRUE
27	Spain	$\lambda_{27}^{E\setminus e}$	1.00	43809	0.61	-1.44	0.10	TRUE
28	Switzerland	$\lambda_{28}^{E\setminus e}$	1.00	43783	0.05	0.12	2.01	TRUE
29	Tunisia	$\lambda_{29}^{E\setminus e}$	1.00	25874	-2.48	0.36	-0.15	TRUE
30	United States	$\lambda_{30}^{E\setminus e}$	1.00	49918	-1.15	0.46	0.49	TRUE
31	Uruguay	$\lambda_{31}^{E\setminus e}$	1.00	51739	0.35	0.73	-0.62	TRUE
32	Wales	$\lambda_{32}^{E\setminus e}$	1.00	53406	-1.43	0.48	1.11	TRUE

Appendix C

Passing

C.1 Negative Binomial Model

Table C.1.1: *Team effects for each national team under the negative binomial model for passing, representing each team's passing quality, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

No.	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-8.66	0.76	-10.11	-7.13
2	Australia	-3.22	0.74	-4.67	-1.78
3	Belgium	-0.47	0.73	-1.89	0.93
4	Brazil	0.45	0.49	-0.55	1.43
5	Cameroon	1.00	0.60	-0.19	2.20
6	Canada	1.83	1.12	-0.31	4.00
7	Costa Rica	-3.55	0.96	-5.46	-1.69
8	Croatia	-0.28	0.66	-1.61	0.99
9	Denmark	-1.50	0.76	-2.97	0.00
10	Ecuador	7.40	1.64	4.36	10.79
11	England	5.90	0.78	4.32	7.43
12	France	-1.81	0.50	-2.82	-0.81
13	Germany	-3.00	0.96	-4.93	-1.17
14	Ghana	1.50	0.80	-0.07	3.08
15	IR Iran	4.25	1.42	1.35	6.97
16	Japan	-3.88	0.97	-5.79	-1.97
17	Korea Republic	1.28	0.72	-0.17	2.68
18	Mexico	-4.15	1.12	-6.35	-1.93
19	Morocco	-1.89	0.56	-2.97	-0.08
20	Netherlands	12.49	1.56	9.25	15.42
21	Poland	-3.62	1.06	-5.69	-1.52
22	Portugal	1.40	0.62	0.18	2.60
23	Qatar	-7.80	1.32	-10.37	-5.28
24	Saudi Arabia	-2.11	1.27	-4.61	0.42
25	Senegal	4.83	1.41	2.17	7.88
26	Serbia	1.57	0.73	0.13	3.02
27	Spain	-0.51	0.75	-1.94	1.01
28	Switzerland	0.72	0.68	-0.63	2.05
29	Tunisia	-2.62	0.84	-4.26	-0.94
30	United States	-4.04	1.26	-6.39	-1.39
31	Uruguay	2.19	0.78	0.64	3.73
32	Wales	6.26	0.92	4.45	8.06

Table C.1.2: *Opponent team effects for each national team under the negative binomial model for passing, representing each team's ability to intercept the passing of its opponents, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

No.	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-7.07	1.29	-9.85	-4.59
2	Australia	-3.43	1.41	-6.17	-0.62
3	Belgium	5.49	2.67	0.28	10.56
4	Brazil	-0.55	0.93	-2.39	1.29
5	Cameroon	0.20	0.98	-1.74	2.14
6	Canada	-1.60	1.26	-4.05	0.86
7	Costa Rica	-2.32	1.46	-5.15	0.65
8	Croatia	-3.02	0.80	-4.61	-1.46
9	Denmark	-4.28	1.64	-7.52	-1.08
10	Ecuador	4.65	1.71	1.32	8.06
11	England	3.48	1.32	0.84	6.07
12	France	-4.54	0.99	-6.46	-2.57
13	Germany	-2.97	1.21	-5.35	-0.63
14	Ghana	1.53	1.13	-0.65	3.77
15	IR Iran	12.62	2.78	7.63	18.25
16	Japan	-0.34	1.02	-2.34	1.66
17	Korea Republic	0.40	2.11	-3.69	4.61
18	Mexico	-1.17	0.81	-2.75	0.46
19	Morocco	-43.52	2.92	-49.47	-37.90
20	Netherlands	-3.19	1.13	-5.48	-0.96
21	Poland	-0.15	0.81	-1.77	1.49
22	Portugal	50.62	3.98	42.64	57.95
23	Qatar	-7.10	1.36	-9.77	-4.42
24	Saudi Arabia	11.64	1.73	8.17	14.94
25	Senegal	-0.92	0.93	-2.76	0.93
26	Serbia	0.78	0.90	-0.98	2.55
27	Spain	-10.72	2.00	-14.62	-6.85
28	Switzerland	0.82	0.72	-0.57	2.26
29	Tunisia	-0.86	0.77	-2.37	0.63
30	United States	-1.37	2.78	-7.07	3.62
31	Uruguay	-0.09	1.40	-2.80	2.67
32	Wales	7.01	1.58	3.95	10.22

Table C.1.3: *Top 10 players under the negative binomial model for passes, ranked by posterior predicted total passes, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Passes	Lower HPD Bound	Upper HPD Bound	Observed Passes	90-minute units
1	Assim Madibo	2029430	73360.90	10347100	58	1.7
2	Jordan Henderson	91686.40	231.20	751517	137	3.0
3	Phil Foden	57364.20	175.33	473425	90	2.9
4	Nicolas Tagliafico	36685.60	1238.94	207369	145	4.1
5	Jhegson Mendez	6057.91	261.12	33552.60	128	2.0
6	Shogo Taniguchi	4350.23	126.50	27806.20	72	2.3
7	Lisandro Martinez	2427.61	189.50	12262.30	133	3.4
8	Marcos Acuna	2385.14	285.58	9578.71	167	4.1
9	Alexis Mac Allister	2029.29	156.07	11094.10	234	6.1
10	Josko Gvardiol	1722.30	285.97	29054.30	154	7.7

Table C.1.4: *Bottom 10 players under the negative binomial model for passes, ranked by posterior predicted total passes, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Passes	Lower HPD Bound	Upper HPD Bound	Observed Passes	90-minute units
1	Yerry Mina	10.27	3.71	24.02	13	1.6
2	Michy Batshuayi	11.75	3.79	30.11	16	1.7
3	Vincent Aboubakar	14.85	5.99	32.74	20	1.6
4	Cyle Larin	16.00	6.55	35.00	21	1.5
5	Jordan Ayew	18.53	8.05	39.50	35	1.5
6	Daniel James	20.58	8.56	45.28	24	1.7
7	Luis Suarez	20.83	8.39	45.85	25	1.6
8	Randal Kolo Muani	23.61	6.41	71.95	25	2.0
9	Karl Toko Ekambi	25.50	10.74	55.60	43	1.8
10	Hector Herrera	26.20	10.10	61.82	64	1.8

Table C.1.5: *Convergence diagnostic results for parameters 1–35, under the negative binomial model for passing. The parameter column reports Δ_i for player effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Aaron Mooy	Δ_1	1.00	177246	0.12	-1.07	0.14	TRUE
2	Aaron Ramsey	Δ_2	1.00	140803	0.73	-1.59	0.43	FALSE
3	Abdelkarim Hassan	Δ_3	1.00	175848	0.01	-1.51	-0.04	TRUE
4	Abdou Diallo	Δ_4	1.00	95756	-1.59	5.31	1.07	FALSE
5	Abdulaziz Hatem	Δ_5	1.00	85042	-0.24	0.62	-3.21	FALSE
6	Abdullelah Al-Amri	Δ_6	1.00	3158	0.02	1.95	-2.34	FALSE
7	Abdullelah Al-Malki	Δ_7	1.00	18198	-2.24	-1.47	1.74	FALSE
8	Achraf Hakimi	Δ_8	1.00	69130	-1.39	0.65	2.10	TRUE
9	Adrien Rabiot	Δ_9	1.00	53918	2.07	-0.01	1.16	FALSE
10	Ahmad Nourollahi	Δ_{10}	1.00	13206	-0.05	-0.77	0.09	FALSE
11	Aissa Laidouni	Δ_{11}	1.00	86249	2.13	-6.81	0.07	FALSE
12	Akram Afif	Δ_{12}	1.00	773866	-0.77	0.47	0.38	TRUE
13	Aleksandar Mitrovic	Δ_{13}	1.00	65115	-0.53	-1.61	1.16	FALSE
14	Alexis Mac Allister	Δ_{14}	1.00	21160	0.85	-1.32	-0.88	FALSE
15	Alexis Vega	Δ_{15}	1.00	12944	-1.11	-3.31	1.38	TRUE
16	Ali Abdi	Δ_{16}	1.00	81992	2.44	-4.99	-2.88	FALSE
17	Ali Al-Bulaihi	Δ_{17}	1.00	71593	0.22	2.03	-5.20	FALSE
18	Ali Gholizadeh	Δ_{18}	1.00	150138	-0.31	0.77	-0.99	FALSE
19	Alidu Seidu	Δ_{19}	1.00	14332	1.75	1.96	-3.96	FALSE
20	Alistair Johnston	Δ_{20}	1.00	168656	0.94	1.04	-1.38	TRUE
21	Almoez Ali	Δ_{21}	1.00	154250	-0.65	-1.24	1.36	FALSE
22	Alphonso Davies	Δ_{22}	1.00	95481	-0.13	-1.57	2.27	FALSE
23	Alvaro Morata	Δ_{23}	1.00	209180	0.28	2.86	0.16	TRUE
24	Andre-Frank Zambo Anguissa	Δ_{24}	1.00	443777	0.05	-1.23	-1.53	TRUE
25	Andre Ayew	Δ_{25}	1.00	191216	-1.64	0.26	1.89	TRUE
26	Andreas Christensen	Δ_{26}	1.00	165128	0.58	0.63	1.36	TRUE
27	Andrej Kramaric	Δ_{27}	1.00	96121	-2.18	-1.17	3.17	TRUE
28	Andrija Zivkovic	Δ_{28}	1.00	165727	0.75	2.24	-0.76	FALSE
29	Angelo Preciado	Δ_{29}	1.00	173245	0.48	1.61	-2.16	TRUE
30	Anis Ben Slimane	Δ_{30}	1.00	4146	0.48	0.74	1.02	TRUE
31	Antoine Griezmann	Δ_{31}	1.00	43791	-2.01	-0.70	0.37	FALSE
32	Antonee Robinson	Δ_{32}	1.00	15847	-1.16	0.04	0.32	TRUE
33	Antonio Rudiger	Δ_{33}	1.00	171064	0.24	-0.06	-0.82	TRUE
34	Assim Madibo	Δ_{34}	1.00	14860	-0.44	-1.35	3.04	FALSE
35	Atiba Hutchinson	Δ_{35}	1.00	28970	0.20	1.45	-2.70	FALSE

Table C.1.6: *Convergence diagnostic results for parameters 36–70, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
36	Aurelien Tchouameni	Δ_{36}	1.00	42970	-1.54	-0.82	0.50	TRUE
37	Axel Witsel	Δ_{37}	1.00	143514	0.75	-1.48	1.39	TRUE
38	Aymeric Laporte	Δ_{38}	1.00	160495	-1.53	-2.96	4.65	TRUE
39	Aziz Behich	Δ_{39}	1.00	175918	0.92	0.06	0.47	TRUE
40	Azzedine Ounahi	Δ_{40}	1.00	77266	-1.87	2.15	1.78	TRUE
41	Baba Rahman	Δ_{41}	1.00	6762	-0.07	1.25	-1.02	TRUE
42	Bartosz Bereszynski	Δ_{42}	1.00	179215	-0.92	0.28	-0.76	TRUE
43	Ben Davies	Δ_{43}	1.00	163672	0.21	1.18	-2.36	TRUE
44	Bernardo Silva	Δ_{44}	1.00	65102	-1.61	-0.22	2.61	TRUE
45	Borna Sosa	Δ_{45}	1.00	7011	0.99	4.87	-3.38	TRUE
46	Boualem Khoukhi	Δ_{46}	1.00	157407	-0.58	-0.74	-1.27	TRUE
47	Boulaye Dia	Δ_{47}	1.00	52027	-0.41	-0.07	2.93	TRUE
48	Breel Embolo	Δ_{48}	1.00	52269	0.32	-0.82	1.63	TRUE
49	Bruno Fernandes	Δ_{49}	1.00	87067	-0.96	0.82	1.24	TRUE
50	Bryan Mbeumo	Δ_{50}	1.00	153918	1.06	-0.89	1.13	TRUE
51	Bryan Oviedo	Δ_{51}	1.00	166205	0.81	-1.19	-1.00	TRUE
52	Bukayo Saka	Δ_{52}	1.00	9882	0.57	0.55	-0.36	TRUE
53	Casemiro	Δ_{53}	1.00	84658	0.32	-1.72	2.30	TRUE
54	Celso Borges	Δ_{54}	1.00	139395	0.68	0.41	-2.93	TRUE
55	Cesar Montes	Δ_{55}	1.00	84891	1.59	-1.25	2.72	TRUE
56	Cho Gue-sung	Δ_{56}	1.00	94752	-1.11	-2.52	2.99	TRUE
57	Chris Mepham	Δ_{57}	1.00	178259	0.15	-1.18	-1.30	TRUE
58	Christian Eriksen	Δ_{58}	1.00	168355	-0.09	-0.55	1.17	TRUE
59	Christian Pulisic	Δ_{59}	1.00	136031	-0.59	-1.71	3.25	TRUE
60	Cody Gakpo	Δ_{60}	1.00	65175	0.19	-1.63	2.67	TRUE
61	Collins Fai	Δ_{61}	1.00	140213	0.28	0.17	2.48	TRUE
62	Connor Roberts	Δ_{62}	1.00	80311	-0.18	1.92	-2.77	TRUE
63	Craig Goodwin	Δ_{63}	1.00	117415	0.49	1.56	-1.23	TRUE
64	Cristian Romero	Δ_{64}	1.00	73845	-0.44	0.92	-2.69	TRUE
65	Cristiano Ronaldo	Δ_{65}	1.00	110315	-1.41	-0.16	3.60	TRUE
66	Cyle Larin	Δ_{66}	1.00	144824	0.25	3.26	-1.69	TRUE
67	Daichi Kamada	Δ_{67}	1.00	124445	-2.52	3.56	-4.65	TRUE
68	Daizen Maeda	Δ_{68}	1.00	31450	-0.59	-0.41	2.05	TRUE
69	Daley Blind	Δ_{69}	1.00	118280	0.57	0.87	-1.27	TRUE
70	Dani Olmo	Δ_{70}	1.00	64944	-0.54	-1.73	1.58	TRUE

Table C.1.7: *Convergence diagnostic results for parameters 71–105, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
71	Daniel Amartey	Δ_{71}	1.00	159790	-0.52	0.23	-0.49	TRUE
72	Daniel James	Δ_{72}	1.00	129128	0.09	2.82	-0.71	TRUE
73	Danilo	Δ_{73}	1.00	14103	0.56	1.02	-1.30	TRUE
74	Darwin Nunez	Δ_{74}	1.00	55895	-1.36	-2.15	2.99	TRUE
75	David Raum	Δ_{75}	1.00	174263	-0.04	-1.26	-0.67	TRUE
76	Dayot Upamecano	Δ_{76}	1.01	7536	-1.27	0.02	-1.44	TRUE
77	Declan Rice	Δ_{77}	1.00	105199	0.26	-1.18	-0.93	TRUE
78	Dejan Lovren	Δ_{78}	1.00	45972	0.02	-1.75	-0.80	TRUE
79	Denzel Dumfries	Δ_{79}	1.00	80571	1.48	-0.67	1.66	TRUE
80	Diego Godin	Δ_{80}	1.00	12663	1.55	0.09	-2.46	TRUE
81	Diogo Dalot	Δ_{81}	1.00	14022	-1.85	-1.22	1.68	TRUE
82	Djibril Sow	Δ_{82}	1.00	136205	1.75	-0.91	0.57	TRUE
83	Dusan Tadic	Δ_{83}	1.00	125064	1.25	-1.21	0.27	TRUE
84	Dylan Bronn	Δ_{84}	1.03	5274	-0.52	3.42	-0.88	TRUE
85	Eder Militao	Δ_{85}	1.00	18675	1.39	-1.15	1.90	TRUE
86	Edson Alvarez	Δ_{86}	1.00	60119	1.09	0.70	-2.44	TRUE
87	Ehsan Hajsafi	Δ_{87}	1.00	158991	-0.22	-1.20	0.82	TRUE
88	Ellyes Skhiri	Δ_{88}	1.00	63160	0.78	-4.48	1.63	TRUE
89	Enner Valencia	Δ_{89}	1.00	136218	-0.40	-0.55	1.09	TRUE
90	Enzo Fernandez	Δ_{90}	1.00	62366	-0.71	0.77	-0.73	TRUE
91	Eric Maxim Choupo-Moting	Δ_{91}	1.00	115079	-0.26	-2.48	3.20	TRUE
92	Ethan Ampadu	Δ_{92}	1.00	186082	0.93	0.56	1.71	TRUE
93	Facundo Pellistri	Δ_{93}	1.00	190697	0.03	0.25	1.64	TRUE
94	Federico Valverde	Δ_{94}	1.00	161200	0.27	-0.95	0.18	TRUE
95	Felix Torres Caicedo	Δ_{95}	1.00	110091	0.76	2.83	-1.61	TRUE
96	Ferran Torres	Δ_{96}	1.00	142153	0.37	1.33	-1.65	TRUE
97	Filip Kostic	Δ_{97}	1.00	70335	-0.44	0.66	-0.36	TRUE
98	Firas Al-Buraikan	Δ_{98}	1.00	111974	0.03	-1.15	2.11	TRUE
99	Francisco Calvo	Δ_{99}	1.00	16690	-1.46	-0.09	1.94	TRUE
100	Frenkie de Jong	Δ_{100}	1.00	139974	1.67	1.26	0.13	TRUE
101	Gareth Bale	Δ_{101}	1.00	105497	0.03	-0.47	1.65	TRUE
102	Gavi	Δ_{102}	1.00	175784	-1.86	1.39	-1.58	TRUE
103	Gonzalo Plata	Δ_{103}	1.00	183539	1.13	-0.18	0.07	TRUE
104	Granit Xhaka	Δ_{104}	1.00	112674	-1.52	-0.41	0.07	TRUE
105	Grzegorz Krychowiak	Δ_{105}	1.00	178879	3.24	0.93	0.72	TRUE

Table C.1.8: *Convergence diagnostic results for parameters 106–140, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
106	Guillermo Varela	Δ_{106}	1.00	112557	0.19	-1.65	-1.45	TRUE
107	Hakim Ziyech	Δ_{107}	1.00	54376	-0.87	0.98	2.72	TRUE
108	Harry Kane	Δ_{108}	1.00	68931	0.03	-2.24	1.89	TRUE
109	Harry Maguire	Δ_{109}	1.00	105436	-0.29	-0.16	-0.62	TRUE
110	Harry Souttar	Δ_{110}	1.00	178350	1.14	-1.67	1.91	TRUE
111	Harry Wilson	Δ_{111}	1.00	73365	0.42	2.11	-1.81	TRUE
112	Hassan Al-Haydos	Δ_{112}	1.00	137807	-1.36	-0.72	-2.03	TRUE
113	Hassan Al Tambakti	Δ_{113}	1.00	99215	-2.36	-0.45	0.53	TRUE
114	Hector Herrera	Δ_{114}	1.00	7810	-1.24	1.44	-0.99	TRUE
115	Hector Moreno	Δ_{115}	1.00	85878	-0.12	-2.23	1.88	TRUE
116	Henry Martin	Δ_{116}	1.00	31397	0.08	0.69	-2.98	TRUE
117	Hidemasa Morita	Δ_{117}	1.00	24453	-0.40	0.49	-0.02	TRUE
118	Hirving Lozano	Δ_{118}	1.00	39016	1.49	-1.89	2.02	TRUE
119	Homam Ahmed	Δ_{119}	1.00	185003	0.32	-3.66	1.04	TRUE
120	Hwang In-beom	Δ_{120}	1.00	154909	-1.16	0.90	-0.04	FALSE
121	Ibrahima Konate	Δ_{121}	1.01	13868	0.59	-2.40	0.53	TRUE
122	Idrissa Gana Gueye	Δ_{122}	1.00	4098	0.49	-0.39	1.10	FALSE
123	Ilkay Gundogan	Δ_{123}	1.00	43812	-0.24	1.43	-3.24	TRUE
124	Inaki Williams	Δ_{124}	1.00	42992	-1.86	-0.63	3.91	TRUE
125	Ismail Jakobs	Δ_{125}	1.00	98570	-0.98	5.82	-1.63	TRUE
126	Ismaila Sarr	Δ_{126}	1.00	50012	-1.19	-0.25	1.76	TRUE
127	Issam Jebali	Δ_{127}	1.00	11134	1.69	-6.04	-0.28	TRUE
128	Ivan Perisic	Δ_{128}	1.00	42502	-1.05	-1.17	2.51	TRUE
129	Jackson Irvine	Δ_{129}	1.00	160469	2.19	-0.42	2.17	TRUE
130	Jakub Kaminski	Δ_{130}	1.00	20154	0.81	0.40	0.04	FALSE
131	Jakub Kiwior	Δ_{131}	1.00	172521	1.65	1.14	0.00	FALSE
132	Jamal Musiala	Δ_{132}	1.00	98066	-0.31	-0.88	1.67	TRUE
133	Jan Vertonghen	Δ_{133}	1.00	151133	0.55	-0.51	-1.13	FALSE
134	Javad El Yamiq	Δ_{134}	1.00	7356	-0.73	1.10	-2.88	TRUE
135	Jean-Charles Castelletto	Δ_{135}	1.00	34865	1.75	-0.50	-0.78	FALSE
136	Jesper Lindstrom	Δ_{136}	1.00	184222	0.31	0.05	1.06	TRUE
137	Jesus Gallardo	Δ_{137}	1.00	55461	-0.93	-1.44	1.73	TRUE
138	Jhegson Mendez	Δ_{138}	1.00	25178	0.32	-2.54	2.74	TRUE
139	Joachim Andersen	Δ_{139}	1.00	165940	-0.65	1.37	2.21	TRUE
140	Joakim Maehle	Δ_{140}	1.00	171724	-1.71	0.06	3.52	TRUE

Table C.1.9: *Convergence diagnostic results for parameters 141–175, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
141	Joao Cancelo	Δ_{141}	1.00	19671	3.25	-0.04	-0.16	TRUE
142	Joao Felix	Δ_{142}	1.00	72120	-0.07	-0.33	1.25	TRUE
143	Joe Rodon	Δ_{143}	1.00	18270	0.60	1.81	1.87	TRUE
144	Joel Campbell	Δ_{144}	1.00	16206	-0.06	0.73	-1.64	TRUE
145	John Stones	Δ_{145}	1.00	79585	0.01	-0.50	-0.07	TRUE
146	Jonathan David	Δ_{146}	1.00	149656	0.13	1.96	-2.32	FALSE
147	Jordan Ayew	Δ_{147}	1.00	47184	-0.89	1.61	-1.61	TRUE
148	Jordan Henderson	Δ_{148}	1.00	28523	0.35	-0.06	1.91	TRUE
149	Jordi Alba	Δ_{149}	1.00	34820	-0.38	2.00	-2.55	TRUE
150	Jorge Eduardo Sanchez	Δ_{150}	1.00	6777	1.15	0.96	-2.99	TRUE
151	Jose Maria Gimenez	Δ_{151}	1.00	146823	2.02	-3.05	3.31	TRUE
152	Joshua Kimmich	Δ_{152}	1.00	16934	-0.15	0.06	-2.57	TRUE
153	Josip Juranovic	Δ_{153}	1.00	52606	0.01	-2.33	-1.51	FALSE
154	Josko Gvardiol	Δ_{154}	1.00	60271	-1.23	-1.76	3.33	TRUE
155	Jude Bellingham	Δ_{155}	1.00	14434	-2.19	-0.71	1.28	FALSE
156	Jules Kounde	Δ_{156}	1.00	54771	2.19	-0.03	1.42	TRUE
157	Julian Alvarez	Δ_{157}	1.00	18076	-0.25	-1.32	0.68	TRUE
158	Jung Woo-young	Δ_{158}	1.00	13856	-1.61	-2.02	1.09	TRUE
159	Junior Hoilett	Δ_{159}	1.00	10321	0.73	0.92	-1.95	FALSE
160	Junya Ito	Δ_{160}	1.00	12866	2.32	2.33	-3.49	TRUE
161	Jurien Timber	Δ_{161}	1.00	60731	-0.67	-1.06	0.37	TRUE
162	Kalidou Koulibaly	Δ_{162}	1.00	81639	-1.18	1.72	2.21	FALSE
163	Kamal Miller	Δ_{163}	1.00	165267	0.12	1.90	0.30	TRUE
164	Kamil Glik	Δ_{164}	1.00	17087	1.54	0.09	3.04	FALSE
165	Karim Boudiaf	Δ_{165}	1.00	42459	-0.45	1.18	-1.12	TRUE
166	Karl Toko Ekambi	Δ_{166}	1.00	58781	0.47	1.30	-1.59	TRUE
167	Kendall Waston	Δ_{167}	1.00	169386	2.05	0.18	-1.15	TRUE
168	Kevin De Bruyne	Δ_{168}	1.00	82785	0.80	-1.57	2.34	FALSE
169	Keysher Fuller	Δ_{169}	1.00	163278	1.69	-0.36	-1.05	TRUE
170	Kieffer Moore	Δ_{170}	1.00	115315	-0.37	-1.43	2.63	TRUE
171	Kim Jin-su	Δ_{171}	1.00	154491	1.29	-2.01	1.88	FALSE
172	Kim Min-jae	Δ_{172}	1.00	107672	-0.55	1.68	-1.88	TRUE
173	Kim Moon-hwan	Δ_{173}	1.00	118813	0.08	-1.74	0.97	TRUE
174	Kim Young-gwon	Δ_{174}	1.00	128446	-1.66	-0.20	0.89	TRUE
175	Ko Itakura	Δ_{175}	1.00	104536	-2.39	-3.70	5.15	TRUE

Table C.1.10: *Convergence diagnostic results for parameters 176–210, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
176	Krepin Diatta	Δ_{176}	1.00	2299	0.80	-4.05	-1.21	TRUE
177	Krystian Bielik	Δ_{177}	1.00	4005	0.29	1.35	-1.76	TRUE
178	Kudus Mohammed	Δ_{178}	1.00	16857	-1.64	-2.04	0.91	TRUE
179	Kye Rowles	Δ_{179}	1.00	16748	0.85	-0.03	0.35	TRUE
180	Kyle Walker	Δ_{180}	1.00	6277	0.26	-0.19	-0.18	TRUE
181	Kylian Mbappe	Δ_{181}	1.00	74057	-0.73	-1.95	1.97	FALSE
182	Leander Dendoncker	Δ_{182}	1.00	8042	-1.19	-1.68	2.70	TRUE
183	Lee Jae-sung	Δ_{183}	1.00	5832	2.22	0.04	-0.80	TRUE
184	Leon Goretzka	Δ_{184}	1.00	13127	1.58	1.13	-1.81	TRUE
185	Lionel Messi	Δ_{185}	1.00	16052	0.13	-1.07	-0.24	FALSE
186	Lisandro Martinez	Δ_{186}	1.00	27248	0.51	0.27	2.89	FALSE
187	Lucas Paqueta	Δ_{187}	1.00	7993	-1.27	-0.71	-0.36	TRUE
188	Luis Chavez	Δ_{188}	1.00	56895	0.74	-1.81	3.42	FALSE
189	Luis Suarez	Δ_{189}	1.00	14916	1.66	-0.20	-1.40	TRUE
190	Luka Modric	Δ_{190}	1.00	6972	1.23	-0.80	3.71	FALSE
191	Luke Shaw	Δ_{191}	1.00	11565	-1.14	0.12	-0.09	TRUE
192	Majid Hosseini	Δ_{192}	1.00	14427	0.47	-1.92	-0.24	TRUE
193	Manuel Akanji	Δ_{193}	1.00	11989	1.54	-0.92	1.93	TRUE
194	Marcelo Brozovic	Δ_{194}	1.00	29285	-0.48	-1.17	-1.61	TRUE
195	Marco Asensio	Δ_{195}	1.00	8993	0.51	1.29	-1.83	TRUE
196	Marcos Acuna	Δ_{196}	1.00	24962	0.45	-0.94	2.14	FALSE
197	Mario Pasalic	Δ_{197}	1.00	33577	-0.72	1.09	-1.07	TRUE
198	Marquinhos	Δ_{198}	1.00	2097	-2.70	3.12	-0.61	TRUE
199	Mateo Kovacic	Δ_{199}	1.00	7024	-2.39	-2.03	2.18	TRUE
200	Mathew Leckie	Δ_{200}	1.00	12929	-1.20	-2.76	2.17	TRUE
201	Mathias Olivera	Δ_{201}	1.00	16256	-0.55	-2.20	1.03	TRUE
202	Matias Vecino	Δ_{202}	1.00	14331	1.15	1.08	-0.50	TRUE
203	Matty Cash	Δ_{203}	1.00	17538	-0.03	0.53	2.74	TRUE
204	Maya Yoshida	Δ_{204}	1.00	8705	3.09	4.97	-5.37	FALSE
205	Mehdi Taremi	Δ_{205}	1.00	9672	-0.20	-1.20	1.79	TRUE
206	Memphis	Δ_{206}	1.00	8576	0.11	1.68	-0.77	TRUE
207	Michael Estrada	Δ_{207}	1.00	16792	0.49	-0.18	2.04	TRUE
208	Michy Batshuayi	Δ_{208}	1.00	1116	0.79	3.72	-1.71	TRUE
209	Mikkel Damsgaard	Δ_{209}	1.00	5984	-0.55	1.49	-2.31	TRUE
210	Milad Mohammadi	Δ_{210}	1.00	15975	0.52	1.35	-1.25	TRUE

Table C.1.11: *Convergence diagnostic results for parameters 211–245, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
211	Milos Degenek	Δ_{211}	1.00	26517	0.67	1.37	-3.25	TRUE
212	Milos Veljkovic	Δ_{212}	1.00	171546	0.39	2.37	-1.17	TRUE
213	Mitchel Duke	Δ_{213}	1.00	190681	3.30	-1.15	2.83	TRUE
214	Mohamed Kanno	Δ_{214}	1.00	135388	-0.76	1.21	-0.36	FALSE
215	Mohammed Salisu	Δ_{215}	1.00	164573	-1.13	-0.24	1.45	TRUE
216	Moises Caicedo	Δ_{216}	1.00	160365	1.77	2.08	-1.48	TRUE
217	Montassar Talbi	Δ_{217}	1.00	68334	0.96	-5.10	1.85	TRUE
218	Morteza Pouraliganji	Δ_{218}	1.00	61386	0.60	-1.66	1.21	TRUE
219	Nahuel Molina	Δ_{219}	1.00	102686	-2.00	-0.39	-1.93	TRUE
220	Nampalys Mendy	Δ_{220}	1.00	65949	0.50	4.59	-1.30	TRUE
221	Nathan Ake	Δ_{221}	1.00	139237	0.66	0.88	1.14	TRUE
222	Nayef Aguerd	Δ_{222}	1.00	33087	-1.69	-0.70	-1.48	TRUE
223	Nico Williams	Δ_{223}	1.00	177305	0.02	1.90	-2.62	TRUE
224	Neymar	Δ_{224}	1.00	36211	3.15	-2.29	1.64	TRUE
225	Nico Elvedi	Δ_{225}	1.00	18443	-2.12	0.90	-2.51	FALSE
226	Nicolas Nkoulou	Δ_{226}	1.00	30146	2.39	-0.45	-1.93	TRUE
227	Nicolas Otamendi	Δ_{227}	1.00	145080	-2.37	-0.39	0.62	TRUE
228	Nicolas Tagliafico	Δ_{228}	1.00	16472	-0.57	-1.55	3.03	TRUE
229	Niklas Sule	Δ_{229}	1.00	171220	0.70	-1.22	-1.10	TRUE
230	Nikola Milenkovic	Δ_{230}	1.00	160409	-0.61	0.16	-0.45	TRUE
231	Nouhou Tolo	Δ_{231}	1.00	135522	-0.34	-0.02	2.58	TRUE
232	Noussair Mazraoui	Δ_{232}	1.00	24329	0.94	-4.39	-2.50	TRUE
233	Olivier Giroud	Δ_{233}	1.00	39053	-0.13	-1.80	2.25	TRUE
234	Oscar Duarte	Δ_{234}	1.00	17004	1.55	0.04	0.19	TRUE
235	Ousmane Dembele	Δ_{235}	1.00	37639	-1.14	-0.20	0.91	TRUE
236	Pedri	Δ_{236}	1.00	65199	-0.65	0.99	0.15	TRUE
237	Pepe	Δ_{237}	1.00	21379	1.07	-0.08	-0.49	TRUE
238	Pervis Estupinan	Δ_{238}	1.00	14593	0.97	3.33	-1.06	TRUE
239	Phil Foden	Δ_{239}	1.00	28012	0.20	-0.63	1.83	TRUE
240	Piero Hincapie	Δ_{240}	1.00	117301	-0.41	2.70	-1.40	TRUE
241	Pierre Hojbjerg	Δ_{241}	1.00	16873	1.31	0.84	1.37	TRUE
242	Piotr Zielinski	Δ_{242}	1.00	18356	0.28	2.46	-0.61	TRUE
243	Przemyslaw Frankowski	Δ_{243}	1.00	18469	-0.47	0.65	-1.98	TRUE
244	Ramin Rezaeian	Δ_{244}	1.00	4522	0.61	1.35	-0.47	TRUE
245	Randal Kolo Muani	Δ_{245}	1.00	6134	-0.65	-0.08	-2.24	TRUE

Table C.1.12: *Convergence diagnostic results for parameters 246–280, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
246	Raphael Guerreiro	Δ_{246}	1.00	77434	-2.05	2.08	-3.54	TRUE
247	Raphael Varane	Δ_{247}	1.00	47492	1.39	0.12	1.25	TRUE
248	Raphinha	Δ_{248}	1.00	28443	-1.86	3.74	-1.30	TRUE
249	Rasmus Nissen	Δ_{249}	1.00	56804	-0.85	0.72	-1.10	TRUE
250	Remo Freuler	Δ_{250}	1.00	12959	0.68	-0.72	1.16	TRUE
251	Ricardo Rodriguez	Δ_{251}	1.00	10139	0.29	-1.77	-0.26	TRUE
252	Richarlison	Δ_{252}	1.00	53678	-0.93	-1.00	2.53	TRUE
253	Richie Laryea	Δ_{253}	1.00	14390	0.48	1.69	-1.55	TRUE
254	Riley McGree	Δ_{254}	1.00	9094	1.86	1.95	-3.38	TRUE
255	Ritsu Doan	Δ_{255}	1.00	8303	2.35	3.61	-4.88	TRUE
256	Ro-Ro	Δ_{256}	1.00	82673	0.12	-0.71	1.03	TRUE
257	Robert Lewandowski	Δ_{257}	1.00	76187	-0.57	-1.02	2.68	TRUE
258	Rodri	Δ_{258}	1.00	59237	0.87	1.01	0.61	TRUE
259	Rodrigo Bentancur	Δ_{259}	1.00	101114	0.19	1.30	-1.90	TRUE
260	Rodrigo De Paul	Δ_{260}	1.00	79222	0.16	1.23	-1.82	TRUE
261	Romain Saiss	Δ_{261}	1.00	13664	-1.06	1.16	-2.53	TRUE
262	Ruben Dias	Δ_{262}	1.00	90139	-0.25	0.88	-0.72	TRUE
263	Ruben Neves	Δ_{263}	1.00	20194	-1.05	0.48	0.49	TRUE
264	Ruben Vargas	Δ_{264}	1.00	14934	1.73	-0.33	-1.56	TRUE
265	Saeid Ezatolahi	Δ_{265}	1.00	44626	-0.81	0.62	-1.76	TRUE
266	Saleh Al-Shehri	Δ_{266}	1.00	17620	-1.16	-0.07	2.28	TRUE
267	Salem Al-Dawsari	Δ_{267}	1.00	62668	2.03	1.29	-1.23	TRUE
268	Sali Abdul Samed	Δ_{268}	1.00	16202	-2.68	0.16	-0.29	TRUE
269	Sasa Lucic	Δ_{269}	1.00	57186	0.53	1.01	-1.07	TRUE
270	Saud Abdulhamid	Δ_{270}	1.00	16845	-0.25	2.18	-1.29	TRUE
271	Sebastian Coates	Δ_{271}	1.00	24248	-0.32	-0.42	0.20	TRUE
272	Selim Amallah	Δ_{272}	1.00	11945	-2.29	0.01	3.08	TRUE
273	Serge Gnabry	Δ_{273}	1.00	10025	0.64	-1.39	1.38	TRUE
274	Sergej Milinkovic-Savic	Δ_{274}	1.00	75443	-0.17	-1.22	0.68	TRUE
275	Sergio Dest	Δ_{275}	1.00	17287	0.10	0.97	-1.80	TRUE
276	Sergio Busquets	Δ_{276}	1.00	3946	0.26	0.52	2.21	TRUE
277	Shogo Taniguchi	Δ_{277}	1.00	1798	-1.00	-0.84	3.00	TRUE
278	Silvan Widmer	Δ_{278}	1.00	1607	-1.78	1.40	-1.80	TRUE
279	Sofiane Boufal	Δ_{279}	1.00	9367	-2.84	0.03	2.65	TRUE
280	Sofyan Amrabat	Δ_{280}	1.00	58099	-1.62	-0.25	2.44	TRUE

Table C.1.13: *Convergence diagnostic results for parameters 281–310, under the negative binomial model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
281	Son Heung-min	Δ_{281}	1.00	51665	-0.42	-1.14	2.32	TRUE
282	Steven Vitoria	Δ_{282}	1.00	171055	1.93	1.08	-0.95	TRUE
283	Strahinja Pavlovic	Δ_{283}	1.00	162573	0.21	0.16	-1.38	TRUE
284	Tajon Buchanan	Δ_{284}	1.00	81153	0.03	-1.30	0.59	TRUE
285	Takehiro Tomiyasu	Δ_{285}	1.00	9600	-0.02	1.39	1.42	TRUE
286	Theo Hernandez	Δ_{286}	1.00	51297	1.76	0.30	0.95	TRUE
287	Thiago Silva	Δ_{287}	1.00	86534	-1.21	-1.23	1.78	TRUE
288	Thomas Meunier	Δ_{288}	1.00	99155	1.43	1.25	-2.63	FALSE
289	Thomas Muller	Δ_{289}	1.00	186716	-1.85	0.22	-0.11	TRUE
290	Thomas Partey	Δ_{290}	1.00	163498	-2.33	-1.02	1.16	TRUE
291	Tim Ream	Δ_{291}	1.00	173576	0.87	-0.63	0.57	TRUE
292	Timothy Castagne	Δ_{292}	1.00	152558	0.88	-1.33	0.15	TRUE
293	Timothy Weah	Δ_{293}	1.00	158203	0.28	-2.11	-0.48	FALSE
294	Toby Alderweireld	Δ_{294}	1.00	146423	1.48	-0.97	-0.94	TRUE
295	Tyler Adams	Δ_{295}	1.00	169890	0.10	-0.43	-2.50	TRUE
296	Vincent Aboubakar	Δ_{296}	1.00	119740	-0.74	0.92	-1.05	TRUE
297	Vinicius Junior	Δ_{297}	1.00	113264	0.47	0.08	1.40	TRUE
298	Virgil van Dijk	Δ_{298}	1.00	123361	1.06	-1.80	-0.53	TRUE
299	Walker Zimmerman	Δ_{299}	1.00	68825	0.11	1.89	-1.80	TRUE
300	Wataru Endo	Δ_{300}	1.00	43218	2.69	3.10	-5.51	FALSE
301	Weston McKennie	Δ_{301}	1.00	141616	0.02	0.64	-3.48	TRUE
302	Yerdhan Shaji	Δ_{302}	1.00	105248	0.88	0.01	-0.41	TRUE
303	Yahya Attiat Allah	Δ_{303}	1.00	51142	0.99	-1.42	-1.34	TRUE
304	Yassine Meriah	Δ_{304}	1.00	74613	2.20	-5.50	1.60	FALSE
305	Yeltsin Tejeda	Δ_{305}	1.00	166703	-1.59	-1.82	0.31	TRUE
306	Youssef En-Nesyri	Δ_{306}	1.00	366122	-0.42	-1.06	2.46	TRUE
307	Youssef Msakni	Δ_{307}	1.00	76303	-1.36	3.93	-0.66	TRUE
308	Youssouf Sabaly	Δ_{308}	1.00	90241	-1.67	2.41	1.34	TRUE
309	Yunus Musah	Δ_{309}	1.00	167735	0.80	-0.93	0.31	TRUE
310	Yuto Nagatomo	Δ_{310}	1.00	22039	1.22	2.16	-5.13	TRUE

Table C.1.14: *Convergence diagnostic results for the team-effect parameters under the negative binomial model for passing. The parameter column reports λ_i^e for team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	λ_1^e	1.04	172	1.46	-0.22	-1.03	TRUE
2	Australia	λ_2^e	1.02	416	-0.90	0.58	0.94	FALSE
3	Belgium	λ_3^e	1.01	732	1.30	0.24	-1.00	TRUE
4	Brazil	λ_4^e	1.01	607	0.73	0.39	-0.73	TRUE
5	Cameroon	λ_5^e	1.01	578	-0.16	1.32	-1.12	TRUE
6	Canada	λ_6^e	1.02	282	1.29	0.30	-1.43	TRUE
7	Costa Rica	λ_7^e	1.01	436	-0.37	-2.62	3.84	FALSE
8	Croatia	λ_8^e	1.04	246	1.08	0.90	-1.22	TRUE
9	Denmark	λ_9^e	1.01	639	-0.83	-1.21	-1.50	TRUE
10	Ecuador	λ_{10}^e	1.05	141	-0.42	0.27	0.41	FALSE
11	England	λ_{11}^e	1.03	302	-1.13	-0.34	0.28	TRUE
12	France	λ_{12}^e	1.02	364	-0.86	1.12	-0.20	TRUE
13	Germany	λ_{13}^e	1.01	366	-0.32	-2.74	3.32	TRUE
14	Ghana	λ_{14}^e	1.01	575	1.57	1.05	-0.91	TRUE
15	IR Iran	λ_{15}^e	1.06	213	-0.94	1.04	0.14	TRUE
16	Japan	λ_{16}^e	1.01	267	-0.20	-2.54	3.30	TRUE
17	Korea Republic	λ_{17}^e	1.02	362	0.92	0.98	-0.59	TRUE
18	Mexico	λ_{18}^e	1.04	288	0.16	0.44	1.10	FALSE
19	Morocco	λ_{19}^e	1.02	380	1.57	-1.56	-0.75	TRUE
20	Netherlands	λ_{20}^e	1.08	79	-0.75	1.20	0.53	TRUE
21	Poland	λ_{21}^e	1.04	201	1.06	0.02	0.33	FALSE
22	Portugal	λ_{22}^e	1.02	490	0.93	1.29	-0.85	TRUE
23	Qatar	λ_{23}^e	1.03	226	-0.16	0.72	-0.22	TRUE
24	Saudi Arabia	λ_{24}^e	1.03	221	1.14	0.33	0.28	FALSE
25	Senegal	λ_{25}^e	1.05	134	-0.63	0.38	0.33	FALSE
26	Serbia	λ_{26}^e	1.01	581	-0.10	1.13	-0.70	TRUE
27	Spain	λ_{27}^e	1.02	427	0.57	-3.06	0.11	TRUE
28	Switzerland	λ_{28}^e	1.01	524	0.40	0.88	-0.79	TRUE
29	Tunisia	λ_{29}^e	1.01	572	-0.71	0.19	-1.01	FALSE
30	United States	λ_{30}^e	1.01	165	-0.31	-0.60	-0.24	TRUE
31	Uruguay	λ_{31}^e	1.01	680	0.57	0.53	-0.04	TRUE
32	Wales	λ_{32}^e	1.04	363	-1.51	0.49	0.28	FALSE

Table C.1.15: *Convergence diagnostic results for the opponent team-effect parameters under the negative binomial model for passing. The parameter column reports* $\lambda_i^{E\setminus e}$ *for opponent team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	$\lambda_{1}^{E\setminus e}$	1.04	424	-0.40	0.36	1.77	FALSE
2	Australia	$\lambda_{2}^{E\setminus e}$	1.00	872	-0.76	-1.15	-1.81	TRUE
3	Belgium	$\lambda_{3}^{E\setminus e}$	1.01	291	1.13	0.38	-1.64	TRUE
4	Brazil	$\lambda_{4}^{E\setminus e}$	1.01	894	0.16	1.43	-0.99	TRUE
5	Cameroon	$\lambda_{5}^{E\setminus e}$	1.01	1000	0.25	1.13	-0.95	TRUE
6	Canada	$\lambda_{6}^{E\setminus e}$	1.00	1303	0.22	0.67	0.01	TRUE
7	Costa Rica	$\lambda_{7}^{E\setminus e}$	1.01	980	-0.06	-4.60	0.90	TRUE
8	Croatia	$\lambda_{8}^{E\setminus e}$	1.01	1017	2.02	-1.35	-0.66	FALSE
9	Denmark	$\lambda_{9}^{E\setminus e}$	1.01	785	-0.32	1.95	0.64	TRUE
10	Ecuador	$\lambda_{10}^{E\setminus e}$	1.03	741	-1.36	2.77	0.75	FALSE
11	England	$\lambda_{11}^{E\setminus e}$	1.05	529	-1.31	1.29	0.27	FALSE
12	France	$\lambda_{12}^{E\setminus e}$	1.01	768	-0.21	-2.04	-1.17	FALSE
13	Germany	$\lambda_{13}^{E\setminus e}$	1.01	1140	-0.30	-5.00	3.35	FALSE
14	Ghana	$\lambda_{14}^{E\setminus e}$	1.00	1409	-0.11	0.39	0.57	TRUE
15	IR Iran	$\lambda_{15}^{E\setminus e}$	1.03	282	-0.34	-2.43	0.29	TRUE
16	Japan	$\lambda_{16}^{E\setminus e}$	1.01	1551	1.60	-3.08	-1.29	TRUE
17	Korea Republic	$\lambda_{17}^{E\setminus e}$	1.02	454	2.25	0.38	-0.37	FALSE
18	Mexico	$\lambda_{18}^{E\setminus e}$	1.02	1029	1.82	1.52	-2.13	TRUE
19	Morocco	$\lambda_{19}^{E\setminus e}$	1.03	160	0.98	0.03	-1.00	TRUE
20	Netherlands	$\lambda_{20}^{E\setminus e}$	1.03	729	0.36	1.10	0.09	TRUE
21	Poland	$\lambda_{21}^{E\setminus e}$	1.01	1095	1.68	0.76	-0.63	TRUE
22	Portugal	$\lambda_{22}^{E\setminus e}$	1.07	143	-0.93	-0.25	0.74	TRUE
23	Qatar	$\lambda_{23}^{E\setminus e}$	1.03	745	0.54	0.43	1.41	FALSE
24	Saudi Arabia	$\lambda_{24}^{E\setminus e}$	1.06	364	-1.09	2.52	0.89	FALSE
25	Senegal	$\lambda_{25}^{E\setminus e}$	1.00	1804	0.20	0.94	-1.38	TRUE
26	Serbia	$\lambda_{26}^{E\setminus e}$	1.01	957	0.99	0.75	-0.30	TRUE
27	Spain	$\lambda_{27}^{E\setminus e}$	1.01	409	-0.68	-1.97	3.44	FALSE
28	Switzerland	$\lambda_{28}^{E\setminus e}$	1.01	1758	-1.28	1.86	-0.96	TRUE
29	Tunisia	$\lambda_{29}^{E\setminus e}$	1.00	2727	-2.43	1.68	0.66	FALSE
30	United States	$\lambda_{30}^{E\setminus e}$	1.06	205	-0.38	1.94	-0.04	TRUE
31	Uruguay	$\lambda_{31}^{E\setminus e}$	1.01	1110	1.46	1.57	-1.67	TRUE
32	Wales	$\lambda_{32}^{E\setminus e}$	1.00	699	-1.31	-1.41	0.53	TRUE

C.2 Poisson Model

Table C.2.1: *Team effects for each national team under the Poisson model for passing, showing each team's ability to complete passing sequences, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-2.15	0.60	-3.20	-1.04
2	Australia	-0.71	0.46	-1.65	0.11
3	Belgium	-0.32	0.65	-1.65	0.93
4	Brazil	0.85	0.43	0.17	1.95
5	Cameroon	-0.12	0.52	-1.15	0.94
6	Canada	1.26	0.61	0.09	2.49
7	Costa Rica	-1.28	0.69	-2.60	-0.05
8	Croatia	-0.99	0.37	-1.72	-0.37
9	Denmark	-1.21	0.62	-2.50	-0.07
10	Ecuador	0.52	0.78	-1.19	1.85
11	England	1.47	0.58	0.47	2.70
12	France	-0.75	0.28	-1.24	-0.27
13	Germany	-0.66	0.60	-1.71	0.60
14	Ghana	0.88	0.73	-0.47	2.33
15	IR Iran	0.27	0.71	-1.02	1.47
16	Japan	-1.35	0.74	-2.87	-0.25
17	Korea Republic	-0.27	0.53	-1.18	1.00
18	Mexico	-0.31	0.50	-1.35	0.51
19	Morocco	-0.10	0.42	-1.80	-0.32
20	Netherlands	1.59	0.86	0.08	3.31
21	Poland	-0.42	0.47	-1.26	0.39
22	Portugal	0.66	0.51	-0.23	1.85
23	Qatar	-1.45	0.84	-3.23	0.03
24	Saudi Arabia	0.98	0.64	-0.02	2.32
25	Senegal	0.95	0.72	-0.15	2.65
26	Serbia	0.31	0.57	-0.75	1.40
27	Spain	0.26	0.37	-0.59	0.87
28	Switzerland	0.05	0.52	-0.79	0.99
29	Tunisia	0.44	0.52	-0.52	1.48
30	United States	0.57	0.78	-1.23	1.85
31	Uruguay	0.63	0.70	-0.31	2.67
32	Wales	1.30	0.61	0.07	2.65

Table C.2.2: *Opponent team effects for each national team under the Poisson model for passing, showing each team's passing ability against opponent defensive pressure, with corresponding posterior means, posterior standard deviations, and lower and upper HPD bounds.*

Rank	Team	Posterior Mean	Posterior SD	Lower HPD Bound	Upper HPD Bound
1	Argentina	-7.13	0.88	-8.83	-5.44
2	Australia	-3.27	1.47	-6.28	-0.21
3	Belgium	6.99	1.95	3.59	10.94
4	Brazil	-5.88	0.82	-6.86	-3.63
5	Cameroon	2.18	0.89	0.55	3.86
6	Canada	1.56	1.67	-1.88	4.30
7	Costa Rica	0.17	0.98	-1.15	2.45
8	Croatia	-2.83	0.55	-3.98	-1.78
9	Denmark	4.93	1.15	2.66	7.18
10	Ecuador	5.18	1.26	2.92	7.62
11	England	-0.93	1.26	-3.36	1.14
12	France	-5.32	0.98	-7.41	-3.77
13	Germany	-0.07	1.26	-2.34	2.48
14	Ghana	0.79	1.36	-1.82	3.48
15	IR Iran	8.51	1.36	5.82	11.04
16	Japan	-0.27	1.00	-2.27	1.51
17	Korea Republic	6.48	1.20	4.65	8.87
18	Mexico	-4.16	0.71	-6.07	-2.73
19	Morocco	-15.44	2.67	-20.41	-10.34
20	Netherlands	0.79	0.97	-0.91	2.55
21	Poland	-3.06	1.02	-5.17	-1.35
22	Portugal	11.24	3.45	4.18	17.30
23	Qatar	0.36	1.09	-1.60	2.54
24	Saudi Arabia	2.32	1.62	-0.47	5.49
25	Senegal	1.73	1.14	-0.17	3.92
26	Serbia	0.09	1.09	-1.70	2.80
27	Spain	-6.70	1.57	-9.79	-4.03
28	Switzerland	-0.17	1.02	-2.66	1.62
29	Tunisia	0.72	1.10	-1.49	2.94
30	United States	-7.57	1.51	-10.81	-5.17
31	Uruguay	1.90	1.24	-0.81	3.96
32	Wales	5.82	1.34	3.70	9.15

Table C.2.3: *Top 10 players under the Poisson model for passes, ranked by posterior predicted total passes, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Passes	Lower HPD Bound	Upper HPD Bound	Observed Passes	90-minute units
1	Rodri	641.76	593.13	692.46	642	4.3
2	Nicolas Otamendi	531.71	487.57	577.69	532	7.7
3	Rodrigo De Paul	469.54	428.12	512.97	470	6.7
4	Luka Modric	465.60	424.13	509.00	465	7.3
5	Marcelo Brozovic	464.73	423.33	507.78	466	6.1
6	Josko Gvardiol	464.68	423.37	507.92	464	7.7
7	John Stones	434.15	394.37	475.98	435	4.8
8	Aurelien Tchouameni	425.54	386.11	466.88	426	6.9
9	Mateo Kovacic	415.43	376.58	456.27	415	7.1
10	Aymeric Laporte	412.91	374.09	453.53	413	3.3

Table C.2.4: *Bottom 10 players under the Poisson model for passes, ranked by posterior predicted total passes, with lower and upper HPD bounds.*

Rank	Player	Predicted Total Passes	Lower HPD Bound	Upper HPD Bound	Observed Passes	90-minute units
1	Daizen Maeda	8.81	4.08	15.34	8	2.0
2	Henry Martin	11.34	5.81	18.75	13	1.6
3	Michy Batshuayi	15.45	8.80	24.00	16	1.7
4	Vincent Aboubakar	19.43	11.88	28.87	20	1.6
5	Cyle Larin	20.40	12.63	30.07	21	1.5
6	Randal Kolo Muani	22.61	14.32	32.77	25	2.0
7	Daniel James	23.52	15.09	33.83	24	1.7
8	Luis Suarez	24.45	15.83	34.97	25	1.6
9	Issam Jebali	25.04	16.29	35.62	25	2.0
10	Alvaro Morata	27.67	18.42	38.78	28	2.1

Table C.2.5: *Convergence diagnostic results for parameters 1–35, under the Poisson model for passing. The parameter column reports Δ_i for player effects.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Aaron Mooy	Δ_1	1.00	550	-0.04	-1.04	-1.05	TRUE
2	Aaron Ramsey	Δ_2	1.00	692	0.18	1.35	-0.47	TRUE
3	Abdelkarim Hassan	Δ_3	1.00	633	-1.03	-2.10	-0.97	TRUE
4	Abdou Diallo	Δ_4	1.00	507	-0.84	1.89	-0.29	TRUE
5	Abdulaziz Hatem	Δ_5	1.01	689	-1.00	-2.15	-1.02	TRUE
6	Abdullelah Al-Amri	Δ_6	1.01	811	2.96	2.52	1.07	TRUE
7	Abdullelah Al-Malki	Δ_7	1.04	148	-0.18	-12.11	0.70	TRUE
8	Achraf Hakimi	Δ_8	1.04	302	-3.06	-0.83	-0.20	TRUE
9	Adrien Rabiot	Δ_9	1.14	163	-1.39	-0.57	-0.93	TRUE
10	Ahmad Nourollahi	Δ_{10}	1.08	1054	-0.42	0.40	-0.51	TRUE
11	Aissa Laidouni	Δ_{11}	1.03	573	0.35	3.09	0.12	TRUE
12	Akram Afif	Δ_{12}	1.06	627	-1.05	-2.16	-1.03	TRUE
13	Alexandre Mitrovic	Δ_{13}	1.02	576	-1.00	-0.58	-0.94	TRUE
14	Alexis Mac Allister	Δ_{14}	1.05	54	-0.32	-1.04	-3.64	TRUE
15	Alexis Vega	Δ_{15}	1.01	662	-2.10	-1.87	1.24	TRUE
16	Ali Abdi	Δ_{16}	1.03	609	0.37	3.12	0.13	TRUE
17	Ali Al-Bulaihi	Δ_{17}	1.00	777	2.90	2.58	1.02	TRUE
18	Ali Gholizadeh	Δ_{18}	1.00	1054	-0.44	0.39	-0.54	TRUE
19	Alidu Seidu	Δ_{19}	1.01	179	2.05	-3.55	-2.10	TRUE
20	Alistair Johnston	Δ_{20}	1.02	567	1.92	-0.43	0.36	TRUE
21	Almoez Ali	Δ_{21}	1.01	725	-1.06	-2.26	-1.07	TRUE
22	Alphonso Davies	Δ_{22}	1.02	582	1.90	-0.49	0.35	TRUE
23	Alvaro Morata	Δ_{23}	1.01	390	0.50	6.44	1.59	TRUE
24	Andre-Frank Zambo Anguissa	Δ_{24}	1.01	681	-0.70	-1.03	1.49	TRUE
25	Andre Ayew	Δ_{25}	1.03	804	-0.16	-0.07	0.28	TRUE
26	Andreas Christensen	Δ_{26}	1.01	423	-1.43	0.18	-1.38	TRUE
27	Andrej Kramaric	Δ_{27}	1.07	103	-1.92	-4.73	-4.53	TRUE
28	Andrija Zivkovic	Δ_{28}	1.02	593	-0.09	-0.56	-0.95	TRUE
29	Angelo Preciado	Δ_{29}	1.01	553	-2.20	-1.05	1.39	TRUE
30	Anis Ben Slimane	Δ_{30}	1.10	43	-6.44	-3.15	1.93	TRUE
31	Antoine Griezmann	Δ_{31}	1.04	155	2.88	-0.46	1.66	TRUE
32	Antonee Robinson	Δ_{32}	1.01	384	0.24	-0.03	0.49	TRUE
33	Antonio Rudiger	Δ_{33}	1.00	402	2.17	0.53	1.57	TRUE
34	Assim Madibo	Δ_{34}	1.13	114	4.13	-0.05	-0.60	TRUE
35	Atiba Hutchinson	Δ_{35}	1.05	628	1.95	-0.42	0.38	TRUE

Table C.2.6: *Convergence diagnostic results for parameters 36–70, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
36	Aurelien Tchouameni	Δ_{36}	1.04	155	2.85	-0.46	1.61	TRUE
37	Axel Witsel	Δ_{37}	1.04	380	0.21	-1.54	-2.79	TRUE
38	Aymeric Laporte	Δ_{38}	1.16	83	2.50	-2.43	-0.44	TRUE
39	Aziz Behich	Δ_{39}	1.11	550	-0.06	-1.09	-1.04	TRUE
40	Azzedine Ounahi	Δ_{40}	1.00	307	-3.01	-0.82	-0.20	TRUE
41	Baba Rahman	Δ_{41}	1.01	739	-0.11	-0.04	0.27	TRUE
42	Bartosz Bereszynski	Δ_{42}	1.00	550	1.62	1.18	-1.03	TRUE
43	Ben Davies	Δ_{43}	1.00	675	0.20	1.33	-0.42	TRUE
44	Bernardo Silva	Δ_{44}	1.02	258	1.96	0.25	7.40	TRUE
45	Bono Sosa	Δ_{45}	1.03	62	1.61	0.05	-1.24	TRUE
46	Boualem Khoukhi	Δ_{46}	1.01	619	-1.01	-2.14	-1.00	TRUE
47	Boulaye Dia	Δ_{47}	1.01	556	-0.91	1.93	-0.31	TRUE
48	Breel Embolo	Δ_{48}	1.03	491	-0.08	-1.54	0.02	TRUE
49	Bruno Fernandes	Δ_{49}	1.00	259	0.35	-1.25	-0.55	TRUE
50	Bryan Mbeumo	Δ_{50}	1.01	730	-0.73	-1.05	1.51	TRUE
51	Bryan Oviedo	Δ_{51}	1.00	776	-0.99	-0.97	2.08	TRUE
52	Bukayo Saka	Δ_{52}	1.12	60	-3.52	-0.28	-0.79	TRUE
53	Casemiro	Δ_{53}	1.02	267	-1.06	-2.45	0.04	TRUE
54	Celso Borges	Δ_{54}	1.00	758	0.98	-0.95	2.14	TRUE
55	Cesar Montes	Δ_{55}	1.01	567	-2.02	-1.70	1.22	TRUE
56	Cho Gue-sung	Δ_{56}	1.00	454	-1.21	0.37	2.41	TRUE
57	Chris Mepham	Δ_{57}	1.00	682	0.19	1.28	-0.44	TRUE
58	Christian Eriksen	Δ_{58}	1.01	421	-1.50	0.17	-1.37	TRUE
59	Christian Pulisic	Δ_{59}	1.01	398	0.24	-0.04	0.43	TRUE
60	Cody Gakpo	Δ_{60}	1.01	285	-0.29	-3.49	0.58	TRUE
61	Collins Fai	Δ_{61}	1.01	690	-0.72	-1.02	1.48	TRUE
62	Connor Roberts	Δ_{62}	1.00	722	0.22	1.34	-0.42	TRUE
63	Craig Goodwin	Δ_{63}	1.00	638	-0.04	-1.00	-1.09	TRUE
64	Cristian Romero	Δ_{64}	1.01	185	-4.60	-1.84	2.38	TRUE
65	Cristiano Ronaldo	Δ_{65}	1.15	274	2.03	0.23	7.60	TRUE
66	Cyle Larin	Δ_{66}	1.01	1066	2.02	-0.47	0.37	TRUE
67	Daichi Kamada	Δ_{67}	1.07	543	-3.22	-0.18	1.66	TRUE
68	Dazen Maeda	Δ_{68}	1.10	149	-1.52	-5.70	-4.39	TRUE
69	Daley Blind	Δ_{69}	1.03	277	-0.30	-3.45	0.61	TRUE
70	Dani Olmo	Δ_{70}	1.01	211	0.39	5.50	1.41	TRUE

Table C.2.7: *Convergence diagnostic results for parameters 71–105, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
71	Daniel Amartey	Δ_{71}	1.01	695	-0.17	-0.07	0.26	TRUE
72	Daniel James	Δ_{72}	1.00	1089	0.19	1.42	-0.38	TRUE
73	Danilo	Δ_{73}	1.05	183	0.85	-4.37	-0.57	TRUE
74	Darwin Nunez	Δ_{74}	1.00	613	-0.82	2.64	2.56	TRUE
75	David Raum	Δ_{75}	1.01	406	2.14	0.54	1.58	TRUE
76	Dayot Upamecano	Δ_{76}	1.07	71	-5.31	-2.72	-1.01	TRUE
77	Declan Rice	Δ_{77}	1.01	203	-1.36	-4.52	-2.07	TRUE
78	Dejan Lovren	Δ_{78}	1.21	81	0.76	3.67	4.15	TRUE
79	Denzel Dumfries	Δ_{79}	1.01	281	-0.29	-3.55	0.59	TRUE
80	Diego Godin	Δ_{80}	1.06	156	-1.08	-0.37	-0.56	TRUE
81	Diogo Dalot	Δ_{81}	1.05	100	-3.82	0.68	0.02	TRUE
82	Djibril Sow	Δ_{82}	1.03	480	-0.05	-1.46	0.04	TRUE
83	Dusan Tadic	Δ_{83}	1.02	562	-1.02	-0.58	-0.93	TRUE
84	Dylan Bronn	Δ_{84}	1.11	142	7.35	5.43	-1.19	TRUE
85	Eder Militao	Δ_{85}	1.03	101	-0.84	-0.94	1.15	TRUE
86	Edson Alvarez	Δ_{86}	1.14	129	11.93	9.24	-0.85	TRUE
87	Ehsan Hajsafi	Δ_{87}	1.03	970	-0.39	0.35	-0.49	TRUE
88	Ellyes Skhiri	Δ_{88}	1.03	563	-0.34	3.11	0.12	TRUE
89	Enner Valencia	Δ_{89}	1.01	581	-2.33	-1.09	1.31	TRUE
90	Enzo Fernandez	Δ_{90}	1.07	182	-4.59	-1.76	2.41	TRUE
91	Eric Maxim Choupo-Moting	Δ_{91}	1.01	729	-0.71	-1.05	1.50	TRUE
92	Ethan Ampadu	Δ_{92}	1.00	685	0.18	1.32	-0.45	TRUE
93	Facundo Pellistri	Δ_{93}	1.00	696	-0.82	2.84	2.71	TRUE
94	Federico Valverde	Δ_{94}	1.00	553	-0.81	2.65	2.61	TRUE
95	Felix Torres Caicedo	Δ_{95}	1.01	544	-2.27	-1.05	1.36	TRUE
96	Ferran Torres	Δ_{96}	1.01	260	0.44	5.59	1.54	TRUE
97	Filip Kostic	Δ_{97}	1.38	188	3.05	1.39	2.98	TRUE
98	Firas Al-Buraikan	Δ_{98}	1.01	809	2.83	2.60	1.02	TRUE
99	Francisco Calvo	Δ_{99}	1.05	179	2.26	1.81	-2.36	TRUE
100	Frenkie de Jong	Δ_{100}	1.01	272	-0.30	-3.35	0.60	TRUE
101	Gareth Bale	Δ_{101}	1.00	797	0.21	1.34	-0.49	TRUE
102	Gavi	Δ_{102}	1.01	218	0.40	5.49	1.40	TRUE
103	Gonzalo Plata	Δ_{103}	1.01	550	-2.21	-1.09	1.35	TRUE
104	Granit Xhaka	Δ_{104}	1.03	445	-0.06	-1.43	0.04	TRUE
105	Grzegorz Krychowiak	Δ_{105}	1.00	560	1.61	1.27	-1.05	TRUE

Table C.2.8: *Convergence diagnostic results for parameters 106–140, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
106	Guillermo Varela	Δ_{106}	1.00	614	-0.08	2.80	2.60	TRUE
107	Hakim Ziyech	Δ_{107}	1.00	301	-3.10	-0.80	-0.20	TRUE
108	Harry Kane	Δ_{108}	1.01	214	-1.37	-4.69	-2.09	TRUE
109	Harry Maguire	Δ_{109}	1.01	202	-1.33	-4.48	-2.07	TRUE
110	Harry Souttar	Δ_{110}	1.01	552	-0.05	-1.03	-1.08	TRUE
111	Harry Wilson	Δ_{111}	1.00	832	0.21	1.40	-0.50	TRUE
112	Hassan Al-Haydos	Δ_{112}	1.01	665	-1.09	-2.16	-1.08	TRUE
113	Hassan Al Tambakti	Δ_{113}	1.14	178	-6.33	1.69	1.52	TRUE
114	Hector Herrera	Δ_{114}	1.03	90	-5.22	-8.55	-2.03	TRUE
115	Hector Moreno	Δ_{115}	1.01	566	-2.01	-1.68	1.21	TRUE
116	Henry Martin	Δ_{116}	1.12	177	13.96	10.39	-0.90	TRUE
117	Hidemasa Morita	Δ_{117}	1.01	137	2.08	-1.44	0.22	TRUE
118	Hirving Lozano	Δ_{118}	1.01	602	-2.06	-1.85	1.21	TRUE
119	Homam Ahmed	Δ_{119}	1.01	634	-1.91	-2.08	-1.03	TRUE
120	Hwang In-beom	Δ_{120}	1.00	410	-1.19	0.43	2.40	TRUE
121	Ibrahima Konate	Δ_{121}	1.09	95	0.30	-1.37	1.05	TRUE
122	Idrissa Gana Gueye	Δ_{122}	1.21	146	2.61	1.97	0.52	TRUE
123	Ilkay Gundogan	Δ_{123}	1.01	414	2.18	0.52	1.62	TRUE
124	Inaki Williams	Δ_{124}	1.01	756	-0.23	-0.17	0.25	TRUE
125	Ismail Jakobs	Δ_{125}	1.01	520	-0.82	1.97	-0.30	TRUE
126	Ismaila Sarr	Δ_{126}	1.01	540	-0.85	1.80	-0.32	TRUE
127	Issam Jebali	Δ_{127}	1.02	707	-0.43	3.04	0.10	TRUE
128	Ivan Perisic	Δ_{128}	1.07	102	-1.85	-4.63	-4.31	TRUE
129	Jackson Irvine	Δ_{129}	1.01	562	-0.06	-1.06	-1.07	TRUE
130	Jakub Kaminski	Δ_{130}	1.00	683	1.76	1.29	-1.10	TRUE
131	Jakub Kiwior	Δ_{131}	1.00	540	1.62	1.18	-1.04	TRUE
132	Jamal Musiala	Δ_{132}	1.01	407	2.21	0.52	1.63	TRUE
133	Jan Vertonghen	Δ_{133}	1.04	376	0.22	-1.55	-2.80	TRUE
134	Javad El Yamiq	Δ_{134}	1.22	146	0.72	8.80	-4.32	TRUE
135	Jean-Charles Castelletto	Δ_{135}	1.05	280	-1.07	-0.08	-2.46	TRUE
136	Jesper Lindstrom	Δ_{136}	1.00	522	-1.47	0.19	-1.45	TRUE
137	Jesus Gallardo	Δ_{137}	1.10	576	-1.93	-1.75	1.23	TRUE
138	Jhegson Mendez	Δ_{138}	1.06	63	13.18	8.11	-7.28	TRUE
139	Joachim Andersen	Δ_{139}	1.01	432	-1.46	0.18	-1.38	TRUE
140	Joakim Maehle	Δ_{140}	1.01	432	-1.41	0.20	-1.40	TRUE

Table C.2.9: *Convergence diagnostic results for parameters 141–175, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
141	Joao Cancelo	Δ_{141}	1.02	167	2.16	-1.45	0.28	TRUE
142	Joao Felix	Δ_{142}	1.00	260	0.37	-1.26	-0.56	TRUE
143	Joe Rodon	Δ_{143}	1.00	669	0.19	1.28	-0.44	TRUE
144	Joel Campbell	Δ_{144}	1.00	748	-0.08	-0.98	2.00	TRUE
145	John Stones	Δ_{145}	1.01	200	-1.35	-4.49	-2.12	TRUE
146	Jonathan David	Δ_{146}	1.01	675	1.99	-0.47	0.41	TRUE
147	Jordan Ayew	Δ_{147}	1.01	981	-0.06	0.06	0.33	TRUE
148	Jordan Henderson	Δ_{148}	1.02	68	1.38	3.87	-0.39	TRUE
149	Jordi Alba	Δ_{149}	1.01	212	0.40	5.39	1.36	TRUE
150	Jorge Eduardo Sanchez	Δ_{150}	1.14	133	11.88	9.56	-0.88	TRUE
151	Jose Maria Gimenez	Δ_{151}	1.00	563	-0.80	2.76	2.61	TRUE
152	Joshua Kimmich	Δ_{152}	1.00	399	2.14	0.53	1.60	TRUE
153	Josip Juranovic	Δ_{153}	1.21	82	0.78	3.64	4.29	TRUE
154	Josko Gvardiol	Δ_{154}	1.08	101	-1.88	-4.59	-4.24	TRUE
155	Jude Bellingham	Δ_{155}	1.01	207	-1.36	-4.47	-2.12	TRUE
156	Jules Kounde	Δ_{156}	1.04	165	-1.40	-0.57	-0.09	TRUE
157	Julian Alvarez	Δ_{157}	1.07	200	-4.67	-2.00	2.46	TRUE
158	Jung Woo-young	Δ_{158}	1.00	410	-1.18	0.42	2.44	TRUE
159	Junior Hoilett	Δ_{159}	1.01	733	1.96	-0.39	0.37	TRUE
160	Junya Ito	Δ_{160}	1.01	558	-3.40	-0.21	1.64	TRUE
161	Jurien Timber	Δ_{161}	1.07	85	11.39	2.77	-2.67	TRUE
162	Kalidou Koulibaly	Δ_{162}	1.01	506	-0.84	1.92	-0.30	TRUE
163	Kamal Miller	Δ_{163}	1.02	563	1.87	-0.44	0.36	TRUE
164	Kamil Glik	Δ_{164}	1.00	550	1.61	1.19	-1.05	TRUE
165	Karim Boudiaf	Δ_{165}	1.00	654	-1.04	-2.05	-0.95	TRUE
166	Karl Toko Ekambi	Δ_{166}	1.00	839	-0.65	-1.04	1.63	TRUE
167	Kendall Waston	Δ_{167}	1.00	797	-0.93	-0.99	2.14	TRUE
168	Kevin De Bruyne	Δ_{168}	1.04	390	0.20	-1.53	-2.90	TRUE
169	Keysher Fuller	Δ_{169}	1.00	808	-1.03	-0.98	2.08	TRUE
170	Kieffer Moore	Δ_{170}	1.00	774	0.18	1.32	-0.47	TRUE
171	Kim Jin-su	Δ_{171}	1.00	411	-1.22	0.43	2.47	TRUE
172	Kim Min-jae	Δ_{172}	1.02	190	-2.17	0.37	1.23	TRUE
173	Kim Moon-hwan	Δ_{173}	1.00	405	-1.17	0.45	2.37	TRUE
174	Kim Young-gwon	Δ_{174}	1.00	406	-1.15	0.42	2.36	TRUE
175	Ko Itakura	Δ_{175}	1.08	169	1.06	5.33	4.94	TRUE

Table C.2.10: *Convergence diagnostic results for parameters 176–210, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
176	Krepin Diatta	Δ_{176}	1.25	69	-1.35	-9.47	-9.46	TRUE
177	Krystian Bielik	Δ_{177}	1.00	607	1.69	1.29	-1.02	TRUE
178	Kudus Mohammed	Δ_{178}	1.01	722	-0.13	-0.08	0.26	TRUE
179	Kye Rowles	Δ_{179}	1.01	553	-0.06	-1.06	-1.06	TRUE
180	Kyle Walker	Δ_{180}	1.43	72	-3.48	0.45	-0.62	TRUE
181	Kylian Mbappe	Δ_{181}	1.04	158	2.93	-0.47	1.61	TRUE
182	Leander Dendoncker	Δ_{182}	1.60	84	5.49	6.19	9.95	TRUE
183	Lee Jae-sung	Δ_{183}	1.21	135	6.57	-0.12	2.13	TRUE
184	Leon Goretzka	Δ_{184}	1.01	496	2.94	0.59	1.64	TRUE
185	Lionel Messi	Δ_{185}	1.07	183	-4.40	-1.79	2.37	TRUE
186	Lisandro Martinez	Δ_{186}	1.09	37	4.34	3.69	-0.88	TRUE
187	Lucas Paqueta	Δ_{187}	1.02	269	-1.20	-2.39	0.03	TRUE
188	Luis Chavez	Δ_{188}	1.01	571	-2.03	-1.70	1.20	TRUE
189	Luis Suarez	Δ_{189}	1.00	866	-0.85	2.97	2.74	TRUE
190	Luka Modric	Δ_{190}	1.08	101	-1.90	-4.61	-4.23	TRUE
191	Luke Shaw	Δ_{191}	1.01	202	-1.39	-4.56	-2.09	TRUE
192	Majid Hosseini	Δ_{192}	1.03	924	-0.42	0.35	-0.53	TRUE
193	Manuel Akanji	Δ_{193}	1.03	441	-0.06	-1.44	0.03	TRUE
194	Marcelo Brozovic	Δ_{194}	1.21	82	0.77	3.54	4.27	TRUE
195	Marco Asensio	Δ_{195}	1.01	227	0.43	5.64	1.36	TRUE
196	Marcos Acuna	Δ_{196}	1.15	74	5.29	12.11	0.80	TRUE
197	Mario Pasalic	Δ_{197}	1.06	128	-2.10	-5.00	-4.41	TRUE
198	Marquinhos	Δ_{198}	1.21	170	0.44	0.52	-2.07	TRUE
199	Mateo Kovacic	Δ_{199}	1.08	101	1.85	-4.55	-4.40	TRUE
200	Mathew Leckie	Δ_{200}	1.01	578	-0.05	-1.05	-1.09	TRUE
201	Mathias Olivera	Δ_{201}	1.00	565	-0.77	2.64	2.60	TRUE
202	Matias Vecino	Δ_{202}	1.00	591	-0.75	2.57	2.57	TRUE
203	Matty Cash	Δ_{203}	1.00	538	-1.67	1.20	-1.05	TRUE
204	Maya Yoshida	Δ_{204}	1.01	525	-3.30	-0.20	1.65	TRUE
205	Mehdi Taremi	Δ_{205}	1.03	948	-0.41	0.37	-0.52	TRUE
206	Memphis	Δ_{206}	1.01	314	-0.28	-3.50	0.62	TRUE
207	Michael Estrada	Δ_{207}	1.01	623	-2.22	-1.19	1.39	TRUE
208	Michy Batshuayi	Δ_{208}	1.34	91	0.76	17.29	-8.18	TRUE
209	Mikkel Damsgaard	Δ_{209}	1.00	526	-1.54	0.27	-1.36	TRUE
210	Milad Mohammadi	Δ_{210}	1.03	1056	-0.38	0.40	-0.55	TRUE

Table C.2.11: *Convergence diagnostic results for parameters 211–245, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
211	Milos Degenek	Δ_{211}	1.01	68	-0.01	-1.01	-1.06	TRUE
212	Milos Veljkovic	Δ_{212}	1.02	569	-0.95	-0.55	-0.90	TRUE
213	Mitchel Duke	Δ_{213}	1.01	672	-0.04	-1.07	-1.08	TRUE
214	Mohamed Kanno	Δ_{214}	1.01	742	2.97	2.52	0.97	TRUE
215	Mohammed Salisu	Δ_{215}	1.01	704	-0.15	-0.07	0.26	TRUE
216	Moises Caicedo	Δ_{216}	1.01	546	-2.58	-1.10	1.37	TRUE
217	Montassar Talbi	Δ_{217}	1.03	563	-0.34	2.99	0.12	TRUE
218	Morteza Pouraliganji	Δ_{218}	1.03	933	0.42	0.36	-0.49	TRUE
219	Nahuel Molina	Δ_{219}	1.07	183	-4.51	-1.75	2.48	TRUE
220	Nampalys Mendy	Δ_{220}	1.01	524	0.85	1.92	-0.29	TRUE
221	Nathan Ake	Δ_{221}	1.01	274	-0.29	-3.37	0.61	TRUE
222	Nayef Aguerd	Δ_{222}	1.13	65	3.42	5.25	0.40	TRUE
223	Nico Williams	Δ_{223}	1.00	724	0.21	1.25	-0.43	TRUE
224	Neymar	Δ_{224}	1.16	159	0.97	-3.24	0.12	TRUE
225	Nico Elvedi	Δ_{225}	1.03	127	-4.53	0.05	-2.56	TRUE
226	Nicolas Nkoulou	Δ_{226}	1.05	285	-1.07	-0.09	-2.39	TRUE
227	Nicolas Otamendi	Δ_{227}	1.07	182	-4.53	-1.84	2.37	TRUE
228	Nicolas Tagliafico	Δ_{228}	1.01	71	2.20	-2.83	0.79	TRUE
229	Niklas Sule	Δ_{229}	1.01	395	2.13	0.52	1.56	TRUE
230	Nikola Milenkovic	Δ_{230}	1.02	562	-0.99	-0.56	-0.90	TRUE
231	Nouhou Tolo	Δ_{231}	1.01	693	-0.67	-1.06	1.53	TRUE
232	Noussair Mazraoui	Δ_{232}	1.13	105	0.41	9.74	0.88	TRUE
233	Olivier Giroud	Δ_{233}	1.04	194	-1.42	-0.63	-0.93	TRUE
234	Oscar Duarte	Δ_{234}	1.00	758	-1.02	-0.97	2.12	TRUE
235	Ousmane Dembele	Δ_{235}	1.04	163	2.87	-0.47	1.63	TRUE
236	Pedri	Δ_{236}	1.01	211	0.41	5.42	1.37	TRUE
237	Pepe	Δ_{237}	1.01	103	-0.56	-1.69	-0.06	TRUE
238	Pervis Estupinan	Δ_{238}	1.01	532	-2.34	-1.09	1.32	TRUE
239	Phil Foden	Δ_{239}	1.02	68	1.37	4.00	-0.40	TRUE
240	Piero Hincapie	Δ_{240}	1.01	539	-2.25	-1.06	1.35	TRUE
241	Pierre Hojbjerg	Δ_{241}	1.01	427	-1.45	0.18	-1.39	TRUE
242	Piotr Zielinski	Δ_{242}	1.01	547	1.62	1.22	-1.03	TRUE
243	Przemyslaw Frankowski	Δ_{243}	1.00	647	1.74	1.19	-1.10	TRUE
244	Ramin Rezaeian	Δ_{244}	1.94	188	2.55	-3.60	3.40	TRUE
245	Randal Kolo Muani	Δ_{245}	1.07	38	11.54	4.34	1.29	TRUE

Table C.2.12: *Convergence diagnostic results for parameters 246–280, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
246	Raphael Guerreiro	Δ_{246}	1.00	262	0.37	-1.26	-0.57	TRUE
247	Raphael Varane	Δ_{247}	1.05	164	-1.42	-0.58	-0.93	TRUE
248	Raphinha	Δ_{248}	1.21	173	0.44	0.53	-2.11	TRUE
249	Rasmus Nissen	Δ_{249}	1.01	438	-1.47	0.20	-1.37	TRUE
250	Remo Freuler	Δ_{250}	1.03	454	-0.05	-1.50	0.05	TRUE
251	Ricardo Rodriguez	Δ_{251}	1.03	448	-0.06	-1.43	0.03	TRUE
252	Richarlison	Δ_{252}	1.02	294	-1.09	-2.67	0.03	TRUE
253	Richie Laryea	Δ_{253}	1.01	449	2.06	-0.44	0.40	TRUE
254	Riley McGree	Δ_{254}	1.01	607	-0.03	-1.10	-1.09	TRUE
255	Ritsu Doan	Δ_{255}	1.00	659	-3.55	-0.16	1.66	TRUE
256	Ro-Ro	Δ_{256}	1.01	638	-1.04	-2.11	-1.03	TRUE
257	Robert Lewandowski	Δ_{257}	1.00	573	1.68	1.23	-1.04	TRUE
258	Rodri	Δ_{258}	1.01	207	0.39	5.54	1.38	TRUE
259	Rodrigo Bentancur	Δ_{259}	1.00	570	-0.82	2.75	2.65	TRUE
260	Rodrigo De Paul	Δ_{260}	1.07	181	-4.50	-1.77	2.36	TRUE
261	Romain Saiss	Δ_{261}	1.22	135	0.70	9.34	-4.24	TRUE
262	Ruben Dias	Δ_{262}	1.00	256	0.36	-1.22	-0.54	TRUE
263	Ruben Neves	Δ_{263}	1.02	253	1.97	0.25	7.62	TRUE
264	Ruben Vargas	Δ_{264}	1.02	474	-0.06	-1.51	0.03	TRUE
265	Saeid Ezatolahi	Δ_{265}	1.03	968	-0.42	0.39	-0.49	TRUE
266	Saleh Al-Shehri	Δ_{266}	1.01	874	2.98	2.37	0.96	TRUE
267	Salem Al-Dawsari	Δ_{267}	1.00	614	2.82	2.51	0.99	TRUE
268	Sali Abdul Samed	Δ_{268}	1.01	679	-0.15	-0.08	0.29	TRUE
269	Sasa Lucic	Δ_{269}	1.02	546	-0.96	-0.57	-0.88	TRUE
270	Saud Abdulhamid	Δ_{270}	1.00	765	2.85	2.60	1.02	TRUE
271	Sebastian Coates	Δ_{271}	1.01	203	0.17	0.40	-0.80	TRUE
272	Selim Amallah	Δ_{272}	1.00	336	-3.28	-0.89	-0.23	TRUE
273	Serge Gnabry	Δ_{273}	1.00	412	2.17	0.52	1.57	TRUE
274	Sergej Milinkovic-Savic	Δ_{274}	1.02	563	-1.00	-0.59	-0.91	TRUE
275	Sergio Dest	Δ_{275}	1.01	390	0.24	-0.02	0.47	TRUE
276	Sergio Busquets	Δ_{276}	1.01	210	0.39	5.34	1.40	TRUE
277	Shogo Taniguchi	Δ_{277}	1.06	81	2.43	-7.29	-6.81	TRUE
278	Silvan Widmer	Δ_{278}	1.13	141	-2.39	2.08	-0.34	TRUE
279	Sofiane Boufal	Δ_{279}	1.00	317	-3.13	-0.85	-0.19	TRUE
280	Sofyan Amrabat	Δ_{280}	1.00	299	-3.04	-0.81	-0.20	TRUE

Table C.2.13: *Convergence diagnostic results for parameters 281–310, under the Poisson model for passing.*

No.	Label	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
281	Son Heung-min	Δ_{281}	1.00	412	-1.92	0.39	2.51	TRUE
282	Steven Vitoria	Δ_{282}	1.02	570	1.35	-0.45	0.39	TRUE
283	Strahinja Pavlovic	Δ_{283}	1.02	574	-0.97	-0.55	-0.88	TRUE
284	Tajon Buchanan	Δ_{284}	1.01	603	1.95	-0.47	0.36	TRUE
285	Takehiro Tomiyasu	Δ_{285}	1.04	94	-1.18	-4.28	-3.55	TRUE
286	Theo Hernandez	Δ_{286}	1.04	167	-1.38	-0.58	-0.92	TRUE
287	Thiago Silva	Δ_{287}	1.02	268	-1.07	-2.44	0.03	TRUE
288	Thomas Meunier	Δ_{288}	1.04	395	0.24	-1.52	-2.76	TRUE
289	Thomas Muller	Δ_{289}	1.01	434	2.29	0.54	1.60	TRUE
290	Thomas Partey	Δ_{290}	1.01	692	-0.14	-0.08	0.26	TRUE
291	Tim Ream	Δ_{291}	1.01	378	0.23	-0.04	0.46	TRUE
292	Timothy Castagne	Δ_{292}	1.04	383	0.22	-1.58	-2.80	TRUE
293	Timothy Weah	Δ_{293}	1.01	397	0.25	-0.03	0.46	TRUE
294	Toby Alderweireld	Δ_{294}	1.04	375	2.08	-1.55	-2.77	TRUE
295	Tyler Adams	Δ_{295}	1.01	381	0.23	-0.04	0.50	TRUE
296	Vincent Aboubakar	Δ_{296}	1.00	1083	-0.71	-1.10	1.58	TRUE
297	Vinicius Junior	Δ_{297}	1.01	285	-1.07	-2.52	0.03	TRUE
298	Virgil van Dijk	Δ_{298}	1.01	274	-0.29	-3.50	0.59	TRUE
299	Walker Zimmerman	Δ_{299}	1.01	382	0.25	-0.02	0.49	TRUE
300	Wataru Endo	Δ_{300}	1.01	524	-3.22	-0.18	1.55	TRUE
301	Weston McKennie	Δ_{301}	1.01	399	0.24	-0.01	0.47	TRUE
302	Yerdhan Shaji	Δ_{302}	1.41	137	4.12	0.47	0.80	TRUE
303	Yahya Attiat Allah	Δ_{303}	1.60	31	-2.95	-14.92	-0.60	TRUE
304	Yassine Meriah	Δ_{304}	1.03	560	-0.35	3.09	0.11	TRUE
305	Yeltsin Tejeda	Δ_{305}	1.00	759	-1.01	-0.99	2.12	TRUE
306	Youssef En-Nesyri	Δ_{306}	1.00	327	-2.81	-0.88	-0.22	TRUE
307	Youssef Msakni	Δ_{307}	1.11	139	7.32	5.84	-1.16	TRUE
308	Youssouf Sabaly	Δ_{308}	1.01	501	-0.85	1.94	-0.29	TRUE
309	Yunus Musah	Δ_{309}	1.01	383	0.24	-0.04	0.47	TRUE
310	Yuto Nagatomo	Δ_{310}	1.01	586	-3.46	-0.13	1.78	TRUE

Table C.2.14: *Convergence diagnostic results for the team-effect parameters under the Poisson model for passing. The parameter column reports λ_i^e for team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	λ_1^e	1.31	6	14.47	13.99	1.47	TRUE
2	Australia	λ_2^e	1.17	36	-3.52	0.41	1.88	TRUE
3	Belgium	λ_3^e	1.44	28	0.44	2.50	-0.84	TRUE
4	Brazil	λ_4^e	1.31	29	3.51	-1.70	-0.48	TRUE
5	Cameroon	λ_5^e	1.50	94	-0.85	0.05	-8.53	TRUE
6	Canada	λ_6^e	1.23	37	-7.04	-3.88	-2.67	TRUE
7	Costa Rica	λ_7^e	1.15	66	1.87	-0.84	2.33	TRUE
8	Croatia	λ_8^e	1.12	11	-12.05	1.88	-3.46	TRUE
9	Denmark	λ_9^e	1.05	39	2.95	4.20	-0.52	TRUE
10	Ecuador	λ_{10}^e	1.19	20	-6.98	-6.42	-6.41	TRUE
11	England	λ_{11}^e	1.63	8	-6.05	-16.04	2.04	TRUE
12	France	λ_{12}^e	1.43	27	-0.42	3.17	-1.20	TRUE
13	Germany	λ_{13}^e	1.14	30	2.45	0.16	3.61	TRUE
14	Ghana	λ_{14}^e	1.26	39	2.69	-3.89	-0.90	TRUE
15	IR Iran	λ_{15}^e	1.57	50	-1.43	-5.75	0.34	TRUE
16	Japan	λ_{16}^e	1.33	38	9.49	-3.36	4.16	TRUE
17	Korea Republic	λ_{17}^e	1.81	36	2.11	-5.23	-2.37	TRUE
18	Mexico	λ_{18}^e	1.10	36	0.84	3.91	4.52	TRUE
19	Morocco	λ_{19}^e	1.07	21	5.86	-0.12	0.73	TRUE
20	Netherlands	λ_{20}^e	1.46	6	-10.06	-11.57	-0.99	TRUE
21	Poland	λ_{21}^e	1.18	26	-1.48	3.62	5.91	TRUE
22	Portugal	λ_{22}^e	1.46	24	0.46	-2.14	-4.51	TRUE
23	Qatar	λ_{23}^e	1.51	24	3.94	4.57	6.01	TRUE
24	Saudi Arabia	λ_{24}^e	1.13	66	-1.21	5.13	4.80	TRUE
25	Senegal	λ_{25}^e	1.03	37	-10.35	0.83	-5.56	TRUE
26	Serbia	λ_{26}^e	1.86	72	-1.35	-0.16	-3.15	TRUE
27	Spain	λ_{27}^e	1.06	18	-1.73	-1.14	1.82	TRUE
28	Switzerland	λ_{28}^e	1.75	32	2.01	-0.01	-2.32	TRUE
29	Tunisia	λ_{29}^e	1.22	54	3.34	4.13	-0.27	TRUE
30	United States	λ_{30}^e	1.15	28	9.66	1.33	2.46	TRUE
31	Uruguay	λ_{31}^e	1.57	57	6.97	-1.62	-1.73	TRUE
32	Wales	λ_{32}^e	1.48	47	-8.33	-7.55	-0.18	TRUE

Table C.2.15: *Convergence diagnostic results for the opponent team-effect parameters under the Poisson model for passing. The parameter column reports $\lambda_i^{E\setminus e}$ for opponent team effects.*

No.	Team	Parameter	$\hat{R}$	ESS	Geweke Chain 1	Geweke Chain 2	Geweke Chain 3	H.W. Stationarity Pass
1	Argentina	$\lambda_{1}^{E\setminus e}$	1.30	18	-2.36	-2.65	3.63	TRUE
2	Australia	$\lambda_{2}^{E\setminus e}$	1.02	10	3.57	3.00	-1.60	TRUE
3	Belgium	$\lambda_{3}^{E\setminus e}$	1.19	9	-0.84	0.29	-0.77	TRUE
4	Brazil	$\lambda_{4}^{E\setminus e}$	1.75	41	-5.16	-11.55	-3.04	TRUE
5	Cameroon	$\lambda_{5}^{E\setminus e}$	1.43	51	1.77	0.41	-1.00	TRUE
6	Canada	$\lambda_{6}^{E\setminus e}$	1.67	13	2.00	7.64	0.56	TRUE
7	Costa Rica	$\lambda_{7}^{E\setminus e}$	1.13	21	1.10	0.26	3.64	TRUE
8	Croatia	$\lambda_{8}^{E\setminus e}$	1.36	29	-0.05	-9.97	2.91	TRUE
9	Denmark	$\lambda_{9}^{E\setminus e}$	1.29	16	-0.27	0.37	0.47	TRUE
10	Ecuador	$\lambda_{10}^{E\setminus e}$	1.46	28	0.39	6.59	3.62	TRUE
11	England	$\lambda_{11}^{E\setminus e}$	1.23	41	-3.59	-0.33	-2.00	TRUE
12	France	$\lambda_{12}^{E\setminus e}$	1.02	27	-2.06	1.63	0.55	TRUE
13	Germany	$\lambda_{13}^{E\setminus e}$	1.03	31	-0.51	-1.44	5.24	TRUE
14	Ghana	$\lambda_{14}^{E\setminus e}$	1.18	47	1.58	0.39	-0.08	TRUE
15	IR Iran	$\lambda_{15}^{E\setminus e}$	1.15	10	-6.51	-6.10	0.36	TRUE
16	Japan	$\lambda_{16}^{E\setminus e}$	1.21	33	-3.19	2.17	1.88	TRUE
17	Korea Republic	$\lambda_{17}^{E\setminus e}$	1.13	34	0.28	6.18	0.48	TRUE
18	Mexico	$\lambda_{18}^{E\setminus e}$	1.33	14	-1.49	-4.02	-6.31	TRUE
19	Morocco	$\lambda_{19}^{E\setminus e}$	1.38	7	10.72	4.90	2.25	TRUE
20	Netherlands	$\lambda_{20}^{E\setminus e}$	1.56	41	3.39	4.41	0.21	TRUE
21	Poland	$\lambda_{21}^{E\setminus e}$	1.43	43	3.84	-2.29	-1.39	TRUE
22	Portugal	$\lambda_{22}^{E\setminus e}$	1.34	7	-7.20	-6.11	-4.87	TRUE
23	Qatar	$\lambda_{23}^{E\setminus e}$	1.08	20	1.76	3.56	3.52	TRUE
24	Saudi Arabia	$\lambda_{24}^{E\setminus e}$	1.18	15	-11.76	-12.91	1.64	TRUE
25	Senegal	$\lambda_{25}^{E\setminus e}$	1.32	42	2.46	0.91	-1.30	TRUE
26	Serbia	$\lambda_{26}^{E\setminus e}$	1.18	48	1.87	0.01	0.10	TRUE
27	Spain	$\lambda_{27}^{E\setminus e}$	1.20	29	5.62	-1.72	0.87	TRUE
28	Switzerland	$\lambda_{28}^{E\setminus e}$	1.17	69	-0.23	-0.46	-0.35	TRUE
29	Tunisia	$\lambda_{29}^{E\setminus e}$	1.06	52	3.46	0.71	0.29	TRUE
30	United States	$\lambda_{30}^{E\setminus e}$	2.19	27	1.67	-0.51	1.57	TRUE
31	Uruguay	$\lambda_{31}^{E\setminus e}$	1.06	27	1.48	-3.25	-0.60	TRUE
32	Wales	$\lambda_{32}^{E\setminus e}$	1.43	28	-1.64	-3.74	0.90	TRUE

Bibliography

Anderson, T. W., & Darling, D. A. (1952). Asymptotic theory of certain "goodness of fit" criteria based on stochastic processes. *The Annals of Mathematical Statistics*, *23*(2), 193–212. https://doi.org/10.1214/aoms/1177729437

Attard, P., Suda, D., & Sammut, F. (2023). Bayesian hierarchical modelling of basketball team performance: An NBA regular season case study. *Proceedings of the 11th International Conference on Sport Sciences Research and Technology Support (icSPORTS 2023)*, 101–111. https://doi.org/10.5220/0012159100003587

Baio, G., & Blangiardo, M. (2010). Bayesian hierarchical model for the prediction of football results. *Journal of Applied Statistics*, *37*(2), 253–264. https://doi.org/10.1080/02664760802684177

Bernardo, J. M., & Smith, A. F. M. (1994). *Bayesian theory*. John Wiley & Sons.

de Finetti, B. (1937). La prévision: Ses lois logiques, ses sources subjectives. *Annales de l'Institut Henri Poincaré*, *7*(1), 1–68.

Dixon, M. J., & Coles, S. G. (1997). Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, *46*(2), 265–280. https://doi.org/10.1111/1467-9876.00065

Dutta, A., Saikia, H., Gogoi, J., & Bhattacharjee, D. (2025). Forecasting the opening goal in second-half of a football match: Bayesian and frequentist perspectives. *Computational Statistics*, *40*, 1959–1984. https://doi.org/10.1007/s00180-024-01558-2

Egidi, L., & Gabry, J. (2018). Bayesian hierarchical models for predicting individual performance in soccer. *Journal of Quantitative Analysis in Sports*, *14*(3), 143–157. https://doi.org/10.1515/jqas-2017-0066

Gabrio, A. (2019). Bayesian hierarchical models for the prediction of volleyball results [arXiv preprint]. *arXiv*. https://arxiv.org/abs/1911.08791

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). Chapman; Hall/CRC. https://sites.stat.columbia.edu/gelman/book/

Gelman, A., Carlin, J. B., Stern, H. S., & Rubin, D. B. (2004). *Bayesian data analysis* (2nd ed.). Chapman; Hall/CRC.

Heidelberger, P., & Welch, P. D. (1983). Simulation run length control in the presence of an initial transient. *Operations Research*, *31*(6), 1109–1144.

Ingram, M. (2019). A point-based bayesian hierarchical model to predict the outcome of tennis matches. *Journal of Quantitative Analysis in Sports*, *15*(4), 313–325. https://doi.org/10.1515/jqas-2018-0008

Kaggle. (2023). Fifa world cup 2022 player dataset [Accessed: 2025-10-26].

Karlis, D., & Ntzoufras, I. (2003). Analysis of sports data by using bivariate poisson models. *The Statistician*, *52*(3), 381–393.

Lee, A. J. (1997). Modeling scores in the premier league: Is manchester united really the best? *CHANCE*, *10*(1), 15–19. https://doi.org/10.1080/09332480.1997.10554791

Maher, M. J. (1982). Modelling association football scores. *Statistica Neerlandica*, *36*(3), 109–118. https://doi.org/10.1111/j.1467-9574.1982.tb00782.x

Neal, R. M. (2011). Mcmc using hamiltonian dynamics. In S. Brooks, A. Gelman, G. Jones, & X.-L. Meng (Eds.), *Handbook of markov chain monte carlo* (pp. 113–162). Chapman & Hall/CRC. https://arxiv.org/abs/1206.1901

O'Hanlon, E. (2022). Using supervised machine learning to predict the final rankings of the 2021 formula one championship msc research project master of science in data analytics. *National College of Ireland Repository (NORMA)*. https://norma.ncirl.ie/6628/1/emmaohanlon.pdf

Patil, A., Huard, D., & Fonnesbeck, C. J. (2010). Pymc: Bayesian stochastic modelling in python. *Journal of Statistical Software*, *35*(4), 1–81. https://doi.org/10.18637/jss.v035.i04

Rivoltini, M. (2025). Estimating tennis player abilities and predicting match outcomes using bayesian methods. *University of Malta*.

Robert, C. P., & Casella, G. (2011). A short history of Markov chain Monte Carlo: Subjective recollections from incomplete data. *Statistical Science*, *26*(1), 102–115. https://doi.org/10.1214/10-STS351

van Kesteren, E.-J., & Bergkamp, T. (2023). Bayesian analysis of formula one race results: Disentangling driver skill and constructor advantage. *Journal of Quantitative Analysis in Sports*, *19*(4), 273–293. https://doi.org/10.1515/jqas-2022-0021

Whitaker, G. A., Silva, R., Edwards, D., & Kosmidis, I. (2021). A bayesian approach for determining player abilities in football. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, *70*(1), 174–201. https://doi.org/10.1111/rssc.12454